



PH202

Physics II



PH202 PHYSICS II

ABET eCOURSEBOOK

2021-2026

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SECTION I

COURSE ASSESSMENTS



TAB A AY2026 COURSE ASSESSMENT

Enclosure 1 – Core Physics Program AY 26 Executive Summary

1. References.

- a. DPOM 5-05, Managing Curricular Change, 08 April 2020.
- b. DPOM 5-07, Assessing and Improving Student Learning in the Academic Program, 24 April 2019.
- c. Office of the Dean, MADN, memorandum (Dean's AY26 Annual Guidance), 27 June 2025.
- d. Core Physics Program AY 25 Executive Summary, September 2025.

2. Purpose. This enclosure summarizes the Core Physics Program assessment, curriculum changes, and assessment process for academic year 2026 (AY 26), in compliance with the annual reporting requirements of references 1a, 1b, and 1c.

3. Status of Curricular Change Initiatives for AY 26. For AY 26, the Core Physics Program implemented minor curriculum change to the PH201/PH251, PH202/PH252. There were no changes to the Student Learning Outcomes (SLOs).

4. Summary of Student Outcome Evaluation for AY 26. Evidence for cadet achievement of the SLOs was obtained from direct indicators (e.g., WPRs, lab reports, TEE, etc.) and indirect indicators (i.e., course end feedback surveys). A summary of the assessment results is given in Table 1 in Appendix A and repeated on the next page. The assessment methodology is given in Appendix B.

Enclosure 1 – Core Physics Program AY 26 Executive Summary

Table 1 - Summary of SLO Outcomes

SLOs	Assessment								
	AY 19	AY 20		AY 21	AY 22	AY 23	AY 24	AY 25	AY 26
Cadets who successfully complete the Core Physics program can:		AY 19 Rubric	AY 20 Rubric						
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds	Meets	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Good
c. Analyze and solve problems of increasing complexity.	Meets	Exceeds	Good	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets	Exceeds	Good	Good	Acceptable	Good	Good	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Meets	Acceptable	Good	Acceptable	Acceptable	Acceptable	Good	Good

a. Cadet performance on SLO a, *comprehend physical concepts relevant to follow-on disciplines and future service as an officer*, for AY 26 was **Acceptable**. This rating is consistent with rating from AY 25.

b. Cadet performance on SLO b, *apply physical principles and mathematical competencies with logical processes*, for AY 26 was **Good**. This rating is an increase from the Acceptable rating from AY 25.

c. Cadet performance on SLO c, *analyze and solve problems of increasing complexity*, for AY 26 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 25.

d. Cadet performance on SLO d, *develop the ability to learn independently*, for AY 26 was **Good**. There is no statistically significant change from AY 24 to AY 25.

e. Cadet performance on SLO e, *examine physical phenomena, analyze data, and communicate results*, for AY 26 was **Good**. This rating is consistent with the Good rating from AY 25.

(1) The Core Physics Program continued the implementation of the two lab writs and the 10 labs that culminate in the written lab reports from AY26. The lab reports improve the written technical communication skills of cadets and better prepare them for more advanced labs in PH2X2 and future STEM courses.

5. Summary of Results of APG-WGCD Responsibility Evaluations Conducted During the Year. The summary of the Core Program APG-WGCD to SLO crosswalk for AY 26 is located in Appendix C. 251, and 252 were assessed with 201 and 202.

6. Significant Planned or Continuing Curricular Changes for AY 26. There are no significant changes planned for the Core Physics Program for AY 26. The Core Physics Program will continue to refine the PH201/PH251 and PH202/PH252 course sequence. Minor changes include continuing AI assisted grading in 201 and creating a test bank in both 2x1 and 2x2. A Core Physics book review was completed and attached in Appendix D. We are drafting an update to the SLAM and lab procedures in 2X1/2X2 to increase continuity between 2X1/2X2.

Enclosure 1 – Core Physics Program AY 26 Executive Summary

7. Proposed Changes to Student Learning Outcomes / Significant Planned or Implemented Assessment Process Changes. There are no planned changes to the SLOs for AY 27.

8. Updated Assessment Schedule and Time-line. Currently, the Core Physics program assess each of its SLOs every year. There are no plans to change this assessment timeline.

9. Point of Contact. The POC for this action is the undersigned at extension 845-938-5895.

3 Appendices

- A. Student Learning Outcomes and Assessment Matrix for Core Physics SLOs
- B. Assessment Methodology for Core Physics Program Student Learning Outcomes
- C. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment
- D. Core Physics textbook review

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Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

1. Core Physics SLOs for AY 26. Cadets who successfully complete the Core Physics program can:

- a. **comprehend** physical concepts relevant to follow-on disciplines and future service as an officer.
- b. **apply** physical principles and mathematical competencies with logical processes.
- c. **analyze** and **solve** problems of increasing complexity.
- d. develop the ability to **learn independently**.
- e. **examine** physical phenomena, **analyze** data, and **communicate** results.

2. The Assessment Matrix for Core Physics SLOs.

a. For AY 26, the Program Director (PD) continued to use the 5-tiered scale for direct indicators that the PD adopted from the Nuclear Engineering program in AY 20 and the associated assessment ratings and color codes.

b. The Assessment Matrix for the Core Physics SLOs is shown in Table 1. Of note, the assessment for AY 20 shows the assessment for AY 20 using both the AY 19 rubric and the AY 20 rubric. The increase (↗), decrease (↘), or equivalency (→) in SLO achievement from AY 19 to AY 20 is indicated graphically with arrows. For example, for SLO b, the AY 19 assessment was Exceeds. The AY 20 Assessment was Meets using the AY 19 rubric, which represents a decrease (↘) in SLO performance compared to AY 19. The AY 20 Assessment was Acceptable using the AY 20 rubric, which is equivalent (→) to Meets using the AY 19 rubric.

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

Table 2 - Summary of SLO Outcomes

SLOs	Assessment								
	AY 19	AY 20		AY 21	AY22	AY23	AY24	AY25	AY26
		AY 19 Rubric	AY 20 Rubric						
Cadets who successfully complete the Core Physics program can:									
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds	Meets	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Good
c. Analyze and solve problems of increasing complexity.	Meets	Exceeds	Good	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets	Exceeds	Good	Good	Acceptable	Good	Good	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Meets	Acceptable	Good	Acceptable	Acceptable	Acceptable	Good	Good

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

1. Evaluation and assessment of each SLO is integrated into each individual core course offering, including the laboratory program. At the end of each semester, Course Directors (CDs) conduct a course assessment to gain situational understanding of immediate changes required for the next semester and to evaluate each SLO. CDs then provide this information to the PD in the form of a Course Assessment (CAT) brief. At the conclusion of each academic year, the PD completes a program review that focuses on assessing the SLOs and the assigned Academic Program Goals – What Graduates Can Do (APG-WGCD) statements.

2. For AY 26, the PD continued using the SLO Assessment Rubric adopted in AY 20. In AY 20, the PD changed the SLO assessment rubric for the program and for the direct indicators from a 4-tiered rating scale (see Table 2) to a 5-tiered rating scale (see Table 3). The PD made this change in order to combine data from both direct and indirect indicators into the SLO assessment for each SLO where possible. The main difference between the AY 19 rubric and the AY 20 rubric is that the AY 20 rubric added an additional tier of Excellent for direct indicators for when the most frequent grade is A and B, and there are no D or F grades. Additionally, the AY 19 rubric did not use a Likert rating for direct indicators (see Table 4). In AY 21-2, the PD also added two additional questions to the Course End Feedback (CEF) survey to provide indirect indicators to assess SLOs c and e.

Table 2. SLO Assessment Rubric, AY 19 and prior.

Assessment Rating	Color Code	Most Frequent Letter Grade	Low Performance Threshold
Exceeds Expectations		B or better	< 10% D or F
Meets Expectations		C or better	< 20% D or F
Marginally Meets Expectations		C or better	< 40% D or F
Fails to Meet Expectations		Any	> 40% D or F

Table 3. Core Physics Program SLO Assessment Rubric (Adopted AY 20).

Assessment Rating	Color Code	Likert Rating	Direct Indicators: Criteria	Indirect Indicators: Level of Agreement
Excellent		5	Most frequent grade is A and B, with no D or F	Strongly agree
Good		4	Most frequent grade is A and B, with some D or F	Agree
Acceptable		3	Most frequent grade is C or better, with < 20% D or F	Neutral
Marginally Acceptable		2	Most frequent grade is C or better, with > 20% D or F but < 40% D or F	Disagree
Unacceptable		1	Most frequent grade is D or F	Strongly disagree

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 4. SLO Assessment Rubric Comparison, AY 19 and AY 20.

AY 19		AY 20	
Assessment Rating	Color Code	Assessment Rating	Color Code
		Excellent	
Exceeds Expectations		Good	
Meets Expectations		Acceptable	
Marginally Meets Expectations		Marginally Acceptable	
Fails to Meet Expectations		Unacceptable	

3. Data from the AY 21-2 PH205 CAT brief will be used to explain the assessment process.

a. For SLO a, for example, the AY 21-2 PH205 CD assessed each direct indicator (WPR 1, WPR 2, and the TEE) and each indirect indicator (CEF Survey question) for AY 21-2 using the assessment rubric from Table 3. The CD's assessment of these direct and indirect indicators is shown in Figure 1. Note that Figure 1 includes assessment data from the previous term, AY 21-1. For the direct indicators, the CD computed a simple average to determine the aggregate assessment of the direct indicators for each semester.

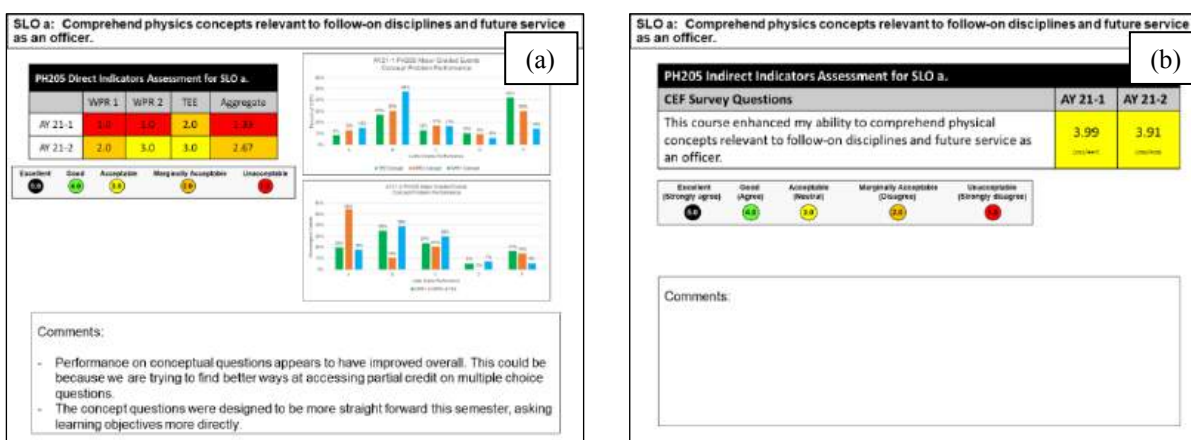


Figure 1. PH205 CD's assessment of (a) direct and (b) indirect indicators for SLO a.

b. The CDs summarize the assessment of the direct and indirect indicators for each SLO for their courses. To determine the overall assessment for each SLO, each CD computes a simple average of the direct and indirect indicator assessment data for each SLO, then rounds down to the nearest 0.5. Figure 2 provides an example from the AY 21-2 PH205 CAT brief.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

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PH205
SLO Assessment AY 21

SLO	AY	Indicators		CD	Comments	
		Direct	Indirect			
a	AY 21-1	1.33	3.99	2.5	2.98 3.53 2.89 4.11 4.31	
	AY 21-2	2.67	3.91			
b	AY 21-1	4.00	4.13	3.5		
	AY 21-2	2.00	4.00			
c	AY 21-1	3.33	N/A	2.5		
	AY 21-2	1.33	4.01			
d	AY 21-1	N/A	4.20	4.0		
	AY 21-2	N/A	4.01			
e	AY 21-1	4.50	N/A	4.0		
	AY 21-2	4.50	3.93			

Direct:

Excellent (5.0)	Good (4.5)	Acceptable (4.0)	Marginally Acceptable (3.5)	Unacceptable (3.0)
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Indirect:

Excellent (Strongly agrees) (5.0)	Good (Agrees) (4.5)	Acceptable (Neutral) (4.0)	Marginally Acceptable (Disagrees) (3.5)	Unacceptable (Strongly disagrees) (3.0)
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Figure 2. PH205 CD's assessment of each SLO, based on a simple average.

4. The PD aggregates the SLO assessments from each CD. To determine the overall program assessment, the PD performs a weighted average of the CD assessments for each SLO and rounds down to the nearest 0.5. The weighting factors are used to ensure that PH255 and PH256, which have lower enrollment and higher performance, do not unfairly skew the overall program assessment. The weighting factors are based on course enrollment. The weighting factor for PH201X and PH205 was normalized to the enrollment for PH255, and the weighting factor for PH202X and PH206 was normalized to the enrollment of PH256.

5. The PD's assessment of the SLOs for AY 23 is shown in Table 5. The weighting factors for AY 23 are shown in Table 6.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 5. Program Director's Assessment of Student Learning Outcomes using Direct and Indirect Indicators.

SLOs	Assessment				
	PH2X1 (AY 26)		PH2X2 (AY 26)		Weighted Average
Cadets who successfully complete the Core Physics program can:					Program Director
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	2.5		1.8		2.6
b. Apply physical principles and mathematical competencies with logical processes.	4.0		3.5		3.6
c. Analyze and solve problems of increasing complexity.	3.5		2.7		3.3
d. Develop the ability to learn independently.	4.0		4.0		4.0
e. Examine physical phenomena, analyze data, and communicate results.	3.8		4.0		4.0
Direct: Indirect:					

Table 6. Weighting Factors

Course	AY 26 Enrollment	Weighting Factor
PH2X1	1218	0.69
PH2X2	553	0.21

Appendix C – Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

1. The Core Physics program APG-WGCD to SLO crosswalk and assessment for AY 25 is shown in Table 7. For WGCD statements that were based on the assessment of multiples SLO, the lower SLO assessment was used for the overall assessment of the WGCD statement. For example, if the assessment of WGCD statement 2.2 is based on the assessment of SLO a (i.e. Marginally Acceptable) and SLO c (i.e. Acceptable). The overall assessment of WGCD statement 2.2 is therefore Marginally Acceptable.

Table 7. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment, AY 26

APG	WGCD Statement and Course Responsibility	SLO Crosswalk and SLO Assessment	WGCD Statement Assessment
1. Communication	1.2 Speak and write using Standard American English. PH206/202 (I)	e. examine physical phenomena, analyze data, and communicate results.	Good
	1.5 Use sound logic and relevant evidence to make convincing arguments. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Good
2. Critical and Creative Thinking	2.2 Integrate knowledge and skills from a variety of disciplines PH205/201 (I)	a. comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	2.4 Reason both quantitatively and qualitatively PH205/201 (I)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
3. Lifelong Learning	3.1 Demonstrate the willingness and ability to learn independently. PH205/201 (I), PH206/202 (R)	d. develop the ability to learn independently.	Good
	3.3 Pursue self-awareness and embrace the responsibility for personal intellectual development PH206/202 (R)	d. develop the ability to learn independently.	Good
5. STEM	5.1 Apply mathematics, science, and computing to model devices, systems, processes or behaviors PH205/201 (R), PH206/202 (R)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	5.2 Apply the scientific method PH205/201 (R), PH206/202 (R)	e. examine physical phenomena, analyze data, and communicate results.	Good
	5.3 Collect and analyze data in support of decision making R—PH205/201 and PH206/202	e. examine physical phenomena, analyze data, and communicate results.	Good

Appendix D – Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

1. The Core Physics program Results Dashboard was used to assess the current E-textbook and possibility of changing textbooks is in Table 8. The 5 assessment buckets were A-Pedagogy & Content, B-Platform & Technology, C-Instructor Workflow, and D-Cost, Reks & Flexibility. The final recommendation was to retain ExpertTA/OpenStax, but explore new text for possible implementation in AY 28-1.

Table 8. Core Physics Program new textbook Assessment, AY 26

RESULTS DASHBOARD					
<i>All scores pull automatically from Scoring Matrix.</i>					
Metric	Pearson	Expert TA	Cengage-Openstax	Cengage - New Text	Kudu
★ OVERALL WEIGHTED SCORE	3.45	4.20	3.59	3.63	3.70
RANK (1 = best)	5	1	4	3	2
Category Weights: Max 1.25					
Bucket A — Pedagogy & Content	1.0	1.0	0.7	0.9	1.2
Bucket B — Platform & Technology	1.1	1.0	1.0	1.0	1.0
Bucket C — Instructor Workflow	0.8	1.1	0.9	0.9	0.8
Bucket D — Cost, Risk & Flexibility	0.6	1.1	1.0	0.9	0.8
RECOMMENDATION & DECISION NOTES					
Recommendation:	Retain Expert TA, explore Kudu (Self-Hosted Content from OpenStax) or Cengage (Webassign with OpenStax) in AY 27-1 [REDACTED] with next target of course change occurring summer of 2027 implementation AY 28-1. The learning outcome benefits from changing books does not exceed the cost to change the course. Instead we should focus on platforms which maximize learning outcomes with Cadet cost and long-term content accessibility for STEM majors in mind. Expert TA and Kudu are the best options to achieve that. Cengage minimizes cost but does not improve material engagement and regresses our course.				
Primary Rationale:	Expert TA is at a good state of maturity with flexibility to implement our own content at a comparable price point for what it offers. They can offer us the flexibility to deliver our course the way we want to with one of the better online homework platforms. Kudu is a close second but not in a mature enough state for a confident transition.				
Key Risks / Concerns:	The cost of Expert TA has risen and with Cengage as an already paid for option. The question is if the \$50 for Expert TA is worth it for hosting our custom course content and its usefulness as a homework platform. Expert TA is slow to deliver on new features (similar to Kudu based on size of company and trends in last 2 years). Expert TA is behind other vendors in terms of interactive content integration and AI / LLM integration, though they have indicated it is on their roadmap.				
Next Steps:	Option 1 (if we want more self-developed interactive content leaning into leveraging LLMs): AY 27-1 [REDACTED] adopts and builds a Kudu course over the 2026 summer (1:1 from our Expert TA content, est ~20 work hours to build). [REDACTED] and [REDACTED] run Kudu for 6 sections and assess faculty workflow, content development, and homework platform feasibility across split sections. Course end feedback measures cadet engagement with the material and achieving course outcomes. If we think the learning outcome benefits are worth it then in AY27-2 we decide to adopt Kudu in AY28-1 ([REDACTED] semester). PH202 transitions in Summer 2027 for an AY28-1 concurrent adoption and NIT is trained on Kudu. Option 2 (if we want to lean into the single source/cost optimized solution and away from our self-developed material integrated in the book): Run a similar pilot course as above [REDACTED] testing Webassign with OpenStax. Concurrently conduct textbook review in AY 27-1 and AY 27-2 for content migration in Summer 2027 for both PH201 and PH202 and adoption during [REDACTED]'s second semester.				



TAB B AY2025 COURSE ASSESSMENT

Enclosure 1 – Core Physics Program AY 25 Executive Summary

1. References.

- a. DPOM 5-05, Managing Curricular Change, 08 April 2020.
- b. DPOM 5-07, Assessing and Improving Student Learning in the Academic Program, 24 April 2019.
- c. Office of the Dean, MADN, memorandum (Dean's AY25 Annual Guidance), 28 June 2024.
- d. Core Physics Program AY 24 Executive Summary, September 2024.

2. Purpose. This enclosure summarizes the Core Physics Program assessment, curriculum changes, and assessment process for academic year 2025 (AY 25), in compliance with the annual reporting requirements of references 1a, 1b, and 1c.

3. Status of Curricular Change Initiatives for AY 25. For AY 25, the Core Physics Program implemented minor curriculum change to the PH201/PH251, PH202/PH252. There were no changes to the Student Learning Outcomes (SLOs).

4. Summary of Student Outcome Evaluation for AY 25. Evidence for cadet achievement of the SLOs was obtained from direct indicators (e.g., WPRs, lab reports, TEE, etc.) and indirect indicators (i.e., course end feedback surveys). A summary of the assessment results is given in Table 1 in Appendix A and repeated on the next page. The assessment methodology is given in Appendix B.

Enclosure 1 – Core Physics Program AY 25 Executive Summary

Table 1 - Summary of SLO Outcomes

SLOs	Assessment							
	AY 19	AY 20		AY 21	AY22	AY23	AY24	AY25
Cadets who successfully complete the Core Physics program can:		AY 19 Rubric	AY 20 Rubric					
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable	Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds	Meets	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets	Exceeds	Good	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets	Exceeds	Good	Good	Acceptable	Good	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Meets	Acceptable	Good	Acceptable	Acceptable	Acceptable	Good

a. Cadet performance on SLO a, *comprehend physical concepts relevant to follow-on disciplines and future service as an officer*, for AY 25 was **Acceptable**. This rating is consistent with rating from AY 24.

b. Cadet performance on SLO b, *apply physical principles and mathematical competencies with logical processes*, for AY 25 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 24.

c. Cadet performance on SLO c, *analyze and solve problems of increasing complexity*, for AY 25 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 24.

d. Cadet performance on SLO d, *develop the ability to learn independently*, for AY 25 was **Good**. There is no statistically significant change from AY 24 to AY 25.

e. Cadet performance on SLO e, *examine physical phenomena, analyze data, and communicate results*, for AY 25 was **Good**. This rating is consistent with the Acceptable rating from AY 24 with a slight improvement from AY25.

(1) The Core Physics Program continued the implementation of the two lab writs and the 10 labs that culminate in the written lab reports from AY25. The lab reports improve the written technical communication skills of cadets and better prepare them for more advanced labs in PH2X2 and future STEM courses.

5. Summary of Results of APG-WGCD Responsibility Evaluations Conducted During the Year. The summary of the Core Program APG-WGCD to SLO crosswalk for AY 25 is located in Appendix C.

6. Significant Planned or Continuing Curricular Changes for AY 26. There are no significant changes planned for the Core Physics Program for AY 26. The Core Physics Program will continue to refine the PH201/PH251 and PH202/PH252 course sequence. Minor changes include continuing AI assisted grading in 201 and creating a test bank in

Enclosure 1 – Core Physics Program AY 25 Executive Summary

both 2x1 and 2x2. A Core Physics Assessment Review will occur to provide further assessment of the curriculum.

7. Proposed Changes to Student Learning Outcomes / Significant Planned or Implemented Assessment Process Changes. There are no planned changes to the SLOs for AY 26.

8. Updated Assessment Schedule and Time-line. Currently, the Core Physics program assess each of its SLOs every year. There are no plans to change this assessment timeline.

9. Point of Contact. The POC for this action is the undersigned at extension 845-938-5895.

3 Appendices

- A. Student Learning Outcomes and Assessment Matrix for Core Physics SLOs
- B. Assessment Methodology for Core Physics Program Student Learning Outcomes
- C. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

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Program Director, Core Physics

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

1. Core Physics SLOs for AY 25. Cadets who successfully complete the Core Physics program can:

- a. **comprehend** physical concepts relevant to follow-on disciplines and future service as an officer.
- b. **apply** physical principles and mathematical competencies with logical processes.
- c. **analyze** and **solve** problems of increasing complexity.
- d. develop the ability to **learn independently**.
- e. **examine** physical phenomena, **analyze** data, and **communicate** results.

2. The Assessment Matrix for Core Physics SLOs.

a. For AY 25, the Program Director (PD) continued to use the 5-tiered scale for direct indicators that the PD adopted from the Nuclear Engineering program in AY 20 and the associated assessment ratings and color codes.

b. The Assessment Matrix for the Core Physics SLOs is shown in Table 1. Of note, the assessment for AY 20 shows the assessment for AY 20 using both the AY 19 rubric and the AY 20 rubric. The increase (↗), decrease (↘), or equivalency (→) in SLO achievement from AY 19 to AY 20 is indicated graphically with arrows. For example, for SLO b, the AY 19 assessment was Exceeds. The AY 20 Assessment was Meets using the AY 19 rubric, which represents a decrease (↘) in SLO performance compared to AY 19. The AY 20 Assessment was Acceptable using the AY 20 rubric, which is equivalent (→) to Meets using the AY 19 rubric.

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

Table 2 - Summary of SLO Outcomes

SLOs	Assessment							
	AY 19	AY 20		AY 21	AY22	AY23	AY24	AY25
		AY 19 Rubric	AY 20 Rubric					
Cadets who successfully complete the Core Physics program can:								
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable	Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds	Meets	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets	Exceeds	Good	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets	Exceeds	Good	Good	Acceptable	Good	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Meets	Acceptable	Good	Acceptable	Acceptable	Acceptable	Good

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

1. Evaluation and assessment of each SLO is integrated into each individual core course offering, including the laboratory program. At the end of each semester, Course Directors (CDs) conduct a course assessment to gain situational understanding of immediate changes required for the next semester and to evaluate each SLO. CDs then provide this information to the PD in the form of a Course Assessment (CAT) brief. At the conclusion of each academic year, the PD completes a program review that focuses on assessing the SLOs and the assigned Academic Program Goals – What Graduates Can Do (APG-WGCD) statements.
2. For AY 25, the PD continued using the SLO Assessment Rubric adopted in AY 20. In AY 20, the PD changed the SLO assessment rubric for the program and for the direct indicators from a 4-tiered rating scale (see Table 2) to a 5-tiered rating scale (see Table 3). The PD made this change in order to combine data from both direct and indirect indicators into the SLO assessment for each SLO where possible. The main difference between the AY 19 rubric and the AY 20 rubric is that the AY 20 rubric added an additional tier of Excellent for direct indicators for when the most frequent grade is A and B, and there are no D or F grades. Additionally, the AY 19 rubric did not use a Likert rating for direct indicators (see Table 4). In AY 21-2, the PD also added two additional questions to the Course End Feedback (CEF) survey to provide indirect indicators to assess SLOs c and e.

Table 2. SLO Assessment Rubric, AY 19 and prior.

Assessment Rating	Color Code	Most Frequent Letter Grade	Low Performance Threshold
Exceeds Expectations		B or better	< 10% D or F
Meets Expectations		C or better	< 20% D or F
Marginally Meets Expectations		C or better	< 40% D or F
Fails to Meet Expectations		Any	> 40% D or F

Table 3. Core Physics Program SLO Assessment Rubric (Adopted AY 20).

Assessment Rating	Color Code	Likert Rating	Direct Indicators: Criteria	Indirect Indicators: Level of Agreement
Excellent		5	Most frequent grade is A and B, with no D or F	Strongly agree
Good		4	Most frequent grade is A and B, with some D or F	Agree
Acceptable		3	Most frequent grade is C or better, with < 20% D or F	Neutral
Marginally Acceptable		2	Most frequent grade is C or better, with > 20% D or F but < 40% D or F	Disagree
Unacceptable		1	Most frequent grade is D or F	Strongly disagree

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 4. SLO Assessment Rubric Comparison, AY 19 and AY 20.

AY 19		AY 20	
Assessment Rating	Color Code	Assessment Rating	Color Code
		Excellent	
Exceeds Expectations		Good	
Meets Expectations		Acceptable	
Marginally Meets Expectations		Marginally Acceptable	
Fails to Meet Expectations		Unacceptable	

3. Data from the AY 21-2 PH205 CAT brief will be used to explain the assessment process.

a. For SLO a, for example, the AY 21-2 PH205 CD assessed each direct indicator (WPR 1, WPR 2, and the TEE) and each indirect indicator (CEF Survey question) for AY 21-2 using the assessment rubric from Table 3. The CD's assessment of these direct and indirect indicators is shown in Figure 1. Note that Figure 1 includes assessment data from the previous term, AY 21-1. For the direct indicators, the CD computed a simple average to determine the aggregate assessment of the direct indicators for each semester.



Figure 1. PH205 CD's assessment of (a) direct and (b) indirect indicators for SLO a.

b. The CDs summarize the assessment of the direct and indirect indicators for each SLO for their courses. To determine the overall assessment for each SLO, each CD computes a simple average of the direct and indirect indicator assessment data for each SLO, then rounds down to the nearest 0.5. Figure 2 provides an example from the AY 21-2 PH205 CAT brief.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

UNITED STATES MILITARY ACADEMY
WEST POINT

PH205
SLO Assessment AY 21

SLO	AY	Indicators		CD	Comments	
		Direct	Indirect			
a	AY 21-1	1.33	3.99	2.5	2.98 3.53 2.89 4.11 4.31	
	AY 21-2	2.67	3.91			
b	AY 21-1	4.00	4.13	3.5		
	AY 21-2	2.00	4.00			
c	AY 21-1	3.33	N/A	2.5		
	AY 21-2	1.33	4.01			
d	AY 21-1	N/A	4.20	4.0		
	AY 21-2	N/A	4.01			
e	AY 21-1	4.50	N/A	4.0		
	AY 21-2	4.50	3.93			

Direct:

Excellent (5.0)	Good (4.5)	Acceptable (3.5)	Marginally Acceptable (2.5)	Unacceptable (1.5)
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Indirect:

Excellent (Strongly agrees) (5.0)	Good (Agrees) (4.5)	Acceptable (Neutral) (3.5)	Marginally Acceptable (Disagrees) (2.5)	Unacceptable (Strongly disagrees) (1.5)
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Figure 2. PH205 CD's assessment of each SLO, based on a simple average.

4. The PD aggregates the SLO assessments from each CD. To determine the overall program assessment, the PD performs a weighted average of the CD assessments for each SLO and rounds down to the nearest 0.5. The weighting factors are used to ensure that PH255 and PH256, which have lower enrollment and higher performance, do not unfairly skew the overall program assessment. The weighting factors are based on course enrollment. The weighting factor for PH201X and PH205 was normalized to the enrollment for PH255, and the weighting factor for PH202X and PH206 was normalized to the enrollment of PH256.

5. The PD's assessment of the SLOs for AY 23 is shown in Table 5. The weighting factors for AY 23 are shown in Table 6.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 5. Program Director's Assessment of Student Learning Outcomes using Direct and Indirect Indicators.

SLOs	Assessment					Weighted Average	Program Director								
	PH201 (AY 25)	PH251 (AY 25)	PH202 (AY 25)	PH252 (AY 25)											
Cadets who successfully complete the Core Physics program can:															
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	2.5	2.5	3.4	0.0	3.1	3									
b. Apply physical principles and mathematical competencies with logical processes.	4.0	4.0	3.9	0.0	3.7	3.5									
c. Analyze and solve problems of increasing complexity.	2.8	4.0	3.4	0.0	3.1	3.5									
d. Develop the ability to learn independently.	4.0	4.0	4.3	0.0	4.1	4									
e. Examine physical phenomena, analyze data, and communicate results.	3.8	4.0	4.5	0.0	3.7	4									
Direct: Excellent 4.0 Good 3.5 Acceptable 3.0 Marginally Acceptable 2.5 Unacceptable 1.0								Indirect: Excellent (throughly agree) 4.0 Good (agree) 3.5 Acceptable (neutral) 3.0 Marginally Acceptable (disagree) 2.5 Unacceptable (strongly disagree) 1.0							

Table 6. Weighting Factors

Course	AY 25 Enrollment	Weighting Factor
PH201	973	0.60
PH251	203	0.13
PH202	433	0.27
PH252	0	0.00

Appendix C – Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

1. The Core Physics program APG-WGCD to SLO crosswalk and assessment for AY 25 is shown in Table 7. For WGCD statements that were based on the assessment of multiples SLO, the lower SLO assessment was used for the overall assessment of the WGCD statement. For example, if the assessment of WGCD statement 2.2 is based on the assessment of SLO a (i.e. Marginally Acceptable) and SLO c (i.e. Acceptable). The overall assessment of WGCD statement 2.2 is therefore Marginally Acceptable.

Table 7. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment, AY 25

APG	WGCD Statement and Course Responsibility	SLO Crosswalk and SLO Assessment	WGCD Statement Assessment
1. Communication	1.2 Speak and write using Standard American English. PH206/202 (I)	e. examine physical phenomena, analyze data, and communicate results.	Good
	1.5 Use sound logic and relevant evidence to make convincing arguments. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Good
2. Critical and Creative Thinking	2.2 Integrate knowledge and skills from a variety of disciplines PH205/201 (I)	a. comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	2.4 Reason both quantitatively and qualitatively PH205/201 (I)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
3. Lifelong Learning	3.1 Demonstrate the willingness and ability to learn independently. PH205/201 (I), PH206/202 (R)	d. develop the ability to learn independently.	Acceptable
	3.3 Pursue self-awareness and embrace the responsibility for personal intellectual development PH206/202 (R)	d. develop the ability to learn independently.	Acceptable
5. STEM	5.1 Apply mathematics, science, and computing to model devices, systems, processes or behaviors PH205/201 (R), PH206/202 (R)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	5.2 Apply the scientific method PH205/201 (R), PH206/202 (R)	e. examine physical phenomena, analyze data, and communicate results.	Good
	5.3 Collect and analyze data in support of decision making R—PH205/201 and PH206/202	e. examine physical phenomena, analyze data, and communicate results.	Good



TAB C AY2024 COURSE ASSESSMENT

Enclosure 1 – Core Physics Program AY 24 Executive Summary

1. References.

- a. DPOM 5-05, Managing Curricular Change, 08 April 2020.
- b. DPOM 5-07, Assessing and Improving Student Learning in the Academic Program, 24 April 2019.
- c. Office of the Dean, MADN, memorandum (Dean's AY24 Annual Guidance), 26 June 2023.
- d. Core Physics Program AY 23 Executive Summary, September 2023.

2. Purpose. This enclosure summarizes the Core Physics Program assessment, curriculum changes, and assessment process for academic year 2024 (AY 24), in compliance with the annual reporting requirements of references 1a, 1b, and 1c.

3. Status of Curricular Change Initiatives for AY 24. For AY 24, the Core Physics Program implemented minor curriculum change to the PH201/PH251, PH202/PH252. There were no changes to the Student Learning Outcomes (SLOs).

4. Summary of Student Outcome Evaluation for AY 23. Evidence for cadet achievement of the SLOs was obtained from direct indicators (e.g., WPRs, lab reports, TEE, etc.) and indirect indicators (i.e., course end feedback surveys). A summary of the assessment results is given in Table 1 in Appendix A and repeated on the next page. The assessment methodology is given in Appendix B.

Enclosure 1 – Core Physics Program AY 24 Executive

Summary Table 1 - Summary of SLO Outcomes

SLOs	Assessment						
	AY 19	AY 20		AY 21	AY22	AY23	AY24
		AY 19 Rubric	AY 20 Rubric				
Cadets who successfully complete the Core Physics program can:							
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds	Meets	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets	Exceeds	Good	Acceptable	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets	Exceeds	Good	Good	Acceptable	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Meets	Acceptable	Good	Acceptable	Acceptable	Acceptable

a. Cadet performance on SLO a, *comprehend physical concepts relevant to follow-on disciplines and future service as an officer*, for AY 24 was **Acceptable**. This rating is consistent with rating from AY 23.

b. Cadet performance on SLO b, *apply physical principles and mathematical competencies with logical processes*, for AY 24 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 23.

c. Cadet performance on SLO c, *analyze and solve problems of increasing complexity*, for AY 24 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 23.

d. Cadet performance on SLO d, *develop the ability to learn independently*, for AY 24 was **Good**. There is no statistically significant change from AY 23 to AY 24.

e. Cadet performance on SLO e, *examine physical phenomena, analyze data, and communicate results*, for AY 24 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 23.

(1) The Core Physics Program continued the implementation of the two lab writs and the 10 labs that culminate in the written lab reports from AY24. The lab reports improve the written technical communication skills of cadets and better prepare them for more advanced labs in PH2X2 and future STEM courses.

5. Summary of Results of APG-WGCD Responsibility Evaluations Conducted During the Year. The summary of the Core Program APG-WGCD to SLO crosswalk for AY 24 is located in Appendix C.

6. Significant Planned or Continuing Curricular Changes for AY 25. There are no significant changes planned for the Core Physics Program for AY 25. The Core Physics Program will continue to refine the PH201/PH251 and PH202/PH252 course sequence.

Enclosure 1 – Core Physics Program AY 24 Executive Summary

7. Proposed Changes to Student Learning Outcomes / Significant Planned or Implemented Assessment Process Changes. There are no planned changes to the SLOs for AY 25.

8. Updated Assessment Schedule and Time-line. Currently, the Core Physics program assess each of its SLOs every year. There are no plans to change this assessment timeline.

9. Point of Contact. The POC for this action is the undersigned at extension 845-938-5895.

3 Appendices

- A. Student Learning Outcomes and Assessment Matrix for Core Physics SLOs
- B. Assessment Methodology for Core Physics Program Student Learning Outcomes
- C. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

JACOB W. CAPPS, PhD
LTC, AV (FA 47)
Program Director, Core Physics

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

1. Core Physics SLOs for AY 24. Cadets who successfully complete the Core Physics program can:

- a. **comprehend** physical concepts relevant to follow-on disciplines and future service as an officer.
- b. **apply** physical principles and mathematical competencies with logical processes.
- c. **analyze** and **solve** problems of increasing complexity.
- d. develop the ability to **learn independently**.
- e. **examine** physical phenomena, **analyze** data, and **communicate** results.

2. The Assessment Matrix for Core Physics SLOs.

a. For AY 24, the Program Director (PD) continued to use the 5-tiered scale for direct indicators that the PD adopted from the Nuclear Engineering program in AY 20 and the associated assessment ratings and color codes.

b. The Assessment Matrix for the Core Physics SLOs is shown in Table 1. Of note, the assessment for AY 20 shows the assessment for AY 20 using both the AY 19 rubric and the AY 20 rubric. The increase (↗), decrease (↘), or equivalency (→) in SLO achievement from AY 19 to AY 20 is indicated graphically with arrows. For example, for SLO b, the AY 19 assessment was Exceeds. The AY 20 Assessment was Meets using the AY 19 rubric, which represents a decrease (↘) in SLO performance compared to AY 19. The AY 20 Assessment was Acceptable using the AY 20 rubric, which is equivalent (→) to Meets using the AY 19 rubric.

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

Table 2 - Summary of SLO Outcomes

SLOs	Assessment						
	AY 19	AY 20		AY 21	AY22	AY23	AY24
		AY 19 Rubric	AY 20 Rubric				
Cadets who successfully complete the Core Physics program can:							
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets →	Marginally Meets →	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds →	Meets →	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets →	Exceeds →	Good	Acceptable	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets →	Exceeds →	Good	Good	Acceptable	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds →	Meets →	Acceptable	Good	Acceptable	Acceptable	Acceptable

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

1. Evaluation and assessment of each SLO is integrated into each individual core course offering, including the laboratory program. At the end of each semester, Course Directors (CDs) conduct a course assessment to gain situational understanding of immediate changes required for the next semester and to evaluate each SLO. CDs then provide this information to the PD in the form of a Course Assessment (CAT) brief. At the conclusion of each academic year, the PD completes a program review that focuses on assessing the SLOs and the assigned Academic Program Goals – What Graduates Can Do (APG-WGCD) statements.
2. For AY 24, the PD continued using the SLO Assessment Rubric adopted in AY 20. In AY 20, the PD changed the SLO assessment rubric for the program and for the direct indicators from a 4-tiered rating scale (see Table 2) to a 5-tiered rating scale (see Table 3). The PD made this change in order to combine data from both direct and indirect indicators into the SLO assessment for each SLO where possible. The main difference between the AY 19 rubric and the AY 20 rubric is that the AY 20 rubric added an additional tier of Excellent for direct indicators for when the most frequent grade is A and B, and there are no D or F grades. Additionally, the AY 19 rubric did not use a Likert rating for direct indicators (see Table 4). In AY 21-2, the PD also added two additional questions to the Course End Feedback (CEF) survey to provide indirect indicators to assess SLOs c and e.

Table 2. SLO Assessment Rubric, AY 19 and prior.

Assessment Rating	Color Code	Most Frequent Letter Grade	Low Performance Threshold
Exceeds Expectations		B or better	< 10% D or F
Meets Expectations		C or better	< 20% D or F
Marginally Meets Expectations		C or better	< 40% D or F
Fails to Meet Expectations		Any	> 40% D or F

Table 3. Core Physics Program SLO Assessment Rubric (Adopted AY 20).

Assessment Rating	Color Code	Likert Rating	Direct Indicators: Criteria	Indirect Indicators: Level of Agreement
Excellent		5	Most frequent grade is A and B, with no D or F	Strongly agree
Good		4	Most frequent grade is A and B, with some D or F	Agree
Acceptable		3	Most frequent grade is C or better, with < 20% D or F	Neutral
Marginally Acceptable		2	Most frequent grade is C or better, with > 20% D or F but < 40% D or F	Disagree
Unacceptable		1	Most frequent grade is D or F	Strongly disagree

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 4. SLO Assessment Rubric Comparison, AY 19 and AY 20.

AY 19		AY 20	
Assessment Rating	Color Code	Assessment Rating	Color Code
		Excellent	
Exceeds Expectations		Good	
Meets Expectations		Acceptable	
Marginally Meets Expectations		Marginally Acceptable	
Fails to Meet Expectations		Unacceptable	

3. Data from the AY 21-2 PH205 CAT brief will be used to explain the assessment process.

a. For SLO a, for example, the AY 21-2 PH205 CD assessed each direct indicator (WPR 1, WPR 2, and the TEE) and each indirect indicator (CEF Survey question) for AY 21-2 using the assessment rubric from Table 3. The CD's assessment of these direct and indirect indicators is shown in Figure 1. Note that Figure 1 includes assessment data from the previous term, AY 21-1. For the direct indicators, the CD computed a simple average to determine the aggregate assessment of the direct indicators for each semester.

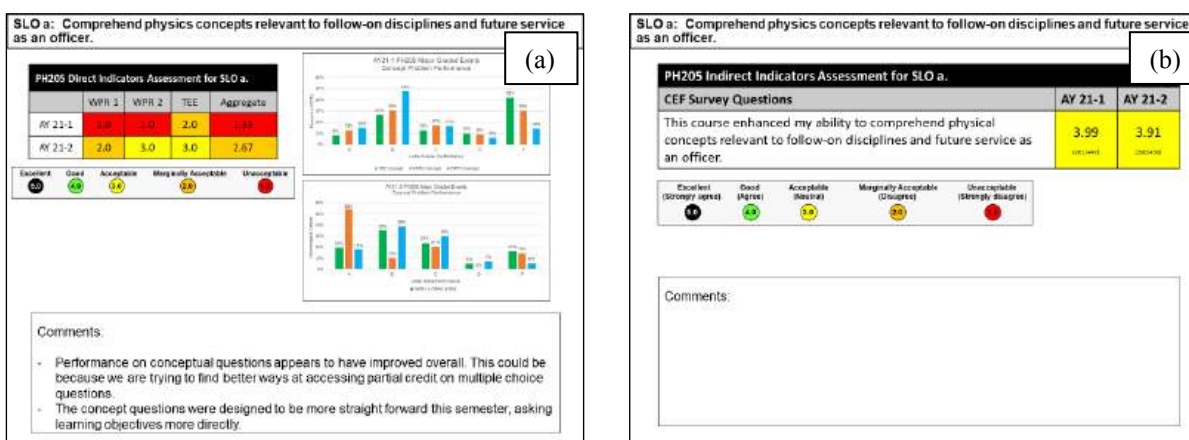


Figure 1. PH205 CD's assessment of (a) direct and (b) indirect indicators for SLO a.

b. The CDs summarize the assessment of the direct and indirect indicators for each SLO for their courses. To determine the overall assessment for each SLO, each CD computes a simple average of the direct and indirect indicator assessment data for each SLO, then rounds down to the nearest 0.5. Figure 2 provides an example from the AY 21-2 PH205 CAT brief.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

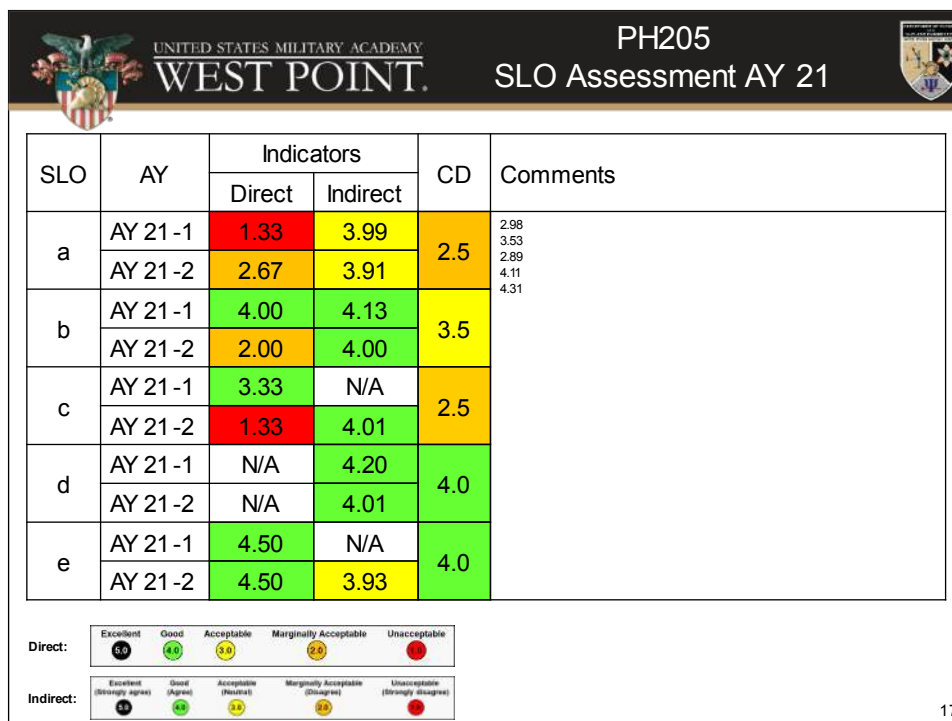


Figure 2. PH205 CD's assessment of each SLO, based on a simple average.

4. The PD aggregates the SLO assessments from each CD. To determine the overall program assessment, the PD performs a weighted average of the CD assessments for each SLO and rounds down to the nearest 0.5. The weighting factors are used to ensure that PH255 and PH256, which have lower enrollment and higher performance, do not unfairly skew the overall program assessment. The weighting factors are based on course enrollment. The weighting factor for PH201X and PH205 was normalized to the enrollment for PH255, and the weighting factor for PH202X and PH206 was normalized to the enrollment of PH256.

5. The PD's assessment of the SLOs for AY 23 is shown in Table 5. The weighting factors for AY 23 are shown in Table 6.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 5. Program Director's Assessment of Student Learning Outcomes using Direct and Indirect Indicators.

SLOs	Assessment						
Cadets who successfully complete the Core Physics program can:	PH201	PH251	PH202	PH252	Weighted Average	Program Director	
	(AY 24)	(AY 24)	(AY 24)	(AY 24)			
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	2.8	4.0	3.4	4.0	3.0	3	
b. Apply physical principles and mathematical competencies with logical processes.	3.5	4.0	3.9	4.0	3.6	3.5	
c. Analyze and solve problems of increasing complexity.	2.8	4.0	3.4	4.0	3.0	3.5	
d. Develop the ability to learn independently.	4.0	4.0	4.3	4.3	4.1	4	
e. Examine physical phenomena, analyze data, and communicate results.	3.3	4.0	4.5	4.5	3.7	3.5	
Direct: Excellent (Strongly agree) Good (Agree) Acceptable (Neutral) Marginally Acceptable (Disagree) Unacceptable (Strongly disagree)							Indirect: Excellent (Strongly agree) Good (Agree) Acceptable (Neutral) Marginally Acceptable (Disagree) Unacceptable (Strongly disagree)

Table 6. Weighting Factors

Course	AY 24 Enrollment	Weighting Factor
PH201	1185	0.68
PH251	50	0.03
PH202	469	0.27
PH252	37	0.02

Appendix C – Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

1. The Core Physics program APG-WGCD to SLO crosswalk and assessment for AY 22 is shown in Table 7. For WGCD statements that were based on the assessment of multiples SLO, the lower SLO assessment was used for the overall assessment of the WGCD statement. For example, the assessment of WGCD statement 2.2 is based on the assessment of SLO a (Marginally Acceptable) and SLO c (Acceptable). The overall assessment of WGCD statement 2.2 is therefore Marginally Acceptable.

Table 7. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment, AY 24

APG	WGCD Statement and Course Responsibility	SLO Crosswalk and SLO Assessment	WGCD Statement Assessment
1. Communication	1.2 Speak and write using Standard American English. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
	1.5 Use sound logic and relevant evidence to make convincing arguments. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
2. Critical and Creative Thinking	2.2 Integrate knowledge and skills from a variety of disciplines PH205 (I)	a. comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Acceptable
		c. analyze and solve problems of increasing complexity.	
	2.4 Reason both quantitatively and qualitatively PH205 (I)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
3. Lifelong Learning	3.1 Demonstrate the willingness and ability to learn independently. PH205 (I), PH206 (R)	d. develop the ability to learn independently.	Acceptable
	3.3 Pursue self-awareness and embrace the responsibility for personal intellectual development PH206 (R)	d. develop the ability to learn independently.	Acceptable
5. STEM	5.1 Apply mathematics, science, and computing to model devices, systems, processes or behaviors PH205 (R), PH206 (R)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	5.2 Apply the scientific method PH205 (R), PH206 (R)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
	5.3 Collect and analyze data in support of decision making R—PH205 and PH206	e. examine physical phenomena, analyze data, and communicate results.	Acceptable



TAB D AY2023 COURSE ASSESSMENT

Enclosure 1 – Core Physics Program AY 23 Executive Summary

1. References.

- a. DPOM 5-05, Managing Curricular Change, 08 April 2020.
 - b. DPOM 5-07, Assessing and Improving Student Learning in the Academic Program, 24 April 2019.
 - c. Office of the Dean, MADN, memorandum (Dean's AY22 Annual Guidance), 02 July 2021.
 - d. Core Physics Program AY 22 Executive Summary, September 2022.
 - e. Department of Physics and Nuclear Engineering, MADN-PNE, memorandum (Core Course Curricular Change Request for PH205 (Physics I) and PH255 (Advanced Physics I), 21 April 2022.
 - f. Department of Physics and Nuclear Engineering, MADN-PNE, memorandum (Core Course Curricular Change Request for PH206 (Physics II) and PH256 (Advanced Physics II), 21 April 2022.
2. Purpose. This enclosure summarizes the Core Physics Program assessment, curriculum changes, and assessment process for academic year 2023 (AY 23), in compliance with the annual reporting requirements of references 1a, 1b, and 1c.
3. Status of Curricular Change Initiatives for AY 23. For AY 23, the Core Physics Program implemented curriculum change to the PH201/PH251, PH202/PH252 sequence approved by the General Committee of the Academic Board on 12 May 2022. There were no changes to the Student Learning Outcomes (SLOs).
4. Summary of Student Outcome Evaluation for AY 22. Evidence for cadet achievement of the SLOs was obtained from direct indicators (e.g., WPRs, lab reports, TEE, etc.) and indirect indicators (i.e., course end feedback surveys). A summary of the assessment results is given in Table 1 in Appendix A and repeated on the next page. The assessment methodology is given in Appendix B.

Enclosure 1 – Core Physics Program AY 23 Executive Summary

Table 1 - Summary of SLO Outcomes

SLOs	Assessment					
	AY 19	AY 20		AY 21	AY22	AY23
		AY 19 Rubric	AY 20 Rubric			
Cadets who successfully complete the Core Physics program can:						
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds	Meets	Acceptable	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets	Exceeds	Good	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets	Exceeds	Good	Good	Acceptable	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Meets	Acceptable	Good	Acceptable	Acceptable

a. Cadet performance on SLO a, *comprehend physical concepts relevant to follow-on disciplines and future service as an officer*, for AY 23 was **Acceptable**. This rating is an improvement over the Marginally Acceptable rating from AY 22. The assessment is based on a weighted average across all courses of 2.98 out of 5, compared to 2.88 in AY 22. There is no statistically significant change from AY 22 to AY 23.

b. Cadet performance on SLO b, *apply physical principles and mathematical competencies with logical processes*, for AY 23 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 22.

c. Cadet performance on SLO c, *analyze and solve problems of increasing complexity*, for AY 23 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 22.

d. Cadet performance on SLO d, *develop the ability to learn independently*, for AY 23 was **Good**. This rating demonstrates an improvement from the Acceptable rating from AY 22. The AY 22 rating was based on weighted average of 3.98, whereas the AY 23 rating was based on a weighted average of 4.00 (see Table 5 in Appendix B). There is no statistically significant change from AY 22 to AY 23.

e. Cadet performance on SLO e, *examine physical phenomena, analyze data, and communicate results*, for AY 23 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 22.

(1) The Core Physics Program continued the implementation of the two lab writs and the 10 labs that culminate in the written lab reports from AY22. The lab reports improve the written technical communication skills of cadets and better prepare them for more advanced labs in PH2X2 and future STEM courses.

Enclosure 1 – Core Physics Program AY 23 Executive Summary

5. Summary of Results of APG-WGCD Responsibility Evaluations Conducted During the Year. The summary of the Core Program APG-WGCD to SLO crosswalk for AY 22 is located in Appendix C.

6. Significant Planned or Continuing Curricular Changes for AY 23. There are no significant changes planned for the Core Physics Program for AY 23. The Core Physics Program will continue to refine the PH201/PH251 and PH202/PH252 course sequence.

7. Proposed Changes to Student Learning Outcomes / Significant Planned or Implemented Assessment Process Changes. There are no planned changes to the SLOs for AY 23.

8. Updated Assessment Schedule and Time-line. Currently, the Core Physics program assess each of its SLOs every year. There are no plans to change this assessment timeline.

9. Point of Contact. The POC for this action is the undersigned at extension 845-938-5895.

3 Appendices

- A. Student Learning Outcomes and Assessment Matrix for Core Physics SLOs
- B. Assessment Methodology for Core Physics Program Student Learning Outcomes
- C. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

For DAVID J. PHILLIPS, PhD
COL, MS (FA 47)
Program Director, Core Physics

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

1. Core Physics SLOs for AY 23. Cadets who successfully complete the Core Physics program can:

- a. **comprehend** physical concepts relevant to follow-on disciplines and future service as an officer.
- b. **apply** physical principles and mathematical competencies with logical processes.
- c. **analyze** and **solve** problems of increasing complexity.
- d. develop the ability to **learn independently**.
- e. **examine** physical phenomena, **analyze** data, and **communicate** results.

2. The Assessment Matrix for Core Physics SLOs.

a. For AY 23, the Program Director (PD) continued to use the 5-tiered scale for direct indicators that the PD adopted from the Nuclear Engineering program in AY 20 and the associated assessment ratings and color codes. The change is discussed in more detail in Appendix B.

b. The Assessment Matrix for the Core Physics SLOs is shown in Table 1. Of note, the assessment for AY 20 shows the assessment for AY 20 using both the AY 19 rubric and the AY 20 rubric. The increase ($\nearrow$), decrease ($\searrow$), or equivalency ($\rightarrow$) in SLO achievement from AY 19 to AY 20 is indicated graphically with arrows. For example, for SLO b, the AY 19 assessment was Exceeds. The AY 20 Assessment was Meets using the AY 19 rubric, which represents a decrease ($\searrow$) in SLO performance compared to AY 19. The AY 20 Assessment was Acceptable using the AY 20 rubric, which is equivalent ($\rightarrow$) to Meets using the AY 19 rubric.

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

Table 2 - Summary of SLO Outcomes

SLOs	Assessment					
	AY 19	AY 20		AY 21	AY22	AY23
		AY 19 Rubric	AY 20 Rubric			
Cadets who successfully complete the Core Physics program can:						
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets →	Marginally Meets →	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds →	Meets →	Acceptable	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets →	Exceeds →	Good	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets →	Exceeds →	Good	Good	Acceptable	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds →	Meets →	Acceptable	Good	Acceptable	Acceptable

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

1. Evaluation and assessment of each SLO is integrated into each individual core course offering, including the laboratory program. At the end of each semester, Course Directors (CDs) conduct a course assessment to gain situational understanding of immediate changes required for the next semester and to evaluate each SLO. CDs then provide this information to the PD in the form of a Course Assessment (CAT) brief. At the conclusion of each academic year, the PD completes a program review that focuses on assessing the SLOs and the assigned Academic Program Goals – What Graduates Can Do (APG-WGCD) statements.
2. For AY 23, the PD continued using the SLO Assessment Rubric adopted in AY 20. In AY 20, the PD changed the SLO assessment rubric for the program and for the direct indicators from a 4-tiered rating scale (see Table 2) to a 5-tiered rating scale (see Table 3). The PD made this change in order to combine data from both direct and indirect indicators into the SLO assessment for each SLO where possible. The main difference between the AY 19 rubric and the AY 20 rubric is that the AY 20 rubric added an additional tier of Excellent for direct indicators for when the most frequent grade is A and B, and there are no D or F grades. Additionally, the AY 19 rubric did not use a Likert rating for direct indicators (see Table 4). In AY 21-2, the PD also added two additional questions to the Course End Feedback (CEF) survey to provide indirect indicators to assess SLOs c and e.

Table 2. SLO Assessment Rubric, AY 19 and prior.

Assessment Rating	Color Code	Most Frequent Letter Grade	Low Performance Threshold
Exceeds Expectations		B or better	< 10% D or F
Meets Expectations		C or better	< 20% D or F
Marginally Meets Expectations		C or better	< 40% D or F
Fails to Meet Expectations		Any	> 40% D or F

Table 3. Core Physics Program SLO Assessment Rubric (Adopted AY 20).

Assessment Rating	Color Code	Likert Rating	Direct Indicators: Criteria	Indirect Indicators: Level of Agreement
Excellent		5	Most frequent grade is A and B, with no D or F	Strongly agree
Good		4	Most frequent grade is A and B, with some D or F	Agree
Acceptable		3	Most frequent grade is C or better, with < 20% D or F	Neutral
Marginally Acceptable		2	Most frequent grade is C or better, with > 20% D or F but < 40% D or F	Disagree
Unacceptable		1	Most frequent grade is D or F	Strongly disagree

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 4. SLO Assessment Rubric Comparison, AY 19 and AY 20.

AY 19		AY 20	
Assessment Rating	Color Code	Assessment Rating	Color Code
		Excellent	
Exceeds Expectations		Good	
Meets Expectations		Acceptable	
Marginally Meets Expectations		Marginally Acceptable	
Fails to Meet Expectations		Unacceptable	

3. Data from the AY 21-2 PH205 CAT brief will be used to explain the assessment process.

a. For SLO a, for example, the AY 21-2 PH205 CD assessed each direct indicator (WPR 1, WPR 2, and the TEE) and each indirect indicator (CEF Survey question) for AY 21-2 using the assessment rubric from Table 3. The CD's assessment of these direct and indirect indicators is shown in Figure 1. Note that Figure 1 includes assessment data from the previous term, AY 21-1. For the direct indicators, the CD computed a simple average to determine the aggregate assessment of the direct indicators for each semester.

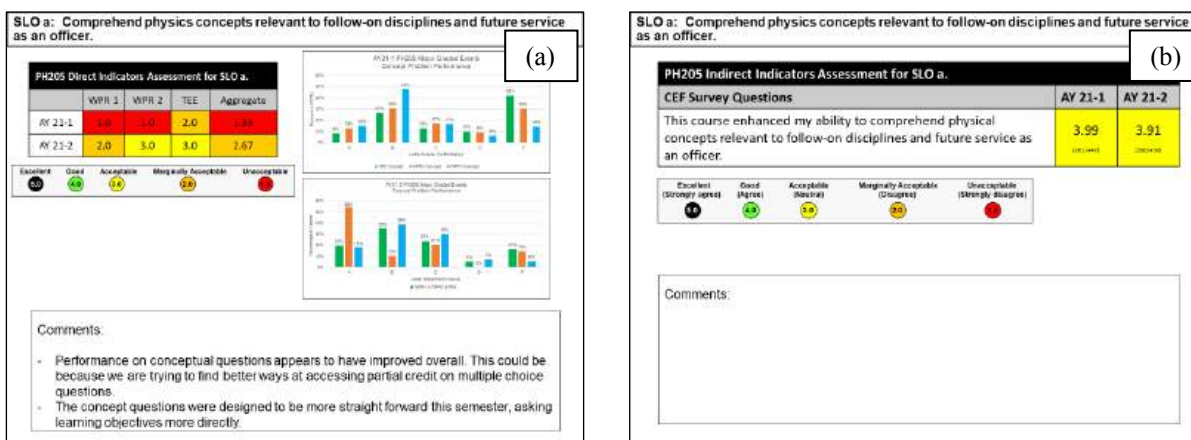


Figure 1. PH205 CD's assessment of (a) direct and (b) indirect indicators for SLO a.

b. The CDs summarize the assessment of the direct and indirect indicators for each SLO for their courses. To determine the overall assessment for each SLO, each CD computes a simple average of the direct and indirect indicator assessment data for each SLO, then rounds down to the nearest 0.5. Figure 2 provides an example from the AY 21-2 PH205 CAT brief.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

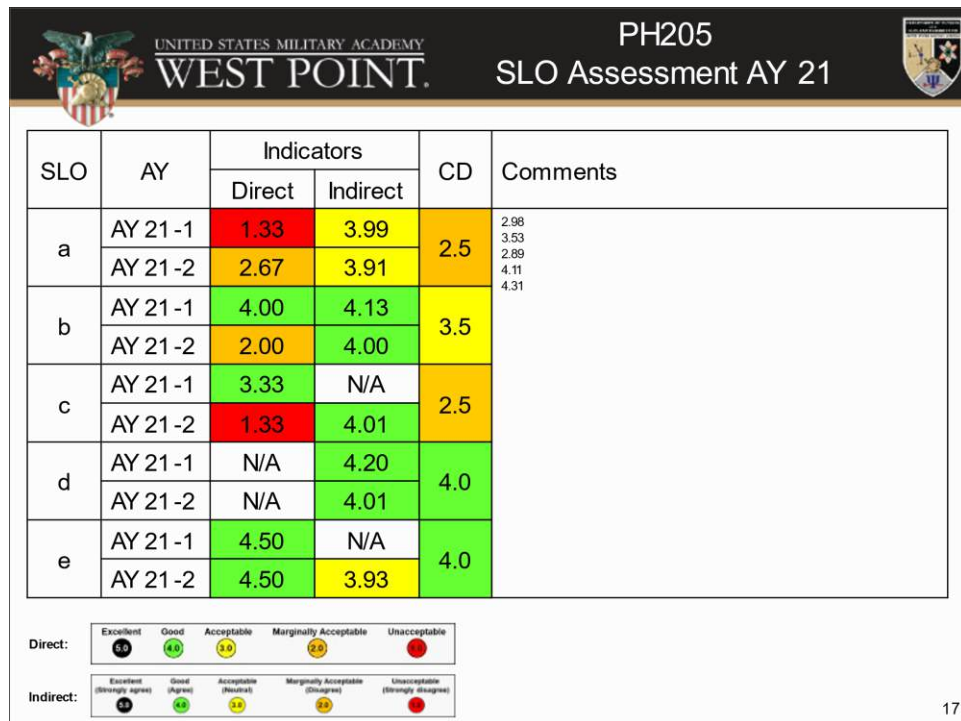


Figure 2. PH205 CD's assessment of each SLO, based on a simple average.

4. The PD aggregates the SLO assessments from each CD. To determine the overall program assessment, the PD performs a weighted average of the CD assessments for each SLO and rounds down to the nearest 0.5. The weighting factors are used to ensure that PH255 and PH256, which have lower enrollment and higher performance, do not unfairly skew the overall program assessment. The weighting factors are based on course enrollment. The weighting factor for PH201X and PH205 was normalized to the enrollment for PH255, and the weighting factor for PH202X and PH206 was normalized to the enrollment of PH256.

5. The PD's assessment of the SLOs for AY 23 is shown in Table 5. The weighting factors for AY 23 are shown in Table 6.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 5. Program Director's Assessment of Student Learning Outcomes using Direct and Indirect Indicators.

SLOs	Assessment					
Cadets who successfully complete the Core Physics program can:	PH201	PH251	PH202	PH252	Weighted Average	Program Director
	(AY 23)	(AY 23)	(AY 23)	(AY 23)		
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	3	4	2.5	3	2.98	3
b. Apply physical principles and mathematical competencies with logical processes.	3.5	4	4	4	3.73	3.5
c. Analyze and solve problems of increasing complexity.	3.5	4	3.5	4	3.6	3.5
d. Develop the ability to learn independently.	4	4	4	4	4	4
e. Examine physical phenomena, analyze data, and communicate results.	3	4	4	4.5	3.5	3.5
Direct: Excellent 5.0 Good 4.0 Acceptable 3.0 Marginally Acceptable 2.0 Unacceptable 1.0 Indirect: Excellent (Strongly agree) 5.0 Good (Agree) 4.0 Acceptable (Neutral) 3.0 Marginally Acceptable (Disagree) 2.0 Unacceptable (Strongly disagree) 1.0						

Table 6. Weighting Factors

Course	AY 23 Enrollment	Weighting Factor
PH201	770	0.54
PH251	168	0.11
PH202	380	0.27
PH252	104	0.07

Appendix C – Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

1. The Core Physics program APG-WGCD to SLO crosswalk and assessment for AY 22 is shown in Table 7. For WGCD statements that were based on the assessment of multiples SLO, the lower SLO assessment was used for the overall assessment of the WGCD statement. For example, the assessment of WGCD statement 2.2 is based on the assessment of SLO a (Marginally Acceptable) and SLO c (Acceptable). The overall assessment of WGCD statement 2.2 is therefore Marginally Acceptable.

Table 7. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment, AY 22

APG	WGCD Statement and Course Responsibility	SLO Crosswalk and SLO Assessment	WGCD Statement Assessment
1. Communication	1.2 Speak and write using Standard American English. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
	1.5 Use sound logic and relevant evidence to make convincing arguments. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
2. Critical and Creative Thinking	2.2 Integrate knowledge and skills from a variety of disciplines PH205 (I)	a. comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Acceptable
		c. analyze and solve problems of increasing complexity.	
	2.4 Reason both quantitatively and qualitatively PH205 (I)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
3. Lifelong Learning	3.1 Demonstrate the willingness and ability to learn independently. PH205 (I), PH206 (R)	d. develop the ability to learn independently.	Acceptable
	3.3 Pursue self-awareness and embrace the responsibility for personal intellectual development PH206 (R)	d. develop the ability to learn independently.	Acceptable
5. STEM	5.1 Apply mathematics, science, and computing to model devices, systems, processes or behaviors PH205 (R), PH206 (R)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	5.2 Apply the scientific method PH205 (R), PH206 (R)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
	5.3 Collect and analyze data in support of decision making R—PH205 and PH206	e. examine physical phenomena, analyze data, and communicate results.	Acceptable



TAB E AY2022 COURSE ASSESSMENT

Enclosure 1 – Core Physics Program AY 22 Executive Summary

1. References.

- a. DPOM 5-05, Managing Curricular Change, 08 April 2020.
- b. DPOM 5-07, Assessing and Improving Student Learning in the Academic Program, 24 April 2019.
- c. Office of the Dean, MADN, memorandum (Dean's AY22 Annual Guidance), 02 July 2021.
- d. Core Physics Program AY 21 Executive Summary, September 2021.
- e. Department of Physics and Nuclear Engineering, MADN-PNE, memorandum (Core Course Curricular Change Request for PH205 (Physics I) and PH255 (Advanced Physics I), 21 April 2022.
- f. Department of Physics and Nuclear Engineering, MADN-PNE, memorandum (Core Course Curricular Change Request for PH206 (Physics II) and PH256 (Advanced Physics II), 21 April 2022.

2. Purpose. This enclosure summarizes the Core Physics Program assessment, curriculum changes, and assessment process for academic year 2022 (AY 22), in compliance with the annual reporting requirements of references 1a, 1b, and 1c.

3. Status of Curricular Change Initiatives for AY 22. For AY 22, the Core Physics Program changed the sequence of content of the core course PH205/PH255 and core Science Depth Option course PH206/256. See references 1e and 1f. The Core Physics Program piloted PH201X in AY 22-1 and PH202X in AY 22-2. The General Committee of the Academic Board voted to approve the curriculum change to the PH201/PH251, PH202/PH252 sequence on 12 May 2022. There were no changes to the Student Learning Outcomes (SLOs).

4. Summary of Student Outcome Evaluation for AY 22. Evidence for cadet achievement of the SLOs was obtained from direct indicators (e.g., WPRs, lab reports, TEE, etc.) and indirect indicators (i.e., course end feedback surveys). A summary of the assessment results is given in Table 1 in Appendix A and repeated on the next page. The assessment methodology is given in Appendix B.

Table 1. Assessment Matrix for Core Physics SLOs.

SLOs	Assessment					
	AY 18	AY 19	AY 20		AY 21	AY 22
			AY 19 Rubric	AY 20 Rubric		
Cadets who successfully complete the Core Physics program can:						
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Fails to Meet	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Meets	Exceeds	Meets	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets	Meets	Exceeds	Good	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets	Meets	Exceeds	Good	Good	Acceptable
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Exceeds	Meets	Acceptable	Good	Acceptable

a. Cadet performance on SLO a, *comprehend physical concepts relevant to follow-on disciplines and future service as an officer*, for AY 22 was **Marginally Acceptable**. This rating is consistent with the Marginally Acceptable rating from AY 21.

b. Cadet performance on SLO b, *apply physical principles and mathematical competencies with logical processes*, for AY 22 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 21.

c. Cadet performance on SLO c, *analyze and solve problems of increasing complexity*, for AY 22 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 21.

d. Cadet performance on SLO d, *develop the ability to learn independently*, for AY 22 was **Acceptable**. This rating demonstrates a downturn from the Good rating from AY 21. The AY 21 rating was based on weighted average of 4.0, whereas the AY 22 rating was based on a weighted average of 3.98 (see Table 5 in Appendix B). The percent difference between the AY 21 and AY 22 rating was 0.5%. This percent difference is less than 1%, and no conclusions may reasonably drawn from this small value.

e. Cadet performance on SLO e, *examine physical phenomena, analyze data, and communicate results*, for AY 22 was **Acceptable**. This rating represents a downturn from the Good rating from AY 21. The change is primarily due to changes to the laboratory program in PH205 and PH201X in AY 22-2.

(1) In AY 21, PH205 had a lab WPR, and the assessment of SLO e for the direct indicator of the lab WPR was Good. In AY 21, the PH205 lab program consisted of 8 labs that included a final design sequence, with lab 6 for design, lab 7 for execution, and lab 8 for presentation. The assessment of SLO e for the direct indicator of the Group

Enclosure 1 – Core Physics Program AY 22 Executive Summary

Lab Grade was Excellent. The aggregate assessment of SLO e for direct indicators for AY 21 was Good.

(2) In AY 22-1, both PH201X and PH205 had a lab WPR, and the assessment of SLO e for the direct indicator of the lab WPR for both courses was Good, consistent with AY 21. In AY 22-1, PH201X had 8 labs without the lab design sequence, whereas PH205 had 8 labs that included the lab design sequence. The assessment of SLO e for the direct indicator of the Group Lab Grade was Good for both courses. It is not known why PH205 had a downturn in performance on the Group Lab Grade. In AY 22-2, PH205 had two lab writs instead of a lab WPR. Cadet performance on lab writ #1 was Acceptable, and performance on lab writ #2 was Marginally Acceptable. The aggregate assessment of SLO e for direct indicators for AY 22 was Acceptable.

(3) The lab program was changed from the lab WPR and the lab design sequence in AY 21 to the lab writs and 10 labs in AY 22-2 to focus on technical writing of lab reports. In AY 21, cadets had limited exposure to writing a full laboratory report. In AY 22, the lab program culminated with the writing of two full lab reports. The change from a lab WPR in AY 21 to the two lab writs in AY 22-2 facilitated the spreading out of teaching and assessing lab material to more logically support the skills necessary to conduct a lab and write a lab report.

(4) The down turn in the assessment of SLO e from AY 21 to AY 22 likely has two causes. First, cadets likely assessed the lab WPR as a higher stakes event than the lab writs and therefore prepared more for the lab WPR, as reflected in the better performance on the lab WPR in AY 21 and AY 22-1 than on the lab writs in AY 22-2. Second, the lab design sequence (labs 6 through 8) was less rigorous than the full lab reports, and cadets scored better on the lab design sequence in AY 21 than on written lab reports in AY 22-2.

(5) The Core Physics Program will continue to implement the two lab writs and the 10 labs that culminate in the written lab reports. The lab reports improve the written technical communication skills of cadets and better prepare them for more advanced labs in PH202X and future STEM courses.

5. Summary of Results of APG-WGCD Responsibility Evaluations Conducted During the Year. The summary of the Core Program APG-WGCD to SLO crosswalk for AY 22 is located in Appendix C.

6. Significant Planned or Continuing Curricular Changes for AY 23. There are no significant changes planned for the Core Physics Program for AY 23. The Core Physics Program will continue to refine the PH201/PH251 and PH202/PH252 course sequence.

7. Proposed Changes to Student Learning Outcomes / Significant Planned or Implemented Assessment Process Changes. There are no planned changes to the SLOs for AY 23.

Enclosure 1 – Core Physics Program AY 22 Executive Summary

8. Updated Assessment Schedule and Time-line. Currently, the Core Physics program assess each of its SLOs every year. There are no plans to change this assessment timeline.

9. Point of Contact. The POC for this action is the undersigned at extension 845-938-5895.

3 Appendices

- A. Student Learning Outcomes and Assessment Matrix for Core Physics SLOs
- B. Assessment Methodology for Core Physics Program Student Learning Outcomes
- C. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

DAVID J. PHILLIPS, PhD
COL, MS (FA 47)
Program Director, Core Physics

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

1. Core Physics SLOs for AY 22. Cadets who successfully complete the Core Physics program can:

- a. **comprehend** physical concepts relevant to follow-on disciplines and future service as an officer.
- b. **apply** physical principles and mathematical competencies with logical processes.
- c. **analyze** and **solve** problems of increasing complexity.
- d. develop the ability to **learn independently**.
- e. **examine** physical phenomena, **analyze** data, and **communicate** results.

2. The Assessment Matrix for Core Physics SLOs.

a. For AY 22, the Program Director (PD) continued to use the 5-tiered scale for direct indicators that the PD adopted from the Nuclear Engineering program in AY 20 and the associated assessment ratings and color codes. The change is discussed in more detail in Appendix B.

b. The Assessment Matrix for the Core Physics SLOs is shown in Table 1. Of note, the assessment for AY 20 shows the assessment for AY 20 using both the AY 19 rubric and the AY 20 rubric. The increase (↗), decrease (↘), or equivalency (→) in SLO achievement from AY 19 to AY 20 is indicated graphically with arrows. For example, for SLO b, the AY 19 assessment was Exceeds. The AY 20 Assessment was Meets using the AY 19 rubric, which represents a decrease (↘) in SLO performance compared to AY 19. The AY 20 Assessment was Acceptable using the AY 20 rubric, which is equivalent (→) to Meets using the AY 19 rubric.

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

Table 1. Assessment Matrix for Core Physics SLOs.

SLOs	Assessment					
	AY 18	AY 19	AY 20		AY 21	AY 22
			AY 19 Rubric	AY 20 Rubric		
Cadets who successfully complete the Core Physics program can:						
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Fails to Meet	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Meets	Exceeds	Meets	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets	Meets	Exceeds	Good	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets	Meets	Exceeds	Good	Good	Acceptable
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Exceeds	Meets	Acceptable	Good	Acceptable

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

1. Evaluation and assessment of each SLO is integrated into each individual core course offering, including the laboratory program. At the end of each semester, Course Directors (CDs) conduct a course assessment to gain situational understanding of immediate changes required for the next semester and to evaluate each SLO. CDs then provide this information to the PD in the form of a Course Assessment (CAT) brief. At the conclusion of each academic year, the PD completes a program review that focuses on assessing the SLOs and the assigned Academic Program Goals – What Graduates Can Do (APG-WGCD) statements.
2. For AY 22, the PD continued using the SLO Assessment Rubric adopted in AY 20. In AY 20, the PD changed the SLO assessment rubric for the program and for the direct indicators from a 4-tiered rating scale (see Table 2) to a 5-tiered rating scale (see Table 3). The PD made this change in order to combine data from both direct and indirect indicators into the SLO assessment for each SLO where possible. The main difference between the AY 19 rubric and the AY 20 rubric is that the AY 20 rubric added an additional tier of Excellent for direct indicators for when the most frequent grade is A and B, and there are no D or F grades. Additionally, the AY 19 rubric did not use a Likert rating for direct indicators (see Table 4). In AY 21-2, the PD also added two additional questions to the Course End Feedback (CEF) survey to provide indirect indicators to assess SLOs c and e.

Table 2. SLO Assessment Rubric, AY 19 and prior.

Assessment Rating	Color Code	Most Frequent Letter Grade	Low Performance Threshold
Exceeds Expectations		B or better	< 10% D or F
Meets Expectations		C or better	< 20% D or F
Marginally Meets Expectations		C or better	< 40% D or F
Fails to Meet Expectations		Any	> 40% D or F

Table 3. Core Physics Program SLO Assessment Rubric (Adopted AY 20).

Assessment Rating	Color Code	Likert Rating	Direct Indicators: Criteria	Indirect Indicators: Level of Agreement
Excellent		5	Most frequent grade is A and B, with no D or F	Strongly agree
Good		4	Most frequent grade is A and B, with some D or F	Agree
Acceptable		3	Most frequent grade is C or better, with < 20% D or F	Neutral
Marginally Acceptable		2	Most frequent grade is C or better, with > 20% D or F but < 40% D or F	Disagree
Unacceptable		1	Most frequent grade is D or F	Strongly disagree

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 4. SLO Assessment Rubric Comparison, AY 19 and AY 20.

AY 19		AY 20	
Assessment Rating	Color Code	Assessment Rating	Color Code
		Excellent	
Exceeds Expectations		Good	
Meets Expectations		Acceptable	
Marginally Meets Expectations		Marginally Acceptable	
Fails to Meet Expectations		Unacceptable	

3. Data from the AY 21-2 PH205 CAT brief will be used to explain the assessment process.

a. For SLO a, for example, the AY 21-2 PH205 CD assessed each direct indicator (WPR 1, WPR 2, and the TEE) and each indirect indicator (CEF Survey question) for AY 21-2 using the assessment rubric from Table 3. The CD's assessment of these direct and indirect indicators is shown in Figure 1. Note that Figure 1 includes assessment data from the previous term, AY 21-1. For the direct indicators, the CD computed a simple average to determine the aggregate assessment of the direct indicators for each semester.

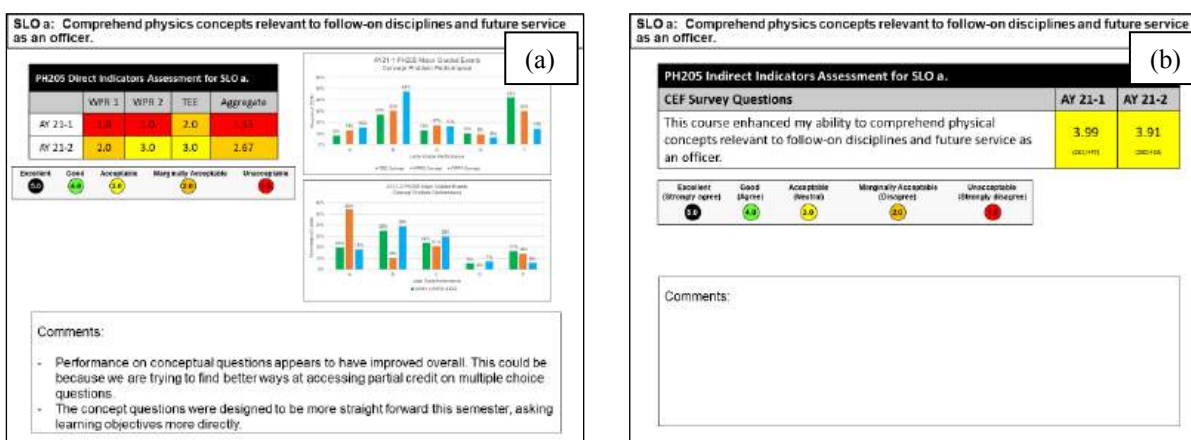


Figure 1. PH205 CD's assessment of (a) direct and (b) indirect indicators for SLO a.

b. The CDs summarize the assessment of the direct and indirect indicators for each SLO for their courses. To determine the overall assessment for each SLO, each CD computes a simple average of the direct and indirect indicator assessment data for each SLO, then rounds down to the nearest 0.5. Figure 2 provides an example from the AY 21-2 PH205 CAT brief.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

UNITED STATES MILITARY ACADEMY
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PH205

SLO Assessment AY 21

SLO	AY	Indicators		CD	Comments	
		Direct	Indirect			
a	AY 21-1	1.33	3.99	2.5	2.98 3.53 2.89 4.11 4.31	
	AY 21-2	2.67	3.91			
b	AY 21-1	4.00	4.13	3.5		
	AY 21-2	2.00	4.00			
c	AY 21-1	3.33	N/A	2.5		
	AY 21-2	1.33	4.01			
d	AY 21-1	N/A	4.20	4.0		
	AY 21-2	N/A	4.01			
e	AY 21-1	4.50	N/A	4.0		
	AY 21-2	4.50	3.93			

Direct:

Excellent (5.0)	Good (4.0)	Acceptable (3.0)	Marginally Acceptable (2.0)	Unacceptable (1.0)
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Indirect:

Excellent (Strategy agreed) (5.0)	Good (Strategy agreed) (4.0)	Acceptable (Results) (3.0)	Marginally Acceptable (Strategy disagreed) (2.0)	Unacceptable (Strategy disagreed) (1.0)
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Figure 2. PH205 CD's assessment of each SLO, based on a simple average.

4. The PD aggregates the SLO assessments from each CD. To determine the overall program assessment, the PD performs a weighted average of the CD assessments for each SLO and rounds down to the nearest 0.5. The weighting factors are used to ensure that PH255 and PH256, which have lower enrollment and higher performance, do not unfairly skew the overall program assessment. The weighting factors are based on course enrollment. The weighting factor for PH201X and PH205 was normalized to the enrollment for PH255, and the weighting factor for PH202X and PH206 was normalized to the enrollment of PH256.

5. The PD's assessment of the SLOs for AY 22 is shown in Table 5. The weighting factors for AY 22 are shown in Table 6.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 5. Core Physics SLOs Assessment Summary, AY 22, Direct and Indirect Indicators.

SLOs	Assessment							
Cadets who successfully complete the Core Physics program can:	PH201X (AY 22-1)	PH202X (AY 22-2)	PH205 (AY 22)	PH255 (AY 22-2)	PH206 (AY 22)	PH256 (AY 22-1)	Weighted Average	Program Director
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	3.0	2.5	3.0	4.0	2.5	2.5	2.88	2.5
b. Apply physical principles and mathematical competencies with logical processes.	4.0	4.0	3.5	4.5	4.0	4.0	3.74	3.5
c. Analyze and solve problems of increasing complexity.	4.0	4.0	3.5	4.5	3.5	4.0	3.62	3.5
d. Develop the ability to learn independently.	4.0	3.5	4.0	4.0	4.0	4.0	3.98	3.5
e. Examine physical phenomena, analyze data, and communicate results.	4.0	4.0	3.5	4.0	4.0	4.0	3.72	3.5
Direct: Excellent (5.0) Good (4.0) Acceptable (3.0) Marginally Acceptable (2.0) Unacceptable (1.0) Indirect: Excellent (Strongly agree) (5.0) Good (Agree) (4.0) Acceptable (Neutral) (3.0) Marginally Acceptable (Disagree) (2.0) Unacceptable (Strongly disagree) (1.0)								

Table 6. Weighting Factors for Core Physics SLOs, AY 22.

Course	Enrollment			Weighting Factor
	AY 22-1	AY 22-2	AY 22 Total	
PH201X	118	---	118	1.7
PH202X	---	84	84	1.2
PH205	505	427	932	13.7
PH255	---	68	68	1.0
PH206	200	230	430	5.9
PH256	73	---	73	1.0

Appendix C – Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

1. The Core Physics program APG-WGCD to SLO crosswalk and assessment for AY 22 is shown in Table 7. For WGCD statements that were based on the assessment of multiples SLO, the lower SLO assessment was used for the overall assessment of the WGCD statement. For example, the assessment of WGCD statement 2.2 is based on the assessment of SLO a (Marginally Acceptable) and SLO c (Acceptable). The overall assessment of WGCD statement 2.2 is therefore Marginally Acceptable.

Table 7. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment, AY 22

APG	WGCD Statement and Course Responsibility ¹	SLO Crosswalk and SLO Assessment	WGCD Statement Assessment
1. Communication	1.2 Speak and write using Standard American English. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
	1.5 Use sound logic and relevant evidence to make convincing arguments. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
2. Critical and Creative Thinking	2.2 Integrate knowledge and skills from a variety of disciplines PH205 (I)	a. comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Acceptable
		c. analyze and solve problems of increasing complexity.	
	2.4 Reason both quantitatively and qualitatively PH205 (I)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
3. Lifelong Learning	3.1 Demonstrate the willingness and ability to learn independently. PH205 (I), PH206 (R)	d. develop the ability to learn independently.	Acceptable
	3.3 Pursue self-awareness and embrace the responsibility for personal intellectual development PH206 (R)	d. develop the ability to learn independently.	Acceptable
5. STEM	5.1 Apply mathematics, science, and computing to model devices, systems, processes or behaviors PH205 (R), PH206 (R)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	5.2 Apply the scientific method PH205 (R), PH206 (R)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
	5.3 Collect and analyze data in support of decision making R—PH205 and PH206	e. examine physical phenomena, analyze data, and communicate results.	Acceptable

¹See Annex A: Academic Program Development Framework, DPOM 5-07.



TAB F AY2021 COURSE ASSESSMENT

Enclosure 1 – Core Physics Program AY 21 Executive Summary

1. References.

- a. DPOM 5-05, Managing Curricular Change, 08 April 2020.
- b. DPOM 5-07, Assessing and Improving Student Learning in the Academic Program, 24 April 2019.
- c. Dean's Annual Guidance, Academic Year 2021 (AY21), 10 July 2020.
- d. Core Physics Program AY 20 Executive Summary, September 2020.

2. Purpose. This memorandum summarizes the Core Physics program assessment, curriculum changes, and assessment process for AY 21, in compliance with the annual reporting requirements of references 1.a, 1.b, and 1.c.

3. Status of Curricular Change Initiatives for AY 21. For AY 21, no curricular changes requiring Academy-level approval were implemented. Internally, for AY 21 we maintained the change to the assessment mechanism for the Student Learning Outcomes (SLOs) that we first implemented in AY 20. The change has allowed us to combine assessment data from both direct and indirect indicators to assess all SLOs using a common 5-tiered system.

4. Summary of Student Outcome Evaluation for AY 21. Evidence for cadet achievement of the SLOs was obtained from direct indicators (e.g., WPRs, Lab reports, TEE, etc.) and indirect indicators (i.e., course end feedback surveys). A summary of the assessment results is given in Table 1 in Appendix A and repeated here. The assessment methodology is given in Appendix B.

Table 1. Assessment Matrix for Core Physics SLOs.

SLOs	Assessment					
	AY 17	AY 18	AY 19	AY 20		AY 21
				AY 19 Rubric	AY 20 Rubric	
Cadets who successfully complete the Core Physics program can:						
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Fails to Meet	Fails to Meet	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Meets	Meets	Exceeds	Meets	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets	Meets	Meets	Exceeds	Good	Acceptable
d. Develop the ability to learn independently.	Meets	Meets	Meets	Exceeds	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Exceeds	Exceeds	Meets	Acceptable	Good

Enclosure 1 – Core Physics Program AY 21 Executive Summary

a. Cadet performance on SLO a, *comprehend physical concepts relevant to follow-on disciplines and future service as an officer*, was **Marginally Acceptable**. This rating is consistent with the Marginally Acceptable rating from AY 20.

b. Cadet performance on SLO b, *apply physical principles and mathematical competencies with logical processes*, was **Acceptable**. This rating is consistent with the Acceptable rating from AY 20.

c. Cadet performance on SLO c, *analyze and solve problems of increasing complexity*, was **Acceptable**. This rating represents a downturn from the Good rating from AY 20. The reason for the downturn is unclear. The lack of clarity is due to the fact that the indicators used to assess this SLO are too broad; the SLO is assessed based on aggregate performance on both conceptual and calculation problems, in addition to indirect indicators from the CEF survey. For AY 22, the indicators need to be more narrowly focused. A best indicator, such as cadet performance on a specific problem on a WPR or TEE, will be considered to replace the current indicators for this SLO.

d. Cadet performance on SLO d, *develop the ability to learn independently*, was **Good**. This rating is consistent with the Good rating from AY 20.

e. Cadet performance on SLO e, *examine physical phenomena, analyze data, and communicate results*, was **Good**. This rating represents an upturn from the Acceptable rating from AY 20. The change is due to better cadet performance in AY 21 on the Signature Writing Event (SWE) and the group laboratory reports. The increased performance on these events can likely be attributed to changes in the lab program for AY 21 that focused on clarifying the requirements for cadets on the SWE and on laboratory reports.

5. Summary of Results of APG-WGCD Responsibility Evaluations Conducted During the Year. The summary of the Core Program APG-WGCD to SLO crosswalk is in Appendix C.

6. Significant Planned or Continuing Curricular Changes for AY 22. There is one planned curricular change for the Core Physics program for AY 22. The planned curricular change is to realign the course content for PH205 and PH206 to the traditional topical coverage of mechanics in PH205 and electricity and magnetism in PH206. The change is currently underway. For AY 22-1, the Core Physics program is piloting 8 sections of PH201X as a replacement for PH205.

7. Proposed Changes to Student Learning Outcomes / Significant Planned or Implemented Assessment Process Changes. There are no changes to the SLOs for the upcoming AY. The assessment process for AY 22 may change to include assessment data from cadet performance on national standardized tests, and best indicators will be considered for assessment of SLO c.

Enclosure 1 – Core Physics Program AY 21 Executive Summary

8. Updated Assessment Schedule and Time-line. Currently, the Core Physics program assess each of its SLOs every year. There are no plans to change this assessment timeline.

9. Point of Contact. The POC for this action is the undersigned at extension 845-938-5895.

4 Appendices

- A. Student Learning Outcomes and Assessment Matrix for Core Physics SLOs
- B. Assessment Methodology for Core Physics Program Student Learning Outcomes
- C. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment
- D. Planned Curricular Changes

DAVID J. PHILLIPS, PhD
COL, MS (FA 47)
Program Director, Core Physics

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

1. Core Physics SLOs for AY 21. Cadets who successfully complete the Core Physics program can:

- a. **comprehend** physical concepts relevant to follow-on disciplines and future service as an officer.
- b. **apply** physical principles and mathematical competencies with logical processes.
- c. **analyze** and **solve** problems of increasing complexity.
- d. develop the ability to **learn independently**.
- e. **examine** physical phenomena, **analyze** data, and **communicate** results.

2. The Assessment Matrix for Core Physics SLOs.

a. For AY 21, the Program Director (PD) continued to use the 5-tiered scale for direct indicators that the PD adopted from the Nuclear Engineering program in AY 20 and the associated assessment ratings and color codes. The change is discussed in more detail in Appendix B.

b. The Assessment Matrix for the Core Physics SLOs is shown in Table 1. Of note, the assessment for AY 20 shows the assessment for AY 20 using both the AY 19 rubric and the AY 20 rubric. The increase ($\nearrow$), decrease ($\searrow$), or equivalency ($\rightarrow$) in SLO achievement from AY 19 to AY 20 is indicated graphically with arrows. For example, for SLO b, the AY 19 assessment was Exceeds. The AY 20 Assessment was Meets using the AY 19 rubric, which represents a decrease ($\searrow$) in SLO performance compared to AY 19. The AY 20 Assessment was Acceptable using the AY 20 rubric, which is equivalent ($\rightarrow$) to Meets using the AY 19 rubric.

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

Table 1. Assessment Matrix for Core Physics SLOs.

SLOs	Assessment					
	AY 17	AY 18	AY 19	AY 20		AY 21
				AY 19 Rubric	AY 20 Rubric	
Cadets who successfully complete the Core Physics program can:						
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Fails to Meet	Fails to Meet	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Meets	Meets	Exceeds	Meets	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets	Meets	Meets	Exceeds	Good	Acceptable
d. Develop the ability to learn independently.	Meets	Meets	Meets	Exceeds	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Exceeds	Exceeds	Meets	Acceptable	Good

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

1. Evaluation and assessment of each SLO is integrated into each individual core course offering, including the laboratory program. At the end of each semester, Course Directors (CDs) conduct a course assessment to gain situational understanding of immediate changes required for the next semester and to evaluate each SLO. CDs then provide this information to the PD in the form of a Course Assessment (CAT) brief. At the conclusion of each academic year, the PD completes a program review that focuses on assessing the SLOs and the assigned Academic Program Goals – What Graduates Can Do (APG-WGCD) statements.

2. For AY 21, the PD continued using the SLO Assessment Rubric adopted in AY 20. In AY 20, the PD changed the SLO assessment rubric for the program and for the direct indicators from a 4-tiered rating scale (see Table 2) to a 5-tiered rating scale (see Table 3). The PD made this change in order to combine data from both direct and indirect indicators into the SLO assessment for each SLO where possible. The main difference between the AY 19 rubric and the AY 20 rubric is that the AY 20 rubric added an additional tier of Excellent for direct indicators for when the most frequent grade is A and B, and there are no D or F grades. Additionally, the AY 19 rubric did not use a Likert rating for direct indicators (see Table 4). For AY 21-2, the PD also added two additional questions to the Course End Feedback (CEF) survey to provide indirect indicators to assess SLOs c and e.

Table 2. SLO Assessment Rubric, AY 19 and prior.

Assessment Rating	Color Code	Most Frequent Letter Grade	Low Performance Threshold
Exceeds Expectations		B or better	< 10% D or F
Meets Expectations		C or better	< 20% D or F
Marginally Meets Expectations		C or better	< 40% D or F
Fails to Meet Expectations		Any	> 40% D or F

Table 3. Core Physics Program SLO Assessment Rubric (Adopted AY 20)

Assessment Rating	Color Code	Likert Rating	Direct Indicators: Criteria	Indirect Indicators: Level of Agreement
Excellent		5	Most frequent grade is A and B, with no D or F	Strongly agree
Good		4	Most frequent grade is A and B, with some D or F	Agree
Acceptable		3	Most frequent grade is C or better, with < 20% D or F	Neutral
Marginally Acceptable		2	Most frequent grade is C or better, with > 20% D or F but < 40% D or F	Disagree
Unacceptable		1	Most frequent grade is D or F	Strongly disagree

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 4. SLO Assessment Rubric Comparison, AY 19 and AY 20.

AY 19		AY 20	
Assessment Rating	Color Code	Assessment Rating	Color Code
		Excellent	
Exceeds Expectations		Good	
Meets Expectations		Acceptable	
Marginally Meets Expectations		Marginally Acceptable	
Fails to Meet Expectations		Unacceptable	

3. Data from the AY 21-2 PH205 CAT brief will be used to explain the assessment process.

a. For SLO a, for example, the AY 21-2 PH205 CD assessed each direct indicator (WPR 1, WPR 2, and the TEE) and each indirect indicator (CEF Survey question) for AY 21-2 using the assessment rubric from Table 3. The CD's assessment of these direct and indirect indicators is shown in Figure 1. Note that Figure 1 includes assessment data from the previous term, AY 21-1. For the direct indicators, the CD computed a simple average to determine the aggregate assessment of the direct indicators for each semester.

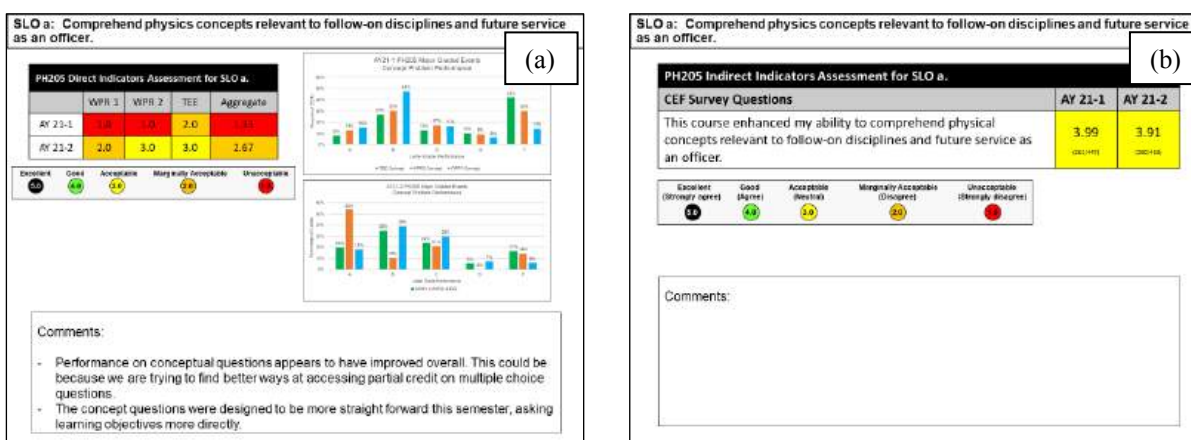


Figure 1. PH205 CD's assessment of (a) direct and (b) indirect indicators for SLO a.

b. The CDs summarize the assessment of the direct and indirect indicators for each SLO for their courses. To determine the overall assessment for each SLO, each CD computes a simple average of the direct and indirect indicator assessment data for each SLO, then rounds down to the nearest 0.5. Figure 2 provides an example from the AY 21-2 PH205 CAT brief.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

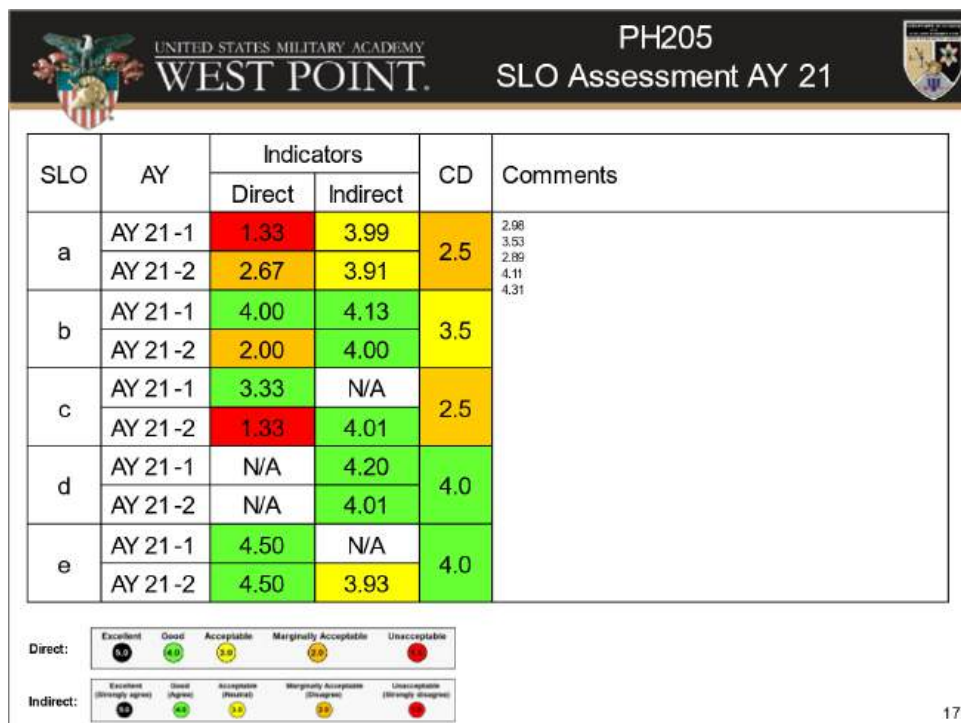


Figure 2. PH205 CD's assessment of each SLO, based on a simple average.

4. The PD aggregates the SLO assessments from each CD. To determine the overall program assessment, the PD performs a weighted average of the CD assessments for each SLO and rounds down to the nearest 0.5. The PD's assessment of the SLOs for AY 21 is shown in Table 5. The weighting factors are used to ensure that PH255 and PH256, which have lower enrollment and higher performance, do not unfairly skew the overall program assessment. The weighting factors are based on course enrollment. The weighting factor for PH205 is normalized to the enrollment for PH255, and the weighting factor for PH256 is normalized to the enrollment of PH256. The weighting factors for AY 21 are shown in Table 6.

Table 5. Core Physics SLOs Assessment Summary, AY 21, Direct and Indirect Indicators

SLOs	Assessment					
Cadets who successfully complete the Core Physics program can:	PH205	PH255	PH206	PH256	Weighted Average	Program Director
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	2.5	4.0	2.5	3.0	2.77	2.5
b. Apply physical principles and mathematical competencies with logical processes.	3.5	4.0	4.0	4.5	3.85	3.5
c. Analyze and solve problems of increasing complexity.	2.5	4.5	4.0	4.5	3.48	3.0
d. Develop the ability to learn independently.	4.0	4.0	4.0	4.0	4.00	4.0
e. Examine physical phenomena, analyze data, and communicate results.	4.0	4.0	4.0	4.0	4.00	4.0

Direct: Excellent (5.0), Good (4.0), Acceptable (3.0), Marginally Acceptable (2.0), Unacceptable (1.0)
Indirect: Excellent (Strongly agree) (5.0), Good (Agree) (4.0), Acceptable (Neutral) (3.0), Marginally Acceptable (Disagree) (2.0), Unacceptable (Strongly disagree) (1.0)

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 6. Weighting Factors for Core Physics SLOs, AY 21

	Enrollment			
Course	AY 21-1	AY 21-2	AY 21 Total	Weighting Factor
PH205	447	436	883	3.2
PH255	144	135	279	1.0
PH206	156	168	324	2.1
PH256	71	82	153	1.0

5. The AY 21 assessment includes assessment of indirect indicators for all SLOs based on CEF surveys. The CEF survey results tend to be high, which has the potential to skew the assessment results. Table 7 shows the Core Physics SLO Assessment, AY 21, considering direct indicators only for all SLOs.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 7. Core Physics SLOs Assessment Summary, AY 21, Direct Indicators Only

SLOs	Assessment					
Cadets who successfully complete the Core Physics program can:	PH205	PH255	PH206	PH256	Weighted Average	Program Director
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	2.0	4.0	1.0	2.0	1.99	1.5
b. Apply physical principles and mathematical competencies with logical processes.	3.0	4.0	4.0	4.5	3.63	3.5
c. Analyze and solve problems of increasing complexity.	2.0	4.5	4.0	4.5	3.26	3.0
d. Develop the ability to learn independently.	N/A	N/A	N/A	N/A	N/A	N/A
e. Examine physical phenomena, analyze data, and communicate results.	4.5	4.5	4.5	4.0	4.43	4.0

Direct:

Excellent
5.0

Good
4.0

Acceptable
3.0

Marginally Acceptable
2.0

Unacceptable
1.0

Indirect:

Excellent (Strongly agree)
5.0

Good (Agree)
4.0

Acceptable (Neutral)
3.0

Marginally Acceptable (Disagree)
2.0

Unacceptable (Strongly disagree)
1.0

6. Table 8 shows a comparison of the SLO assessment based on direct and indirect indicators vs. direct indicators only.

Table 8. Core Physics SLOs Assessment Summary, AY 21, Comparison of Indicators

SLOs	Direct and Indirect Indicators		Direct Indicators Only	
Cadets who successfully complete the Core Physics program can:	Weighted Average	Program Director	Weighted Average	Program Director
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	2.77	2.5	1.99	1.5
b. Apply physical principles and mathematical competencies with logical processes.	3.85	3.5	3.63	3.5
c. Analyze and solve problems of increasing complexity.	3.48	3.0	3.26	3.0
d. Develop the ability to learn independently.	4.00	4.0	N/A	N/A
e. Examine physical phenomena, analyze data, and communicate results.	4.00	4.0	4.43	4.0

Direct:

Excellent
5.0

Good
4.0

Acceptable
3.0

Marginally Acceptable
2.0

Unacceptable
1.0

Indirect:

Excellent (Strongly agree)
5.0

Good (Agree)
4.0

Acceptable (Neutral)
3.0

Marginally Acceptable (Disagree)
2.0

Unacceptable (Strongly disagree)
1.0

7. Examination of SLO a in Table 8 shows a weighted average of 2.77 for when considering both direct and indirect indicators compared to a weighted average of 1.99 when only considering direct indicators. This comparison shows that inclusion of indirect indicators slightly skews the assessment higher. However, the weighted average of 1.99 is very close to 2.00, which would have resulted in an overall assessment of Marginally Acceptable. Therefore, the inclusion of indirect indicators on the assessment is minimal, and the assessment in Table 5 will be used.

Appendix C – Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

1. The Core Physics program APG-WGCD to SLO crosswalk and assessment for AY 21 is shown in Table 9. For WGCD statements that were based on the assessment of multiples SLO, the lower SLO assessment was used for the overall assessment of the WGCD statement. For example, the assessment of WGCD statement 2.2 is based on the assessment of SLO a (Marginally Acceptable) and SLO c (Acceptable). The overall assessment of WGCD statement 2.2 is therefore Marginally Acceptable.

Table 9. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment, AY 20

APG	WGCD Statement and Course Responsibility	SLO Crosswalk and SLO Assessment	WGCD Statement Assessment
1. Communication	1.2 Speak and write using Standard American English. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Good
	1.5 Use sound logic and relevant evidence to make convincing arguments. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Good
2. Critical and Creative Thinking	2.2 Integrate knowledge and skills from a variety of disciplines PH205 (I)	a. comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Acceptable
		c. analyze and solve problems of increasing complexity.	
	2.4 Reason both quantitatively and qualitatively PH205 (I)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
3. Lifelong Learning	3.1 Demonstrate the willingness and ability to learn independently. PH205 (I), PH206 (R)	d. develop the ability to learn independently.	Good
	3.3 Pursue self-awareness and embrace the responsibility for personal intellectual development PH206 (R)	d. develop the ability to learn independently.	Good
5. STEM	5.1 Apply mathematics, science, and computing to model devices, systems, processes or behaviors PH205 (R), PH206 (R)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	5.2 Apply the scientific method PH205 (R), PH206 (R)	e. examine physical phenomena, analyze data, and communicate results.	Good
	5.3 Collect and analyze data in support of decision making R—PH205 and PH206	e. examine physical phenomena, analyze data, and communicate results.	Good

Appendix D – Planned Curricular Changes

1. There is one planned curricular change for AY 22. The change is to realign the course content for PH205 and PH206 to the traditional topical coverage of mechanics in PH205 and electricity and magnetism in PH206 and to increase the time spent in laboratory. This will allow for more in-depth and cohesive treatment of mechanics in PH205 and electricity and magnetism in PH206.
2. The curricular change is currently underway. In AY 22-1, the Core Physics program is piloting 8 sections of PH201X. PH201X is being taught on a 40 lesson by 55-minute schedule, with 120-minute labs. In comparison, PH205 is taught on a 30 lesson by 75-minute schedule, with 75-minute labs.
3. In AY 22-2, the Core Physics program will pilot 6 sections of PH202X. PH202X will be taught on a 40 lesson by 55-minute schedule, with 120-minute labs. In comparison, PH206 is taught on a 30 lesson by 75-minute schedule, with 75-minute labs



SECTION II

EXAMINATIONS



TAB E WPR#1



UNITED STATES MILITARY ACADEMY
WEST POINT.

WPR #1 HANDOUT W/SOLUTION

Time of completion _____
19/20 February 2026

Cadet _____

Section _____
Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
PH2X2, Physics II
Written Partial Review I

1. Instructions:

- Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. No phones, cameras, or smartwatches are allowed in the examination room.
- Check your exam for 9 problems. Check for the Physics Reference Card provided with the examination.
- Show all work**; partial credit may be given for correct work shown.
- For multiple choice questions **bubble in** your selection and choose the **one** answer that best answers each question.
- Take up to 75 minutes to complete the examination.
- An instructor is in the hall for any questions.

2. Exam Security:

- I will not discuss this exam with anyone until it is released from academic security on _____ (date) at _____ (time).
- Cadet Signature: _____
- Cadet Name (print): _____

PROBLEM	WEIGHT	SCORE
1	15	
2	20	
3	50	
4	50	
5	65	
6	50	
7	40	
8	5	
9	5	
	Score:	
	Total:	300

A: 26 (9%)

B: 93 (31%)

D: 181 (60%)

Total: 300

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A: 0 (0%)
B: 0 (0%)
D: 0 (0%)
Total: 0

Wt. No.

Cadet: _____

- 15 1. A current is passed through a very long wire. As a result, a 325 μT induced magnetic field is measured a distance of 4.11 μm from the wire. **Calculate** the current.

$$|\vec{B}| = \frac{\mu_o I}{2\pi R} \quad \boxed{9}$$

3 Amp's Law, Biot Savart Law, or solenoid

$$I = \frac{|\vec{B}| 2\pi R}{\mu_o} \quad \textcircled{3} \text{ Attempts to solve for current}$$

$$I = \frac{(325 \times 10^{-6} \text{ T})(2\pi)(4.11 \times 10^{-6} \text{ m})}{4\pi \times 10^{-7} \frac{\text{N}}{\text{A}^2}}$$

$$I = 6.68 \times 10^{-3} \text{ A} \quad \begin{matrix} \textcircled{1} \text{ Sig Figs} \\ \textcircled{1} \text{ Units} \end{matrix}$$

1 Correct ANS Report your answer in this box.	Report units in this box.
---	---

- 20 2. A long, straight wire with a length of 1.65 m is carrying a current of 127 A. The wire is placed in an external magnetic field. The magnetic field is perpendicular to the wire as shown in the figure. The magnitude of the magnetic force applied to the wire is 4.76 N. **Calculate** the magnitude of the magnetic field.

$$\vec{F}_b = I \vec{l} \times \vec{B} \quad \boxed{12}$$

$$|\vec{F}_b| = |I| |\vec{l}| |\vec{B}| \sin \theta$$

$$|\vec{F}_b| = |I| |\vec{l}| |\vec{B}| \sin(90^\circ)$$

$$F_b = IlB$$

$$B = \frac{F_b}{Il}$$

$$B = \frac{4.76 \text{ N}}{(1.65 \text{ m})(127 \text{ A})}$$

$$B = 0.0227 \text{ T}$$

2 Correct Cross Product to include Sin=90

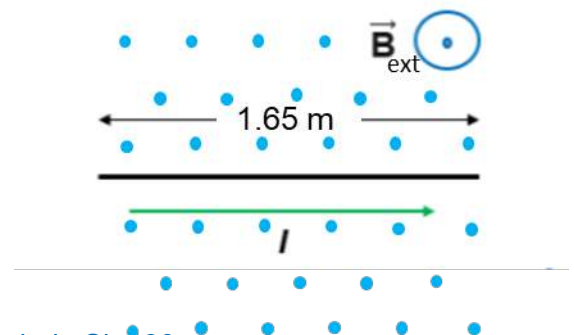
1 Subs F_b

1 Subs I

1 Subs L

1 Sig Figs Report your answer in this box.	Report units in this box.
--	---

1 Correct ANS



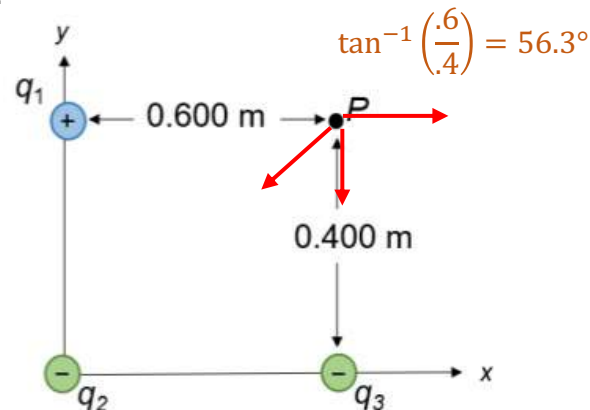
A: 4 (11%)
B: 10 (29%)
D: 21 (60%)
Total: 35

Wt. No.

Cadet: _____

- 50 3. Three charged particles are arranged as shown in the figure below, where $q_1 = +3.00 \text{ nC}$, $q_2 = -6.35 \text{ nC}$, and $q_3 = -3.45 \text{ nC}$.

- (40) a. **Calculate** the net electric field at point P due to q_1 , q_2 , and q_3 . Report your answer in unit-vector notation.



$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

24 6 Any Electric Field EQN

$$\vec{E}_{net} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$$

$$\vec{E}_{net,x} = k \frac{q_1}{r_1^2} - k \frac{q_2}{r_2^2} \sin\theta$$

4 Attempts to solve for $E_{net,x}$

$$\vec{E}_{net,y} = -k \frac{q_3}{r_3^2} - k \frac{q_2}{r_2^2} \cos\theta$$

4 Attempts to solve for $E_{net,y}$

$$\vec{E}_{net,x} = k \left(\frac{(3.00 \times 10^{-9} \text{ C})}{(0.600 \text{ m})^2} - \frac{(6.35 \times 10^{-9} \text{ C})}{(0.721)^2} \sin(\theta) \right) = -16.44$$

2 Correct $E_{net,x}$

$$\vec{E}_{net,y} = -k \left(\frac{(3.45 \times 10^{-9} \text{ C})}{(0.400 \text{ m})^2} + \frac{(6.35 \times 10^{-9} \text{ C})}{(0.721 \text{ m})^2} \cos(\theta) \right) = -254.7$$

2 Correct $E_{net,y}$

$$\vec{E}_{net} = (-16.4 \hat{i} - 255 \hat{j}) \frac{\text{N}}{\text{C}}$$

2 Correctly combines into unit-vector notation

1 Units	
1 Sig Figs	
Report your answer in this box.	Report units in this box.

- (10) b. A charge, q_4 , of 4.25 nC is now placed at point P. **Calculate** the electrostatic force on q_4 . Report your answer in unit-vector notation.

$$\vec{F} = q\vec{E}$$

6

$$\vec{F} = (4.25 \times 10^{-9} \text{ C}) (-16.4 \hat{i} - 255 \hat{j}) \frac{\text{N}}{\text{C}}$$

2 Replaces E and q with correct values, including signs

$$\vec{F} = (-6.98 \times 10^{-8} \hat{i} - 1.08 \times 10^{-6} \hat{j}) \text{ N}$$

1 Correct answer in unit-vector notation	
1 Units	
Report your answer in this box.	Report units in this box.

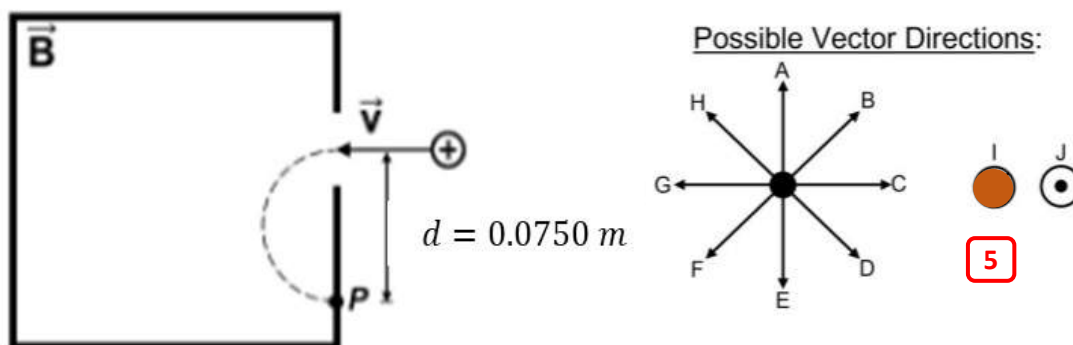
A: 6 (12%)
B: 14 (28%)
D: 30 (60%)
Total: 50

Wt. No.

Cadet: _____

- 50 4. A positively-charged particle of charge $2.12 \mu\text{C}$ and mass $3.52 \times 10^{-14} \text{ kg}$ enters a stationary box at a speed 255 m/s as shown in the figure below. Inside the box there is a uniform magnetic field of unknown magnitude or direction. The particle travels in a circular path once it enters the box and collides with the box wall at point P . Point P is located a distance of 0.0750 m directly below where the particle enters the box.

- (5) a. **Circle** the vector (A through J) which best describes the direction of the magnetic field inside the box necessary for the positively charged particle to collide with the box at point P .



- (45) b. **Calculate** the magnitude of the uniform magnetic field.

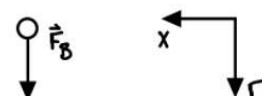
$$\sum \vec{F} = m\vec{a} \quad \vec{F}_B = q\vec{v} \times \vec{B} \quad \vec{a}_c = \frac{v^2}{r}$$

8

15

5

Solve:



$$\vec{F}_B = q\vec{v} \times \vec{B} \quad \vec{F}_B = q|\vec{v}||\vec{B}|\sin\theta \quad \vec{F}_B = q|\vec{v}||\vec{B}|\sin(90)$$

$$\sum \vec{F} = \vec{F}_B = m\vec{a}_c \quad \vec{F}_B = \frac{mv^2}{r}$$

$$q|\vec{v}||\vec{B}| = \frac{mv^2}{r} \quad |\vec{B}| = \frac{mv}{rq}$$

$$|\vec{B}| = \frac{(3.52 \times 10^{-14})(255)}{(\frac{.075}{2})(2.12 \times 10^{-6})}$$

$$\vec{B} = 1.13 \times 10^{-4} \text{ T}$$

Report your answer in this box.

Report units in this box.

2 Answer

1 Sig figs

1 Units

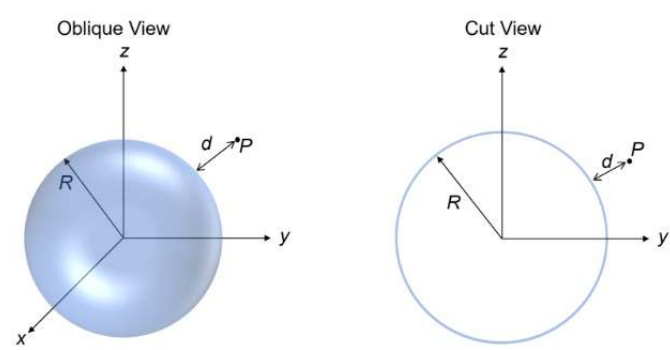
5

A: 4 (8%)
B: 13 (26%)
D: 33 (66%)
Total: 50

Wt. No.

Cadet: _____

- 65 5. A spherical conductive shell with a radius of 10.2 cm and a uniform surface charge density is shown in the figure below. Point P is located at a radial distance of 5.08 cm from the spherical shell as shown in the figure. The magnitude of the electric field at point P is $4.71 \times 10^6 \frac{N}{C}$. **Calculate** the surface charge density.



Identifies Distance from center to point P
 $r = R + d$

4 IDs distance to point P

26 $\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$

$\Phi_E = \int \vec{E} \cdot d\vec{A}$ 10 Concept points

12 $\sigma = \frac{Q}{A}$

$\vec{E} \cdot \vec{A}_{GS} = \frac{q_{enc}}{\epsilon_0}$

$|\vec{E}| |\vec{A}_{GS}| \cos 0^\circ = \frac{q_{enc}}{\epsilon_0}$

$A = 4\pi R^2$

$q_{enc} = \sigma 4\pi R^2$

$|\vec{E}| |\vec{A}_{GS}| = \frac{\sigma 4\pi R^2}{\epsilon_0}$

$|\vec{E}| 4\pi r^2 = \frac{\sigma 4\pi R^2}{\epsilon_0}$

$\sigma = \frac{\epsilon_0 |\vec{E}| (R + d)^2}{R^2} = \frac{(8.854 \times 10^{-12})(4.71 \times 10^6)(.102 + .0508)^2}{(.102)^2}$

$\sigma = 9.36 \times 10^{-5} \frac{C}{m^2}$

2 Answer	
2 Sig Figs	
2 units	

Report your answer in this box. Report units in this box.

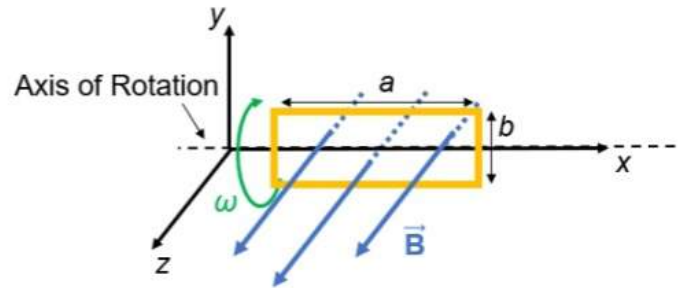
- 3 correctly resolves dot product
- 4 Correct equation for A shell
- 2 Attempts to solve for q enclosed
- 2 correctly solves for q enclosed
- 4 Attempts to combine Electric flux with charge enclosed
- 3 Attempts to solve for σ

A: 5 (8%)
B: 22 (34%)
D: 38 (58%)
Total: 65

Wt. No.

Cadet: _____

- 50 6. A rectangular loop is located in a uniform magnetic field with a strength of 1.25 T that is perpendicular to the axis of rotation as shown in the figure below. The loop has sides of length $a = 17.1 \text{ cm}$ and $b = 12.3 \text{ cm}$. The loop is rotated about the x-axis at a constant angular frequency of 4.88 rad/s . At time $t = 0$, the loop is in the xy-plane. Calculate the magnitude of the emf induced in the loop at time $t = 0.235 \text{ s}$. Show all work.



$$\Phi_B = \int \vec{B} \cdot d\vec{A} \quad \boxed{15}$$

$$\varepsilon = -N \frac{d\Phi_B}{dt} \quad \boxed{15}$$

$$\Phi_B = \int |\vec{B}| |d\vec{A}| \cos\theta$$

$$\Phi_B = BA \cos\theta$$

$$\varepsilon = -N \frac{d}{dt} (BA \cos(\omega t))$$

$$\varepsilon = NBA \omega \sin\omega t$$

$$\varepsilon = Bab \omega \sin\omega t$$

③ Correctly resolves dot product

③ Attempts to integrate

③ Correct Flux

② IDs theta=wt

② Attempts derivative

③ Attempts to solve for emf

$$\varepsilon = (1.25 \text{ T})(0.171 \text{ m})(0.123 \text{ m}) \left(4.88 \frac{\text{rad}}{\text{s}} \right) \sin \left(\left(4.88 \frac{\text{rad}}{\text{s}} \right) (0.235 \text{ s}) \right)$$

$$\varepsilon = 0.117 \text{ V}$$

② Answer

① Units

① SF

Report your answer in this box.	Report units in this box.
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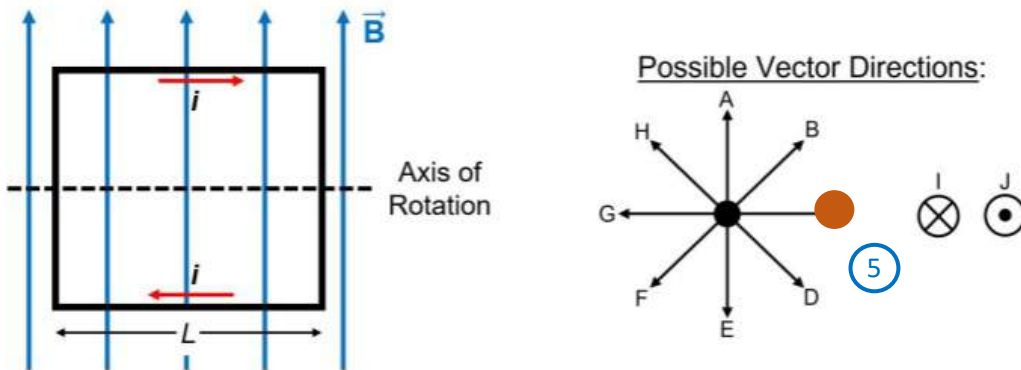
A: 4 (8%)
B: 16 (32%)
D: 30 (60%)
Total: 50

Wt. No.

Cadet: _____

- 40 7. A 250.-turn square wire loop carries a current of 1.63 A as shown in the figure below. The loop is in a uniform magnetic field of 2.15 mT directed up. The current-carrying loop experiences a maximum torque of magnitude 2.75 Nm.

- (5) a. **Circle** the vector direction (A through J) which best describes the direction of the torque on the current-carrying loop at the position shown.



- (35) b. **Calculate** the length L of one of the sides of the current-carrying square-loop.

$$\vec{\tau} = \vec{\mu} \times \vec{B} \quad \boxed{12}$$

$$\vec{\mu} = NIA\hat{n} \quad \boxed{12}$$

$$\vec{\tau} = \vec{\mu} \times \vec{B} = |\vec{\mu}| |\vec{B}| \sin\theta \quad \theta = 90^\circ \quad \textcircled{2} \text{ Correct cross product}$$

$$\vec{\tau} = |NIA| |\vec{B}| \quad \textcircled{2} \text{ IDs } \theta=90^\circ \text{ (if they get to correct algebraic expression, they get all 4 points)}$$

$$\tau = NIl^2B$$

$$\textcircled{1} \text{ Attempts to solve for length}$$

$$L = \sqrt{\frac{\tau}{NIB}}$$

$$\textcircled{3} \text{ IDs } A = l^2$$

$$L = \sqrt{\frac{(2.75 \text{ Nm})}{(250)(1.63 \text{ A})(2.15 \times 10^{-3} \text{ T})}}$$

$$L = 1.77 \text{ m}$$

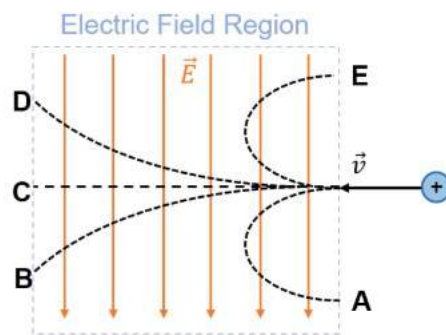
1 Sig Figs	1 Units
Report your answer in this box.	Report units in this box.

A: 3 (8%)
B: 13 (32%)
D: 24 (60%)
Total: 40

Wt. **No.**

Cadet: _____

- 5 8. A proton is fired into a region with a uniform electric field directed toward the bottom of the page as shown in the figure below. **Determine** which of the dashed lines represents the most likely path for the proton.



☐ A

☒ B

☐ C

☐ D

☐ E

- 5 9. The figure below shows a loop of wire moving through a region of uniform magnetic field, $\vec{B}$, at a constant speed v as shown in the figure below. The direction of motion of the loop is perpendicular to the field. **Determine** the direction of the induced current in the wire loop.

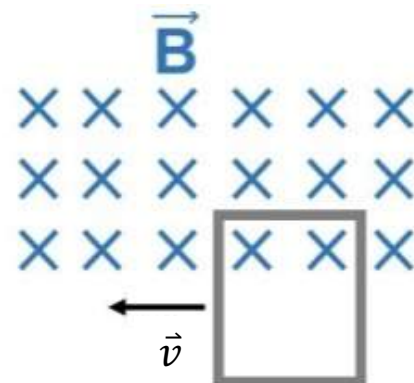
☐ Clockwise

☐ Counterclockwise

☐ Into the page

☐ Out of the page

☒ No current is induced



A: 0 (0%)
B: 5 (50%)
D: 5 (50%)
Total: 10

A: 0 (0%)
B: 0 (0%)
D: 0 (0%)
Total: 0



UNITED STATES MILITARY ACADEMY
WEST POINT.

WPR #2 HANDOUT

Time of completion _____
15 April 2026

Cadet _____

Section _____
Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
PH2X2, Physics II
Written Partial Review II

1. Instructions:

- a. Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. No phones, cameras, or smartwatches are allowed in the examination room.
- b. Check your exam for 10 problems. Check for the Physics Reference Card provided with the examination.
- c. **Show all work**; partial credit may be given for correct work shown.
- d. For multiple choice questions **bubble in** your selection and choose the **one** answer that best answers each question.
- e. Take up to 75 minutes to complete the examination.
- f. An instructor is in the hall for any questions.

2. Exam Security:

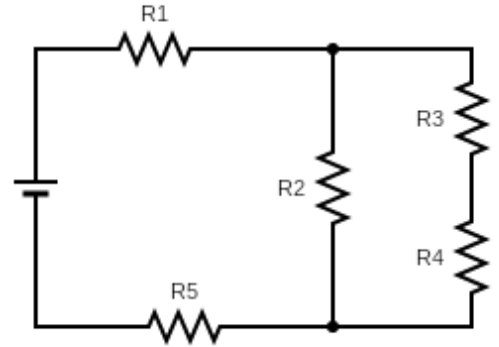
- a. I will not discuss this exam with anyone until it is released from academic security on _____ (date) at _____ (time).
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

PROBLEM	WEIGHT	SCORE
1	20	
2	20	
3	50	
4	45	
5	50	
6	50	
7	45	
8	10	
9	5	
10	5	
	Score:	
	Total:	300

Wt. **No.**

Cadet: _____

- 20 1. In the circuit shown, $R_1 = 175 \, \Omega$, $R_2 = 205 \, \Omega$, $R_3 = 125 \, \Omega$, $R_4 = 355 \, \Omega$, and $R_5 = 215 \, \Omega$. **Calculate** the equivalent resistance of the circuit.



Report your answer in this box.	Report units in this box.
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- 20 2. A parallel plate capacitor has square plates with a side length of $25.0 \, \text{mm}$. The plates are $1.05 \, \text{mm}$ apart with an unknown material between them. The capacitance of the capacitor is $18.5 \, \text{pF}$. **Calculate** the dielectric constant of the material.

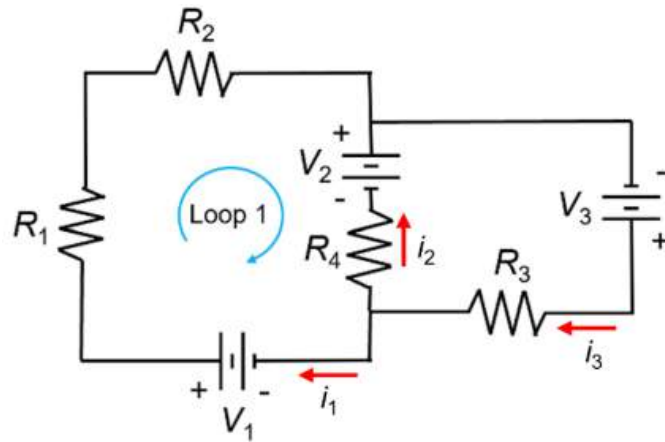
Report your answer in this box.



Wt. **No.**

Cadet: _____

- 50 3. In the multiloop circuit shown below, $R_1 = 64.0 \, \Omega$, $R_2 = 31.0 \, \Omega$, $R_3 = 15.0 \, \Omega$, $R_4 = 35.0 \, \Omega$, $V_1 = 115 \, \text{V}$, $V_2 = 62.5 \, \text{V}$, and $i_1 = 2.55 \, \text{A}$.



- (25) a. Using loop 1, **calculate** the current i_2 .

Report your answer in this box.	Report units in this box.
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- (25) b. **Calculate** the rate at which R_3 dissipates energy.

Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

Wt. **No.**

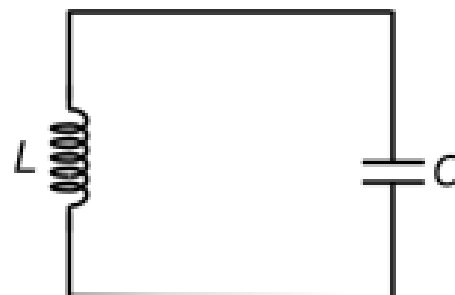
Cadet: _____

45

4.

A fully charged capacitor with a capacitance of $785 \mu F$ is placed in a circuit with an inductor as shown below. The potential difference across the fully charged capacitor is $2.75 V$. The maximum current through the inductor is $1.12 A$.

Calculate the inductance of the inductor.

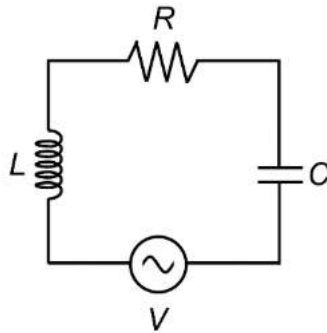


Report your answer in this box.	Report units in this box.
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Wt. **No.**

Cadet: _____

- 50 5. A series RLC circuit consists of a $25.0\ \Omega$ resistor, a $515\ \mu F$ capacitor, and a $33.6\ mH$ inductor. The AC power source for the circuit operates at a frequency of $51.3\ Hz$ and the rms current through the circuit is $1.17\ A$.



- (10) a. **Calculate** the capacitive reactance of the circuit.

Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

- (10) b. **Calculate** the inductive reactance of the circuit.

Report your answer in this box.	Report units in this box.
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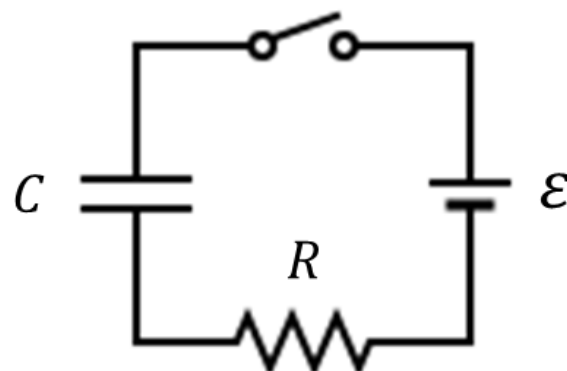
- (30) c. **Calculate** the peak voltage across the power source.

Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

Wt. **No.**

Cadet: _____

- 50 6. An initially uncharged capacitor with a capacitance of $212 \mu F$ is placed in series with a 1250Ω resistor and a 48.0 V battery. The switch is initially open, and at $t = 0 \text{ s}$ it is switched to the closed position. **Calculate** the magnitude of the current through the resistor after 0.355 s .

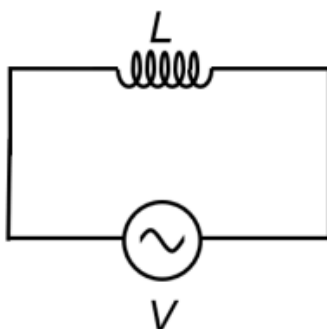


Report your answer in this box.	Report units in this box.
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Wt. **No.**

Cadet: _____

- 45 7. A 705 mH inductor is connected to an AC power source that operates at a frequency of 62.5 Hz . The power source has a peak voltage of 518 V . **Calculate** the peak current in the circuit.



Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

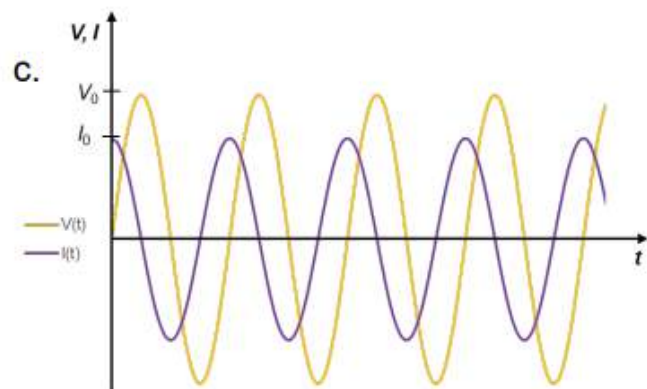
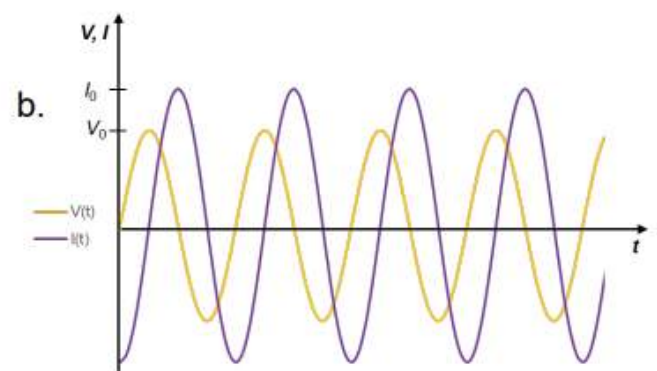
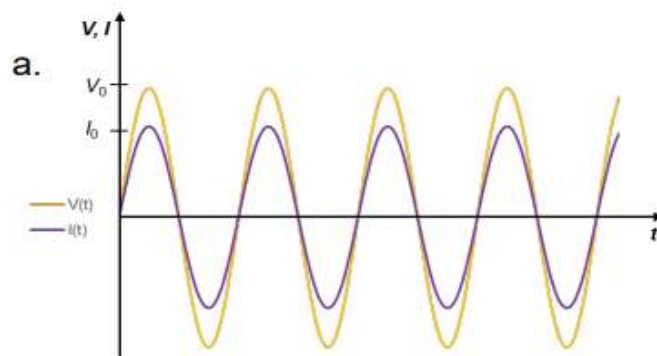
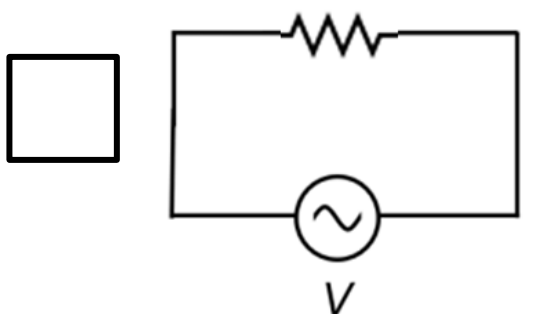
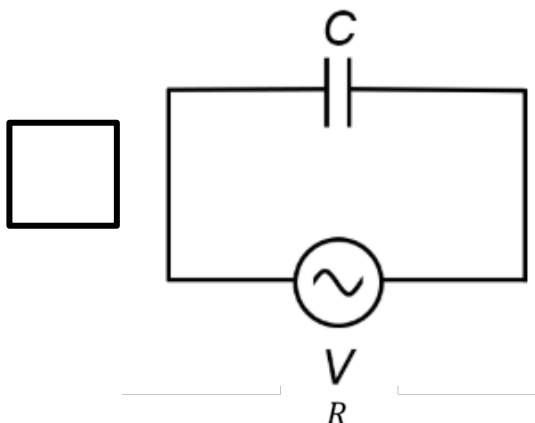
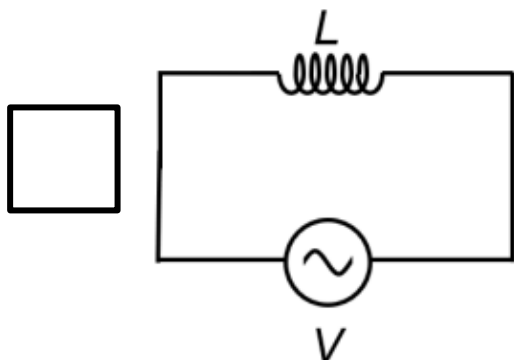
Wt. **No.**

Cadet: _____

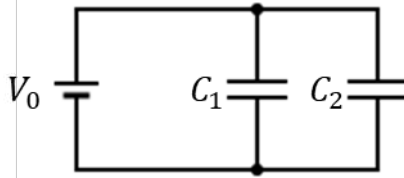
10

8.

Below are plots depicting both the current, $I(t)$, and potential difference, $V(t)$, across the power source for the three simple AC circuits. **Match** each circuit with the most appropriate graph. Write the letter corresponding to your choice in the box to the left of the circuits below.



- 5 9. Consider the circuit below. **Select** the statement that best describes the charge on and the potential differences across the capacitors. The capacitors have different capacitances. Assume the capacitors are fully charged.



- ☐ The charge on and the potential differences across the capacitors are both the same.
- ☐ The charge on and the potential differences across the capacitors are both different.
- ☐ The charge on the capacitors is different and the potential differences across the capacitors are the same.
- ☐ The charge on the capacitors is the same and the potential differences across the capacitors are different.
-
- 5 10. The current passing through the primary side of an ideal transformer is twice the value as the current on the secondary side. Select the answer which best describes the potentials across the primary and secondary sides of the transformer.
- ☐ The potential across the primary side is double the potential across the secondary side
- ☐ The potential across the primary side is half the potential across the secondary side.
- ☐ The potentials across the primary and secondary sides of the transformers is the same.
- ☐ Not enough information is provided.



UNITED STATES MILITARY ACADEMY
WEST POINT.

WPR #2 SOLUTION

Time of completion _____
15 April 2026

Cadet _____

Section _____
Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
PH2X2, Physics II
Written Partial Review II

1. Instructions:

- a. Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. No phones, cameras, or smartwatches are allowed in the examination room.
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- c. **Show all work**; partial credit may be given for correct work shown.
- d. For multiple choice questions **bubble in** your selection and choose the **one** answer that best answers each question.
- e. Take up to 75 minutes to complete the examination.
- f. An instructor is in the hall for any questions.

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on _____ (date) at _____ (time).
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

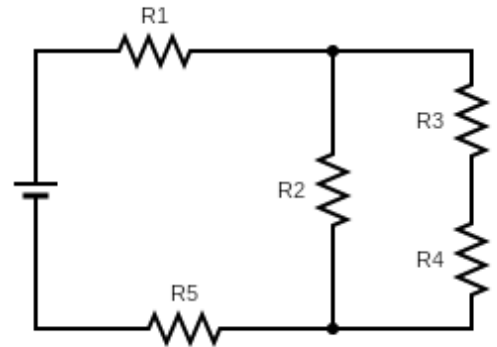
PROBLEM	WEIGHT	SCORE
1	20	
2	20	
3	50	
4	45	
5	50	
6	50	
7	45	
8	10	
9	5	
10	5	
	Score:	
	Total:	300

Wt. No.

Cadet: _____

- 20 1. In the circuit shown, $R_1 = 175 \, \Omega$, $R_2 = 205 \, \Omega$, $R_3 = 125 \, \Omega$, $R_4 = 355 \, \Omega$, and $R_5 = 215 \, \Omega$. **Calculate** the equivalent resistance of the circuit.

$$R_{eq} = 534 \, \Omega$$



Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

- 20 2. A parallel plate capacitor has square plates with a side length of $25.0 \, \text{mm}$. The plates are $1.05 \, \text{mm}$ apart with an unknown material between them. The capacitance of the capacitor is $18.5 \, \text{pF}$. **Calculate** the dielectric constant of the material.

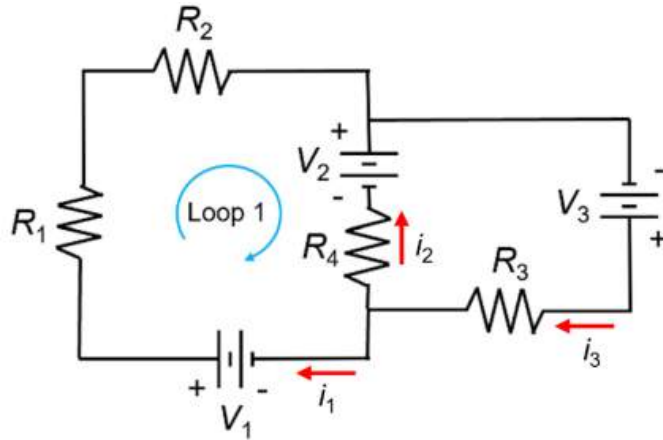
$$\kappa = 3.51$$

Report your answer in this box.

Wt. **No.**

Cadet: _____

- 50 3. In the multiloop circuit shown below, $R_1 = 64.0 \, \Omega$, $R_2 = 31.0 \, \Omega$, $R_3 = 15.0 \, \Omega$, $R_4 = 35.0 \, \Omega$, $V_1 = 115 \, \text{V}$, $V_2 = 62.5 \, \text{V}$, and $i_1 = 2.55 \, \text{A}$.



- (25) a. Using loop 1, **calculate** the current i_2 .

$$I_2 = 5.42 \, \text{A}$$

Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

- (25) b. **Calculate** the rate at which R_3 dissipates energy.

$$P_{R3} = 953 \, \text{W}$$

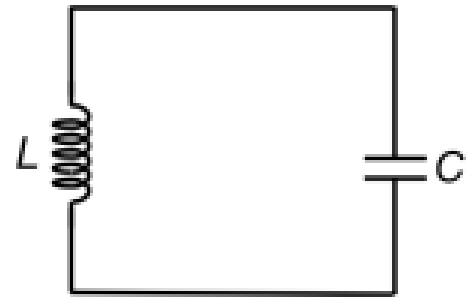
Report your answer in this box.	Report units in this box.
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Wt. **No.**

Cadet: _____

- 45 4. A fully charged capacitor with a capacitance of $785 \mu F$ is placed in a circuit with an inductor as shown below. The potential difference across the fully charged capacitor is $2.75 V$. The maximum current through the inductor is $1.12 A$. **Calculate** the inductance of the inductor.

$$L = 4.73 * 10^{-3} \text{ H}$$

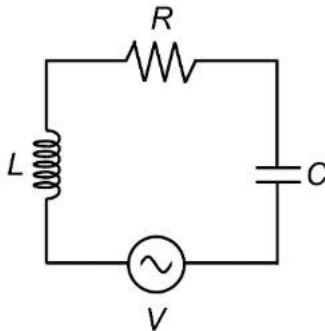


Report your answer in this box.	Report units in this box.
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Wt. **No.**

Cadet: _____

- 50 5. A series RLC circuit consists of a $25.0\ \Omega$ resistor, a $515\ \mu F$ capacitor, and a $33.6\ mH$ inductor. The AC power source for the circuit operates at a frequency of $51.3\ Hz$ and the rms current through the circuit is $1.17\ A$.



- (10) a. **Calculate** the capacitive reactance of the circuit.

$$X_C = 6.02\ \Omega$$

Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

- (10) b. **Calculate** the inductive reactance of the circuit.

$$X_L = 10.8\ \Omega$$

Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

- (30) c. **Calculate** the peak voltage across the power source.

$$V_0 = 42.1\ V$$

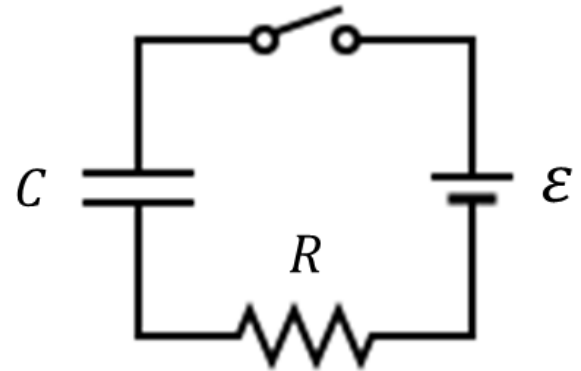
Report your answer in this box.	Report units in this box.
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Wt. **No.**

Cadet: _____

- 50 6. An initially uncharged capacitor with a capacitance of $212 \mu F$ is placed in series with a 1250Ω resistor and a 48.0 V battery. The switch is initially open, and at $t = 0 \text{ s}$ it is switched to the closed position. **Calculate** the magnitude of the current through the resistor after 0.355 s .

$I(t) = 0.0101 \text{ A or } 10.1 \text{ mA}$

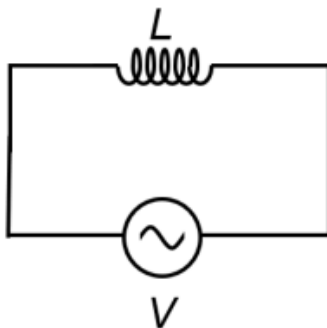


Report your answer in this box.	Report units in this box.
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Wt. **No.**

Cadet: _____

- 45 7. A 705 mH inductor is connected to an AC power source that operates at a frequency of 62.5 Hz . The power source has a peak voltage of 518 V . **Calculate** the peak current in the circuit.



$$I_o = 1.87\text{ A}$$

Report your answer in this box.	Report units in this box.
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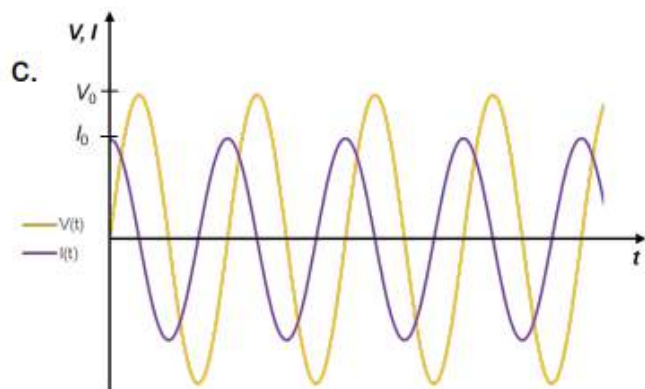
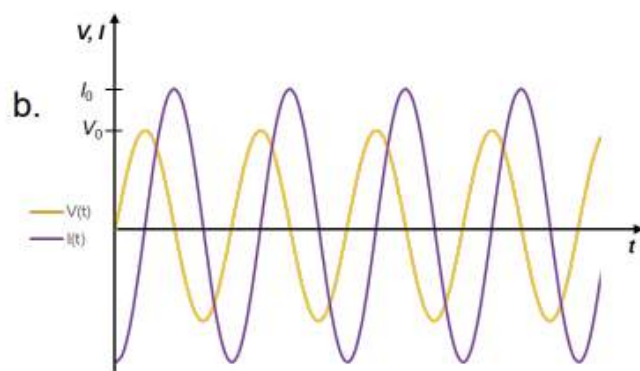
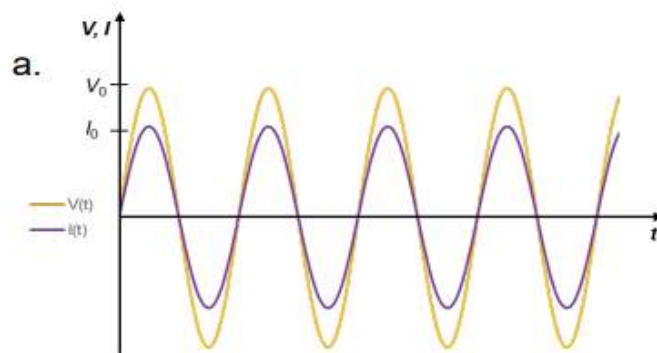
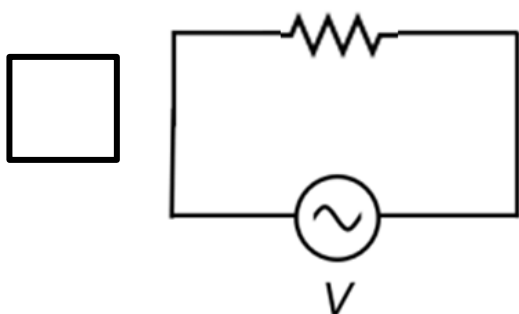
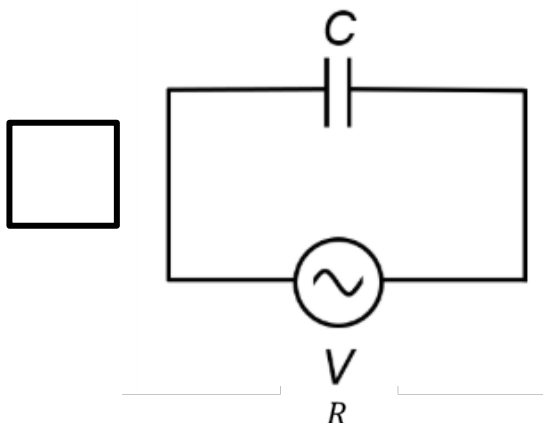
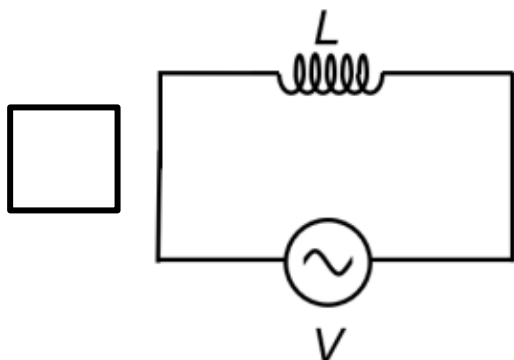
Wt. **No.**

Cadet: _____

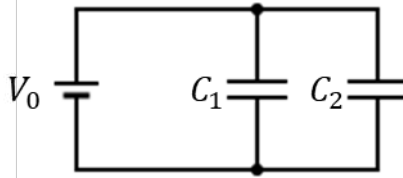
10

8.

Below are plots depicting both the current, $I(t)$, and potential difference, $V(t)$, across the power source for the three simple AC circuits. **Match** each circuit with the most appropriate graph. Write the letter corresponding to your choice in the box to the left of the circuits below.



- 5 9. Consider the circuit below. **Select** the statement that best describes the charge on and the potential differences across the capacitors. The capacitors have different capacitances. Assume the capacitors are fully charged.



- ☐ The charge on and the potential differences across the capacitors are both the same.
- ☐ The charge on and the potential differences across the capacitors are both different.
- ☐ The charge on the capacitors is different and the potential differences across the capacitors are the same.
- ☐ The charge on the capacitors is the same and the potential differences across the capacitors are different.
-
- 5 10. The current passing through the primary side of an ideal transformer is twice the value as the current on the secondary side. Select the answer which best describes the potentials across the primary and secondary sides of the transformer.
- ☐ The potential across the primary side is double the potential across the secondary side
- ☐ The potential across the primary side is half the potential across the secondary side.
- ☐ The potentials across the primary and secondary sides of the transformers is the same.
- ☐ Not enough information is provided.



TAB G

TEE



UNITED STATES MILITARY ACADEMY
WEST POINT.

TEE HANDOUT

Time of completion _____
May 2026

Cadet _____

Section _____
Version 2

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
PH2X2, Physics II
Term End Examination

1. Instructions:

- a. Take up to 3 hours and 30 minutes to complete the examination. If you finish before the time expires, record your time of completion above.
- b. Bring only your issued calculator, a straight-edge, and pencil(s) into the examination room. No phones, cameras, or smart watches are allowed in the examination room.
- c. Check your exam for eighteen problems on fifteen numbered pages. Check for the Physics Reference Card provided with the examination.
- d. For calculation problems, show all work.
- e. You may assume that all components in the circuits on this examination are ideal.
- f. An instructor is in the hall if you have any questions.

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on _____ (date) at _____ (time).
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

PROBLEM	WEIGHT	SCORE	PROBLEM	WEIGHT	SCORE
1	40		12	5	
2	60		13	5	
3	60		14	5	
4	60		15	5	
5	60		16	5	
6	60		17	5	
7	55		18	5	
8	50				
9	55				
10	60			Score:	
				Total:	600

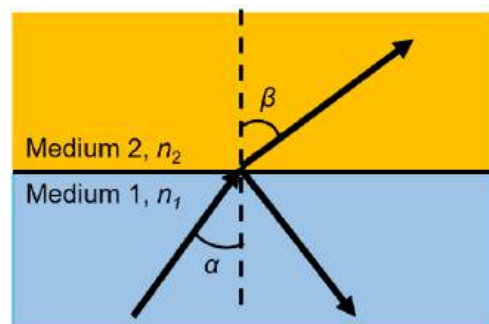
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Wt. **No.**

Cadet: _____

40 1. In the diagram shown, $n_1 = 1.33$ and $n_2 = 1.20$.

(25) a. **Calculate** the refracted angle, β , such that the incident angle, α , is equal to 35.5° .



Report your answer in this box.	Report units in this box.
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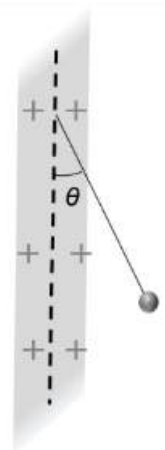
(15) b. **Calculate** the critical angle for the scenario shown above.

Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

Wt. **No.**

Cadet: _____

- 60 2. An infinite sheet of charge has a uniform charge density of $12.7 \mu\text{C}/\text{m}^2$. A small sphere with a mass of $6.75 \times 10^{-5} \text{ kg}$ hangs from a massless string. The string makes an angle of 23.8° with the sheet. **Calculate** the charge of the sphere.



Report your answer in this box.	Report units in this box.
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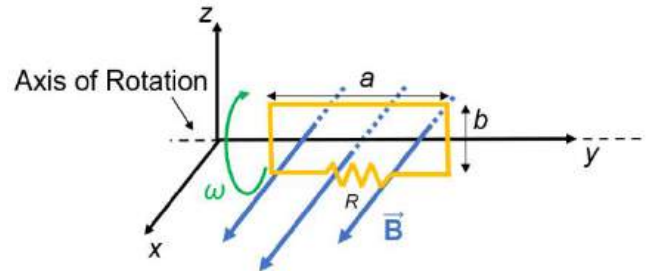
Wt. **No.**

Cadet: _____

60

3.

A rectangular wire loop has a resistance of $15.9 \, \Omega$. The loop is located in a $3.85 \, \text{T}$ uniform magnetic field directed in the positive x -direction. The loop has sides of length $a = 0.573 \, \text{m}$ and $b = 0.321 \, \text{m}$. The loop is rotated about the y -axis at a constant angular frequency, $\omega = 9.32 \, \text{rad/s}$. At time $t = 0$, the loop is in the yz -plane. **Calculate** the magnitude of the current induced in the loop at time $t = 0.751 \, \text{s}$.



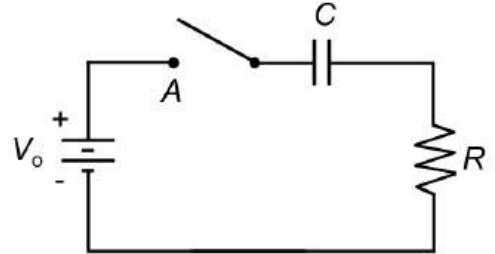
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Wt. **No.**

Cadet: _____

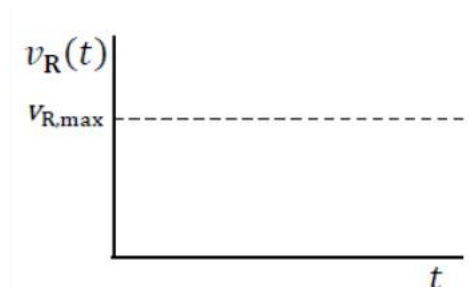
60 4. In the RC circuit shown below, the power supply has a potential difference of 62.6 V, the resistance is 135 Ω , and the capacitance is 476 μF . Initially, the capacitor is fully discharged. At time $t = 0.00$ s, the switch is closed in position A.

(50) a. **Calculate** the potential difference across the resistor at time $t = 0.525$ s.



Report your answer in this box.	Report units in this box.
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(10) b. **Draw** the appropriate graph for the potential difference across the resistor with respect to time.



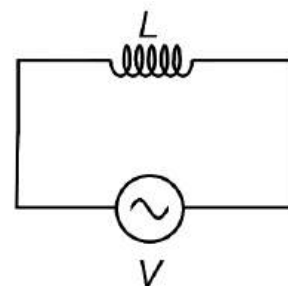
Wt. **No.**

Cadet: _____

60

5.

An 8.46-mH ideal inductor is connected to an AC power source that operates at a frequency of 75.4 Hz with a peak emf of 114 V. **Calculate** the magnitude of the peak current through the inductor.



Report your answer in this box.

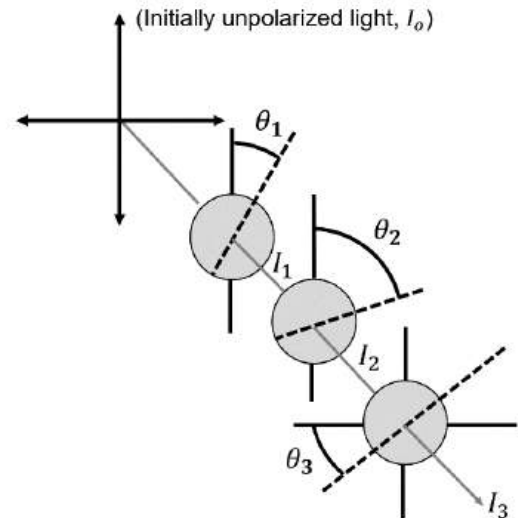
Report units in this box.

Wt. **No.**

Cadet: _____

60

6. Initially, unpolarized light with an unknown intensity is incident upon the first of a series of three polarizing sheets. The polarizing direction of the first two sheets makes angles of $\theta_1 = 30.0^\circ$ and $\theta_2 = 60.0^\circ$ with respect to the vertical, as shown in the figure. The last sheet makes an angle of $\theta_3 = 50.0^\circ$ with respect to the horizontal. The light that emerges from the third polarizing sheet has an intensity of 208 W/m^2 . **Calculate** the intensity of the incident light.

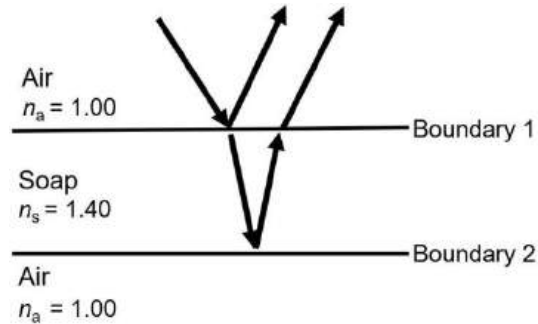


Report your answer in this box.	Report units in this box.
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Wt. **No.**

Cadet: _____

- 55 7. A soap bubble ($n_s = 1.40$) floats in air ($n_a = 1.00$). The soap bubble is illuminated from above by sunlight at the normal incidence.



- (5) a. **Circle** the choice that describes the phase shift of the reflected wave that occurs at Boundary 1 between air and the soap.

Phase shift by $\pi/2$ radians

No phase shift

Phase shift by π radians

- (5) b. **Circle** the choice that describes the phase shift of the reflected wave that occurs at Boundary 2 between the soap and the air.

Phase shift by $\pi/2$ radians

No phase shift

Phase shift by π radians

- (45) c. **Calculate** the second thinnest thickness of soap that minimizes the reflection of incident light with wavelength $\lambda = 623$ nm.

Report your answer in this box.	Report units in this box.
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Wt. **No.**

Cadet: _____

50 8. An object is located on the central axis of a concave mirror. The object is 5.10 cm tall and is located 26.8 cm from the mirror. The distance from the mirror to the focal point is 10.2 cm.

(20) a. **Calculate** the distance from the mirror to the image.

Report your answer in this box.	Report units in this box.
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(25) b. **Calculate** the height of the image.

Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

(5) c. **State** whether the image is real or virtual. _____

Wt. **No.**

Cadet: _____

- 55 9. Red light ($\lambda = 653 \text{ nm}$) incident on two narrow slits that are $d = 0.225 \text{ mm}$ apart, causing an interference pattern on a screen a distance L away. The second maxima for the interference pattern of the red light is a distance $y = 1.13 \text{ mm}$ from the central maxima. **Calculate** the distance, L , from the two slits to the screen.

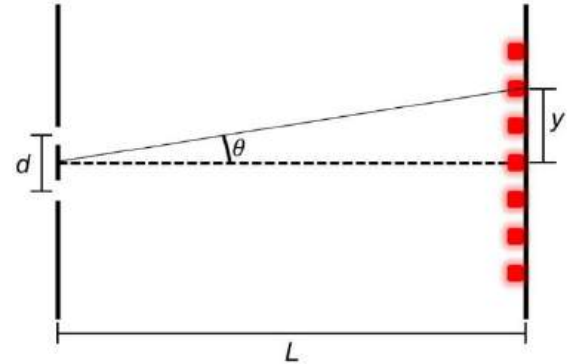


Figure not to scale

Report your answer in this box.

Report units in this box.

Wt. **No.**

Cadet: _____

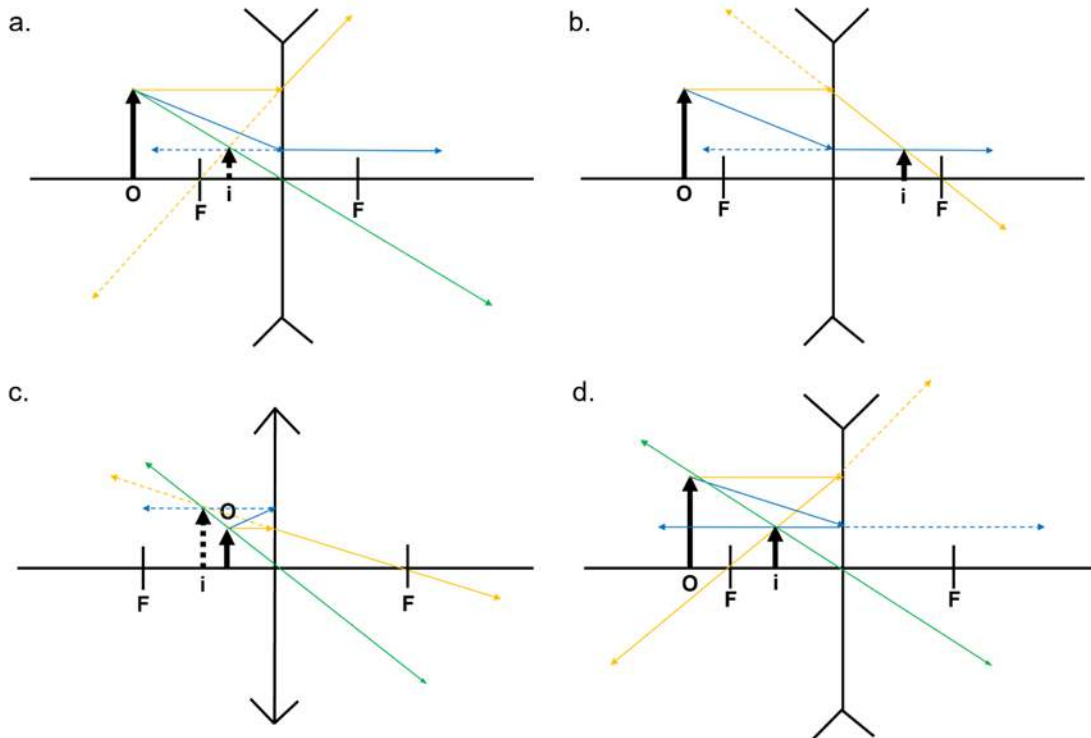
- 60 10. In a demonstration of the photoelectric effect, a photon with a frequency of 3.61×10^{15} Hz is incident on a metallic material with a work function of 7.25 eV.
- (10) a. **Calculate** the energy of the incident photons. Express your answer in joules.

Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

- (50) b. **Calculate** the maximum speed of the electrons ejected from this material.

Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

- 5 11. Consider the figures shown below. **Circle** the figure that corresponds to the correct 3-ray diagram for a diverging lens.



- 5 12. The magnetic field of an electromagnetic wave oscillates along the y-axis and is described by $B_y(z, t) = B_o \sin(kz - \omega t)$. **Select** the answer below which best describes the wave's direction of travel.

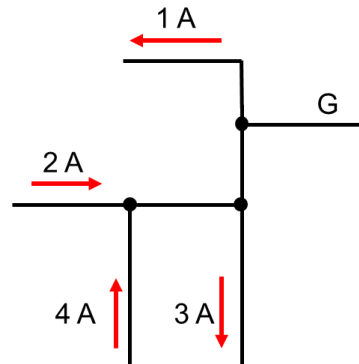
- ☐ + x-direction.
- ☐ - x-direction.
- ☐ + y-direction.
- ☐ - y-direction.
- ☐ + z-direction.
- ☐ - z-direction.

Wt. **No.**

Cadet: _____

- 5 13. A portion of a steady state circuit is shown below. **Select** the statement which best represents the magnitude and direction of the flow of current in the section of wire labeled G.

- ☐ 2 A to the left.
☐ 2 A to the right.
☐ 4 A to the left.
☐ 4 A to the right.



- 5 14. The three equations below represent three waves traveling through the same medium.

$$y(x, t) = 2\sin(4x - 4t) \quad (1)$$

$$y(x, t) = 5\sin(3x - 5t) \quad (2)$$

$$y(x, t) = 3\sin(2x - 6t) \quad (3)$$

Rank the three waves according to their wavelength, longest first.

_____ > _____ > _____

- 5 15. Monochromatic light of wavelength λ is used in Young's double-slit experiment. A bright spot will be observed on the screen when the difference in path length from the slits to the screen is:

- ☐ 1λ
☐ 1.22λ
☐ 1.5λ
☐ All of the above.
☐ Not enough information to determine.

Wt. **No.**

Cadet: _____

- 5 16. In an oscillating LC circuit, the charge in the capacitor changes from having maximum positive charge on one plate to having maximum positive charge on the other plate in time t . **Select** the relationship between the time t and the period of oscillation T .
- ☐ $t = T$
- ☐ $t = \frac{1}{2}T$
- ☐ $t = \frac{1}{4}T$
- ☐ $t = 2T$
- ☐ Not enough information about the circuit is known
-
- 5 17. A basic law of geometric optics states that the angle of incidence is always equal to the angle of refraction.
- ☐ True
- ☐ False
-
- 5 18. **Select** the electromagnetic wave below with the shortest wavelength.
- ☐ Radio waves ($f = 3.0 \times 10^6$ Hz)
- ☐ Infrared waves ($f = 3.0 \times 10^9$ Hz)
- ☐ Visible light ($f = 3.0 \times 10^{12}$ Hz)
- ☐ Gamma rays ($f = 3.0 \times 10^{19}$ Hz)
- ☐ All have equal wavelengths



UNITED STATES MILITARY ACADEMY
WEST POINT.

TEE SOLUTION

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
PH2X2, Physics II
Term End Examination

1. Instructions:
- a. Take up to 3 hours and 30 minutes to complete the examination. If you finish before the time expires, record your time of completion above.

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c. Check your exam for eighteen problems on fifteen numbered pages. Check for the Physics Reference Card provided with the examination.

d. For calculation problems, show all work.

e. You may assume that all components in the circuits on this examination are ideal.

f. An instructor is in the hall if you have any questions.
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- a. I will not discuss this exam with anyone until it is released from academic security on _____ (date) at _____ (time).

b. Cadet Signature: _____

c. Cadet Name (print): _____

PROBLEM	WEIGHT	SCORE	PROBLEM	WEIGHT	SCORE
1	40		12	5	
2	60		13	5	
3	60		14	5	
4	60		15	5	
5	60		16	5	
6	60		17	5	
7	55		18	5	
8	50				
9	55				
10	60				
				Score:	
				Total:	

A: 57 (9%)
600

B: 167 (28%)

D: 376 (63%)

Total: 600

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Wt. No.

Cadet: _____

40 1. In the diagram shown, $n_1 = 1.33$ and $n_2 = 1.20$.

(25) a. **Calculate** the refracted angle, β , such that the incident angle, α , is equal to 35.5° .

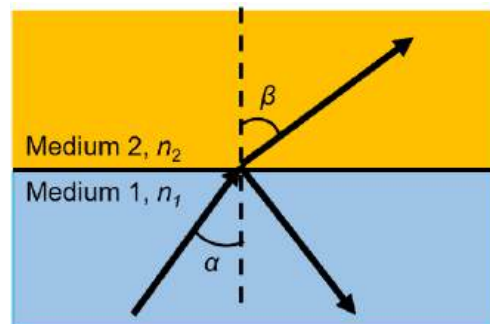
14 Snell's Law

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\theta_1 = \alpha$$

$$\theta_2 = \beta$$

$$n_1 \sin \alpha = n_2 \sin \beta$$



5 Attempts to solve for β or θ_2

$$\beta = \sin^{-1}\left(\frac{n_1}{n_2} \sin \alpha\right)$$

3 Subs or IDs $\alpha = \theta_1$,
 n_1 , or, n_2

$$\beta = \sin^{-1}\left(\frac{1.33}{1.20} \sin(35.5^\circ)\right)$$

1 Answer

1 Units

$$\beta = 40.1^\circ$$

1 Sig figs

Report your answer in this box.	Report units in this box.
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(15) b. **Calculate** the critical angle for the scenario shown above.

10 Critical angle equation or Snell's law with $\theta_2 = 90^\circ$

$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)$$

2 Subs n_1 and n_2

$$\theta_c = \sin^{-1}\left(\frac{1.20}{1.33}\right)$$

1 Answer

1 Units

$$\theta_c = 64.5^\circ$$

1 Sig Figs

Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

A: 6 (15%)
B: 10 (25%)
D: 24 (60%)
Total: 40

Wt. No.

Cadet: _____

60

2. An infinite sheet of charge has a uniform charge density of $12.7 \mu\text{C}/\text{m}^2$. A small sphere with a mass of $6.75 \times 10^{-5} \text{ kg}$ hangs from a massless string. The string makes an angle of 23.8° with the sheet. **Calculate** the charge of the sphere.

14

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{enc}}}{\epsilon_0}$$

10

$$\vec{F}_E = q\vec{E}$$

6

$$\sum \vec{F}_{\text{ext}} = m\vec{a}$$

6

$$\sigma = \frac{Q}{A}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{enc}}}{\epsilon_0}$$

2 Attempts to break into multiple surfaces

2 Correct dot product (only sides)

$$\int_{\text{side}} E \cos(0^\circ) dA + \int_{\text{wall}} E \cos(90^\circ) dA + \int_{\text{side}} E \cos(0^\circ) dA = \frac{q_{\text{enc}}}{\epsilon_0}$$

2 Correct integration

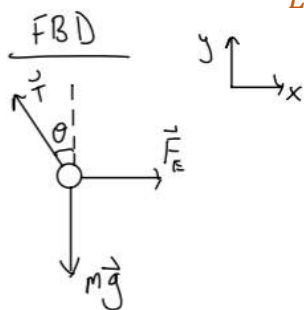
$$EA + EA = \frac{\sigma A}{\epsilon_0}$$

$$2EA = \frac{\sigma A}{\epsilon_0}$$

$$E = \frac{\sigma}{2\epsilon_0}$$

2 Correct q_enc

1 Correct E Equation



$$\sum \vec{F}_{\text{ext}} = m\vec{a}$$

$$\sum \vec{F}_x = m\vec{a}_x$$

$$\sum \vec{F}_y = m\vec{a}_y$$

2 Attempts to break into x and y components

$$\vec{F}_E - T \sin \theta = m\vec{a}_x$$

$$T \cos \theta - mg = m\vec{a}_y$$

2 Correct Fx

$$T \sin \theta = \vec{F}_E$$

$$T = \frac{mg}{\cos \theta}$$

2 Correct Fy

$$mg \tan \theta = \vec{F}_E$$

2 Attempt to combine x and y equations

$$\vec{F}_E = q\vec{E}$$

$$\vec{F}_E = \frac{q\sigma}{2\epsilon_0}$$

2 Attempts to combining F_E and E

$$mg \tan \theta = \frac{q\sigma}{2\epsilon_0}$$

$$q = \frac{2\epsilon_0 mg \tan \theta}{\sigma}$$

1 Attempts to solve for m

$$q = \frac{2(8.854 \times 10^{-12})(6.75 \times 10^{-5} \text{ kg})(9.807 \frac{\text{m}}{\text{s}^2}) \tan(23.8^\circ)}{(12.7 \times 10^{-6} \frac{\text{C}}{\text{m}^2})}$$

2 Answer

1 Units

1 Sig Figs

$$q = 4.07 \times 10^{-10} \text{ C}$$

Report your answer in this box.

Report units in this box.

A: 4 (7%)
B: 20 (33%)
D: 36 (60%)
Total: 60

- 60 3. A rectangular wire loop has a resistance of $15.9 \, \Omega$. The loop is located in a $3.85 \, \text{T}$ uniform magnetic field directed in the positive x-direction. The loop has sides of length $a = 0.573 \, \text{m}$ and $b = 0.321 \, \text{m}$. The loop is rotated about the y-axis at a constant angular frequency, $\omega = 9.32 \, \text{rad/s}$. At time $t = 0$, the loop is in the yz-plane. **Calculate** the magnitude of the current induced in the loop at time $t = 0.751 \, \text{s}$.

$$\Phi_B = \int \vec{B} \cdot d\vec{A} \quad \boxed{15}$$

$$\varepsilon = -N \frac{d\Phi_B}{dt} \quad \boxed{15}$$

$$\Phi_B = \int |\vec{B}| |d\vec{A}| \cos\theta \quad \textcircled{4} \text{ Correctly resolves dot product}$$

$$\Phi_B = BA \cos(\omega t) \quad \textcircled{2} \text{ Identifies } \theta = \omega t \quad \textcircled{4} \text{ Correctly integrates}$$

$$\varepsilon = -N \frac{d}{dt} (BA \cos(\omega t)) \quad \textcircled{4} \text{ Attempts derivative}$$

$$\varepsilon = NBA\omega \sin\omega t \quad \textcircled{2} \text{ Correct derivative}$$

$$V = IR \text{ or } \varepsilon = IR \quad \boxed{6} \text{ V=IR}$$

$$|I| = \frac{|\varepsilon|}{R} \quad \textcircled{2} \text{ Combines } \varepsilon \text{ and Ohm's Law}$$

$$|I| = \frac{NBA\omega \sin\omega t}{R} \quad \textcircled{2} \text{ Attempts to solve for I}$$

$$|I| = \frac{(1)(3.85 \, \text{T})(0.573 \, \text{m})(0.321 \, \text{m})(9.32 \, \frac{\text{rad}}{\text{s}}) \sin((9.32 \, \frac{\text{rad}}{\text{s}})(0.751 \, \text{s}))}{15.9 \, \Omega}$$

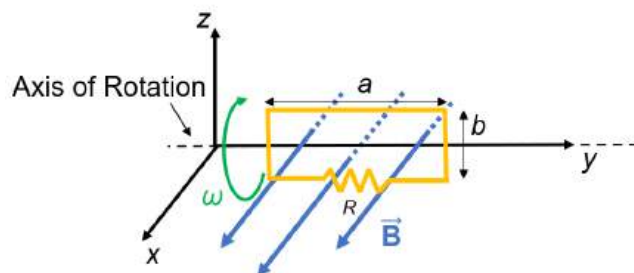
$$I = 0.272 \, \text{A}$$

$\textcircled{2}$ Answer

$\textcircled{1}$ units

$\textcircled{1}$ Sig figs

Report your answer in this box.	Report units in this box.
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Wt. No.

Cadet: _____

60 4. In the RC circuit shown below, the power supply has a potential difference of 62.6 V, the resistance is 135 Ω , and the capacitance is 476 μF . Initially, the capacitor is fully discharged. At time $t = 0.00$ s, the switch is closed in position A.

(50) a. **Calculate** the potential difference across the resistor at time $t = 0.525$ s.

6

10

10

$$\tau_c = RC \quad I(t) = \frac{dQ}{dt}$$

$$q(t) = C\varepsilon(1 - e^{-\frac{t}{\tau_c}})$$

$$q(t) = C\varepsilon(1 - e^{-\frac{t}{\tau_c}})$$

$$I(t) = \frac{d}{dt} [C\varepsilon(1 - e^{-\frac{t}{\tau_c}})]$$

$$I(t) = \frac{C\varepsilon}{\tau_c} (e^{-\frac{t}{\tau_c}})$$

④ Attempts derivative

$$V = IR \quad \text{5}$$

$$V_R(t) = I(t)R$$

④ Applies Ohm's law

$$I(t) = \frac{C\varepsilon}{RC} (e^{-\frac{t}{\tau_c}})$$

$$V_R(t) = \frac{\varepsilon}{R} (e^{-\frac{t}{\tau_c}}) R$$

② Combines q(t) with τ_c

$$I(t) = \frac{\varepsilon}{R} (e^{-\frac{t}{\tau_c}})$$

③ Correct Derivative

$$V_R(t) = \varepsilon e^{-\frac{t}{RC}}$$

② Attempts to solve for $V_R(t)$

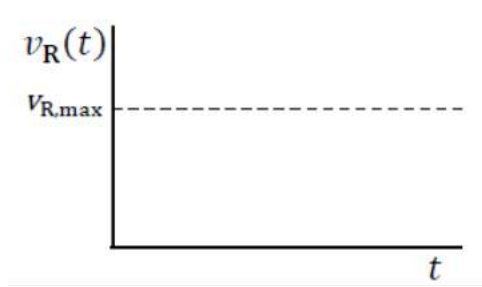
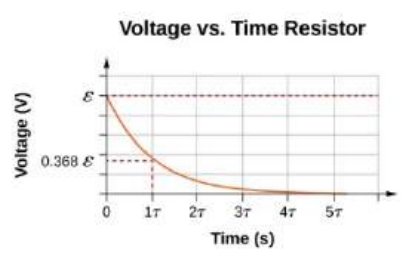
$$V_R(0.525s) = (62.6V) e^{-\frac{(0.525s)}{(135\Omega)(476 \times 10^{-6}F)}}$$

Report your answer in this box.	Report units in this box.
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$$V_R(t) = 1.77 \times 10^{-2} V$$

- ② Answer
- ① Units
- ① Sig Figs

(10) b. **Draw** the appropriate graph for the potential difference across the resistor with respect to time.



5 Exponential Curve

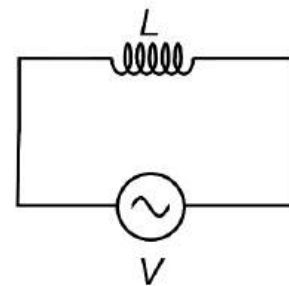
5 Decreasing

A: 4 (7%)
B: 15 (25%)
D: 41 (68%)
Total: 60

Wt. No.

Cadet: _____

- 60 5. An 8.46-mH ideal inductor is connected to an AC power source that operates at a frequency of 75.4 Hz with a peak emf of 114 V. **Calculate** the magnitude of the peak current through the inductor.



$$V_{rms} = I_{rms}Z \quad \boxed{10}$$

$$V_{rms} = \frac{V_o}{\sqrt{2}} \quad \boxed{4}$$

$$I_{rms} = \frac{I_o}{\sqrt{2}} \quad \boxed{4}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \quad \boxed{10}$$

$$X_L = \omega L \quad \boxed{4}$$

$$\omega = 2\pi f \quad \boxed{4}$$

$$Z = X_L = 2\pi fL$$

$$\frac{V_o}{\sqrt{2}} = \frac{I_o}{\sqrt{2}}(2\pi fL)$$

$$I_o = \frac{V_o}{2\pi fL}$$

$$I_o = 28.4 A$$

③ IDs Z is only X_L

③ Combines V_{rms} and Ohms Law

③ Combines I_{rms} and Ohms Law

④ Combines ω and X_L

④ Combines Z and Ohms Law

③ Subs V_o , f , C

② Answer

① Units

① Sig Figs

Report your answer in this box.	Report units in this box.
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Wt. No.

Cadet: _____

60

6.

Initially, unpolarized light with an unknown intensity is incident upon the first of a series of three polarizing sheets. The polarizing direction of the first two sheets makes angles of $\theta_1 = 30.0^\circ$ and $\theta_2 = 60.0^\circ$ with respect to the vertical, as shown in the figure. The last sheet makes an angle of $\theta_3 = 50.0^\circ$ with respect to the horizontal. The light that emerges from the third polarizing sheet has an intensity of 208 W/m^2 . **Calculate** the intensity of the incident light.

18

18

$$I = \frac{1}{2} I_o \quad I = I_o \cos^2 \theta$$

$$I_1 = \frac{1}{2} I_o$$

$$I_2 = I_1 \cos^2 \theta$$

$$I_2 = \frac{1}{2} I_o \cos^2(\theta_2 - \theta_1)$$

$$I_2 = \frac{1}{2} I_o \cos^2(30^\circ)$$

$$I_3 = I_2 \cos^2 \theta$$

$$I_2 = \frac{1}{2} I_o \cos^2(30^\circ) \cos^2[(90 - \theta_3) - \theta_2]$$

$$I_3 = \frac{1}{2} I_o \cos^2(30^\circ) \cos^2(20^\circ)$$

$$I_o = \frac{2I_3}{\cos^2(30^\circ) \cos^2(20^\circ)}$$

$$I_o = \frac{2(208 \frac{\text{W}}{\text{m}^2})}{\cos^2(30^\circ) \cos^2(20^\circ)}$$

$$I_o = 628 \frac{\text{W}}{\text{m}^2}$$

2

Answer

1

Units

1

Sig Figs

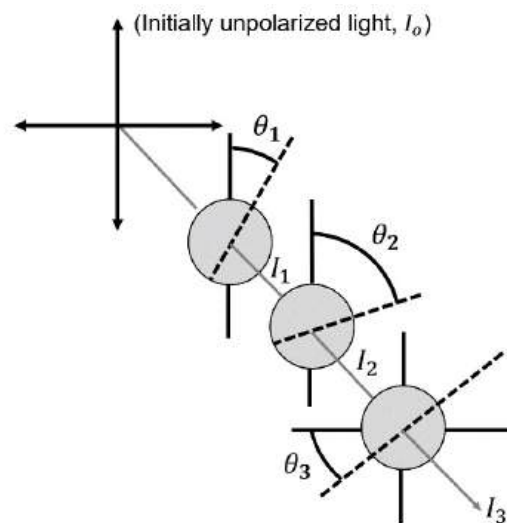
4 Attempts to combine I_1 and I_2

4 Correctly combines I_1 and I_2 using θ

4 Attempts to combine I_2 and I_3

4 Correctly combines I_1 and I_2 using θ

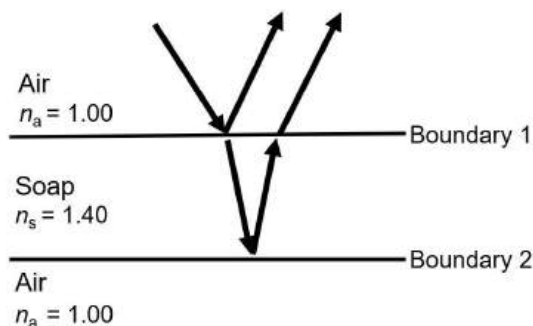
4 Attempts to solve for I_o



Wt. No.

Cadet: _____

- 55 7. A soap bubble ($n_s = 1.40$) floats in air ($n_a = 1.00$). The soap bubble is illuminated from above by sunlight at the normal incidence.



- (5) a. **Circle** the choice that describes the phase shift of the reflected wave that occurs at Boundary 1 between air and the soap.

Phase shift by $\pi/2$ radians

No phase shift

Phase shift by π radians

5

- (5) b. **Circle** the choice that describes the phase shift of the reflected wave that occurs at Boundary 2 between the soap and the air.

Phase shift by $\pi/2$ radians

No phase shift

Phase shift by π radians

5

- (45) c. **Calculate** the second thinnest thickness of soap that minimizes the reflection of incident light with wavelength $\lambda = 623$ nm.

25 $2t = m \frac{\lambda}{n_2} \quad m = 2 \quad n_2 = n_s$

$$2t = 2 \frac{\lambda}{n_s}$$

$$t = \frac{\lambda}{n_s}$$

8 IDs $m=2$ or $m=1$ if $2t = (m + \frac{1}{2}) \frac{\lambda}{n_2}$ is used

7 Attempts to solve for t

1 Answer

1 Units

1 Sig Figs

$t = \frac{623 \text{ nm}}{1.40}$	Report your answer in this box.	Report units in this box.
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2 Subs λ and n

$$t = 445 \text{ nm}$$

A: 3 (5%)
B: 17 (31%)
D: 35 (64%)
Total: 55

Wt. No.

Cadet: _____

- 50 8. An object is located on the central axis of a concave mirror. The object is 5.10 cm tall and is located 26.8 cm from the mirror. The distance from the mirror to the focal point is 10.2 cm.

- (20) a. **Calculate** the distance from the mirror to the image.

$$\boxed{14} \quad \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$$

$$d_i = \left(\frac{1}{f} - \frac{1}{d_o} \right)^{-1} = \left(\frac{1}{10.2 \text{ cm}} - \frac{1}{26.8 \text{ cm}} \right)^{-1} = \mathbf{16.5 \text{ cm}}$$

② Attempts to solve for di

② Subs for f and do

① answer

① Units

Report your answer in this box.	Report units in this box.
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- (25) b. **Calculate** the height of the image.

$$\boxed{8} \quad m = -\frac{d_i}{d_o} \quad m = \frac{h_i}{h_o} \quad \boxed{8}$$

③ Combines magnification equations

$$-\frac{d_i}{d_o} = \frac{h_i}{h_o}$$

$$h_i = -\frac{d_i}{d_o} h_o = -\frac{16.5 \text{ cm}}{26.8 \text{ cm}} (5.10 \text{ cm}) = \mathbf{-3.13 \text{ cm}}$$

① answer

① Units

① Attempts to solve for hi

③ Subs di from part a, f and ho

Report your answer in this box.	Report units in this box.
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- (5) c. **State** whether the image is real or virtual. real

⑤

Wt. **No.**

Cadet: _____

- 55 9. Red light ($\lambda = 653 \text{ nm}$) incident on two narrow slits that are $d = 0.225 \text{ mm}$ apart, causing an interference pattern on a screen a distance L away. The second maxima for the interference pattern of the red light is a distance $y = 1.13 \text{ mm}$ from the central maxima. **Calculate** the distance, L , from the two slits to the screen.

17

$$d \sin \theta = \left(m + \frac{1}{2}\right) \lambda$$

34

$$d \sin \theta = m \lambda$$

Small angle approximation

4

$$\sin \theta \approx \tan \theta$$

$$\tan \theta = \frac{y}{L}$$

3

$$\text{Subs } \sin \theta = \frac{y}{L}$$

$$\frac{dy}{L} = m \lambda$$

$$m = 2$$

4

IDs $m=2$ or $m=1$ if
 $d \sin \theta = \left(m + \frac{1}{2}\right) \lambda$
is used

$$\frac{dy}{L} = 2 \lambda$$

$$L = \frac{dy}{2 \lambda}$$

4

Attempts to solve
for L

3

Subs d, y and λ

$$L = \frac{(0.225 \times 10^{-3})(1.13 \times 10^{-3} \text{ m})}{2(653 \times 10^{-9} \text{ m})}$$

1

Answer

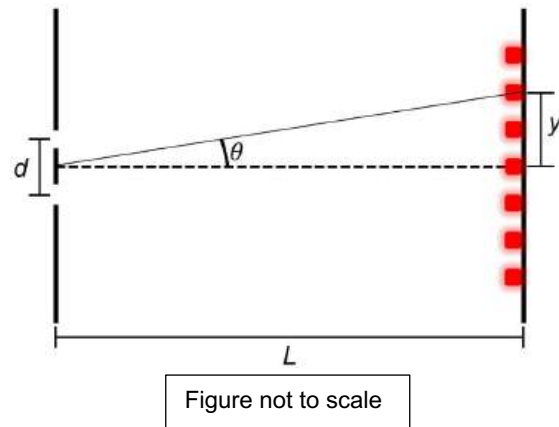
1

Units

1

Sig Figs

$$L = 0.195 \text{ m}$$



Report your answer in this box.

Report units in this box.

Wt. No.

Cadet: _____

60 10. In a demonstration of the photoelectric effect, a photon with a frequency of 3.61×10^{15} Hz is incident on a metallic material with a work function of 7.25 eV.

(10) a. **Calculate** the energy of the incident photons. Express your answer in joules.

$$E = hf \quad \boxed{6}$$

$$E = (6.626 \times 10^{-34} \text{ Js})(3.61 \times 10^{15} \text{ Hz}) \quad \textcircled{2} \text{ Subs } h \text{ and } f$$

$$E = 2.39 \times 10^{-18} \text{ J} \quad \textcircled{1} \text{ Answer}$$

$\textcircled{1}$ Units

Report your answer in this box.	Report units in this box.
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(50) b. **Calculate** the maximum speed of the electrons ejected from this material.

$$K_{max} = hf - \Phi \quad \boxed{20}$$

$$K_T = \frac{1}{2}mv^2 \quad \boxed{12}$$

$$\frac{1}{2}m_e v_{max}^2 = hf - \Phi \quad \textcircled{6} \text{ Combines } K_{max} \text{ with } \frac{1}{2}mv^2$$

$$v_{max} = \sqrt{\frac{2}{m_e}(hf - \Phi)} \quad \textcircled{5} \text{ Attempts to solve for } v_{max}$$

$$= \sqrt{\frac{2}{(9.109 \times 10^{-31} \text{ Js})((6.626 \times 10^{-34} \text{ Js})(3.61 \times 10^{15} \text{ Hz}) - (7.25 \text{ eV})(1.602 \times 10^{-19} \text{ J}))}}$$

$\textcircled{2}$ Answer

$\textcircled{1}$ Units

$\textcircled{1}$ Sig Figs

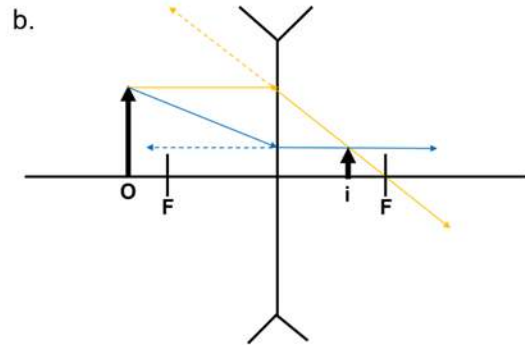
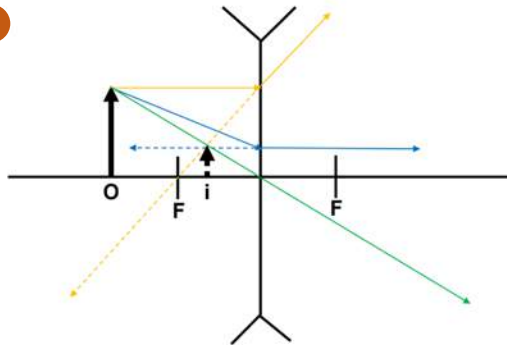
Report your answer in this box.	Report units in this box.
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$$v_{max} = 1.64 \times 10^6 \frac{\text{m}}{\text{s}}$$

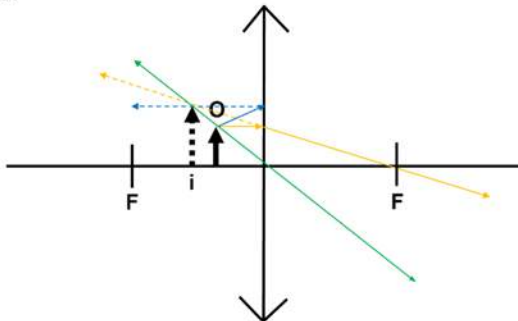
$\textcircled{3}$ convert Φ to J

- 5 11. Consider the figures shown below. **Circle** the figure that corresponds to the correct 3-ray diagram for a diverging lens.

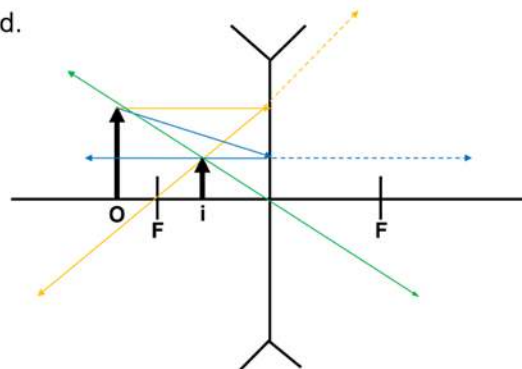
5



c.



d.



- 5 12. The magnetic field of an electromagnetic wave oscillates along the y-axis and is described by $B_y(z, t) = B_o \sin(kz - \omega t)$. **Select** the answer below which best describes the wave's direction of travel.

☐ + x-direction.

☐ - x-direction.

☐ + y-direction.

☐ - y-direction.

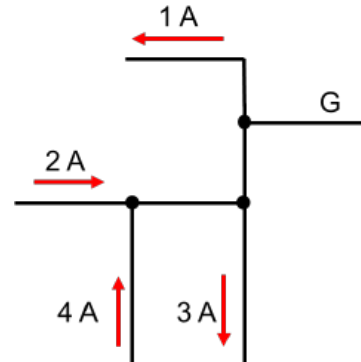
5

☒ + z-direction.

☐ - z-direction.

- 5 13. A portion of a steady state circuit is shown below. **Select** the statement which best represents the magnitude and direction of the flow of current in the section of wire labeled G.

- ☐ 2 A to the left.
- ☒ 2 A to the right.
- ☐ 4 A to the left.
- ☐ 4 A to the right.



- 5 14. The three equations below represent three waves traveling through the same medium.

$$y(x, t) = 2\sin(4x - 4t) \quad (1)$$

$$y(x, t) = 5\sin(3x - 5t) \quad (2)$$

$$y(x, t) = 3\sin(2x - 6t) \quad (3)$$

Rank the three waves according to their wavelength, longest first.

☒ 3 > 2 > 1

- 5 15. Monochromatic light of wavelength λ is used in Young's double-slit experiment. A bright spot will be observed on the screen when the difference in path length from the slits to the screen is:

- ☒ 1λ
- ☐ 1.22λ
- ☐ 1.5λ
- ☐ All of the above.
- ☐ Not enough information to determine.

A: 5 (33%)
B: 0 (0%)
D: 10 (67%)
Total: 15

- 5 16. In an oscillating LC circuit, the charge in the capacitor changes from having maximum positive charge on one plate to having maximum positive charge on the other plate in time t . **Select** the relationship between the time t and the period of oscillation T .

☐ $t = T$



☒ $t = \frac{1}{2}T$

☐ $t = \frac{1}{4}T$

☐ $t = 2T$

☐ Not enough information about the circuit is known

- 5 17. A basic law of geometric optics states that the angle of incidence is always equal to the angle of refraction.

☐ True



☒ False

- 5 18. **Select** the electromagnetic wave below with the shortest wavelength.

☐ Radio waves ($f = 3.0 \times 10^6$ Hz)

☐ Infrared waves ($f = 3.0 \times 10^9$ Hz)

☐ Visible light ($f = 3.0 \times 10^{12}$ Hz)



☒ Gamma rays ($f = 3.0 \times 10^{19}$ Hz)

☐ All have equal wavelengths

A: 0 (0%)
B: 0 (0%)
D: 0 (0%)
Total: 0



TAB H FINAL COURSE GRADES

57	Applegate, Colter	2028	MAJ Jenna Vercollone	1	ECN2	A	92.47%	1849	2000	3.53%	70.60	92.20%	95.67%	87.33%	93.00%
58	Wissemann, Christopher	2028	CPT Thomas O'Connell	1	BIO0	A	92.38%	1848	2000	3.62%	72.40	98.40%	89.33%	82.33%	91.33%
59	Schoenfeld, Roman	2028	LTC Mark Haseman	2	BEA0	A	92.35%	1847	2000	3.65%	73.00	92.80%	88.67%	91.00%	94.00%
60	Keyes, Kelly	2027	CPT Mariah Caid-Loos	3	ASD0	A	92.34%	1836	1988	3.66%	72.76	90.74%	91.33%	95.00%	94.17%
61	Lambino, Zoe	2028	CPT Thomas O'Connell	1	BEA0	A	92.33%	1847	2000	3.67%	73.40	94.10%	94.33%	98.67%	84.67%
62	Mendez Guzman, Armand	2028	COL Pete Chapman	5	ECN2	A	92.28%	1823	1975	3.72%	73.47	89.00%	91.00%	94.33%	94.50%
63	Chau, Paul	2028	CPT Thomas O'Connell	1	CEN2 BE	A	92.24%	1845	2000	3.76%	75.20	92.40%	85.33%	89.33%	92.33%
64	Ward, Liam	2028	LTC Mark Haseman	2	ORE2	A	92.17%	1843	2000	3.83%	76.60	92.30%	93.00%	97.00%	86.67%
65	McDowell, Phinehas	2028	MAJ Jenna Vercollone	1	CEN2	A	92.04%	1841	2000	3.96%	79.20	91.40%	81.67%	98.67%	90.17%
66	Gimino, Sophia	2028	CPT Thomas O'Connell	1	KIN1	A-	91.97%	1839	2000	0.03%	0.60	94.80%	87.67%	98.33%	84.83%
67	Wick, Ashton	2028	CPT Thomas O'Connell	1	CEN2	A-	91.93%	1839	2000	0.07%	1.40	98.40%	93.00%	92.67%	84.17%
68	Crowley, Cecelia	2028	CPT Thomas O'Connell	1	PIFO TST	A-	91.91%	1838	2000	0.09%	1.80	91.30%	91.67%	88.00%	90.17%
69	Holtman, Ethan	2028	MAJ Jenna Vercollone	1	CVN3	A-	91.90%	1838	2000	0.10%	2.00	93.60%	85.33%	87.33%	90.83%
70	Plagens, Abby	2028	CPT Thomas O'Connell	1	NEN1	A-	91.57%	1831	2000	0.43%	8.60	92.50%	83.67%	88.67%	90.83%
71	Thomson, Micah	2028	CPT Thomas O'Connell	1	NEN1	A-	91.51%	1830	2000	0.49%	9.80	96.20%	94.33%	84.00%	88.50%
72	Khan, Ayesha	2028	CPT William Cooperider	4	CHM2	A-	91.18%	1810	1985	0.82%	16.28	96.00%	90.00%	96.33%	81.67%
73	Stroud, Nicole	2028	CPT Thomas O'Connell	1	BIO0	A-	91.06%	1821	2000	0.94%	18.80	91.10%	92.67%	98.33%	84.83%
74	Chao, Austin	2028	MAJ Jenna Vercollone	1	SEN2	A-	90.96%	1819	2000	1.04%	20.80	93.20%	94.00%	91.33%	82.33%
75	Johnson, Jackson	2029	CPT Mariah Caid-Loos	3	AEN0	A-	90.91%	1804	1984	1.09%	21.63	87.43%	92.33%	99.00%	88.00%
76	Brown, Parker	2028	CPT William Cooperider	4	EEN1	A-	90.90%	1818	2000	1.10%	22.00	91.40%	88.67%	95.33%	87.50%
77	Clancy, Jameson	2028	CPT Thomas O'Connell	1	CVN3	A-	90.85%	1817	2000	1.15%	23.00	92.00%	86.33%	95.33%	91.17%
78	Rodgers, Sonny	2028	CPT William Cooperider	4	ECN2	A-	90.75%	1815	2000	1.25%	25.00	91.00%	86.67%	98.00%	88.67%
79	Hart, Jenna	2028	CPT Thomas O'Connell	1	BIO0	A-	90.60%	1812	2000	1.40%	28.00	90.70%	82.33%	94.67%	88.67%
80	Liu, Jada	2028	CPT Mariah Caid-Loos	3	BIO0	A-	90.41%	1790	1980	1.59%	31.48	93.58%	77.67%	95.33%	91.50%
81	Smith, Brady	2028	MAJ Jenna Vercollone	1	BIO0	A-	90.38%	1808	2000	1.62%	32.40	85.60%	97.00%	96.00%	83.50%
82	Sindler, Allan	2028	MAJ Jenna Vercollone	1	PAP3	A-	90.34%	1807	2000	1.66%	33.20	85.40%	91.67%	93.33%	92.33%
83	Fields, Thomas	2028	CPT William Cooperider	4	ECN2	A-	90.28%	1806	2000	1.72%	34.40	91.80%	81.67%	97.33%	83.83%
84	Vair, Eliana	2028	LTC Mark Haseman	2	MEN3	A-	90.24%	1805	2000	1.76%	35.20	93.30%	89.67%	97.33%	86.67%
85	Pippin, Lois	2029	CPT Thomas O'Connell	1	FA52	A-	90.22%	1804	2000	1.78%	35.60	90.30%	76.67%	88.00%	94.33%
86	Hodges, Zachary	2028	CPT Thomas O'Connell	1	BIO0	A-	90.16%	1803	2000	1.84%	36.80	94.30%	94.00%	90.33%	83.17%
87	Cueto, Nicholas	2028	CPT Thomas O'Connell	1	NEN1	A-	90.12%	1802	2000	1.88%	37.60	93.30%	95.00%	100.00%	73.83%
88	Ham, John	2028	CPT William Cooperider	4	ECN2	A-	90.08%	1802	2000	1.92%	38.40	86.20%	92.00%	91.67%	87.67%
89	Edmondson, Josiah	2028	CPT William Cooperider	4	SEN2	A-	89.88%	1798	2000	2.12%	42.40	92.30%	84.33%	72.33%	92.00%
90	Haynes, Christopher	2028	COL Pete Chapman	5	EEN1	A-	89.70%	1794	2000	2.30%	46.00	92.60%	87.33%	95.00%	79.33%
91	Hoag, Jonathan	2029	CPT Thomas O'Connell	1	MEN3	A-	89.62%	1792	2000	2.38%	47.60	95.60%	85.67%	78.00%	87.17%
92	Quijije, Aaron	2028	MAJ Jenna Vercollone	1	BIO0	A-	89.61%	1792	2000	2.39%	47.80	91.60%	81.00%	90.00%	86.17%
93	Myers, Alexa	2028	CPT Thomas O'Connell	1	ORE2	A-	89.60%	1792	2000	2.40%	48.00	94.60%	93.00%	92.67%	78.50%
94	Jang, Hudson	2028	CPT William Cooperider	4	ORE2	A-	89.40%	1788	2000	2.60%	52.00	87.20%	85.67%	94.00%	85.17%
95	Przekora, Alexander	2029	CPT William Cooperider	4	MEN3	A-	89.25%	1785	2000	2.75%	55.00	86.00%	90.00%	78.67%	91.00%
96	O'Keefe, Brendon	2028	CPT Mariah Caid-Loos	3	ORE2	A-	89.00%	1759	1976	3.00%	59.28	85.26%	82.33%	94.33%	94.33%
97	Millan, Romin	2029	CPT Thomas O'Connell	1	PHY1	A-	88.97%	1779	2000	3.03%	60.60	87.70%	83.33%	96.33%	86.33%
98	Chun, Jake	2027	CPT William Cooperider	4	LLS3	A-	88.90%	1778	2000	3.10%	62.00	89.40%	83.67%	91.67%	88.33%
99	Bachmann, Christian	2028	COL Pete Chapman	5	SEN2	A-	88.90%	1778	2000	3.10%	62.00	86.00%	81.00%	97.33%	88.17%
100	Jublou, Connor	2028	MAJ Jenna Vercollone	1	ECN2	A-	88.90%	1778	2000	3.10%	62.00	89.60%	92.67%	86.67%	83.17%
101	Pablo, Sydney	2029	MAJ Jenna Vercollone	1	NEN1	A-	88.86%	1777	2000	3.14%	62.80	82.40%	93.00%	95.00%	84.00%
102	Leandres, Antonio	2028	CPT Mariah Caid-Loos	3	CVN3	A-	88.71%	1771	1996	3.29%	65.67	89.50%	99.33%	99.00%	75.17%
103	Henderson, Danielle	2028	LTC Mark Haseman	2	CEN2	A-	88.43%	1769	2000	3.57%	71.40	92.70%	93.33%	75.67%	84.83%
104	Malik, Mohammad	2028	MAJ Jenna Vercollone	1	CVN3	A-	88.34%	1767	2000	3.66%	73.20	89.80%	92.00%	87.00%	81.33%
105	Pamnani, Krish	2029	CPT Thomas O'Connell	1	MEN3	A-	88.34%	1767	2000	3.66%	73.20	90.40%	93.00%	92.00%	83.33%
106	Braeger, Simon	2029	LTC Mark Haseman	2	EEN1	A-	88.11%	1762	2000	3.89%	77.80	91.00%	71.67%	91.00%	86.17%
107	Torres, Collin	2028	LTC Mark Haseman	2	HUS3T	B+	87.96%	1759	2000	0.04%	0.80	85.20%	85.00%	93.00%	87.00%
108	Medrano, Maya	2028	CPT William Cooperider	4	EEN1	B+	87.95%	1759	2000	0.05%	1.00	91.60%	87.33%	95.33%	74.17%
109	Ferrell, Levi	2028	CPT William Cooperider	4	CEN2	B+	87.83%	1757	2000	0.17%	3.40	79.20%	89.67%	92.00%	87.50%
110	Griffin, Cameron	2028	LTC Mark Haseman	2	ENM1	B+	87.82%	1756	2000	0.18%	3.60	90.70%	75.67%	93.00%	85.00%
111	Adams, Lucas	2028	CPT Thomas O'Connell	1	NEN1	B+	87.80%	1756	2000	0.20%	4.00	87.70%	87.67%	90.00%	82.67%
112	Wray, Alyssa	2028	COL Pete Chapman	5	BIO0	B+	87.54%	1729	1975	0.46%	9.09	89.60%	86.00%	90.33%	82.67%
113	Christian, Caleb	2028	CPT Mariah Caid-Loos	3	SEN2	B+	87.53%	1751	2000	0.47%	9.40	87.30%	87.33%	91.33%	81.67%
114	Deady, Connor	2029	LTC Mark Haseman	2	NEN1	B+	87.43%	1749	2000	0.57%	11.40	89.80%	72.67%	96.67%	85.17%
115	Nguyen, Ta	2028	COL Pete Chapman	5	ORE2	B+	87.30%	1746	2000	0.70%	14.00	84.20%	90.67%	99.67%	75.33%
116	Clemons, Jacob	2029	CPT Mariah Caid-Loos	3	NEN1	B+	87.13%	1743	2000	0.87%	17.40	87.65%	74.67%	95.67%	84.50%

117	Galvin, Evan	2028	CPT William Cooperider	4	SEN2	B+	87.00%	1740	2000	1.00%	20.00	94.20%	86.33%	94.67%	71.00%
118	Salah, Sarra	2026	LTC Mark Haseman	2	CSC1	B+	86.98%	1740	2000	1.02%	20.40	91.30%	86.00%	96.00%	76.50%
119	Biron, Jacob	2028	CPT William Cooperider	4	ENM1	B+	86.98%	1740	2000	1.02%	20.40	89.10%	91.33%	88.00%	76.17%
120	Austin, Davis	2028	CPT William Cooperider	4	PHY1	B+	86.90%	1738	2000	1.10%	22.00	90.80%	74.33%	88.67%	83.50%
121	Tran, Susan	2028	CPT Mariah Caid-Loos	3	EEN1	B+	86.83%	1737	2000	1.17%	23.40	86.70%	62.33%	99.67%	85.17%
122	Yaklich, Kaleb	2028	MAJ Jenna Vercollone	1	SEN2 CYB1N	B+	86.74%	1735	2000	1.26%	25.20	93.20%	90.33%	66.00%	82.00%
123	Pavlak, Benjamin	2028	MAJ Jenna Vercollone	1	ENM1	B+	86.40%	1728	2000	1.60%	32.00	90.20%	90.33%	82.00%	77.17%
124	Sherrod, Samuel	2028	COL Pete Chapman	5	MSC1	B+	86.25%	1704	1975	1.75%	34.56	81.80%	81.33%	93.00%	86.67%
125	Gu, Caleb	2028	MAJ Jenna Vercollone	1	KIN1	B+	86.13%	1723	2000	1.87%	37.40	89.40%	80.67%	94.00%	77.33%
126	Sheddy, Paul	2028	LTC Mark Haseman	2	SEN2H E	B+	86.09%	1722	2000	1.91%	38.20	92.20%	70.00%	96.33%	78.83%
127	Minicozzi, John	2028	CPT William Cooperider	4	MGN0	B+	85.93%	1702	1980	2.07%	40.99	84.70%	86.67%	87.67%	82.00%
128	Aemisegger, Hannah	2028	LTC Mark Haseman	2	ORE2 RS	B+	85.76%	1715	2000	2.24%	44.80	93.30%	63.00%	90.00%	81.67%
129	Thome, John	2028	LTC Mark Haseman	2	ENM1 RS	B+	85.75%	1715	2000	2.25%	45.00	89.10%	74.00%	81.67%	90.50%
130	Rogers, Kennedy	2028	CPT William Cooperider	4	CEN2	B+	85.68%	1714	2000	2.32%	46.40	87.40%	81.67%	94.67%	78.17%
131	Staunton, James	2028	LTC Mark Haseman	2	CVN3	B+	85.52%	1710	2000	2.48%	49.60	89.60%	70.67%	80.00%	86.33%
132	Hester, Jeremiah	2028	MAJ Jenna Vercollone	1	AEN0	B+	85.52%	1710	2000	2.48%	49.60	88.40%	80.33%	86.00%	78.50%
133	Alapati, Hanorah	2028	MAJ Jenna Vercollone	1	CEN2	B+	85.40%	1708	2000	2.60%	52.00	91.20%	78.00%	83.33%	78.33%
134	Schwartz, Nicole	2028	LTC Mark Haseman	2	ENM1	B+	85.26%	1705	2000	2.74%	54.80	89.80%	90.67%	82.00%	76.67%
135	Dohl, Chad	2028	CPT William Cooperider	4	ENM1	B+	85.09%	1698	1995	2.91%	58.05	85.70%	87.00%	92.67%	76.67%
136	Bolon, Cole	2028	CPT Mariah Caid-Loos	3	MEN3 SP	B+	85.04%	1701	2000	2.96%	59.20	85.30%	82.67%	89.00%	77.67%
137	Weir, Caitlyn	2029	CPT Mariah Caid-Loos	3	MEN3	B+	84.83%	1680	1980	3.17%	62.76	82.41%	79.67%	83.67%	84.67%
138	Tanner, Sam	2028	COL Pete Chapman	5	CVN3	B+	84.69%	1652	1950	3.31%	64.55	82.00%	80.00%	93.67%	77.17%
139	Justus, JasyN	2028	MAJ Jenna Vercollone	1	KIN1	B+	84.49%	1690	2000	3.51%	70.20	90.60%	64.67%	86.67%	80.83%
140	Bender, Zachary	2028	CPT Mariah Caid-Loos	3	ENM1	B+	84.45%	1689	2000	3.55%	71.00	80.91%	78.67%	93.00%	81.00%
141	Ryder, Paul	2028	CPT William Cooperider	4	CVN3	B+	84.43%	1689	2000	3.57%	71.40	92.60%	84.67%	83.00%	71.00%
142	Rhyme, William	2027	LTC Mark Haseman	2	EVE2	B+	84.38%	1688	2000	3.62%	72.40	90.30%	62.33%	94.00%	81.83%
143	Sinkovic, Max	2029	CPT Mariah Caid-Loos	3	PHY1	B+	84.30%	1686	2000	3.70%	74.00	87.23%	80.67%	91.00%	73.67%
144	Kim, Heewon	2028	CPT William Cooperider	4	PYL2	B+	84.09%	1665	1980	3.91%	77.42	85.20%	69.67%	88.67%	83.50%
145	Megery, Lorelai	2028	COL Pete Chapman	5	EVE2	B+	84.03%	1626	1935	3.97%	76.82	80.40%	89.67%	95.33%	76.50%
146	Beckwith, Owen	2028	CPT Thomas O'Connell	1	CEN2	B+	84.02%	1680	2000	3.98%	79.60	93.10%	71.67%	93.33%	75.17%
147	Abalos, Leilani	2027	LTC Mark Haseman	2	GIS2H	B	83.97%	1679	2000	0.03%	0.60	92.40%	69.67%	86.00%	77.00%
148	Yelmat, Samyu	2029	CPT Mariah Caid-Loos	3	CHM2A	B	83.76%	1658	1980	0.24%	4.75	86.83%	71.33%	88.00%	79.83%
149	McDonald, Jonathan	2028	COL Pete Chapman	5	NEN1	B	83.72%	1620	1935	0.28%	5.42	91.80%	62.00%	86.67%	79.00%
150	Hemmen, George	2028	COL Pete Chapman	5	ENM1	B	83.09%	1641	1975	0.91%	17.97	85.00%	78.67%	91.67%	71.17%
151	Bridgforth, Ross	2028	MAJ Jenna Vercollone	1	CVN3	B	83.01%	1660	2000	0.99%	19.80	91.60%	88.33%	79.00%	70.83%
152	Puri, Sakshat	2028	CPT Thomas O'Connell	1	NEN1	B	82.81%	1656	2000	1.19%	23.80	87.40%	83.33%	70.33%	81.50%
153	Applegate, Logan	2028	CPT Thomas O'Connell	1	ORE2	B	82.78%	1656	2000	1.22%	24.40	94.00%	91.67%	86.00%	69.33%
154	Spilling, Camdyn	2028	CPT William Cooperider	4	CEN2	B	82.73%	1655	2000	1.27%	25.40	81.80%	59.33%	99.00%	80.50%
155	Tran, Katie	2028	CPT Mariah Caid-Loos	3	EEN1	B	82.69%	1654	2000	1.31%	26.20	87.25%	65.33%	72.00%	84.50%
156	Hofbauer, Connor	2028	LTC Mark Haseman	2	ECN2	B	82.64%	1653	2000	1.36%	27.20	88.90%	91.33%	79.33%	76.83%
157	Campion, Connor	2028	CPT Mariah Caid-Loos	3	ECN2 AIN	B	82.42%	1615	1960	1.58%	30.97	79.11%	68.00%	77.33%	86.00%
158	Steadman, Matthias	2028	LTC Mark Haseman	2	ENM1	B	82.40%	1648	2000	1.60%	32.00	92.40%	78.00%	80.33%	72.00%
159	Srisukwattananan, Radhati	2028	CPT Mariah Caid-Loos	3	EEN1	B	82.25%	1629	1980	1.75%	34.65	79.47%	88.67%	86.33%	74.50%
160	MacDougall, Declan	2028	CPT William Cooperider	4	EEN1	B	82.10%	1642	2000	1.90%	38.00	85.80%	78.00%	96.33%	71.83%
161	Banzuela, Joshua	2028	CPT William Cooperider	4	CVN3 RS	B	82.03%	1641	2000	1.97%	39.40	92.70%	87.67%	50.67%	79.17%
162	Thorsteinson, Tyler	2028	LTC Mark Haseman	2	MGN0	B	82.00%	1640	2000	2.00%	40.00	88.10%	87.67%	94.67%	67.17%
163	Otteson, Jaxxon	2028	COL Pete Chapman	5	CHM2	B	81.97%	1619	1975	2.03%	40.09	82.00%	73.00%	74.67%	86.67%
164	Lefford, Peyton	2028	LTC Mark Haseman	2	ENM1	B	81.90%	1638	2000	2.10%	42.00	90.30%	61.33%	93.00%	76.17%
165	Apthorpe, Rex	2028	CPT Thomas O'Connell	1	PHY1	B	81.66%	1633	2000	2.34%	46.80	86.50%	88.33%	77.33%	74.50%
166	Allen, Shelby	2026	LTC Mark Haseman	2	CVN3	B	81.38%	1628	2000	2.62%	52.40	89.50%	83.67%	84.33%	71.50%
167	Francis, Benjamin	2028	LTC Mark Haseman	2	ECN2	B	81.35%	1627	2000	2.65%	53.00	92.70%	81.33%	75.00%	89.67%
168	Payne, Brenna	2028	CPT William Cooperider	4	NEN1 RS	B	81.33%	1627	2000	2.67%	53.40	88.90%	84.67%	93.00%	62.33%
169	Emery, Gabriel	2028	CPT Mariah Caid-Loos	3	CVN3	B	81.14%	1623	2000	2.86%	57.20	80.95%	75.33%	84.67%	78.67%
170	Urias, Karina	2028	LTC Mark Haseman	2	ECN2	B	81.09%	1622	2000	2.91%	58.20	89.40%	81.67%	83.33%	65.17%
171	McFadden, Carson	2028	COL Pete Chapman	5	MEN3	B	80.91%	1598	1975	3.09%	61.03	89.00%	81.33%	96.33%	63.83%
172	Campbell, Charlotte	2028	CPT Mariah Caid-Loos	3	CVN3	B	80.87%	1617	2000	3.13%	62.60	77.60%	63.67%	91.00%	82.50%
173	Leclair, Ryan	2028	COL Pete Chapman	5	ENM1	B	80.85%	1617	2000	3.15%	63.00	84.40%	58.33%	82.33%	78.50%

174	Ross, Noah	2028	CPT Mariah Caid-Loos	3	ASD0	B	80.77%	1599	1980	3.23%	63.96	87.20%	64.67%	81.33%	79.50%	
175	Hekimian, Mary Ray	2028	CPT Mariah Caid-Loos	3	CVN3	B	80.77%	1615	2000	3.23%	64.60	82.58%	87.00%	91.33%	67.33%	
176	Sam, Connor	2028	CPT William Cooperider	4	ENM1	B	80.76%	1599	1980	3.24%	64.15	90.60%	60.00%	92.00%	73.67%	
177	Alvarez, Michael	2027	CPT Mariah Caid-Loos	3	ESC2	B	80.70%	1614	2000	3.30%	66.00	89.52%	65.00%	86.33%	72.83%	
178	Baszucki, Peter	2029	MAJ Jenna Vercollone	1	CVN3	B	80.28%	1606	2000	3.72%	74.40	89.40%	79.67%	78.00%	64.00%	
179	Houston, Robert	2028	LTC Mark Haseman	2	ENM1	B	80.26%	1605	2000	3.74%	74.80	83.00%	82.67%	89.00%	70.83%	
180	Choi, Dahyeon	2029	LTC Mark Haseman	2	CVN3	B	80.15%	1603	2000	3.85%	77.00	92.50%	82.67%	85.67%	57.83%	
181	De Los Reyes, Ian	2028	CPT William Cooperider	4	CEN2	B	80.13%	1603	2000	3.87%	77.40	88.60%	71.33%	86.67%	71.83%	
182	Summerlin, Marissa	2028	CPT Mariah Caid-Loos	3	MEN3	B-	79.83%	1577	1976	0.17%	3.36	83.10%	83.67%	83.66%	68.67%	
183	Heng, Kaynen	2028	CPT Thomas O'Connell	1	NEN1	B-	79.62%	1592	2000	0.38%	7.60	90.10%	69.33%	88.67%	66.67%	
184	Brady, Ned	2026	COL Pete Chapman	5	SDS0	B-	79.59%	1572	1975	0.41%	8.10	86.00%	85.00%	84.33%	67.50%	
185	Ince, Nathaniel	2028	LTC Mark Haseman	2	MEN3	B-	79.56%	1591	2000	0.44%	8.80	85.60%	73.67%	68.67%	75.17%	
186	Bae, Seongpyo	2028	CPT Mariah Caid-Loos	3	ENM1	B-	79.41%	1572	1980	0.59%	11.68	80.73%	81.33%	77.67%	77.67%	
187	Cataldo, Justin	2028	CPT William Cooperider	4	ENM1	B-	79.38%	1588	2000	0.62%	12.40	84.40%	88.67%	84.66%	61.00%	
188	Kisiel, Samuel	2028	CPT William Cooperider	4	SEN2	B-	79.13%	1583	2000	0.87%	17.40	88.50%	45.67%	87.33%	75.00%	
189	Jones, Jaylen	2029	CPT Thomas O'Connell	1	SEN2	B-	78.98%	1580	2000	1.02%	20.40	91.60%	78.00%	81.00%	70.17%	
190	Maestas, Nathan	2028	COL Pete Chapman	5	CHM2A	B-	78.65%	1573	2000	1.35%	27.00	83.40%	53.67%	85.33%	80.83%	
191	Hershfield, Joseph	2029	MAJ Jenna Vercollone	1	PHY1	B-	78.41%	1568	2000	1.59%	31.80	79.80%	69.67%	71.67%	78.83%	
192	Maughan, Thaddeus	2028	CPT Mariah Caid-Loos	3	SEN2	B-	78.39%	1562	1992	1.61%	32.07	91.22%	76.00%	70.33%	69.33%	
193	Lorek, Anthony	2028	CPT William Cooperider	4	SEN2	B-	78.15%	1563	2000	1.85%	37.00	81.50%	73.33%	88.00%	68.83%	
194	Lu, Jerry	2027	CPT William Cooperider	4	SEN1	B-	78.15%	1563	2000	1.85%	37.00	88.90%	78.67%	75.33%	58.50%	
195	Hull, Brandon	2028	LTC Mark Haseman	2	ESC3	B-	78.09%	1562	2000	1.91%	38.20	87.90%	53.33%	72.00%	83.00%	
196	Schmidt, Alexander	2028	CPT Mariah Caid-Loos	3	ENM1	B-	77.87%	1551	1992	2.13%	42.43	86.08%	68.67%	88.00%	66.17%	
197	Bardales, Elizabeth	2026	LTC Mark Haseman	2	CYS0	B-	77.84%	1557	2000	2.16%	43.20	91.80%	69.33%	76.33%	85.83%	
198	McCurdy, Coleby	2028	LTC Mark Haseman	2	AEN0	B-	77.80%	1556	2000	2.20%	44.00	91.60%	59.67%	78.00%	72.67%	
199	Villarreal, Andrea	2028	CPT Mariah Caid-Loos	3	AEN0	B-	77.71%	1539	1980	2.29%	45.34	79.65%	80.33%	85.33%	68.83%	
200	Plunkett, Hill	2028	CPT William Cooperider	4	ENM1	B-	77.70%	1554	2000	2.30%	46.00	84.70%	61.00%	72.00%	75.50%	
201	Reed, Amelia	2028	MAJ Jenna Vercollone	1	ENM1	B-	77.59%	1536	1980	2.41%	47.72	83.20%	81.67%	84.00%	57.33%	
202	Smith, Caleb	2027	LTC Mark Haseman	2	CHM2 A	B-	77.32%	1546	2000	2.68%	53.60	91.20%	79.67%	70.33%	65.17%	
203	Champlin, Everett	2028	LTC Mark Haseman	2	CVN3	B-	77.23%	1545	2000	2.77%	55.40	87.40%	45.00%	89.67%	71.17%	
204	Harmesen, Kelsey	2028	MAJ Jenna Vercollone	1	CEN2	B-	77.14%	1543	2000	2.86%	57.20	78.00%	66.67%	71.67%	76.67%	
205	Foster, Dylan	2028	CPT Mariah Caid-Loos	3	CEN2	B-	77.14%	1534	1988	2.86%	56.86	80.20%	78.33%	93.00%	63.67%	
206	Lueders, Benjamin	2028	COL Pete Chapman	5	SEN2 RSL	B-	77.12%	1500	1945	2.88%	56.02	79.00%	74.00%	83.33%	72.00%	
207	Davis, Trystin	2028	CPT Mariah Caid-Loos	3	BIO0	B-	76.87%	1538	2000	3.13%	62.60	90.28%	69.67%	85.33%	62.83%	
208	Prabhakaran, Jake	2028	CPT Mariah Caid-Loos	3	ECN2	B-	76.86%	1534	1996	3.14%	62.67	86.98%	57.33%	94.67%	62.67%	
209	Kim, Rebekah	2028	MAJ Jenna Vercollone	1	BIO0	B-	76.80%	1521	1980	3.20%	63.36	90.80%	87.00%	60.33%	62.50%	
210	Sargent, Reese	2028	CPT William Cooperider	4	SEN2	B-	76.65%	1533	2000	3.35%	67.00	88.10%	83.67%	65.00%	65.50%	
211	Gosda, Colson	2028	CPT Mariah Caid-Loos	3	ECN2	B-	76.42%	1522	1992	3.58%	71.31	72.80%	73.67%	65.00%	84.17%	
212	Jordan, Lilian	2028	CPT Thomas O'Connell	1	NEN1	B-	76.37%	1527	2000	3.63%	72.60	87.10%	64.00%	79.67%	67.83%	
213	Jorgensen, Charles	2027	LTC Mark Haseman	2	ASD0	B-	76.21%	1524	2000	3.79%	75.80	82.90%	60.33%	81.00%	69.33%	
214	Riley, Sebastian-Seann	2029	CPT William Cooperider	4	SEN2	B-	76.18%	1524	2000	3.82%	76.40	86.30%	64.33%	85.67%	63.67%	
215	Liu, Melanie	2029	COL Pete Chapman	5	ENM1	B-	76.10%	1503	1975	3.90%	77.03	87.20%	77.67%	82.67%	57.67%	
216	Acevedo, Angel	2028	MAJ Jenna Vercollone	1	CEN2	C+	75.97%	1519	2000	0.03%	0.60	89.05%	82.00%	61.67%	60.17%	
217	Deditch, Tristen	2028	CPT Mariah Caid-Loos	3	ENM1	C+	75.91%	1518	2000	0.09%	1.80	77.20%	56.67%	90.33%	69.33%	
218	Crank, Oliver	2028	CPT Mariah Caid-Loos	3	ECN2	C+	75.89%	1518	2000	0.11%	2.20	77.92%	83.00%	52.00%	77.00%	
219	Hendry, Emma	2028	COL Pete Chapman	5	ENM1	C+	75.87%	1468	1935	0.13%	2.52	75.60%	82.67%	84.67%	56.67%	
220	De Leon, Chloe	2028	CPT William Cooperider	4	ESC3	C+	75.78%	1501	1980	0.22%	4.36	83.00%	72.00%	78.00%	65.83%	
221	Young, Guyon	2028	CPT William Cooperider	4	AEN0	C+	75.73%	1515	2000	0.27%	5.40	84.20%	77.33%	54.67%	73.67%	
222	Monahan, William	2027	LTC Mark Haseman	2	SEN1	C+	75.70%	1514	2000	0.30%	6.00	86.20%	71.33%	77.00%	61.50%	
223	Parrinello, Nicholas	2028	CPT William Cooperider	4	ECN2	C+	75.55%	1511	2000	0.45%	9.00	76.40%	75.33%	91.00%	63.50%	
224	Hong, Brian	2028	CPT Mariah Caid-Loos	3	CHM2	C+	75.15%	1503	2000	0.85%	17.00	89.50%	61.33%	73.67%	68.33%	
225	Diven, Jacob	2028	CPT William Cooperider	4	GSC0	C+	75.15%	1503	2000	0.85%	17.00	87.80%	61.33%	84.33%	58.67%	
226	Andrel, Linda	2028	CPT William Cooperider	4	SEN2	C+	75.15%	1503	2000	0.85%	17.00	87.20%	54.67%	87.00%	58.00%	
227	Sparks, Tyson	2028	CPT Mariah Caid-Loos	3	ECN2	C+	75.03%	1471	1960	0.97%	19.01	86.23%	77.00%	74.33%	63.50%	
228	Wiser, David	2028	COL Pete Chapman	5	ENM1	C+	74.80%	1496	2000	1.20%	24.00	73.40%	71.67%	72.00%	71.67%	
229	Boozar, Curtis	2028	MAJ Jenna Vercollone	1	SEN2	C+	74.49%	1490	2000	1.51%	30.20	82.40%	83.67%	62.00%	59.33%	
230	Corcino, Aaron Job	2028	CPT Mariah Caid-Loos	3	ECN2	C+	74.47%	1448	1945	1.53%	29.76	73.75%	74.67%	86.00%	73.00%	
231	Morehouse, Sean	2028	MAJ Jenna Vercollone	1	MEN3	C+	74.36%	1487	2000	1.64%	32.80	84.60%	69.00%	71.00%	58.50%	

232	McKenna, Liam	2028	COL Pete Chapman	5	CEN2	C+	74.35%	1487	2000	1.65%	33.00	82.80%	79.67%	78.00%	59.50%
233	Black, Garylane	2028	LTC Mark Haseman	2	SEN2	C+	74.31%	1486	2000	1.69%	33.80	90.90%	45.67%	68.67%	74.50%
234	Hawkins, Reid	2028	CPT William Cooperider	4	GSC0	C+	73.78%	1476	2000	2.22%	44.40	83.00%	81.33%	74.00%	59.17%
235	Powers, Huston	2028	LTC Mark Haseman	2	SDS1	C+	73.70%	1474	2000	2.30%	46.00	90.40%	65.00%	46.67%	67.00%
236	Mac Phee, Lucas	2028	COL Pete Chapman	5	EEN1	C+	73.37%	1449	1975	2.63%	51.94	78.00%	76.67%	88.67%	57.00%
237	Moretti, Paige	2028	CPT William Cooperider	4	ENM1	C+	73.10%	1462	2000	2.90%	58.00	92.80%	80.33%	82.00%	37.00%
238	Boettcher, Blake	2028	COL Pete Chapman	5	ECN2	C+	73.06%	1443	1975	2.94%	58.07	83.40%	63.67%	77.67%	65.00%
239	Griffin, Joseph	2028	CPT Mariah Caid-Loos	3	EEN1	C+	73.03%	1461	2000	2.97%	59.40	84.50%	61.33%	77.33%	60.50%
240	Rodgers, Samantha	2028	COL Pete Chapman	5	CHM2	C+	72.86%	1439	1975	3.14%	62.02	86.20%	57.00%	68.67%	65.00%
241	Golden, Aurora	2028	CPT William Cooperider	4	MGN0	C+	72.85%	1457	2000	3.15%	63.00	82.40%	52.00%	84.33%	63.00%
242	Spain, Jack	2028	CPT William Cooperider	4	NEN1	C+	72.70%	1454	2000	3.30%	66.00	77.80%	84.00%	72.67%	57.83%
243	Kim, Isaac	2028	MAJ Jenna Vercollone	1	ENM1	C+	72.67%	1453	2000	3.33%	66.60	84.80%	78.33%	57.67%	54.33%
244	Wilson, Jordan	2028	CPT William Cooperider	4	MGN0 G	C+	72.58%	1452	2000	3.42%	68.40	82.90%	61.33%	73.33%	58.67%
245	Prather, Cannon	2028	COL Pete Chapman	5	ECN2 RS	C+	72.15%	1425	1975	3.85%	76.04	83.60%	68.67%	62.67%	62.17%
246	Murans, Dylan	2028	CPT William Cooperider	4	ENM1	C+	72.15%	1443	2000	3.85%	77.00	81.60%	69.00%	89.33%	51.50%
247	Bosarge, Jaret	2028	MAJ Jenna Vercollone	1	CVN3	C+	72.01%	1440	2000	3.99%	79.80	81.60%	66.00%	65.67%	57.33%
248	Cox, Elijah	2028	CPT William Cooperider	4	ENM1	C	71.70%	1434	2000	0.30%	6.00	79.80%	71.00%	60.33%	65.33%
249	Ross, Lilah	2028	LTC Mark Haseman	2	ENM1	C	71.51%	1430	2000	0.49%	9.80	88.30%	49.33%	80.33%	54.67%
250	Boerema, Katherine	2028	CPT Mariah Caid-Loos	3	EVE2	C	71.38%	1398	1959	0.62%	12.15	77.23%	73.00%	69.67%	60.83%
251	Johnston, Jane	2028	LTC Mark Haseman	2	ENM1	C	71.10%	1422	2000	0.90%	18.00	87.30%	46.67%	82.00%	56.33%
252	Bae, Jacob	2027	LTC Mark Haseman	2	ENM1	C	71.03%	1421	2000	0.97%	19.40	89.60%	62.00%	62.67%	56.00%
253	Jeong, Yewon	2028	CPT William Cooperider	4	SEN2	C	70.73%	1415	2000	1.27%	25.40	80.60%	75.00%	89.67%	41.67%
254	Schmidlin, Walter	2028	CPT Mariah Caid-Loos	3	ENM1	C	70.49%	1407	1996	1.51%	30.14	74.30%	59.67%	75.00%	61.83%
255	Grimes, Tatum	2028	CPT William Cooperider	4	ENM1	C	70.03%	1401	2000	1.97%	39.40	86.20%	57.33%	70.00%	62.17%
256	Montras, George	2028	MAJ Jenna Vercollone	1	SEN2	C	69.65%	1393	2000	2.35%	47.00	80.40%	77.00%	69.67%	50.50%
257	Hosea, Bryson	2028	MAJ Jenna Vercollone	1	ENM1	C	69.59%	1392	2000	2.41%	48.20	85.80%	57.67%	63.00%	53.50%
258	Geldore, Franz Ionardz	2028	CPT Mariah Caid-Loos	3	ORE2	C	69.54%	1388	1996	2.46%	49.10	74.30%	52.67%	49.67%	76.17%
259	Bruchon, Xavier	2028	CPT Mariah Caid-Loos	3	ENM1	C	69.51%	1373	1976	2.49%	49.20	74.51%	52.33%	81.34%	62.00%
260	Delph, Michael	2028	COL Pete Chapman	5	SEN2	C	69.39%	1360	1960	2.61%	51.15	82.00%	41.00%	74.33%	60.83%
261	Greengas, Joshua	2028	LTC Mark Haseman	2	SEN2	C	68.48%	1370	2000	3.52%	70.40	85.90%	34.00%	54.67%	69.50%
262	Diop, Khadija	2028	COL Pete Chapman	5	CVN3	C	68.45%	1369	2000	3.55%	71.00	68.40%	39.33%	87.00%	63.67%
263	Gilfoil, Jakob	2028	LTC Mark Haseman	2	MEN3	C	68.43%	1369	2000	3.57%	71.40	89.60%	14.33%	86.67%	56.83%
264	Damodar, Sivin	2028	COL Pete Chapman	5	CVN3	C	68.43%	1352	1975	3.57%	70.51	82.00%	65.67%	86.33%	46.33%
265	Lavine, David	2028	CPT William Cooperider	4	SEN2	C	68.33%	1367	2000	3.67%	73.39	77.70%	48.67%	74.67%	61.17%
266	Stober, Maximus	2028	CPT Mariah Caid-Loos	3	SEN2	C	68.16%	1350	1980	3.84%	76.03	78.98%	60.33%	71.33%	62.83%
267	Baker, Grayson	2028	CPT William Cooperider	4	CVN3	C-	67.43%	1349	2000	0.57%	11.40	85.50%	70.33%	66.67%	42.33%
268	Auci, Aaron	2028	CPT Mariah Caid-Loos	3	MEN3	C-	67.38%	1348	2000	0.62%	12.40	74.25%	61.33%	74.00%	53.33%
269	Gonzalez-Abreu, Sophia	2028	COL Pete Chapman	5	CEN2	C-	67.10%	1342	2000	0.90%	18.00	69.60%	78.33%	55.67%	52.83%
270	Gedlekirstos, Zerayacob	2028	CPT Mariah Caid-Loos	3	CVN3	C-	66.87%	1337	2000	1.13%	22.60	75.02%	56.00%	53.33%	65.67%
271	Aird, Zander	2028	COL Pete Chapman	5	ENM1	C-	66.67%	1290	1935	1.33%	25.73	85.00%	52.67%	11.00%	78.00%
272	Fuentes, Aaron	2028	LTC Mark Haseman	2	ECN2	C-	66.51%	1330	2000	1.49%	29.80	90.30%	58.67%	55.67%	56.33%
273	Waltman, Jhett	2028	MAJ Jenna Vercollone	1	FLG2	C-	66.41%	1328	2000	1.59%	31.80	82.40%	65.33%	56.00%	58.17%
274	Scherra, Jacob	2028	LTC Mark Haseman	2	NEN1	C-	66.26%	1325	2000	1.74%	34.80	85.20%	62.33%	60.67%	51.83%
275	Knox, Derrick	2028	LTC Mark Haseman	2	ENM1	C-	66.13%	1323	2000	1.87%	37.40	83.20%	77.33%	41.67%	48.33%
276	McDonald, Davis	2028	CPT Mariah Caid-Loos	3	ECN2	C-	64.87%	1279	1972	3.13%	61.72	80.00%	38.00%	39.33%	62.17%
277	Sorenson, Samuel	2028	LTC Mark Haseman	2	ENM1	C-	64.27%	1285	2000	3.73%	74.60	88.90%	63.67%	64.33%	39.33%
278	Osgood, Trista	2028	LTC Mark Haseman	2	CVN3	D	63.15%	1263	2000	0.85%	17.00	85.90%	23.33%	57.00%	59.00%
279	Berend, Evelyn	2028	LTC Mark Haseman	2	CEN2	D	62.76%	1255	2000	1.24%	24.80	90.00%	65.33%	60.67%	47.50%
#N/A										0.00%	0.00				
#N/A										0.00%	0.00				
#N/A										0.00%	0.00				
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#N/A										0.00%	0.00				
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SECTION III

ASSIGNMENTS



TAB I LAB QUIZ-SIGNATURE WRITING EVENT (LAB7)



UNITED STATES MILITARY ACADEMY
WEST POINT.

LAB QUIZ HANDOUT-SEE LAB 7



UNITED STATES MILITARY ACADEMY
WEST POINT.

LAB QUIZ SOLUTION-SEE LAB 7



TAB J **PLW (QUIZ)**



UNITED STATES MILITARY ACADEMY
WEST POINT.

PLW HANDOUT-WITH SOLUTION

Time of completion _____
29-30 January 2026

Cadet _____ **Cutsheet**

Section _____
Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
PH2X2, Physics II
Block 1 Phase Line Writ

1. Instructions:

- a. Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. No phones, cameras, or smartwatches are allowed in the examination room.
- b. Check your exam for 3 problems. Check for the Physics Reference Card provided with the examination.
- c. **Show all work**; partial credit may be given for correct work shown.
- d. For multiple choice questions **bubble in** your selection and choose the **one** answer that best answers each question.
- e. Take up to 30 minutes to complete the examination.
- f. An instructor is in the hall for any questions.

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on _____ (date) at _____ (time).
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

PROBLEM	WEIGHT	SCORE
1	5	
2	15	
3	30	
	Score:	
	Total:	50

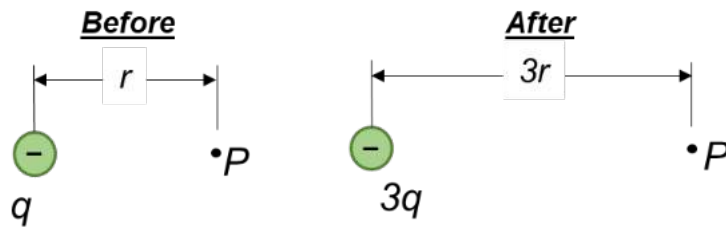
A: 7 (0%)

B: 15 (0%)

D: 28 (0%)

Total: 50

- 5 1. A charge q is a distance of r from point P . **Determine** how the electric field at point P will change after the distance is increased to $3r$ and the charge is increased to $3q$.



- ☐ The magnitude of the electric field decreases by a factor of 9.
- ☒ **5** The magnitude of the electric field decreases by a factor of 3.
- ☐ The magnitude of the electric field remains the same.
- ☐ The magnitude of the electric field increases by a factor of 3.
- ☐ The magnitude of the electric field increases by a factor of 9.

$$\vec{E} = k \frac{q}{r^2} \hat{r}$$

$$q \rightarrow 3q$$

$$r \rightarrow 3r$$

$$k \frac{q}{r^2} \sim k \frac{3q}{9r^2}$$

$$E \rightarrow \frac{1}{3} E$$

A: 0 (0%)
 B: 0 (0%)
 D: 5 (100%)
 Total: 5

Wt. **No.**

15

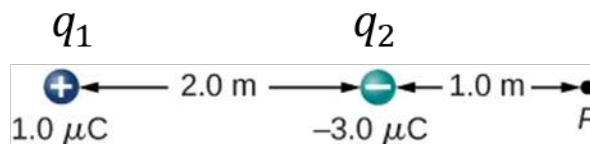
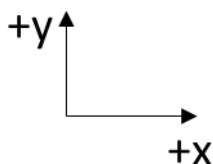
2.

Two charged particles are arranged as shown in the figure below. q_1 has a charge of $1.0 \mu\text{C}$ and q_2 has a charge of $-3.0 \mu\text{C}$. **Calculate** the electric potential at point P due to q_1 and q_2 .

Cadet: _____

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

5



$$V_{net} = V_1 + V_2$$

3

Break into terms

$$V_{net} = k \frac{q_1}{r} + k \frac{q_2}{r}$$

2 Attempts to solve for V_1

2 Attempts to solve for V_2

$$V_{net} = \left(8.988 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2} \right) \frac{(1.0 \times 10^{-6} \text{C})}{3.0 \text{ m}} + \left(8.988 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2} \right) \frac{(-3.0 \times 10^{-6} \text{C})}{1.0 \text{ m}}$$

$$V_{net} = 2996 - 26964$$

$$V_{net} = -2.4 \times 10^4 \text{ V}$$



Correct sign, reported as scalar



Correct answer



Units

Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

A: 3 (20%)
B: 4 (27%)
D: 8 (53%)
Total: 15

Wt. **No.**

Cadet: _____

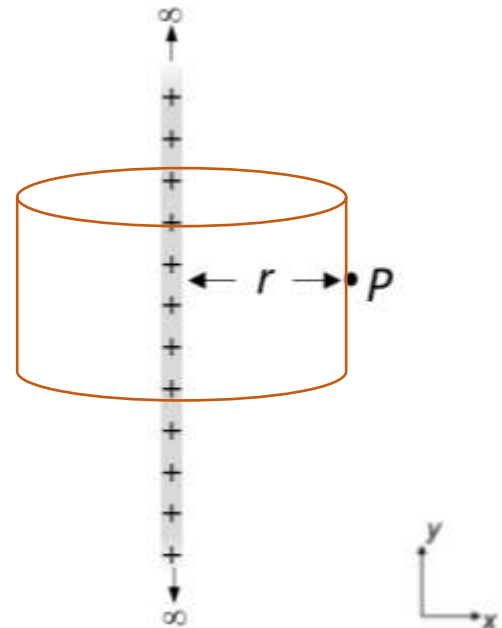
30 3. An infinite, thin, line of charge has a linear charge density $15.4 \times 10^{-7} \text{ C/m}$. Point P is located 25.3 cm from the line of charge.

(5) a. On the figure shown below, **draw** the most appropriate Gaussian Surface.

(25) b. **Calculate** the magnitude of the electric field at point P. Show all work!

3 Draws a Cylinder

2 Cylinder oriented properly



8

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

4

$$\lambda = \frac{Q}{L}$$

3 Identifies which surfaces are zero and which remains

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

$$\int_{Top} E \cos 90^\circ dA + \int_{Wall} E \cos 0^\circ dA + \int_{Bottom} E \cos 90^\circ dA = \frac{\lambda L}{\epsilon_0}$$

4 Replaces q_{enc} with λL

2 Attempts integral

$$EA = \frac{\lambda L}{\epsilon_0}$$

$$E(2\pi rL) = \frac{\lambda L}{\epsilon_0}$$

$$E = 109000 \text{ N/C}$$

$$E = \frac{\lambda}{2\pi r \epsilon_0}$$

2 Correct ANS

1 3 sig figs

1 Units

Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

A: 4 (13%)
B: 11 (37%)
D: 15 (50%)
Total: 30

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TAB K

LAB



UNITED STATES MILITARY ACADEMY
WEST POINT.

LAB#1 HANDOUT

Each appendix, from A to E, guides the experimenter in creating a better narrative for the written lab report. Appendices will be included in each Lab Report. Per the SLAM, Chapter 4, p. 4-3, the only appendix that may be referenced is the Propagation of Uncertainty Table (Appendix D) in the Result section, but the completed calculations must still be present. All other information from appendices must be woven into the narrative of the lab report.

Example Task: The objective is to determine the spring constant, k , of a spring using a cart and mass by varying the mass, m , (independent variable) and measuring the changes in maximum speed, v (dependent variable).

Linearization Table

APPENDIX A

Step 1: Identify fundamental physics. Conservation of Energy: $W_{nc} = E_f - E_o$ Elastic Potential Energy: $U_s = \frac{1}{2}kx^2$ Translational Kinetic Energy: $K_T = \frac{1}{2}mv^2$	Step 2: Apply physics in order to build a model equation.
Step 3: Identify independent and dependent variables. Independent Variable: Dependent Variable:	Step 4: Rearrange equation to $y = mx + b$ form to build the linearized/ model equation.
Step 5: Identify independent and dependent quantities. Independent Quantity: Dependent Quantity:	Step 6: Identify the objective quantity and control variables. Objective Quantity: Control Variables:
Step 7: Write the expression for the slope of your equation	8: Rearrange your slope so that you solve for your objective quantity equation

Measurement Table

APPENDIX B

For more guidance on reporting measurements and using the Measurement Table, refer to the SLAM, p. 3-4.

Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
Mass of the Cart, m [kg]	See App. C	See App. C	Digital Scale
	Measurement Procedure: Zero the scale. Place the cart and any additional masses on the scale. Record the measurement.		
	Justification of Uncertainty: (Reference: SLAM, p. 3-1: Factor , Cause , Absolute Uncertainty) The uncertainty in the mass is due to the quality of the measuring device. The manufacturer specifies that $\delta m = \pm 0.05g$		

Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
Maximum Speed of Cart, v [m/s]	See App. C	See App. C	Smart Cart
	Measurement Procedure: Place any additional mass on top of the cart. Position the compressed spring against the stationary rail. Release the spring as smoothly as possible. Record the velocity in PASCO Capstone. Record the maximum value.		
	Justification of Uncertainty: (Reference: SLAM, p. 3-1: Factor , Cause , Absolute Uncertainty) The uncertainty in the maximum speed is due to the conditions under which the measurement is made. PASCO was set to a low data acquisition rate, which may not have recorded the true maximum speed. Also, there is friction present that may remove energy from the cart when pressing the spring. Therefore, $\delta v = \pm 0.05 \text{ m/s}$.		

Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
Spring Displacement, x [m]	$(0.044 \pm 0.001) \text{ m}$	2.3%	Tape measure
	Measurement Procedure: Measure the distance from the cart to the end of the plunger when the plunger is in both the relaxed and fully compressed position. Subtract measurements from each other to obtain the displacement.		
	Justification of Uncertainty: (Reference: SLAM, p. 3-1: Factor , Cause , Absolute Uncertainty) The uncertainty in the spring displacement is due to the nature of the quality being measured. The displacement of the spring is not something we can measure directly, and taking two measurements increases the uncertainty. We estimate the uncertainty to be $\delta m = \pm 0.001 \text{ m}$.		

Linear Regression

APPENDIX C

Review Lab 1 attachments on the Course Website and find the excel document “AY 24-1 PH2X2 Lab 1 Appendix C Data for Example.” Complete the remainder of the table and conduct a linear regression analysis (Review the SLAM, Appendix H – Graphing, Linear Regression, and Curve Fitting). Remember, the absolute uncertainties for mass and speed are in the measurement tables. See Figure 1 as an example of the setup of Appendix C.

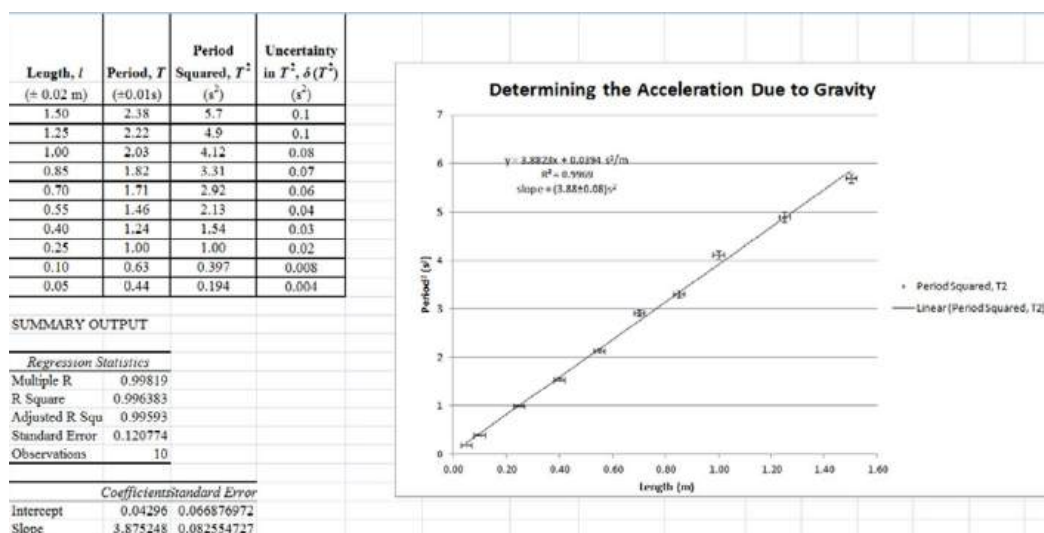


Figure 1. Example Setup of Linear Regression Table (not specific to this example).

Propagation of Uncertainty

APPENDIX D

Objective Quantity:		Objective Quantity Equation:	
Variable	Measured Value	Absolute Uncertainty	Relative Uncertainty
Calculation of Objective Quantity's Magnitude:			
Calculation of Objective Quantity's Absolute Uncertainty:			
Objective Quantity Confidence Interval:			

Error Analysis

APPENDIX E

Experimental Value CI:	Accepted Value CI (given):
Number Line:	
Dominant Form of Error Present (Random or Systematic):	
Acceptable Level of Precision (given):	Acceptable Level of Accuracy (given):
Calculation of Relative Uncertainty:	Calculation of Percent Difference:



UNITED STATES MILITARY ACADEMY
WEST POINT.

LAB#1 SOLUTION

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Linearization Table

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Step 1: Identify fundamental physics. Conservation of Energy: $W_{nc} = E_f - E_o$ Elastic Potential Energy: $U_s = \frac{1}{2}kx^2$ Translational Kinetic Energy: $K_T = \frac{1}{2}mv^2$	Step 2: Apply physics in order to build a model equation. $\frac{1}{2}mv^2 = \frac{1}{2}kx^2$
Step 3: Identify independent and dependent variables. Independent Variable: mass, m Dependent Variable: speed, v	Step 4: Rearrange equation to $y = mx + b$ form to build the linearized/ model equation. $v^2 = kx^2 \frac{1}{m}$
Step 5: Identify independent and dependent quantities. Independent Quantity: $\frac{1}{m}$ Dependent Quantity: v^2	Step 6: Identify the objective quantity and control variables. Objective Quantity: k Control Variables: x
Step 7: Write the expression for the slope of your equation $\text{slope} = kx^2$	8: Rearrange your slope so that you solve for your objective quantity equation $k = \frac{\text{slope}}{x^2}$

Measurement Table

APPENDIX B

For more guidance on reporting measurements and using the Measurement Table, refer to the SLAM, p. 3-4.

Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
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Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
Maximum Speed of Cart, v [m/s]	See App. C	See App. C	Smart Cart
	Measurement Procedure: Place any additional mass on top of the cart. Position the compressed spring against the stationary rail. Release the spring as smoothly as possible. Record the velocity in PASCO Capstone. Record the maximum value.		
	Justification of Uncertainty: (Reference: SLAM, p. 3-1: Factor , Cause , Absolute Uncertainty) The uncertainty in the maximum speed is due to the conditions under which the measurement is made. PASCO was set to a low data acquisition rate, which may not have recorded the true maximum speed. Also, there is friction present that may remove energy from the cart when pressing the spring. Therefore, $\delta v = \pm 0.05 \text{ m/s}$.		

→ could also be quality of the measuring device. Using PASCO capstone software for data collection.

Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
Spring Displacement, x [m]	$(0.044 \pm 0.001) \text{ m}$	2.3%	Tape measure
	Measurement Procedure: Measure the distance from the cart to the end of the plunger when the plunger is in both the relaxed and fully compressed position. Subtract measurements from each other to obtain the displacement.		
	Justification of Uncertainty: (Reference: SLAM, p. 3-1: Factor , Cause , Absolute Uncertainty) The uncertainty in the spring displacement is due to the nature of the quality being measured. The displacement of the spring is not something we can measure directly, and taking two measurements increases the uncertainty. We estimate the uncertainty to be $\delta m = \pm 0.001 \text{ m}$.		

Linear Regression

APPENDIX C

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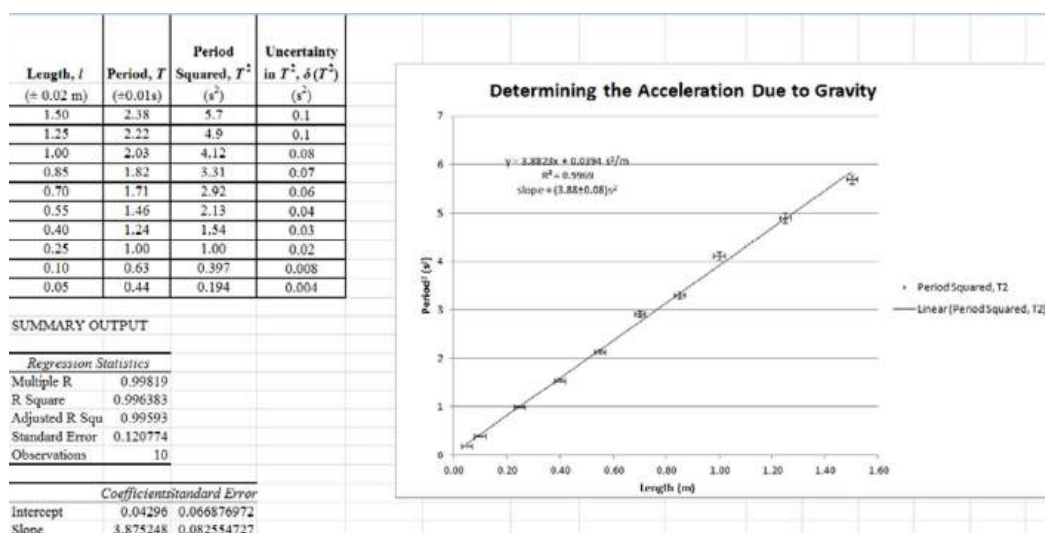


Figure 1. Example Setup of Linear Regression Table (not specific to this example).

See App. C Data for instructors on Share Point.

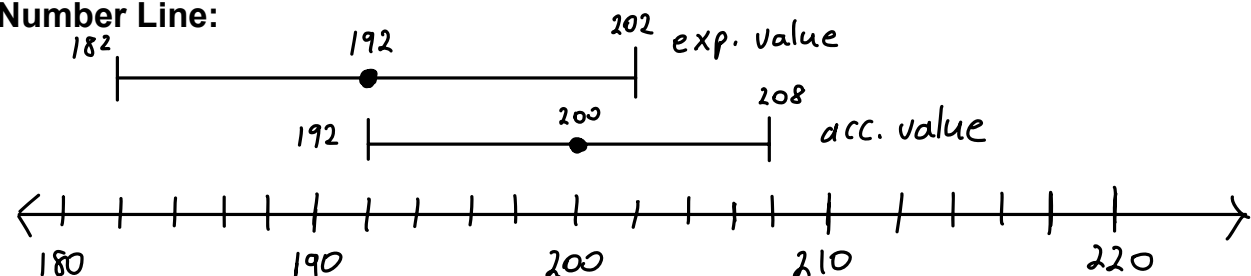
Propagation of Uncertainty

APPENDIX D

Objective Quantity: k		Objective Quantity Equation: $k = \frac{\text{slope}}{x^2}$	
Variable	Measured Value	Absolute Uncertainty	Relative Uncertainty
slope	$0.373 \frac{\text{m}^2 \text{kg}}{\text{s}^2}$	$0.009 \frac{\text{m}^2 \text{kg}}{\text{s}^2}$	2.4%
x	0.044 m	0.001 m	2.3%
Calculation of Objective Quantity's Magnitude: $k = \frac{\text{slope}}{x^2}$ $k = \frac{0.373 \frac{\text{m}^2 \text{kg}}{\text{s}^2}}{(0.044 \text{ m})^2} \left(\frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right)$ Conversion $k = 192.665 \text{ N/m}$			
Calculation of Objective Quantity's Absolute Uncertainty: $\delta k = \sqrt{\left(\frac{\partial k}{\partial \text{slope}} \delta \text{slope} \right)^2 + \left(\frac{\partial k}{\partial x} \delta x \right)^2}$ $\delta k = \sqrt{\left(\frac{1}{x^2} \delta \text{slope} \right)^2 + \left(-\frac{2 \text{slope}}{x^3} \delta x \right)^2}$ $\delta k = \sqrt{\left(\frac{1}{(0.044 \text{ m})^2} (0.009 \frac{\text{m}^2 \text{kg}}{\text{s}^2}) \right)^2 + \left(-\frac{2(0.373 \frac{\text{m}^2 \text{kg}}{\text{s}^2})}{(0.044 \text{ m})^3} (0.001 \text{ m}) \right)^2}$ $\delta k = 9.91 \frac{\text{kg}}{\text{s}^2} \left(\frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right) \longrightarrow \delta k = 9.91 \text{ N/m}$			
Objective Quantity Confidence Interval: $k = (192 \pm 10) \text{ N/m}$			

Error Analysis

APPENDIX E

Experimental Value CI: $k = (192 \pm 10) \text{ N/m}$	Accepted Value CI (given): $k = (200 \pm 8) \text{ N/m}$
Number Line:  physical constant: k units: N/m	
Dominant Form of Error Present (Random or Systematic): Random	
Acceptable Level of Precision (given): 2.0%	Acceptable Level of Accuracy (given): 2.0%
Calculation of Relative Uncertainty: $RU = \frac{\delta k}{ k }$ $RU = \frac{10 \text{ N/m}}{192 \text{ N/m}}$ $RU = 5.2\%$	Calculation of Percent Difference: $PD = \frac{ \text{exp. value} - \text{acc. value} \times 100}{ \text{acc. value} }$ $PD = \frac{ 192 \text{ N/m} - 200 \text{ N/m} \times 100}{ 200 \text{ N/m} }$ $PD = 4.0\%$



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LAB#2 HANDOUT

Lab Worksheet

I. Introduction

All charged objects produce electric fields, $\vec{E}$, in the space surrounding them. Knowing the shape, direction, and magnitude of an electric field is necessary to determine how a charged particle will interact with the field. Visualization is often helpful when analyzing fields and field forces. A convenient way of representing an electric field is using electric field lines. However, they are not something we can see because they are an imaginary description of a gradient. By understanding how to find the lines of equal electric potential between electrodes, you will be able to identify the magnitude and direction of electric field lines. In this exploratory lab, you will identify the lines of equal electric potential surrounding different pairs of charged electrodes and then use them as guides to draw electric field lines and identify the magnitude and direction of the electric field.

II. Theory

Electric field lines are imaginary lines drawn through a region of empty space so that its tangent is at any point in the direction of the electric field vector at that point. The density of lines (the number of lines in each area) represents the magnitude of the field strength. These rules are written more formally as follows:

- The electric field vector is tangent to the electric field line at each point.
- The number of lines per unit area through a surface perpendicular to the lines is proportional to the strength of the electric field in that region.
- Electric field lines must begin on a positive charge and terminate on a negative charge.
- The number of electric field lines drawn leaving a positive charge or approaching a negative charge is proportional to the magnitude of the charge.
- No two electric field lines can cross.
- Electric field lines travel perpendicularly across constant electric potential lines, or equipotential lines, as shown in Figure 1.

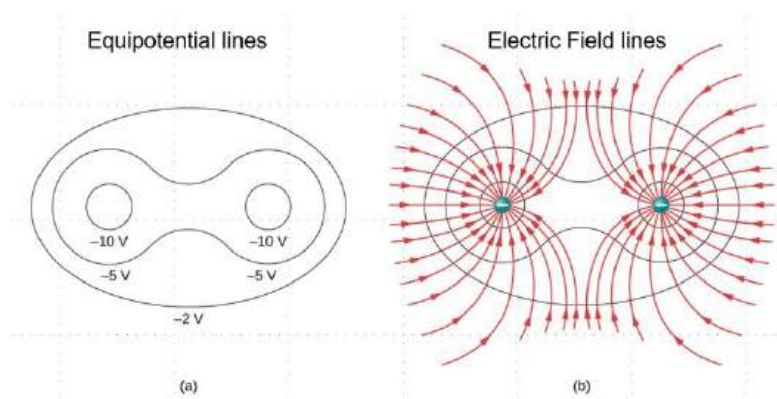


Figure 1 (a) The equipotential lines can be measured with a voltmeter in this laboratory experiment. (b) The corresponding electric field lines are found by drawing them perpendicular to the equipotential lines. Note that these fields are consistent with two equal negative charges. (Figure 7.32, S. Ling, J. Sanny, and W. Moebis, *University Physics*, vol. 2, OpenStax, 2017).

Electric fields like the ones presented in Figure 1 can be graphed in the laboratory setting using the following equipment: a power supply to oppositely charged electrodes (conductors in which electricity enters or leaves the object); a digital multimeter (DMM) to measure the electric potential difference between electrodes; and conductive paper. This paper allows enough current to measure the electric potential difference between the electrodes and graph with a marker to identify equipotential lines.

III. Procedure

The following equipment and materials are required for this lab:

- a) PASCO 850 Universal Interface Box
- b) 4 x PASCO Conductive Papers
- c) 1 x Circular metal component
- d) 2 x Straight metal bar components
- e) 10 x Metal pushpins
- f) 2 x Screws
- g) 2 x Nuts
- h) 1 x Corkboard
- i) 1 x DMM to measure potential difference
- j) 1 x pair of DMM test probes (1 x red, 1 x black)
- k) 3 x banana plug patch cord alligator clips (2 x red, 1 x black)
- l) 1 x pencil or marker to mark equipotential lines.

To Setup your PASCO 850 Universal Interface and provide an electric potential difference (or voltage):

- a. Select the "Signal Generator" option in the tools section on your screen's left-hand side.
- b. If you do not see the option, select the "eye" icon on the top left side of the tools section, pick "Signal Generator" so it is visible in the tools section, and then deselect the "eye" icon.
- c. In the "Signal Generator" option, select the "DC" or Direct-Current "Waveform" option, as shown in Figure 2.
- d. You can now change the voltage in the "DC Voltage" section and turn it on or off.
- e. Ensure the voltage is off for the moment.
- f. Connect two patch cord alligator clips in your PASCO box's red and black ports. To avoid confusion, ensure the cords follow the same color patterns.

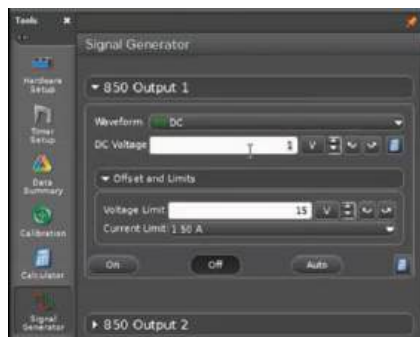


Figure 2 PASCO Signal Generator 850 Output 1

To set up your DMM (Fluke Multimeter):

- Turn the dial to DC Voltage ($\overline{V}$).
- Insert the black test probe into the COM jack (a common return path for electric current).
- Insert the red test probe into the $V\Omega$ jack.
- Test your DMM by turning on the "Signal Generator" in PASCO, and set the DC Voltage to 1 Volt, and measuring the difference in electric potential between the red and black ports in the PASCO box.
- Turn off the Signal Generator once you finish the test.

Below are the electrode patterns you will need to find the equipotential and electric field lines.

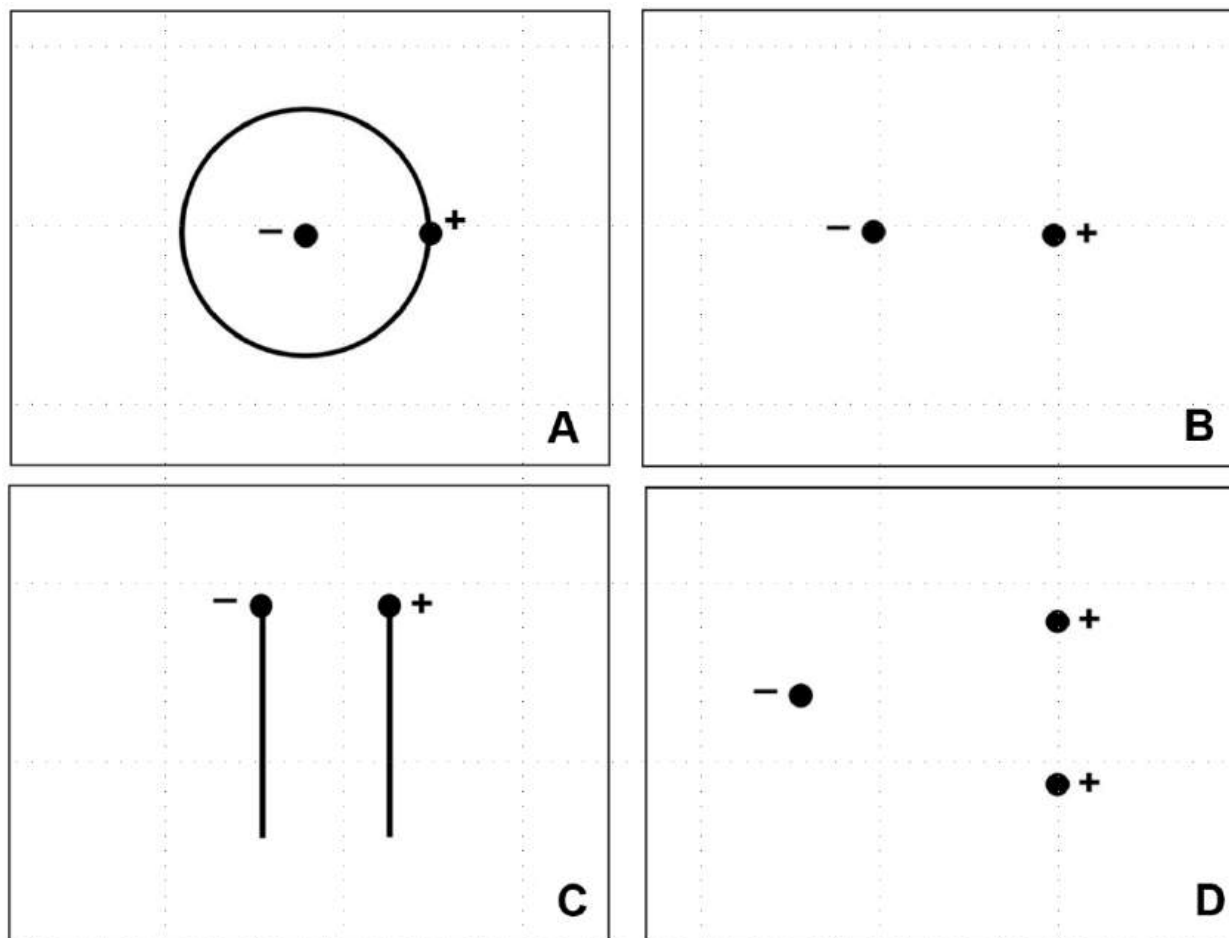


Figure 3 Electrode Patterns – The dots represent the position of the pins or bolts/nuts to which the power supply leads are attached.

To begin the setup:

- Start with Pattern A in Figure 3 using the circular metal component.
- Pin each corner of the conductive paper to the cork board using the metal pushpins. You will have to push the two screws through the paper.

- c. Attach the circular metal component to the cork board by aligning the screws with the holes in the component and tightening the nuts on top.
- d. Attach one metal push pin as shown in the center of the component. The dots in Figure 3 represent the position of the electrodes (push pins or screws/nuts).
- e. Ensure the push pin is firmly pressed into the conductive paper and has good contact.
- f. Connect the two PASCO Box cords with alligator clips to the electrodes: the positive terminal (red) needs to be with the positive electrode and the negative terminal (black) with the negative electrode, as shown in Figure 3 for consistency.
- g. Connect the black test probe of your DMM to the negative electrode. This negative electrode will be your reference point when measuring the potential difference.
- h. Review Figure 4 to validate your setup.

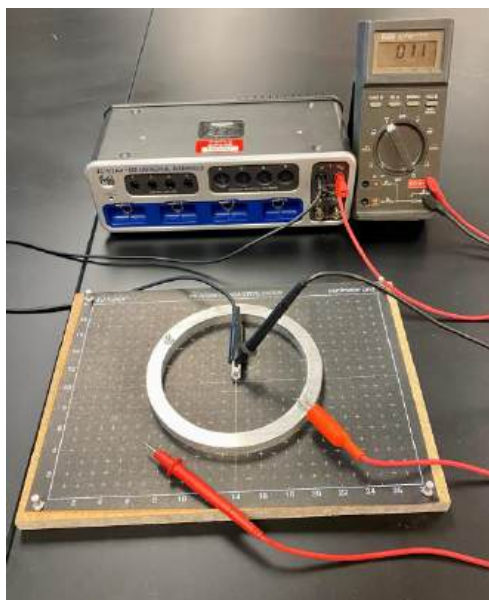


Figure 4 Experimental Setup for Pattern A.

To find the equipotential lines:

- a. Turn on the power supply through the "Signal Generator" and adjust the potential difference to 1 Volt.
- b. Using the red test probe of your DMM, touch any black space surrounding either electrode on the conductive paper and observe the potential difference measurement.
- c. If the DMM does not show different electric potential measurements, do a connections check and retest.
- d. Probe a position relatively close to an electrode using your DMM.
- e. Use a pencil to mark the position.
- f. Continue measuring the same voltage and marking the different positions surrounding the electrode using the DMM. Mark each new point until there are

enough points to accurately draw a smooth line connecting the equipotential line.

- g. Label the line with the voltage you chose and its proper unit of "V."
- h. Repeat the previous data collection steps by finding four or more equipotential lines. Ensure to encompass the equipotential lines.
- i. Once you are complete, turn off the power supply.

To find the electric field lines:

- a. Use a pencil and draw an arrow starting from any point on the surface of the positive electrode and extending it to the nearest equipotential line.
- b. The electric field line must intersect the equipotential line at a right angle.
- c. Continue drawing until reaching the negative electrode.
- d. Draw an arrow indicating the direction of the electric field.
- e. Repeat the process 7 to 9 times until enough electric field lines are shown.
- f. You may review your answers using the [Electric Field Lines PHET Simulator](#) (Note: Check the "Voltage" box and press the "pencil" icon to create equipotential lines).

Once complete with Pattern A, continue with the rest of the patterns until complete.

IV. Submission Requirements

- a. Complete the Results and Discussion sections found in pages 6 through 9 of the lab worksheet.
- b. Submit as a single PDF and label "Lab 3_X Hour_Last Names" and include a properly formatted cover sheet.

V. Results

a. Cut and paste a landscape picture of your field map for Pattern A.

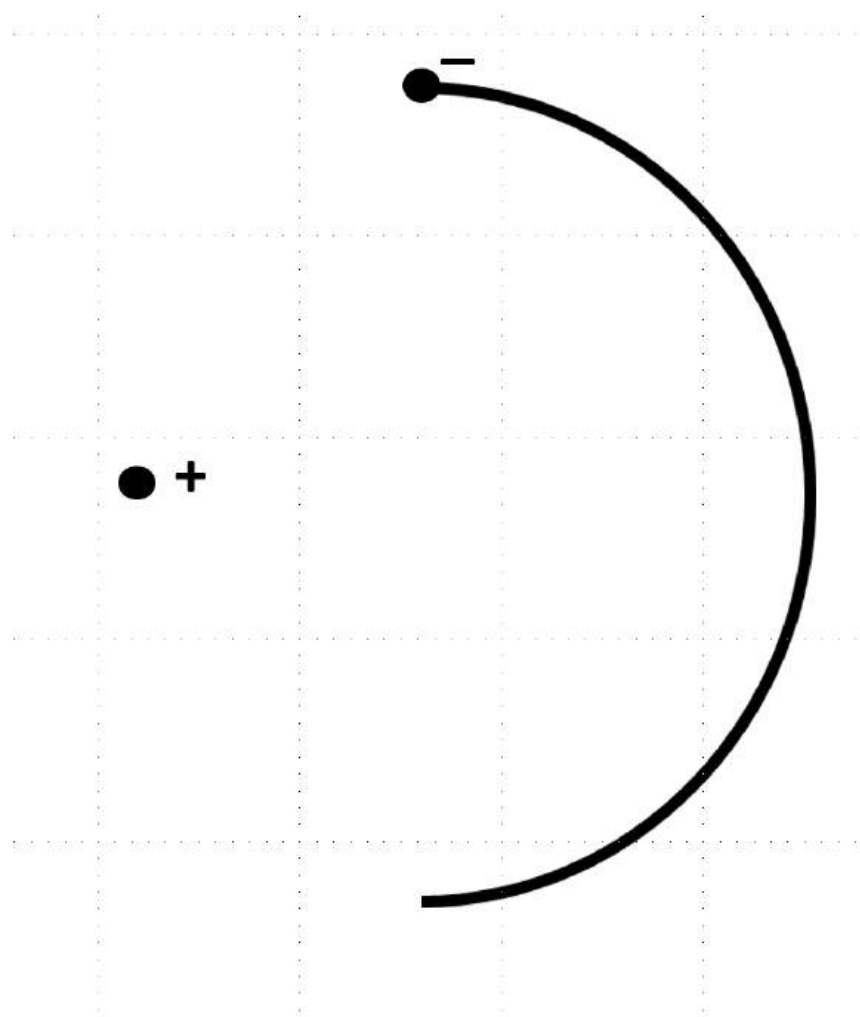
b. Cut and paste a landscape picture of your field map for Pattern B.

c. Cut and paste a landscape picture of your field map for Pattern C.

d. Cut and paste a landscape picture of your field map for Pattern D.

- What are the primary similarities and differences between the electric fields surrounding each electrode configuration?
- Where was the electric field strongest in each electrode configuration? Justify your answer.
- What can be said about the voltage at each point of an equipotential line?
- What do electric field lines represent?
- How would the shape of the electric field lines in each configuration change if you increased the potential difference between the electrodes? Justify your answer.

- f. Based on your experience in this lab activity, estimate the shape of the electric field surrounding the electrodes shown below. Draw equipotential lines and enough electric field lines surrounding the electrodes to distinguish the shape of the field corresponding to your prediction. Assume that the amount of charge on both electrodes is equal and opposite.



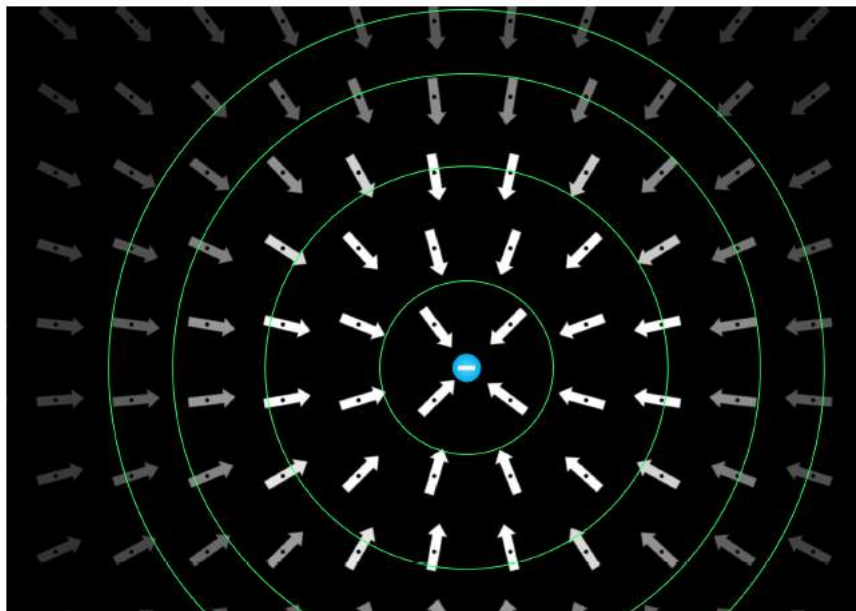


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LAB#2 SOLUTION

V. Results

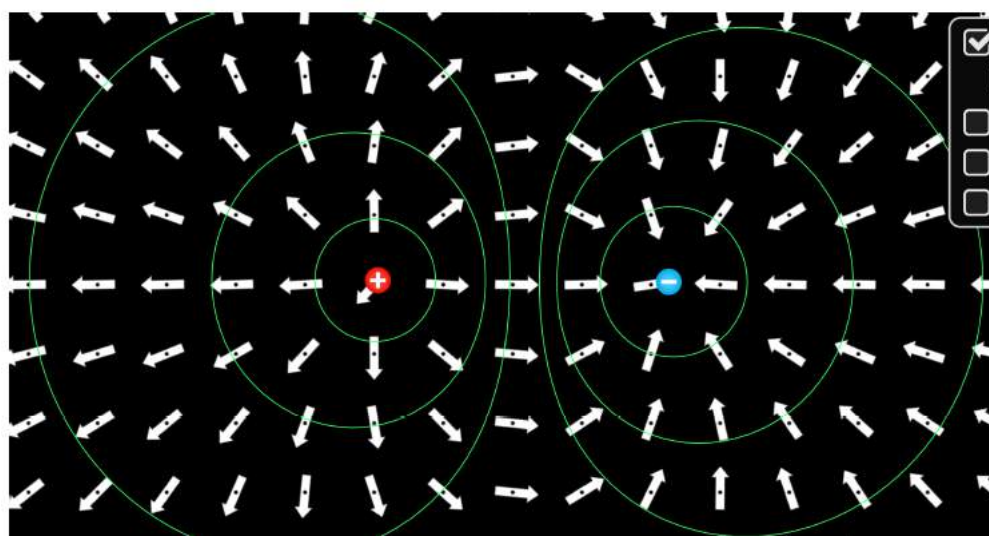
a. Cut and paste a landscape picture of your field map for Pattern A.



4	106
13	32.56
23	57.56

- 2 - At least 4 equipotential lines
- △ - Equipotential lines labeled in volt units
- 2 - At least 7 $\vec{E}$ field lines
- ① - $\vec{E}$ field lines $\perp$ to equipotential lines
- ① - Relatively correct field map

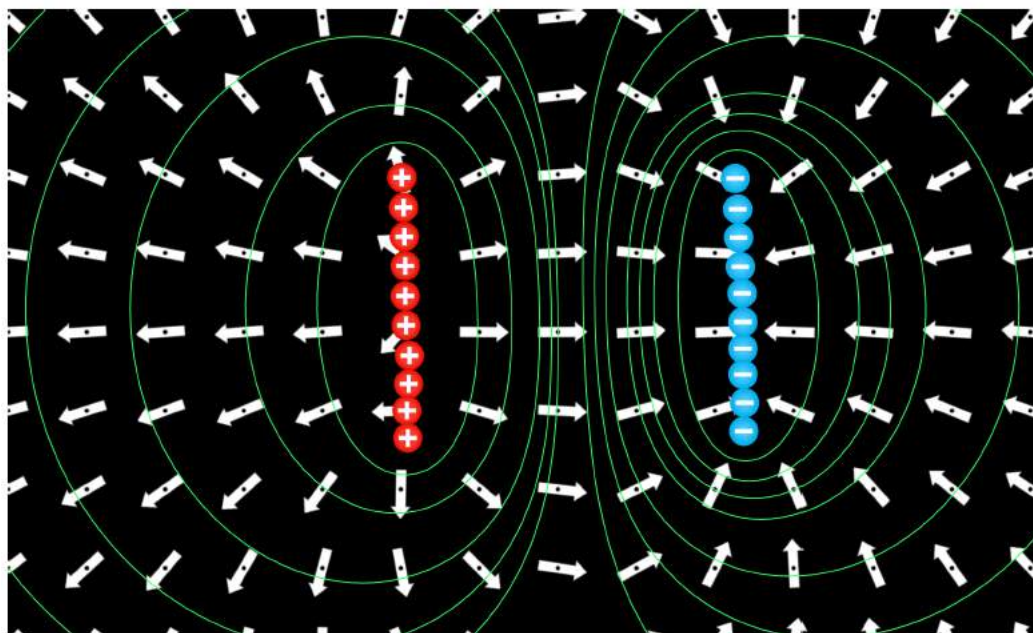
b. Cut and paste a landscape picture of your field map for Pattern B.



- 2 - At least 4 equipotential lines
- △ - Equipotential lines labeled in volt units
- 2 - At least 7 $\vec{E}$ field lines
- ① - $\vec{E}$ field lines $\perp$ to equipotential lines
- ① - Relatively correct field map

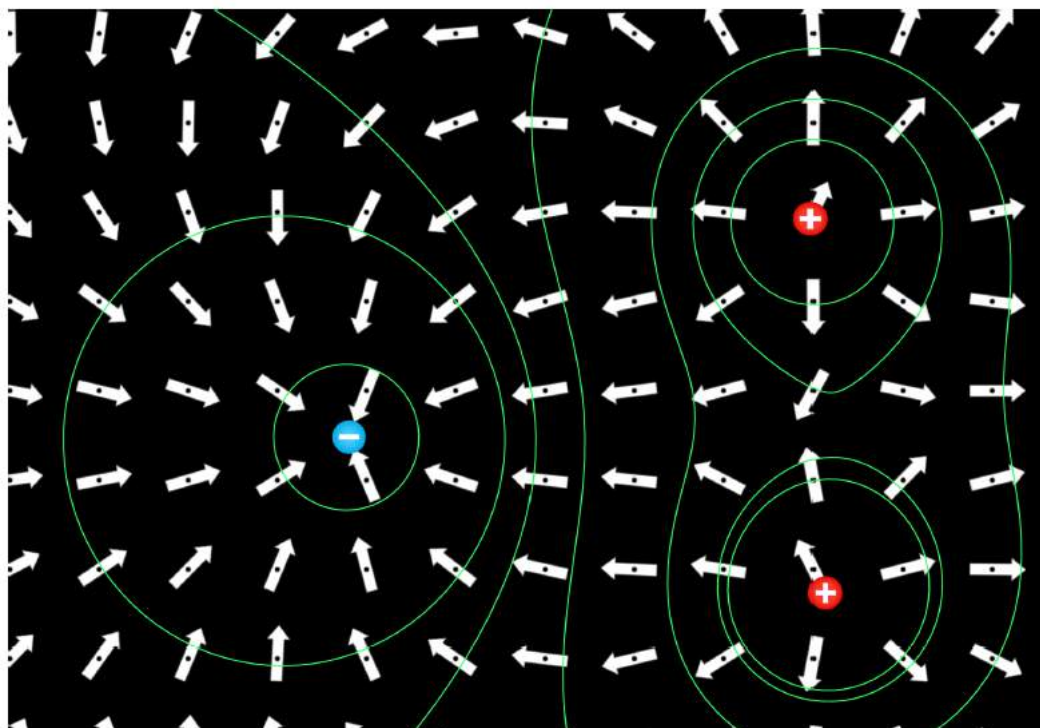
c. Cut and paste a landscape picture of your field map for Pattern C.

pos. & neg
plates
should be
switched →



- 2 - At least 4 equipotential lines
- Δ - Equipotential lines labeled in volt units
- 2 - At least 7 $\vec{E}$ field lines
- ① - $\vec{E}$ field lines $\perp$ to equipotential lines
- ① - Relatively correct field map

d. Cut and paste a landscape picture of your field map for Pattern D.



- 2 - At least 4 equipotential lines
- Δ - Equipotential lines labeled in volt units
- 2 - At least 7 $\vec{E}$ field lines
- ① - $\vec{E}$ field lines $\perp$ to equipotential lines
- ① - Relatively correct field map

VI. Discussion

- a. What are the primary similarities and differences between the electric fields surrounding each electrode configuration?

- ① Similarity: Electric field lines always travel from high to low "voltage" or from positive charge to negative charge.
- ① Difference: Electric field lines curved much sooner in Patterns B and D due to their equipotential lines.

- b. Where was the electric field strongest in each electrode configuration? Justify your answer.

- ② Strongest near the positive or negative electrodes due to a higher density of electric field lines closer to the charge. The attraction or repulsion (strength of electric field) is strongest closest to the electrode.

- c. What can be said about the voltage at each point of an equipotential line?

- ① The voltage at each point on the equipotential line is the same.

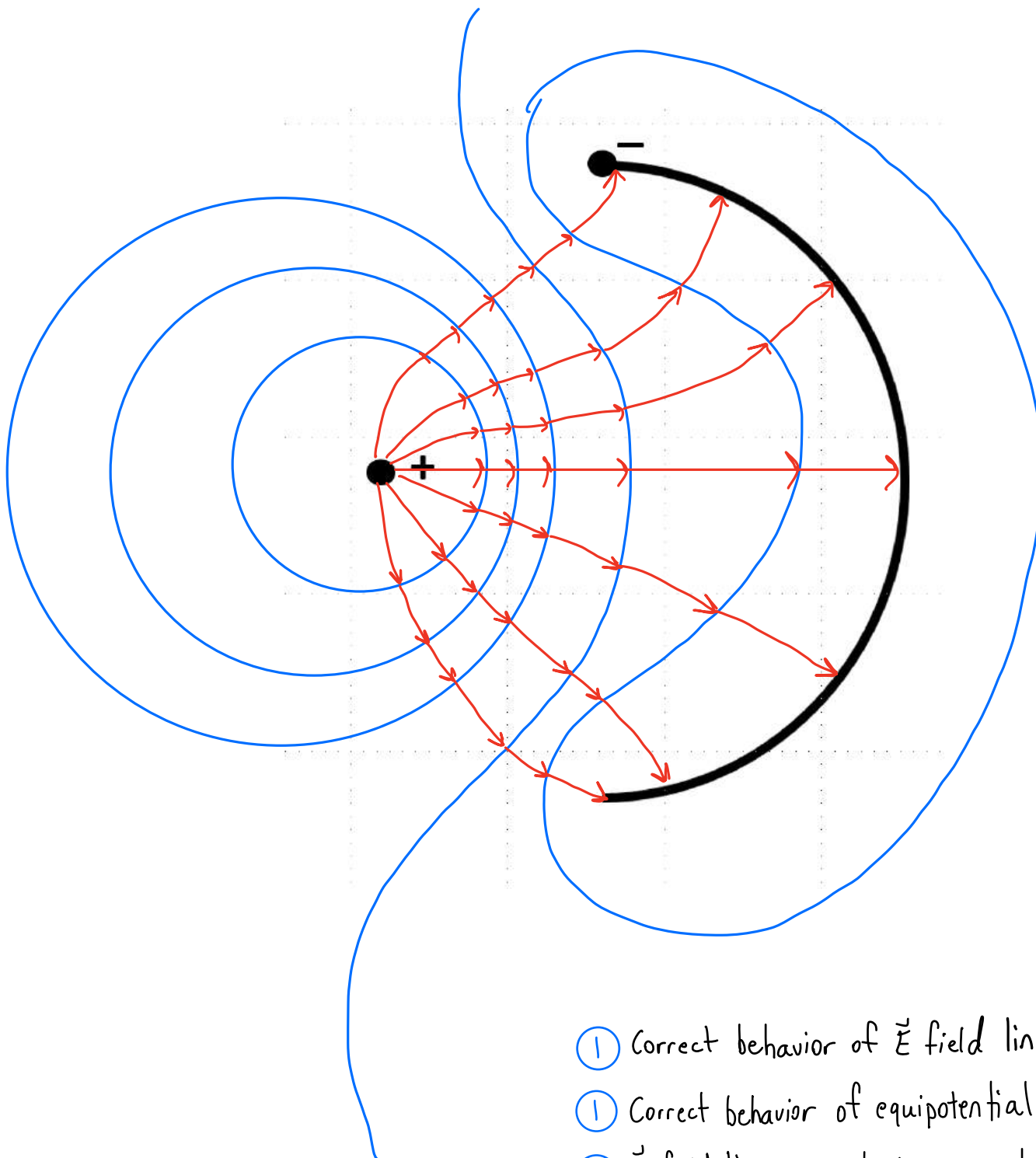
- d. What do electric field lines represent?

- ② They represent the direction of the electric field vector at any point along the line (tangent to the line). The electric field line density is proportional to the magnitude of the electric field in a given area.

- e. How would the shape of the electric field lines in each configuration change if you increased the potential difference between the electrodes? Justify your answer.

- ② The shape would not change. Voltage would increase but the distribution of the equipotential lines would remain the same, hence electric field lines' shapes would remain the same.

- f. Based on your experience in this lab activity, estimate the shape of the electric field surrounding the electrodes shown below. Draw equipotential lines and enough electric field lines surrounding the electrodes to distinguish the shape of the field corresponding to your prediction. Assume that the amount of charge on both electrodes is equal and opposite.



- ① Correct behavior of $\vec{E}$ field lines
- ① Correct behavior of equipotential lines
- ① $\vec{E}$ field lines are $\perp$ to equipotential lines



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LAB#3 HANDOUT

Pre-Lab Tasks

I. Lab Overview

In this lab, you will use a charge-to-mass (q/m) apparatus that provides a simple way to find the q/m of the electron by varying the current, I , through the apparatus' Helmholtz coils and measuring the diameter, d , of the electron's circular path.

It works by accelerating a beam of electrons from rest through an electric potential difference, ΔV , to determine the electron's velocity. Then, a pair of Helmholtz coils produce a uniform and measurable magnetic field, B , at right angles to the electron beam. As result, the magnetic field deflects the electron beam in a circular path. The q/m apparatus is shown in Figures 1 and 2. See TP 11-32 in Lesson 18 for a similar example.

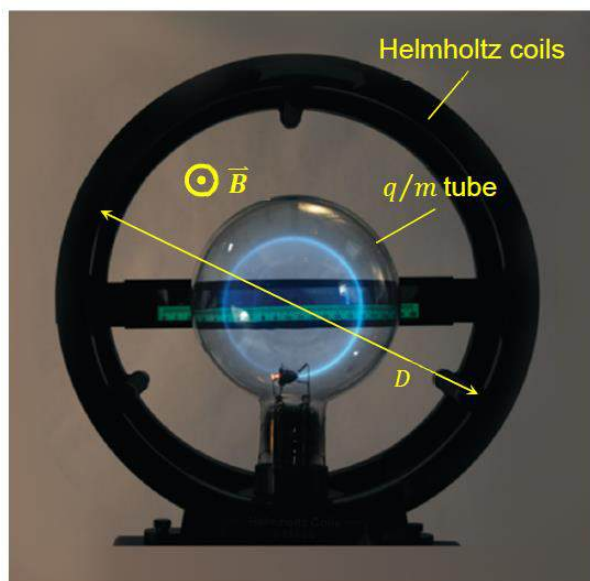


Figure 1. Electron Charge-to-Mass Apparatus (Adapted from "Electron Charge-to-Mass Apparatus," PASCO, <https://www.pasco.com>, Accessed 11 October 2022).

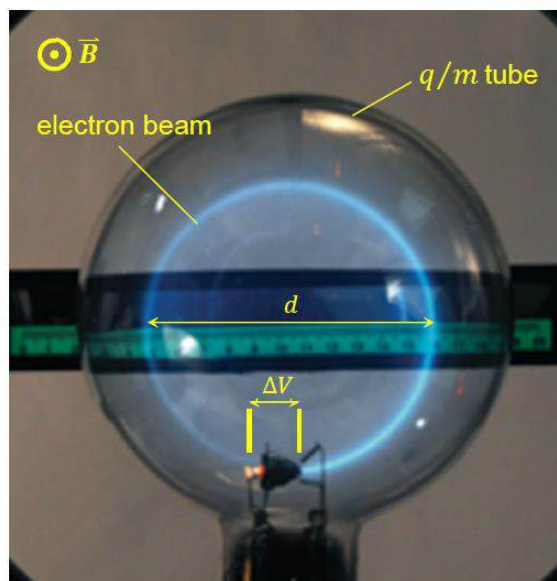


Figure 2. q/m tube. (Adapted from "Electron Charge-to-Mass Apparatus," PASCO, <https://www.pasco.com>, Accessed 11 October 2022).

In the helium-filled electron tube the electrons can move along a circular path in a magnetic field. The tube also contains helium gas at a precisely set pressure. The gas atoms are then ionized along the length of the circular path due to collisions with the electrons, get excited, and as a result, make the electrons' circular path visible.

II. Physics Overview

Recall that if an electron with a charge, q , and a velocity, $\vec{v}$, experiences a magnetic field, $\vec{B}$, perpendicular to its direction, it also experiences a Lorentz force, $\vec{F}$, in a direction perpendicular to both its velocity and the magnetic field:

$$\vec{F} = q\vec{v} \times \vec{B} \quad (1)$$

This gives rise to a centripetal force on the electron in a circular path with a diameter, d . The velocity, $\vec{v}$, also depends on the accelerating voltage, ΔV , of the electron, which initially starts from rest. Therefore, recall the equation for the electric potential energy U_E ,

$$U_E = q\Delta V \quad (2)$$

Lastly, a pair of Helmholtz coils can produce a uniform magnetic field. The two parallel circular coils are separated by a distance equal to half the diameter, D , of the coils, and as such, the magnetic field at a point midway between the coils is given by:

$$B = \mu_o \left(\frac{4}{5}\right)^{3/2} \frac{2NI}{D} \quad (3)$$

where μ_o is the permeability of free space, N is the number of turns of wire on each coil, and I is the current through the coils. Since the magnetic field generated from the Helmholtz coils is proportional to the current through it, it is possible to vary the current to change the magnetic field affecting the electrons in the circular path.

III. Process of Linearization

1. Derive an algebraic expression for the electron's charge-to-mass ratio (q/m) from Equations 1 through 3. (a). To begin, use the theory of conservation of energy to derive an algebraic expression for the final speed, v , of the electrons after they are accelerated through the potential difference. This expression should be expressed in terms of q , m , and ΔV . (b). Using a FBD, Newton's Second Law analysis, and the result from part (a), show that $\frac{q}{m} = \frac{8\Delta V}{B^2 d^2}$. (c). The magnitude of the magnetic field at a point midway between the Helmholtz coils is given by Equation 3 above. Using this equation and the result from part (b), show that $\frac{q}{m} = \frac{2D^2 \Delta V}{\frac{4^3}{5} (\mu_o N I d)^2}$.

- a. The objective is to experimentally determine the q/m of an electron. You will vary the current, I , going through the Helmholtz Coils and measure the diameter, d , of the electron's circular path. Complete the Linearization Table.

Linearization Table

APPENDIX A

Step 1: Identify fundamental physics	Step 2: Apply physics in order to build a model equation
Step 3: Identify independent and dependent variables	Step 4: Rearrange equation to $y = mx + b$ form to build the linearized model equation.
Step 5: Identify independent and dependent quantities	Step 6: Identify the objective quantity and control variables O.Q.: q/m C.V.: $D, \Delta V$ (We can assume that N and μ_o are constants).
Step 7: Write the expression for the slope of your equation	8: Rearrange your slope so that you solve for your objective quantity equation

Lab Tasks

I. Lab Overview

In this lab, you will use a charge-to-mass (q/m) apparatus that provides a simple way to find the q/m of the electron by varying the current, I , through the apparatus' Helmholtz coils and measuring the diameter, d , of the electron's circular path.

II. Lab Set-Up

Important Note: High Voltage is applied to the q/m bulb. Therefore, ensure the power supply is OFF before making any cable connections and do not touch the q/m tube. Only turn on power supplies when ready and after the instructor has reviewed your wiring. If you are disorganized with the placement of your wires, the instructor may ask you to repeat the process. **SAFETY IS PRIORITY.**

As shown in Figure 1 and Figure 2, the items required for this experiment are:

1. 1 x q/m apparatus (q/m tube, Helmholtz Coils, and Base)
2. 1 x universal power supply OR 1 x constant voltage power supply and 1 x constant current power supply
3. 2 x digital multimeter (DMM)
4. 11 x banana plug-patch cords

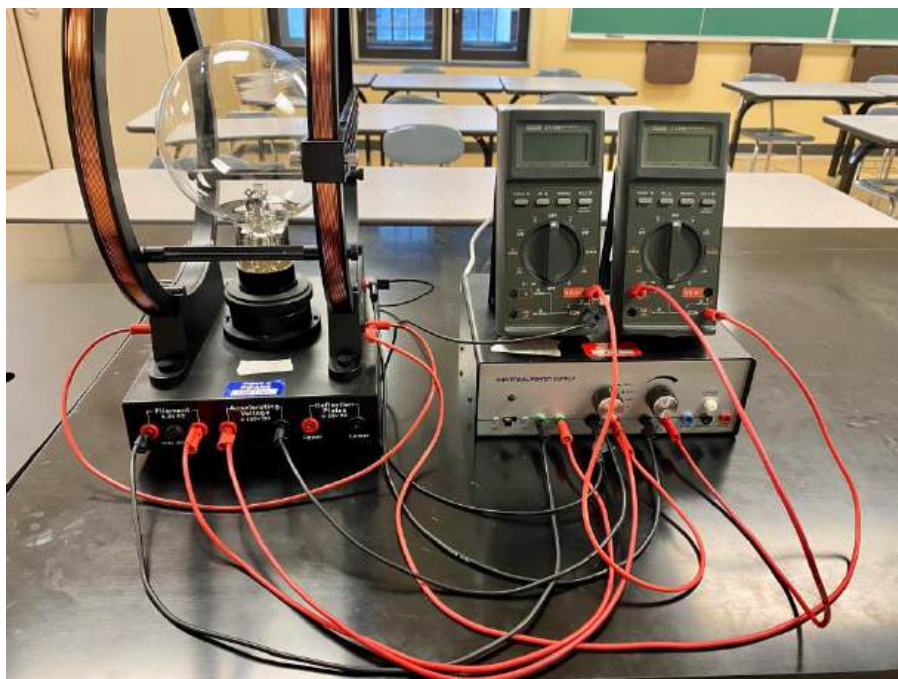


Figure 1 From left to right, q/m apparatus containing q/m tube, Helmholtz Coils, and Base; universal power supply that provides 6 V from 6 Vac ports to Filament terminals in the q/m apparatus, "High Voltage" ranging from 125 to 500 V, and "Regulated" voltage ranging from 0 to 5 V; one DMM measuring voltage in the "High Voltage" port and one DMM measuring Current in the "Regulated" port.

For a lab set up in Figure 1:

- a. Ensure the knob of the universal power supply is at 125 V and the "Regulated" Knob is completely rotated counterclockwise. Turn "ON."
- b. Allow filament to heat up for about 1 minute.
- c. With the "Regulated" knob, slowly increase the current to the Helmholtz coils. Watch the electron beam and check that the electron beam begins to form a circle.
 - If it does not deflect, turn everything off, and reverse the polarity of one of the Helmholtz coils so that the current passes through both in the same direction.
 - If the electron beam forms a spiral, ask your instructor to rotate the tube on the platform until a closed circle is formed.

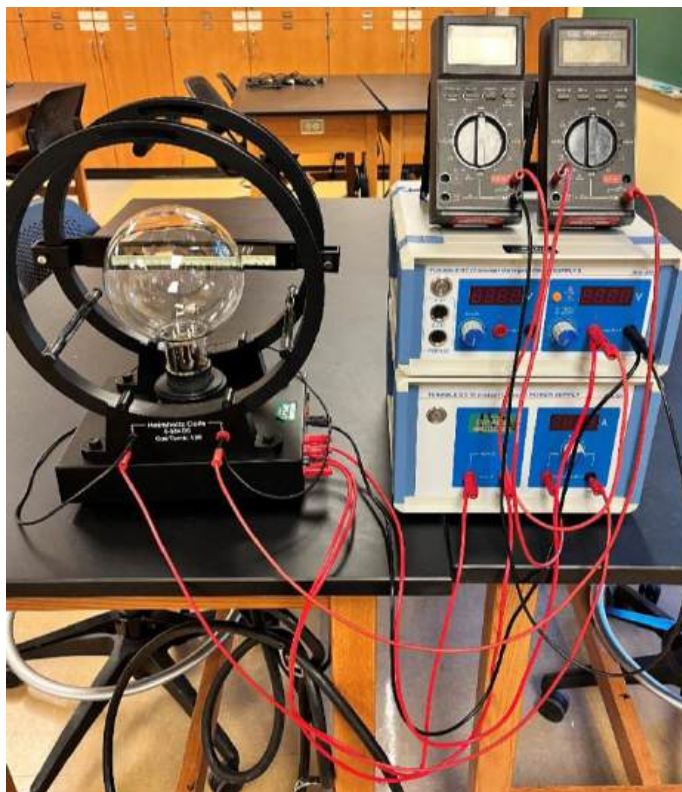


Figure 2 From left to right, q/m apparatus containing q/m tube, Helmholtz Coils, and Base; constant voltage power supply; constant current power supply; one DMM measuring voltage and one DMM measuring Current

For a lab set up in Figure 2:

- a. Turn on "constant voltage power supply" and ensure the knob is at 125 V and the knob on the "constant current power supply" is set to 0.00 Amps.
- b. Allow filament to heat up for about 1 minute.

c. With the knob on the “constant current power supply”, slowly increase the current to the Helmholtz coils. Watch the electron beam and check that the electron beam begins to form a circle.

- If it does not deflect, turn everything off, and reverse the polarity of one of the Helmholtz coils so that the current passes through both in the same direction.
- If the electron beam forms a spiral, ask your instructor to rotate the tube on the platform until a closed circle is formed.

III. Record Data

a. Read the current through the Helmholtz Coils displayed on your DMM and record your data in Appendix C. Calculate its confidence interval as shown in the [SLAM](#), pages 2-14 to 2-16.

b. Measure the diameter of the electron's circular path with the ruler attached to the q/m apparatus. Vary the current to measure the diameter and record your data in Appendix C. Record at least 7 points.

c. Measure the "Accelerating Voltage" using a Digital Multimeter (DMM) and calculate its confidence interval as shown in the [SLAM](#), pages 2-14 to 2-16.

d. Turn "OFF" all power supply boxes.

e. Measure the diameter of the Helmholtz Coils and determine its absolute uncertainty.

IV. Submission Requirements

- Derive an algebraic expression for the electron's charge-to-mass ratio (q/m) from Equations 1 through 3 found in the pre-lab. Express your answer in terms of ΔV , d , N , I , D , and μ_0 (Should have already done in pre-lab – This is separate from Appendix A).
- Complete the Linearization Table (Appendix A).
- Complete the Measured Quantities Tables (Appendix B) for d , D , ΔV , and I . Treat μ_0 and N as constants.
- Conduct a linear regression by completing a full Appendix C.
- Complete the confidence interval using a Propagation of Uncertainty Table (Appendix D). Remember, you already have the electron's charge and mass in the PRC.
- Conduct an error analysis using a Precision and Accuracy Table (Appendix E). Assume that a historically acceptable level of accuracy for this lab is 2.5% and a historically acceptable level of precision is 2.5%.
- Complete the worksheet for this lab to include parts a through f in your submission. Submit as a single PDF and label "Lab 3_X Hour_Last Names."



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LAB#3 SOLUTION

Lab 3 Cutsheet

Read: Derive an algebraic expression for the electron's charge-to-mass (q/m) ratio.

Task: Solve for q/m

Physics: $\vec{F}_b = q\vec{v} \times \vec{B}$

$$F_c = ma_c, a_c = \frac{v^2}{r}, r = \frac{d}{2}$$

$$U_E = q\Delta V$$

$$B = \mu_0 \left(\frac{4}{5}\right)^{3/2} \frac{2NI}{D} \rightarrow B \text{ field between coils}$$

Given: $\Delta V, d, D, N, I$, and μ_0

Assumptions: Energy is conserved, e^- undergoes uniform circular motion, and the e^- initially starts from rest.

Solve:

Step 1

$$\vec{F}_b = q\vec{v} \times \vec{B}$$

$$F_b = qvB \sin 90^\circ$$

$$F_b = qvB$$

$$ma_c = qvB$$

$$m \frac{v^2}{d/2} = qvB$$

$$\frac{2mv}{d} = qB$$

$$q/m = \frac{2v}{dB} \quad (1)$$

Step 2

$$U_E = q\Delta V$$

$$-\Delta U_E = W_c, W_c = \Delta K$$

$$-\Delta U_E = \Delta K$$

$$-(U_{Ef} - U_{Ei}) = K_f - K_i$$

$$-U_{Ef} = K_f \quad (\text{negative charge on } e^- \text{ cancels negative sign on } U_{Ef})$$

$$q\Delta V = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2q\Delta V}{m}} \quad (2)$$

2 Applies CoE to get v eqn

Step 3

Plug equation (2) into equation (1)

$$q/m = \frac{2\sqrt{\frac{2q\Delta V}{m}}}{dB} \quad (0.5) \text{ subs v into } q/m \text{ eqn}$$

$$(q/m)^2 = \left(\frac{2\sqrt{\frac{2q\Delta V}{m}}}{dB} \right)^2$$

$$q^2/m^2 = \frac{8q\Delta V}{d^2B^2}$$

$$\frac{q}{m} = \frac{8\Delta V}{d^2B^2} \quad (3)$$

Step 4

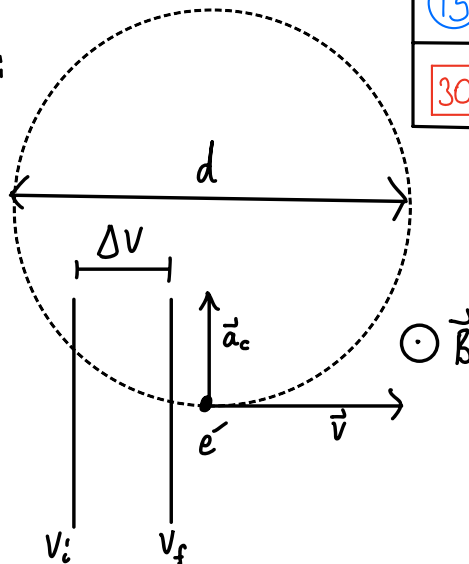
Plug equation for B field between coils into equation (3)

$$\frac{q}{m} = \frac{8\Delta V}{d^2 \left[\mu_0 \left(\frac{4}{5}\right)^{3/2} \frac{2NI}{D} \right]^2} \rightarrow \frac{q}{m} = \frac{2\Delta V D^2}{d^2 \mu_0^2 \left(\frac{4}{5}\right)^3 N^2 I^2}$$

(0.5) Substituted B between coils into q/m equation

3	10.0%
15	30.0%
30	60.0%

Draw:



1 Attempts to solve for $\frac{q}{m}$

1 Answer

4/2/1

Linearization Table

APPENDIX A

Step 1: Identify fundamental physics $\vec{F}_B = q\vec{v} \times \vec{B}$ $f_c = ma_c, a_c = \frac{v^2}{r}, r = \frac{d}{2}$ $U_E = q\Delta V$ $B = \mu_0 \left(\frac{4}{3}\right)^{1/2} \frac{2NI}{D}$	Step 2: Apply physics in order to build a model equation $\frac{q}{m} = \frac{2\Delta V D^2}{d^2 \mu_0^2 \left(\frac{4}{3}\right)^3 N^2 I^2}$
Step 3: Identify independent and dependent variables Independent Variable: I ☐ Dependent Variable: d ☐	Step 4: Rearrange equation to $y = mx + b$ form to build the linearized/ model equation. $\frac{1}{d^2} = \frac{qN^2 \left(\frac{4}{3}\right)^3 \mu_0^2}{2m \Delta V D^2} I^2$ ☐ could also report IQ as $\frac{1}{I^2}$ and DQ as d^2 or could have taken
Step 5: Identify independent and dependent quantities Independent Quantity: I^2 ☐ Dependent Quantity: $1/d^2$ ☐	Step 6: Identify the objective quantity and control variables Objective Quantity: q/m Control Variables: $D, \Delta V$ (N and μ_0 are assumed constant)
Step 7: Write the expression for the slope of your equation $\text{slope} = \frac{qN^2 \left(\frac{4}{3}\right)^3 \mu_0^2}{2m \Delta V D^2}$ ☐	8: Rearrange your slope so that you solve for your objective quantity equation $\frac{q}{m} = \frac{\text{slope } 2\Delta V D^2}{N^2 \left(\frac{4}{3}\right)^3 \mu_0^2}$ ☐

square root of eqn.

Measurement Table

APPENDIX B

See App. C See App. C

Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
	($\pm$) m	% (2 sig fig)	Measuring Device
d, Diameter [m] of the electron's circular path	Measurement Procedure: Detailed description of how they measured the diameter of the electron's path. Must be detailed enough that the process could be repeated using only this description and obtain a similar result.		
	Justification of Uncertainty: Must name one of the four factors and <u>logically explain why</u> that is the largest contributor. Most likely <u>nature of the quantity being measured</u> . Must also include a <u>logical value</u> that must match the absolute uncertainty in the confidence interval.		

correct sig figs and CI properly formatted ①

Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
	($\pm$) m	% (2 sig fig)	Measuring Device
D, Diameter [m] of the Helmholtz Coils	Measurement Procedure: Detailed description of how they measured the diameter of the Helmholtz coils. Must be detailed enough that the process could be repeated using only this description and obtain a similar result.		
	Justification of Uncertainty: Must name one of the four factors and <u>logically explain why</u> that is the largest contributor. Most likely <u>quality of measuring device</u> . Must also include a <u>logical value</u> that must match the absolute uncertainty in the confidence interval.		

4 / 1 / 0

Measurement Table

APPENDIX B

Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
Current [A], I	See App. C	See App. C	DMM
	Measurement Procedure: Detailed description of how they measured the current through the coils. Must be detailed enough that the process could be repeated using only this description and obtain a similar result.		
	Justification of Uncertainty: Must name one of the four factors and <u>logically explain why</u> that is the largest contributor. Most likely <u>quality of measuring device</u> . Must also include a logical <u>value</u> that must match the absolute uncertainty in the confidence interval.		
correct sig figs and CI properly formatted 1			
Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
Potential Difference (Accelerating Voltage) [V], ΔV	($\pm$) V	% (2 sig fig)	DMM
	Measurement Procedure: Detailed description of how they measured the accelerating voltage. Must be detailed enough that the process could be repeated using only this description and obtain a similar result.		
	Justification of Uncertainty: Must name one of the four factors and <u>logically explain why</u> that is the largest contributor. Most likely <u>quality of measuring device</u> . Must also include a logical <u>value</u> that must match the absolute uncertainty in the confidence interval.		

4 / 1 / 0

Linear Regression

APPENDIX C

Current I (A)	δI (A)	diameter d (m)	bd (m)	IQ (I^2)	DQ ($1/d^2$)
1.67	0.03	0.1	0.005	2.789	75.614
1.90	0.03	0.105	0.005	3.610	90.703
2.20	0.04	0.09	0.005	4.840	123.457
2.35	0.04	0.08	0.005	5.523	156.250
2.45	0.04	0.075	0.005	6.003	177.778
2.85	0.04	0.07	0.005	8.123	204.082

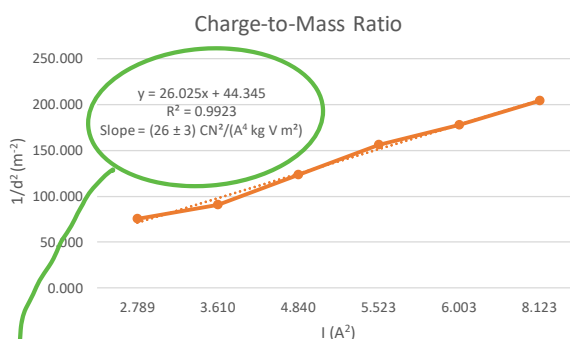
SUMMARY OUTPUT

Regression Statistics

Multiple R	0.9757507
R Square	0.9520895
Adjusted R Square	0.9401119
Standard Error	12.295884
Observations	6

Coefficients

	Coefficients	Standard Error
Intercept	4.010814	15.84263649
X Variable 1	26.025002	2.919021385



2 Columns present for IV , DV , IQ , and DQ

1 All absolute uncertainties 1 sig fig
and all measured values rounded to match

1 Absolute uncertainties correct according to
the SLAM 2-17.

1 Linear regression attempted (table present)

2 Plot w/ IQ on x-axis and
 DQ on y-axis

1 Axes labeled with units

1 slope matches regression stats
and trendline

0.5 correct format

0.5 correct units

5/3/2

Propagation of Uncertainty

APPENDIX D

Objective Quantity: q/m		Objective Formula: $\frac{q}{m} = \frac{\text{slope } 2\Delta V D^2}{N^2 (4/5)^3 \mu_0^2}$	
Variable	Measured Value	Absolute Uncertainty	Relative Uncertainty
slope	$63 \frac{\text{C N}^3}{\text{A}^4 \text{ kg V m}^2} \text{ or } \frac{\text{A}^2 \cdot \text{m}^2}{\text{A}^4 \text{ kg V m}^2}$	$4 \frac{\text{C N}^3}{\text{A}^4 \text{ kg V m}^2} \text{ or } \frac{\text{A}^2 \cdot \text{m}^2}{\text{A}^4 \text{ kg V m}^2}$	12%
D	0.32 m	0.01 m	3.1%
ΔV	129.6 V	0.1 V	0.077%
Calculation of Objective Quantity's Magnitude: $\frac{q}{m} = \frac{\text{slope } 2\Delta V D^2}{N^2 (4/5)^3 \mu_0^2}$ $\frac{q}{m} = \frac{63(2)(129.6)(0.32)^2}{130^2 (4/5)^3 (4\pi \times 10^{-7})^2} \rightarrow \frac{q}{m} = 1.224 \times 10^{11} \text{ C/kg}$			
Calculation of Objective Quantity's Absolute Uncertainty: $\delta \frac{q}{m} = \sqrt{\left(\frac{\partial \frac{q}{m}}{\partial \text{slope}} \delta \text{slope}\right)^2 + \left(\frac{\partial \frac{q}{m}}{\partial D} \delta D\right)^2 + \left(\frac{\partial \frac{q}{m}}{\partial \Delta V} \delta \Delta V\right)^2}$ $\delta \frac{q}{m} = \sqrt{\left(\frac{2\Delta V D^2}{N^2 (4/5)^3 \mu_0^2} \delta \text{slope}\right)^2 + \left(\frac{\text{slope } 4\Delta V D}{N^2 (4/5)^3 \mu_0^2} \delta D\right)^2 + \left(\frac{\text{slope } 2 D^2}{N^2 (4/5)^3 \mu_0^2} \delta \Delta V\right)^2}$ $\delta \frac{q}{m} = 0.1 \times 10^{11} \text{ C/kg}$			
Objective Quantity Confidence Interval: $(1.2 \pm 0.1) \times 10^{11} \text{ C/kg}$			

correct sig figs and units

①

①

①

correct variables present

①

①

①

correct values present ①

① Attempts the intermediate method

① correct partial derivatives

① Answer / units / C ± formatted properly

Experimental Value CI: $(1.2 \pm 0.1) \times 10^{-18} \text{ C/kg}$	Accepted Value CI (given): $1.8 \times 10^{-18} \text{ C/kg}$
Number Line: values plotted correctly exp. value 0.5 acc. value 0.5 ② Numberline formatted correctly (numbers/physical quantity/units) Physical quantity: charge per mass, q/m Units: C/kg	
Dominant Form of Error Present (Random or Systematic): systematic 1 Follows number line	
Acceptable Level of Precision (given): 2.5 %	Acceptable Level of Accuracy (given): 2.5%
Calculation of Relative Uncertainty: $RU = \frac{\delta q/m}{q/m}$ $RU = \frac{0.1}{1.2}$ $RU = 0.083 \text{ or } 8.3\%$ 0.5 correct / 2 sig figs	Calculation of Percent Difference: $\% \text{ diff.} = \frac{ \text{exp. value} - \text{acc. value} }{ \text{acc. value} } \times 100$ $\% \text{ diff.} = \frac{ 1.2 - 1.8 }{ 1.8 } \times 100$ $\% \text{ diff.} = 33\%$ 0.5 correct / 2 sig figs



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LAB#4 HANDOUT

Pre-Lab Tasks

I. Pre-Lab Overview

An essential skill beginning in Block 3 is the ability to analyze a circuit diagram and properly construct the circuit using basic components and wires. You will begin by understanding the basics of a circuit and its terminology.

II. Background

a. Resistance

Resistance is a measure of how strongly the material resists electrical current. Resistance also depends on the shape of the material. In Figure 1, a resistor is modeled as a cylinder.

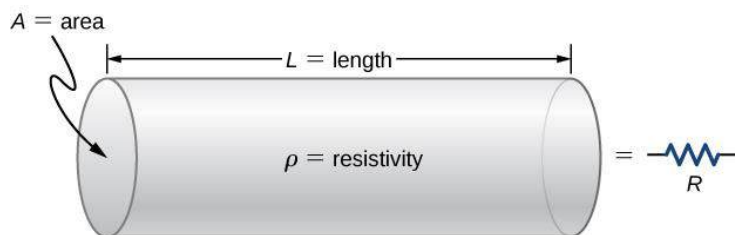


Figure 1 A model of a resistor as a uniform cylinder of length L and cross-sectional area A (Figure 9.15, S. Ling, J. Sanny, and W. Moebs, *University Physics*, vol. 2, OpenStax, 2017).

The equation for resistance is as follows:

$$R = \rho \frac{L}{A} \quad (1)$$

where A is the cross-sectional area of the material and L is its length. The resistance of current flow is analogous to the flow of water in a pipe. The longer the cylinder the larger the resistance. Inversely, the larger the cross-sectional area, the smaller the resistance.

b. Ohm's Law

Current through a metal wire is directly proportional to the difference in electric potential applied, which is the basis of Ohm's Law. Ohm's Law states that there is a linear relationship between voltage and current, with resistance being its slope. The equation for Ohm's Law is as follows:

$$V = IR \quad (2)$$

where V is the measurement of the difference in electric potential across the material in volts (V), I is the current through the material in amps (A), and R is the resistance in ohms (Ω). The term "potential difference", is also known as "voltage." In your experiment, voltage is measured between two points in your circuit when using a digital multimeter (DMM). Current is the rate that charge flows as it travels through a circuit from a positive terminal to a negative terminal.

III. Pre-Lab Activity #1 – Direct Current (DC) Circuits PHET Simulation

Practice circuitry basics using a PHET simulation through the following steps:

- Go to <https://phet.colorado.edu/en/simulations/circuit-construction-kit-dc> and download the simulator.
- Choose the “lab” option, and select “Conventional” on the top right, where the “Show Current” box is located.
- Build a simple circuit by connecting wires from a battery (power source) to a resistor as shown in Image PL 1.
- Measure the voltage across the resistor using a voltmeter and measure the current through the circuit using an ammeter.
- Notice the differences between measuring the voltage and measuring the current. To measure voltage, it is measuring the difference in electric potential between two points in the circuit. In this case, the two points are in opposite sides of the resistor.
- Connect the ammeter to complete the circuit. When measuring current, the ammeter is always in series with the circuit. When measuring the voltage, the voltmeter is acting in parallel with the circuit.

Determine the resistance of the resistor using Ohm’s Law with your measured values of voltage and current. Check your answer by clicking on the resistor component. Increase the voltage across the battery and observe its effects, if any, on the current through the circuit and the resistance of the resistor.

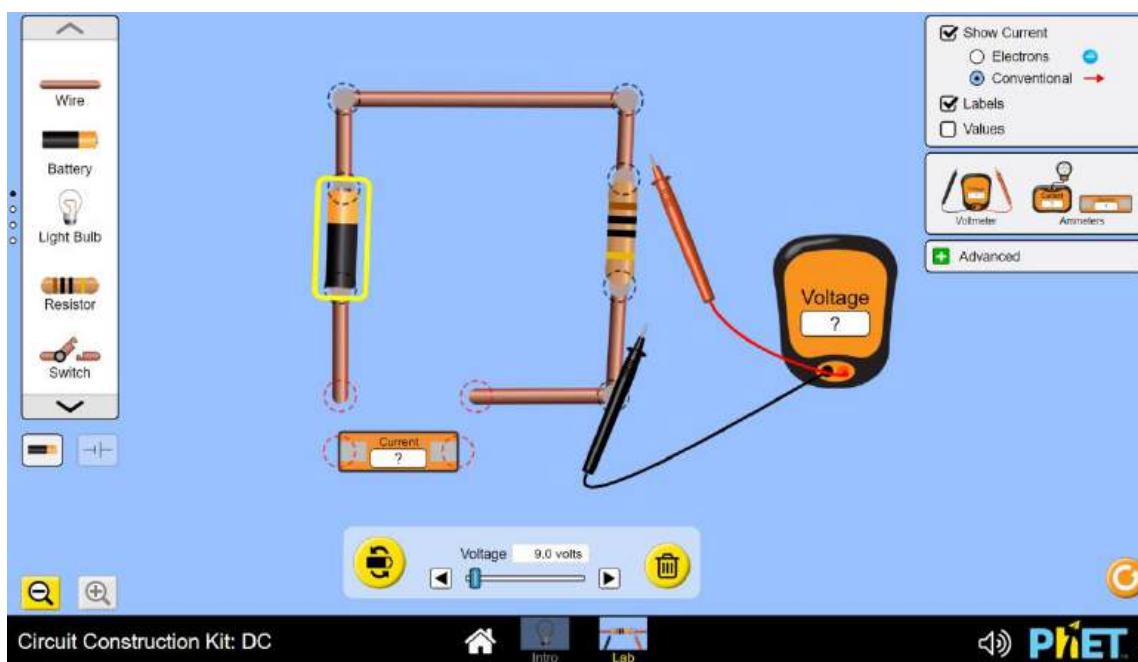


Image PL 1: https://phet.colorado.edu/sims/html/circuit-construction-kit-dc/latest/circuit-construction-kit-dc_en.html retrieved August 27, 2022.

IV. Pre-Lab Activity #2 – Linearization Process

The objective of this lab is to experimentally determine the resistivity, ρ , of a manganin wire using Ohm's Law. To obtain a model equation for resistivity, combine the Resistance Equation (1) with Ohm's Law Equation (2). Consequently, we will use PASCO to vary the voltage, V , across the wire, and measure its resultant current, I .

Complete the Linearization Table for Lab 4. Please note that you will measure the diameter, d , of the manganin wire to obtain its cross-sectional area. Also note the objective quantity and control variables are already listed.

Linearization Table

APPENDIX A

Step 1: Identify fundamental physics	Step 2: Apply physics in order to build a model equation
Step 3: Identify independent and dependent variables	Step 4: Rearrange equation to $y = mx + b$ form to build the linearized model equation.
Step 5: Identify independent and dependent quantities	Step 6: Identify the objective quantity and control variables O.Q.: ρ C.V.: L, d
Step 7: Write the expression for the slope of your equation	8: Rearrange your slope so that you solve for your objective quantity equation

Lab Worksheet

I. Lab Overview

You will experimentally determine/verify a manganin wire's resistivity, ρ , the characteristic property of a material that helps determine the resistance of an object. Using a simple direct current (DC) circuit, you will vary voltage, V , to measure the resultant current, I . The accepted value for manganin wire is $\rho = 48.20 \times 10^{-8} \Omega\text{m}$. Assume a 10% relative uncertainty in the accepted value.

II. Setup

The following equipment and materials are required for this lab:

- PASCO 850 Universal Interface Box
- 2 x Digital Multimeters (DMM) (1 to measure voltage and 1 to measure current)
- 1 x pair of DMM test probes (1 x red, 1 x black)
- 1 x Black Banana plug patch cord
- 2 x banana plug patch cords w/ alligator clips (1 x red, 1 x black)
- 1 x piece of manganin wire provided by instructor.

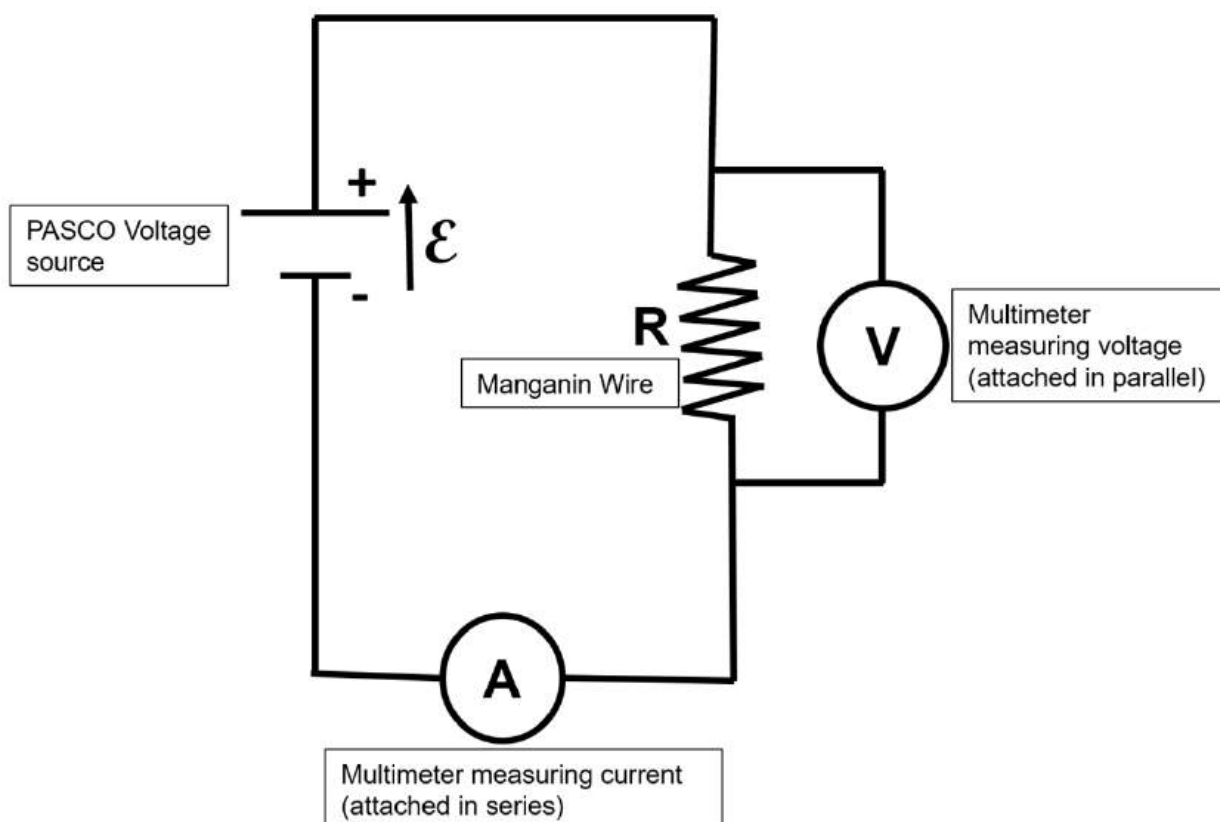


Figure 1 Circuit Diagram Setup for Experiment

To set up your DMMs:

- Review the [SLAM](#), pages 2-14 to 2-17, for guidance on DMM usage and uncertainty measurements.
- To use the voltmeter in the DMM, review Lab 3 for additional instruction, the pre-lab, Figure 1, the SLAM.
- To use the ammeter and measure current, grab the additional DMM and turn its dial to DC amps ($\overline{A}$).
- Instead of measuring between two points, such as when measuring the potential difference, the ammeter must be in “series” with the circuit to measure current.
- A series circuit is defined as having only one path through which current can flow. Therefore, current has to flow through the ammeter to measure current.
- Connect an electric wire in port A (Amps) of your DMM and another to the COM.
- If your amperage readings are too small, switch the cable port from A to mA (milliamps).
- Review the pre-lab, Figure 1, and the SLAM for additional instruction.

To setup your circuit:

- Review the circuit diagram in Figure 1.
- Begin by attaching the cords with alligator clips on each side of the manganin wire.
- Connect the two DMMs where appropriate in the circuit.
- Connect two cords to the PASCO 850 Interface Box, select Signal Generator through the PASCO program, and choose the “DC” waveform. The box will be your voltage source. Review Lab 2 for additional instruction in the setup.
- Keep the Signal Generator off until you are ready to record data.

Once you have your circuit setup complete, turn on the “Signal Generator” on your PASCO computer application. Use PASCO to vary the voltage (between 0 and 2.5 volts) being applied across the manganin wire. Use Excel to record at least 12 iterations of the voltage, the independent variable, and current, the dependent variable.

III. Submission Requirements

- Complete the Measured Quantities Tables (Appendix B) for all measurements conducted in this lab.
- Conduct a linear regression of the data by completing a full Appendix C.
- Complete the confidence interval for the experimental value of resistivity using a Propagation of Uncertainty Table (Appendix D).
- Conduct an error analysis using a Precision and Accuracy Table (Appendix E). Assume that a historically acceptable level of accuracy for this lab is 1.5% and historically acceptable level of precision is 1.5%.
- Complete a full lab report for this lab to include Appendices A-E in your report. Submit as a single PDF and label “Lab 4_X Hour_Last Names.”

As a checklist, your full lab report should include the following:

- Complete Laboratory Report
 - Cover Sheet (Signed)
 - Introduction
 - Theory
 - Procedure
 - Results
 - Discussion
 - Conclusion
 - Works Cited Page (in accordance with Encl. 5-Academic Integrity Brief)
- Appendices
 - Appendix A – Linearization Table
 - Appendix B – Measurement Tables
 - Appendix C – Excel table with plot and linear regression
 - Appendix D – Propagation of Uncertainty Table
 - Appendix E – Precision and Accuracy Table



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LAB#4 SOLUTION

Linearization Table

APPENDIX A

Step 1: Identify fundamental physics Ohm's Law Resistance	Step 2: Apply physics in order to build a model equation $V = IR, R = \rho \frac{L}{A}, A = \pi r^2, r = \frac{d}{2}$ $\frac{V}{I} = \frac{\rho L}{\pi d^2}$
Step 3: Identify independent and dependent variables Independent Variable: Voltage (V) Dependent Variable: Current (I)	Step 4: Rearrange equation to $y = mx + b$ form to build the linearized/ model equation. $I = \frac{\pi d^2}{\rho L} V$
Step 5: Identify independent and dependent quantities Independent Quantity: V Dependent Quantity: I	Step 6: Identify the objective quantity and control variables Objective Quantity: Resistivity (ρ) Control Variables: diameter (d) length (L)
Step 7: Write the expression for the slope of your equation $\text{slope} = \frac{\pi d^2}{\rho L}$	8: Rearrange your slope so that you solve for your objective quantity equation $\rho = \frac{\pi d^2}{\text{slope} L}$

Measurement Table

APPENDIX B

Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
Diameter [m]	($\pm$) m	% (2 sig fig)	Calipers
	Measurement Procedure: Detailed description of how they measured the diameter of the wire. Must be detailed enough that the process could be repeated using only this description and obtain a similar result.		
	Justification of Uncertainty: Must name one of the four factors and <u>logically explain why</u> that is the largest contributor. Most likely <u>quality of measuring device</u> . Must also include a logical <u>value</u> that must match the absolute uncertainty in the confidence interval.		

Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
Length [m]	($\pm$) m	% (2 sig fig)	Tape Measure of meter stick
	Measurement Procedure: Detailed description of how they measured the length of the wire. Must be detailed enough that the process could be repeated using only this description and obtain a similar result.		
	Justification of Uncertainty: Must name one of the four factors and <u>logically explain why</u> that is the largest contributor. Most likely <u>quality of measuring device</u> . Must also include a logical <u>value</u> that must match the absolute uncertainty in the confidence interval.		

Measurement Table

APPENDIX B

Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
Voltage (V)	See Appendix C	See Appendix C	DMM (Voltmeter)
	Measurement Procedure: Detailed description of how they measured the voltage across the wire. Must be detailed enough that the process could be repeated using only this description and obtain a similar result.		
	Justification of Uncertainty: Must name one of the four factors and <u>logically explain why</u> that is the largest contributor. Most likely <u>quality of the measuring device</u> . Must also include a logical <u>value</u> that must match the absolute uncertainty in the confidence interval.		

Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
Current (A)	See Appendix C	See Appendix C	DMM (Ammeter)
	Measurement Procedure: Detailed description of how they measured the current across the wire. Must be detailed enough that the process could be repeated using only this description and obtain a similar result.		
	Justification of Uncertainty: Must name one of the four factors and <u>logically explain why</u> that is the largest contributor. Most likely <u>quality of the measuring device</u> . Must also include a logical <u>value</u> that must match the absolute uncertainty in the confidence interval.		

Uncertainty in a Voltage Measurement

$$\delta V = \left[(V) \times (\text{relative uncertainty}) \right] + \left[(\text{resolution}) \times (\# \text{ of least significant digits}) \right] \quad (2-2)$$

Function	Upper Limit	Relative Uncertainty	Resolution	# of Least Significant Digits
DC mV	320.0 mV	0.001	0.1 mV	1
DC V	3.200 V	0.001	0.001 V	1
	32.00 V	0.001	0.01 V	1
	320.0 V	0.001	0.1 V	1
	1000 V	0.001	1 V	1

Table 2-3. Specifications for Multimeter used as a Voltmeter

Uncertainty in a Current Measurement

$$\delta i = \left[(i) \times (\text{relative uncertainty}) \right] + \left[(\text{resolution}) \times (\# \text{ of least significant digits}) \right] \quad (2-3)$$

Function	Upper Limit	Relative Uncertainty	Resolution	# of Least Significant Digits
DC μA	320.0 μA	0.0075	0.1 μA	2
	3200 μA	0.0075	1 μA	2
DC mA	32.00 mA	0.0075	0.01 mA	2
	320.0 mA	0.0075	0.1 mA	2
DC A	10.00 A	0.0075	0.01 A	2

Table 2-4. Specifications for Multimeter used as an Ammeter

Linear Regression

APPENDIX C

Voltage [V]	δV [V]	Current [A]	δI [A]
0.477	0.001	0.14	0.02
0.715	0.002	0.21	0.02
0.954	0.002	0.28	0.02
1.193	0.002	0.35	0.02
1.431	0.002	0.42	0.02
1.670	0.003	0.50	0.02
1.915	0.003	0.57	0.02
2.154	0.003	0.64	0.02
2.392	0.003	0.71	0.03
2.632	0.004	0.77	0.03
2.872	0.004	0.84	0.03

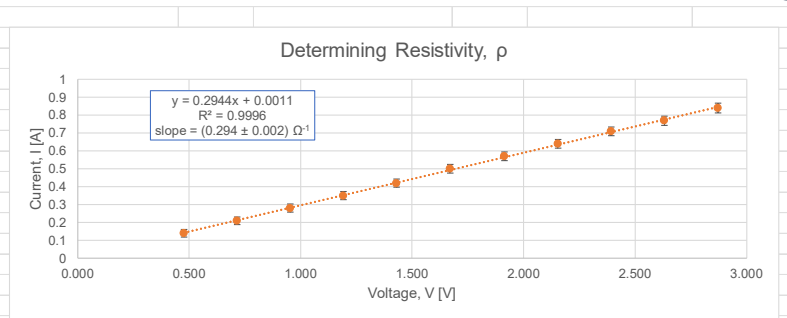
SUMMARY OUTPUT

Regression Statistics

Multiple R	0.9997967
R Square	0.9995935
Adjusted R Square	0.9995483
Standard Error	0.0049737
Observations	11

Coefficients

	Coefficients	Standard Error
Intercept	0.0011336	0.003634491
X Variable 1	0.294351	0.001978681

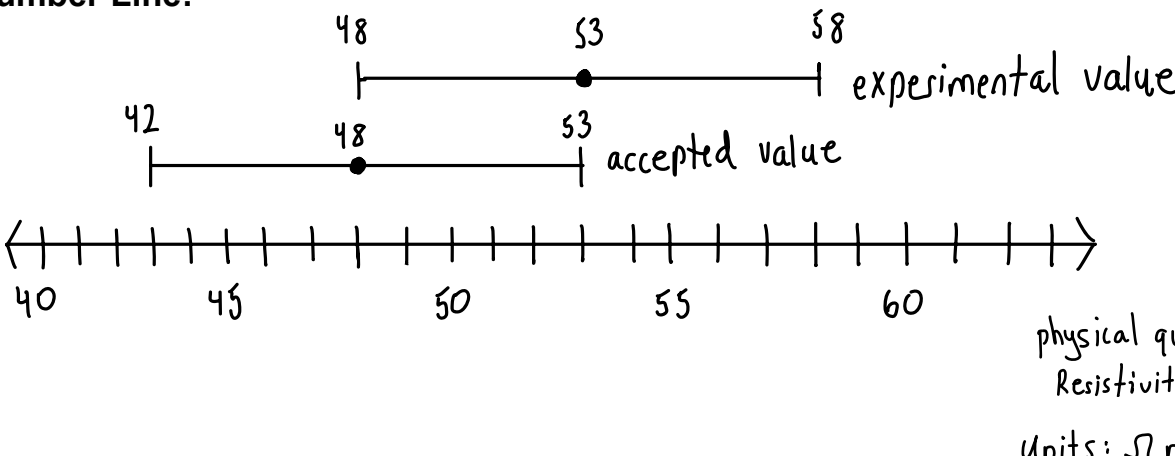


Length of Wire, L [m]	δL [m]	Diameter of Wire, d [m]	δd [m]	Resistivity, ρ [Ωm]	$\delta \rho$ [Ωm]
2.06	0.02	0.00064	0.00003	5.30539E-07	5.0131E-08
				$\partial \rho / \partial d$	4.9738E-08
				$\partial \rho / \partial L$	5.15086E-09
				$\partial \rho / \partial \text{Slope}$	3.56638E-09

Propagation of Uncertainty

APPENDIX D

Objective Quantity: Resistivity, ρ		Objective Formula: $\rho = \frac{\pi d^2}{\text{slope} L^4}$	
Variable	Measured Value	Absolute Uncertainty	Relative Uncertainty
d	0.00064 m	0.00003 m	4.7%
L	2.06 m	0.02 m	0.97%
slope	0.294 Ω^{-1}	0.002 Ω^{-1}	0.68%
Calculation of Objective Quantity's Magnitude: $\rho = \frac{\pi d^2}{\text{slope} L^4}$ $\rho = \frac{\pi (0.00064 \text{ m})^2}{(0.294 \Omega^{-1})(2.06 \text{ m})^4} \quad \rho = 53 \times 10^{-8} \Omega \text{ m}$			
Calculation of Objective Quantity's Absolute Uncertainty: $\delta \rho = \sqrt{\left(\frac{\partial \rho}{\partial d} \delta d\right)^2 + \left(\frac{\partial \rho}{\partial L} \delta L\right)^2 + \left(\frac{\partial \rho}{\partial \text{slope}} \delta \text{slope}\right)^2}$ $\delta \rho = \sqrt{\left(\frac{2\pi d}{4L \text{slope}} \delta d\right)^2 + \left(-\frac{\pi d^2}{4L^3 \text{slope}}\right)^2 + \left(-\frac{\pi d^2}{4L \text{slope}^2}\right)^2}$ $\delta \rho = 5 \times 10^{-8} \Omega \text{ m}$			
Objective Quantity Confidence Interval: $\rho = (53 \pm 5) \times 10^{-8} \Omega \text{ m}$			

Experimental Value CI: $(53 \pm 5) \times 10^{-8} \Omega m$	Accepted Value CI (given): $(48 \pm 5) \times 10^{-8} \Omega m$
Number Line:  physical quantity: Resistivity, ρ units: Ωm	
Dominant Form of Error Present (Random or Systematic): Random	
Acceptable Level of Precision (given): 1.5 %	Acceptable Level of Accuracy (given): 1.5%
Calculation of Relative Uncertainty: $RU = \frac{\delta \rho}{\rho}$ $RU = \frac{5}{53}$ $RU = 0.094 \text{ or } 9.4\%$	Calculation of Percent Difference: $\% \text{ diff.} = \frac{ \text{exp. value} - \text{acc. value} }{ \text{acc. value} } \times 100$ $\% \text{ diff.} = \frac{ 53 - 48 }{48} \times 100$ $\% \text{ diff.} = 10.4\%$

$\triangle$	14.36
18	25.76
42	606

1. Appendices (5 Points)

- Linearization Table (App. A).
- Measurement Tables (App. B). Four App. Bs: L , d , V , I .
- Linear Regression (App. C).
- Propagation of Uncertainty (App. D).
- Error Analysis (App. E).

5 all present + complete

2. Lab Report (55 Points)

I. Introduction

- Provides a section dedicated to the introduction. 1
- States the objective of the experiment is to determine the resistivity. 3
- Provides a succinct summary on how the objective was accomplished (varied the voltage and measured resulting current). 1

II. Theory

- Provides a section dedicated to the theory. 1
- Addresses Ohm's Law and the Resistance Equation used in the experiment. 3
- Provides and correctly explains the appropriate physics equations. 2
- Assumptions made are valid and reasonable. 1
- States model equation and explains how it was derived. 1
- States independent and dependent variables. 2
- States linearized model equation and resulting independent and dependent quantities. 2
- States objective quantity and control variables. 2
- States equation for slope. 1
- States equation for objective quantity. 1
- States method of analysis (linearization) 1

III. Procedures

- Provides a section dedicated to procedures. 1
- Explains the experimental set up/refers to a picture or diagram provided. 3
- States physical quantities being measured and methods to obtain them. (L , d , V , I). 2
- Presents uncertainty of measured quantities, justification of uncertainties, and how they were determined. 2

IV. Results

- a. Provides section dedicated to results. 1
- b. Presents all measured data to include, independent and dependent variables and control variables or physical constants in a logical manner with correct formatting of confidence intervals. 3
- c. Presents plot of independent quantity vs. dependent quantity and reports slope and its uncertainty 2
- d. Shows or references App. D the objective quantity calculation. 2
- e. Shows or references App. D the calculation of the absolute uncertainty of the objective quantity. 2
- f. Values for objective quantity and its absolute uncertainty are reasonable. 1
- g. States correctly formatted final confidence interval for objective quantity. 1

V. Discussion

- a. Provides a section dedicated to discussion. 1
- b. States confidence intervals for experimental and accepted values. 2
- c. Presents a correctly formatted number line. 3
- d. States the relative uncertainty of the experimental value and compares it to the acceptable level of precision. 1
- e. States the percent difference of the experimental value and compares it to the acceptable level of accuracy. 1
- f. States dominant form of error and matches number line. 2
- g. Presents and discusses logical evidence that led to types of error. Largest quadrature term in calculation of objective quantity's absolute uncertainty for random error and high/low chart for systematic error. 1

VI. Conclusion

- a. Provides a section dedicated to the conclusion. 1
- b. Assesses whether experimental objective was met. 2
- c. Concisely communicates lessons learned from the experimental procedure. 1
- d. Provides details of how experiment could be done differently from the lessons learned. 1

1 Overall writing style, presentation, + formatting (up to instructor)



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LAB#5 HANDOUT

Pre-Lab Tasks

I. Pre-Lab Overview

As we delve into laboratory 5, it is essential to understand how direct current (DC) circuits work. We will use the same PHET simulation from the previous lab. Go to [PhET circuit construction kit](#), download the simulator, and select “Conventional” on the top right, where the “Show Current” box is located.

II. Resistors in Series

- Using the simulation, construct a series resistor network using two resistors. Include a switch, an ammeter measuring the current through the circuit, and a voltmeter to measure the potential difference across R_1 , R_2 , and two batteries, in series.
- Adjust the voltage of each battery to 1.5 V, and the resistances of the resistors to $R_1 = 33 \, \Omega$ and $R_2 = 100 \, \Omega$.
- Use the voltmeter and ammeter to record the following parameters shown in Table 4.1.

Parameter	Theoretical Value
Potential difference across the two battery sources, V_1	
Potential difference R_1 , V_{R1}	
Potential difference R_2 , V_{R2}	
Current through entire circuit, I_1	

Table 4.1 Voltmeter and Ammeter Measurements

- Cut and paste a picture of your circuit below.

III. Resistors in Parallel

- Using the simulation, **construct** a parallel resistor network using your two resistors. Include a switch and ammeters measuring the current, I_2 , flowing through R_1 , the current, I_3 , flowing through R_2 , and the current flowing out of the batteries, I_1 . Use a voltmeter to measure the potential difference across R_1 , R_2 , and the two batteries in series.
- Adjust the voltage of each battery to 1.5 V, and the resistances of the resistors to $R_1 = 33\ \Omega$ and $R_2 = 100\ \Omega$.
- Use the voltmeter and ammeter to record the following parameters shown in Table 4.2.

Parameter	Theoretical Value
Potential difference across the two battery sources, V_1	
Potential difference across R_1 , V_{R1}	
Potential difference across R_2 , V_{R2}	
Current out of battery, I_1	
Current into battery, _____	
Current through R_2 , I_3	
Current through R_1 , I_2	

Table 4.2 Voltmeter and Ammeter Measurements

- Include a screen capture of your circuit below.

Lab Worksheet

I. Lab Overview

In this lab, you will practice the concepts learned on resistance (R), resistor networks (in series and parallel), Ohm's Law, and Kirchhoff's Circuit Laws. You will continue to use the two Fluke 27/FM Digital multimeters (DMM) as you did in your previous laboratory project to measure the current, the potential difference (V) across a circuit, the potential difference across resistors, and the potential difference across power sources.

a. Measuring Resistance with the DMM

You will use the DMM to verify the resistance of each resistor used in the circuit. To do this, ensure no current is flowing through the resistor. Rotate the switch on the DMM to the Ohm symbol, Ω . Plug in one electric cable to the COM port and another into the red-colored port labeled Ω . Attach the cables on each side of the resistor to measure the resistance. Review the [SLAM](#), pages 2-14 to 2-17, for guidance on DMM usage and uncertainty measurements.

b. Modular Circuits Kits

You will be given a PASCO Basic Modular Circuits Kit. To connect two square modules for the circuit, students will insert a jumper clip, which emphasizes that an electrical connection has been made. To review how to use the kit, please view PASCO's Modular Circuits Overview [here](#).

c. Equipment

1. 2 x DMMs
2. 1 x PASCO Basic Modular Circuit Kit
3. 1 x PASCO 850 Interface Box
4. 2 x pair of banana plug patch cords w/ alligator clips (1 x red, 1 x black)
5. Banana plug patch cords as needed

d. Submission Requirements

Complete pages 2 to 9 of the worksheet and submit as a single PDF. Label "Lab 5_X Hour_Last Names."

II. Activity #1 - Resistors in Series

- a. **Record** the value and the confidence interval of the measured resistance of the two resistors provided in Table 4.1.

Resistor	Labeled Resistance	Measured Resistance
R_1	33 Ω	($\pm$) Ω
R_2	100 Ω	($\pm$) Ω

Table 4.1 DMM Resistance Measurements

- b. **Build** the circuit you constructed digitally with the PHET simulator for “Resistors in Series” in the Pre-Lab using the two resistors given, a switch, an ammeter (1st DMM) measuring the current (I_1) through the circuit, a voltmeter (2nd DMM) measuring the potential difference (V_{R1}) across R_1 , the potential difference (V_{R2}) across R_2 , and the two batteries from the circuit kit placed in series. If you do not have two batteries, you may use the PASCO 850 Interface. Ensure you have a voltage output of 3.0 V in your interface.
- c. **Take a picture** of your circuit and place it below. Ensure you include your electromotive forces (emf) and the placement of your DMMs to measure the current (I_1) through the circuit and the potential difference (V_{R1}) across R_1 .

- d. Theoretically, the potential difference, V_1 , across the two batteries in a series is 3.0 V. The same applies for the PASCO 850 Interface. **Calculate** the theoretical values for the equivalent resistance (R_{eq}), current through the entire circuit (I_1), potential difference (V_{R1}) across R_1 , and potential difference (V_{R2}) across R_2 . You must **show all your work for each parameter and include a circuit diagram with components labeled.**

- e. Use Table 4.2 to **record** your theoretical values, and **measure** and **record** the appropriate measurements using your DMM. Note: Your theoretical and measured values should be relatively close in number. If they are not, review how you used the DMM and your calculations above. **Check your answers with your instructor before moving on.**

Parameter	Theoretical Value	Measured Value
Potential difference across source V_1	3.0 V	($\pm$) V
Potential difference R_1 , V_{R1}		($\pm$) V
Potential difference R_2 , V_{R2}		($\pm$) V
Current through entire circuit, I_1		($\pm$) A

Table 4.2 DMM Measurements for Resistors in Series

- f. For the series circuit you tested in Activity #1, **compare** and **explain** the sum of the voltage drops across each individual resistor ($V_{R1} + V_{R2}$) with the battery voltage (V_1). Hint: Use Kirchhoff's Circuit Laws in your comparison.
- g. **Determine** if changing the order of resistors in series would have affected the measurements you made in Activity #1. Explain your reasoning.
- h. **Calculate** the percent difference (SLAM Equation 3-8, pg. 3-17) between the theoretical and measured values for the current I_1 . Show your work.

III. Activity #2 - Resistors in Parallel

- a. **Build** a circuit like the one you constructed digitally with the PHET simulator in Part 2 of the Pre-Lab. It is a parallel resistor network using the two resistors given, a switch, and the two batteries in series. If you do not have two batteries, you may use the PASCO 850 Interface. Ensure you have a voltage output of 3.0 V in your interface. **Take a picture** of your circuit and place it below. Ensure you include your electromotive forces (emf), and the placement of your DMMs to show measurements of current (I_3) flowing through R_2 and the potential difference (V_{R1}) through R_1 .

- b. Theoretically, the potential difference, V_1 , across the two batteries in a series is 3.0 V. The same applies for the PASCO 850 Interface. **Calculate** the theoretical values for the equivalent resistance (R_{eq}), the current (I_1) out of the emf source, the potential difference (V_{R1}) across R_1 , the potential difference (V_{R2}) across R_2 , the current (I_2) through R_1 , and the current (I_3) through R_2 based on the circuit you built. You must **show all your work for each parameter and include a circuit diagram with components labeled.**

- c. Use Table 4.3 to **record** your theoretical values, and **measure** and **record** the appropriate measurements using your DMM. Note: Your theoretical and measured values should be relatively close in number. If they are not, review how you used the DMM and your calculations above. **Check your answers with your instructor before moving on.**

Parameter	Theoretical Value	Measured Value
Potential difference across source V_1	3.0 V	($\pm$) V
Potential difference across R_1 , V_{R1}		($\pm$) V
Potential difference across R_2 , V_{R2}		($\pm$) V
Current out of battery, I_1		($\pm$) A
Current into battery, _____		($\pm$) A
Current through R_2 , I_3		($\pm$) A
Current through R_1 , I_2		($\pm$) A

Table 4.3 DMM Measurements for Resistors in Parallel

- d. **Compare** and **explain** the current you measured coming out of the batteries to the current going back into the batteries. Does your data support the conservation of charge? Explain why or why not.
- e. For the parallel circuit you tested in Activity #2, **compare** the sum of the currents through each individual resistor ($I_2 + I_3$) to the total current leaving the battery I_1 . (Hint: Use Kirchhoff's Circuit Laws in your comparison.)
- f. **Calculate** the percent difference (SLAM Equation 3-8) between the theoretical and measured values for the current I_1 . Show your work.

IV. Activity #3 - Find the Resistance of the Unknown Resistor

In this activity, your instructor will give you an unknown resistor (R_3), and you will determine its resistance using Kirchhoff's Loop Rule and Junction Rule. To begin, you will build a circuit that includes R_3 . You will then use the PASCO 850 Interface and increase the potential difference (V_1) to 10 V. The circuit will also include the two batteries (V_2 and V_3) and two resistors (R_1 and R_2) used in the previous activities, as seen in Figure 4.2.

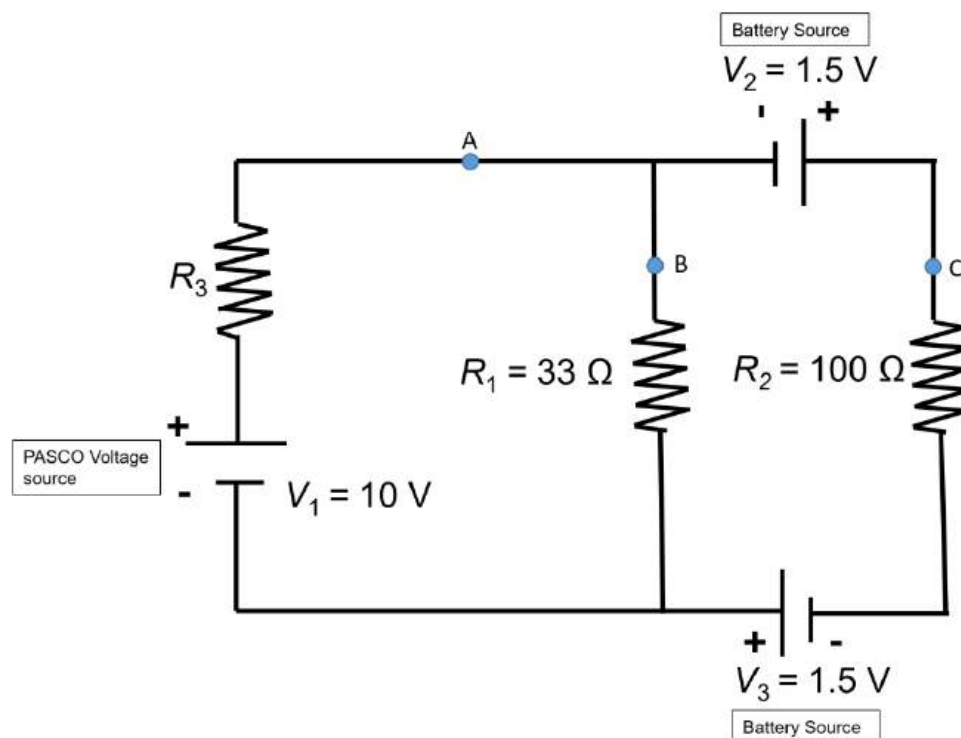


Figure 4.1 Activity #3 Circuit Diagram.

- a. **Measure** the current at either point A, B, or C (do not measure more than one) and verify the potential differences across each component already provided in Figure 4.1 to build three equations and three unknown variables. Use Table 4.4 to organize your measured values.

Parameter	Theoretical Value	Measured Value
Potential difference across source V_1	10.0 V	($\pm$) V
Potential difference across source V_2	1.50 V	($\pm$) V
Potential difference across source V_3	1.50 V	($\pm$) V
R_1	33 Ω	($\pm$) Ω
R_2	100 Ω	($\pm$) Ω
Current (I) measured in A, B, or C.		($\pm$) A

Table 4.4 DMM Measurements for Activity #3.

- b. **Calculate** R_3 based on your current and voltage measurements. You must **show all work and use only the measured potential differences from the PASCO Interface, batteries, and the measured resistances from each resistor.**

- c. **Measure** the resistance of R_3 and **record** both your calculated and measured value in the table below.

Resistor	Calculated Resistance	Measured Resistance
R_3		($\pm$) Ω

Table 4.5 DMM Measurement of R_3 .



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LAB#5 SOLUTION

	10%
	30%
	60%

c. Equipment

- 2 x DMMs
- 1 x PASCO Basic Modular Circuit Kit
- 1 x PASCO 850 Interface Box
- 2 x pair of banana plug patch cords w/ alligator clips (1 x red, 1 x black)
- Banana plug patch cords as needed

d. Submission Requirements

Complete pages 2 to 9 of the worksheet and submit as a single PDF. Label "Lab 4_X Hour_Last Names."

II. Activity #1 - Resistors in Series

- a. **Record** the value and the confidence interval of the measured resistance of the two resistors provided in Table 4.1.

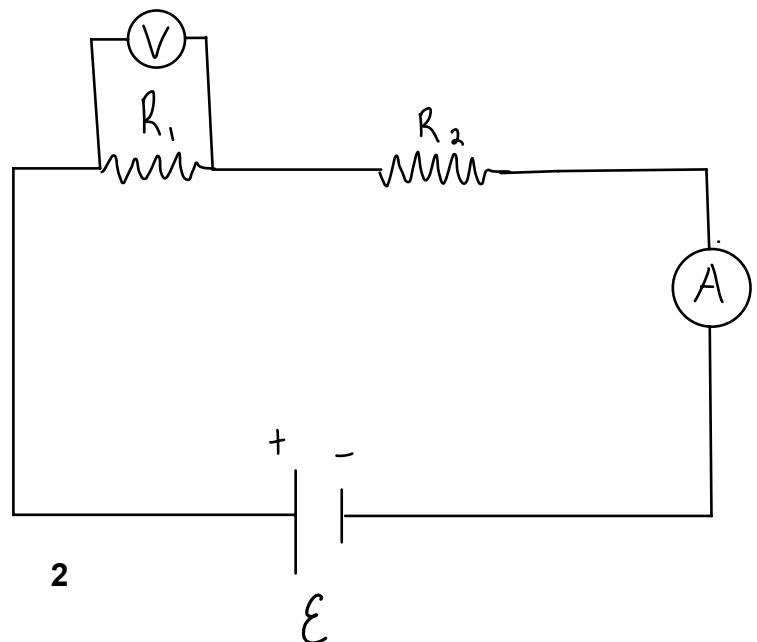
Resistor	Labeled Resistance	Measured Resistance
R_1	33Ω	$(32.5 \pm 0.1) \Omega$
R_2	100Ω	$(100.5 \pm 0.3) \Omega$

Table 4.1 DMM Resistance Measurements

- b. **Build** the circuit you constructed digitally with the PHET simulator for "Resistors in Series" in the Pre-Lab using the two resistors given, a switch, an ammeter (1st DMM) measuring the current (I_1) through the circuit, a voltmeter (2nd DMM) measuring the potential difference (V_{R1}) across R_1 , the potential difference (V_{R2}) across R_2 , and the two batteries from the circuit kit placed in series. If you do not have two batteries, you may use the PASCO 850 Interface. Ensure you have a voltage output of 3.0 V in your interface.
- c. **Take a picture** of your circuit and place it below. Ensure you include your electromotive forces (emf) and the placement of your DMMs to measure the current (I_1) through the circuit and the potential difference (V_{R1}) across R_1 .

 Resistors in Series

 DMMs in correct positions



- d. Theoretically, the potential difference, V_1 , across the two batteries in a series is 3.0 V. The same applies for the PASCO 850 Interface. **Calculate** the theoretical values for the equivalent resistance (R_{eq}), current through the entire circuit (I_1), potential difference (V_{R1}) across R_1 , and potential difference (V_{R2}) across R_2 . You must **show all your work for each parameter and include a circuit diagram with components labeled.**

1 Equivalent resistance Potential across R_1 **1**

$$R_{eq} = R_1 + R_2$$

$$R_{eq} = 33\Omega + 100\Omega$$

$$\underline{\underline{R_{eq} = 133\Omega}}$$

$$V_{R1} = I_1 R_1$$

$$V_{R1} = 0.0226\text{ A} (33\Omega)$$

$$\underline{\underline{V_{R1} = 0.744\text{ V}}}$$

1 Current through circuit Potential across R_2 **1**

$$V_1 = I_1 R_{eq}$$

$$I_1 = \frac{V_1}{R_{eq}}$$

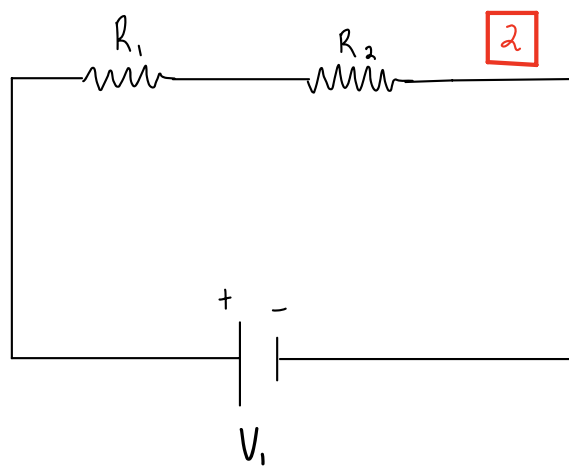
$$I_1 = \frac{3.0\text{ V}}{133\Omega}$$

$$\underline{\underline{I_1 = 0.0226\text{ A}}}$$

$$V_{R2} = I_1 R_2$$

$$V_{R2} = 0.0226\text{ A} (100\Omega)$$

$$\underline{\underline{V_{R2} = 2.256\text{ V}}}$$



- e. Use Table 4.2 to **record** your theoretical values, and **measure** and **record** the appropriate measurements using your DMM. Note: Your theoretical and measured values should be relatively close in number. If they are not, review how you used the DMM and your calculations above. **Check your answers with your instructor before moving on.**

Parameter	Theoretical Value	Measured Value
Potential difference across source V_1	3.0 V	(2.969 ± 0.001) V
Potential difference R_1 , V_{R1}	0.744 V	(0.692 ± 0.001) V
Potential difference R_2 , V_{R2}	2.256 V	(2.040 ± 0.001) V
Current through entire circuit, I_1	0.0226 A	(0.021 ± 0.001) A

Table 4.2 DMM Measurements for Resistors in Series

② CIs properly formatted and correct

- f. For the series circuit you tested in Activity #1, **compare** and **explain** the sum of the voltage drops across each individual resistor ($V_{R1} + V_{R2}$) with the battery voltage (V_1). Hint: Use Kirchhoff's Circuit Laws in your comparison.

$$V_1 = V_{R1} + V_{R2} \rightarrow \text{Loop Rule: } \sum V = 0$$

①

$$V_{R1} + V_{R2} - V_1 = 0$$

The voltage drops across each resistor should add to equal the battery voltage supplied to the circuit.

- g. **Determine** if changing the order of resistors in series would have affected the measurements you made in Activity #1. Explain your reasoning.

① The order of the resistors does not matter, only the equivalent resistance. The equivalent resistance will not change regardless of the order due to the resistors being placed in series.

- h. **Calculate** the percent difference (SLAM Equation 3-8, pg. 3-17) between the theoretical and measured values for the current I_1 . Show your work.

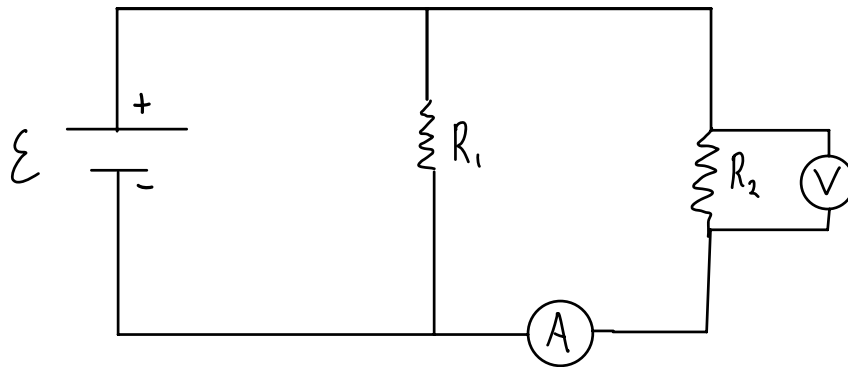
$$\% \text{ difference} = \frac{|\text{exp. value} - \text{acc. value}|}{|\text{acc. value}|} \times 100 \quad \text{① correct equation}$$

$$\% \text{ difference} = \frac{|0.021 \text{ A} - 0.0226 \text{ A}|}{|0.0226 \text{ A}|} \times 100$$

$$\% \text{ difference} = 7.1\%$$

III. Activity #2 - Resistors in Parallel

- a. **Build** a circuit like the one you constructed digitally with the PHET simulator in Part 2 of the Pre-Lab. It is a parallel resistor network using the two resistors given, a switch, and the two batteries in series. If you do not have two batteries, you may use the PASCO 850 Interface. Ensure you have a voltage output of 3.0 V in your interface. **Take a picture** of your circuit and place it below. Ensure you include your electromotive forces (emf), and the placement of your DMMs to show measurements of current (I_3) flowing through R_2 and the potential difference (V_{R1}) through R_1 .



2 Resistors in parallel

2 DMMs in correct positions

- b. Theoretically, the potential difference, V_1 , across the two batteries in a series is 3.0 V. The same applies for the PASCO 850 Interface. **Calculate** the theoretical values for the equivalent resistance (R_{eq}), the current (I_1) out of the emf source, the potential difference (V_{R1}) across R_1 , the potential difference (V_{R2}) across R_2 , the current (I_2) through R_1 , and the current (I_3) through R_2 based on the circuit you built. You must **show all your work for each parameter and include a circuit diagram with components labeled**.

1 Equivalent resistance

$$R_{eq} = \left(\frac{1}{R_1} + \frac{1}{R_2} \right)^{-1}$$

$$R_{eq} = \left(\frac{1}{33\Omega} + \frac{1}{100\Omega} \right)^{-1}$$

$$R_{eq} = 24.81\Omega$$

Potential across R_1

$$V_{R1} = 3.0V \text{ b/c resistors in parallel}$$

Potential across R_2

$$V_{R2} = 3.0V \text{ b/c resistors in parallel}$$

1 Current through R_2

$$V_{R2} = I_3 R_2$$

$$I_3 = \frac{V_{R2}}{R_2}$$

$$I_3 = \frac{3.0V}{100\Omega}$$

$$I_3 = 0.030A$$

or

$$\Sigma I_{in} = \Sigma I_{out}$$

$$I_1 = I_2 + I_3$$

$$I_3 = I_1 - I_2$$

$$I_3 = 0.121A - 0.091A$$

$$I_3 = 0.030A$$

1 Current out of emf source

$$V_1 = I_1 R_{eq}$$

$$I_1 = \frac{V_1}{R_{eq}}$$

$$I_1 = \frac{3.0V}{24.81\Omega}$$

$$I_1 = 0.121A$$

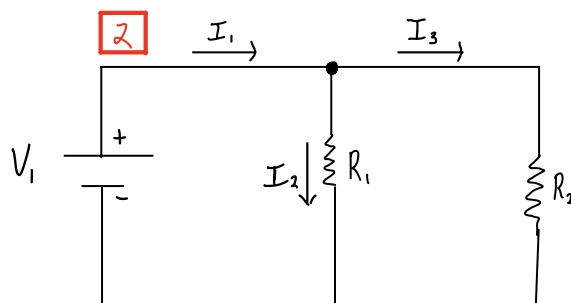
1 Current through R_1

$$V_{R1} = I_2 R_1$$

$$I_2 = \frac{V_{R1}}{R_1}$$

$$I_2 = \frac{3.0V}{33\Omega}$$

$$I_2 = 0.091A$$



- c. Use Table 4.3 to **record** your theoretical values, and **measure** and **record** the appropriate measurements using your DMM. Note: Your theoretical and measured values should be relatively close in number. If they are not, review how you used the DMM and your calculations above. **Check your answers with your instructor before moving on.**

Parameter	Theoretical Value	Measured Value
Potential difference across source V_1	3.0 V	(2.715 ± 0.004) V
Potential difference across R_1 , V_{R1}	3.0 V	(2.681 ± 0.004) V
Potential difference across R_2 , V_{R2}	3.0 V	(2.683 ± 0.004) V
Current out of battery, I_1	0.121 A	(0.110 ± 0.001) A
Current into battery, I_1	0.121 A	(0.110 ± 0.001) A
Current through R_2 , I_3	0.030 A	(0.033 ± 0.001) A
Current through R_1 , I_2	0.091 A	(0.089 ± 0.001) A

Table 4.3 DMM Measurements for Resistors in Parallel

② CIs properly formatted and correct

- d. **Compare** and **explain** the current you measured coming out of the batteries to the current going back into the batteries. Does your data support the conservation of charge? Explain why or why not.

① The current coming out of the batteries is equal to the current going back in, $I_1 = 0.121 \text{ A}$.

This does support conservation of charge, because if $I_{in} = I_{out}$ then no charge is lost from the circuit. (Current is just charged particles that are moving)

- e. For the parallel circuit you tested in Activity #2, **compare** the sum of the currents through each individual resistor ($I_2 + I_3$) to the total current leaving the battery I_1 . (Hint: Use Kirchhoff's Circuit Laws in your comparison.)

① $I_1 = I_2 + I_3 \rightarrow$ follows Kirchhoff's junction rule,
 $0.121 \text{ A} = 0.091 \text{ A} + 0.030 \text{ A} \quad \sum I_{in} = \sum I_{out}$

- f. **Calculate** the percent difference (SLAM Equation 3-8) between the theoretical and measured values for the current I_1 . Show your work.

① correct equation

$$\% \text{ difference} = \frac{|\text{exp. value} - \text{acc. value}|}{|\text{acc. value}|} \times 100$$

$$\% \text{ difference} = \frac{|0.110 \text{ A} - 0.121 \text{ A}|}{0.121 \text{ A}} \times 100$$

$$\underline{\underline{\% \text{ difference} = 9.1\%}}$$

IV. Activity #3 - Find the Resistance of the Unknown Resistor

In this activity, your instructor will give you an unknown resistor (R_3), and you will determine its resistance using Kirchhoff's Loop Rule and Junction Rule. To begin, you will build a circuit that includes R_3 . You will then use the PASCO 850 Interface and increase the potential difference (V_1) to 10 V. The circuit will also include the two batteries (V_2 and V_3) and two resistors (R_1 and R_2) used in the previous activities, as seen in Figure 4.2.

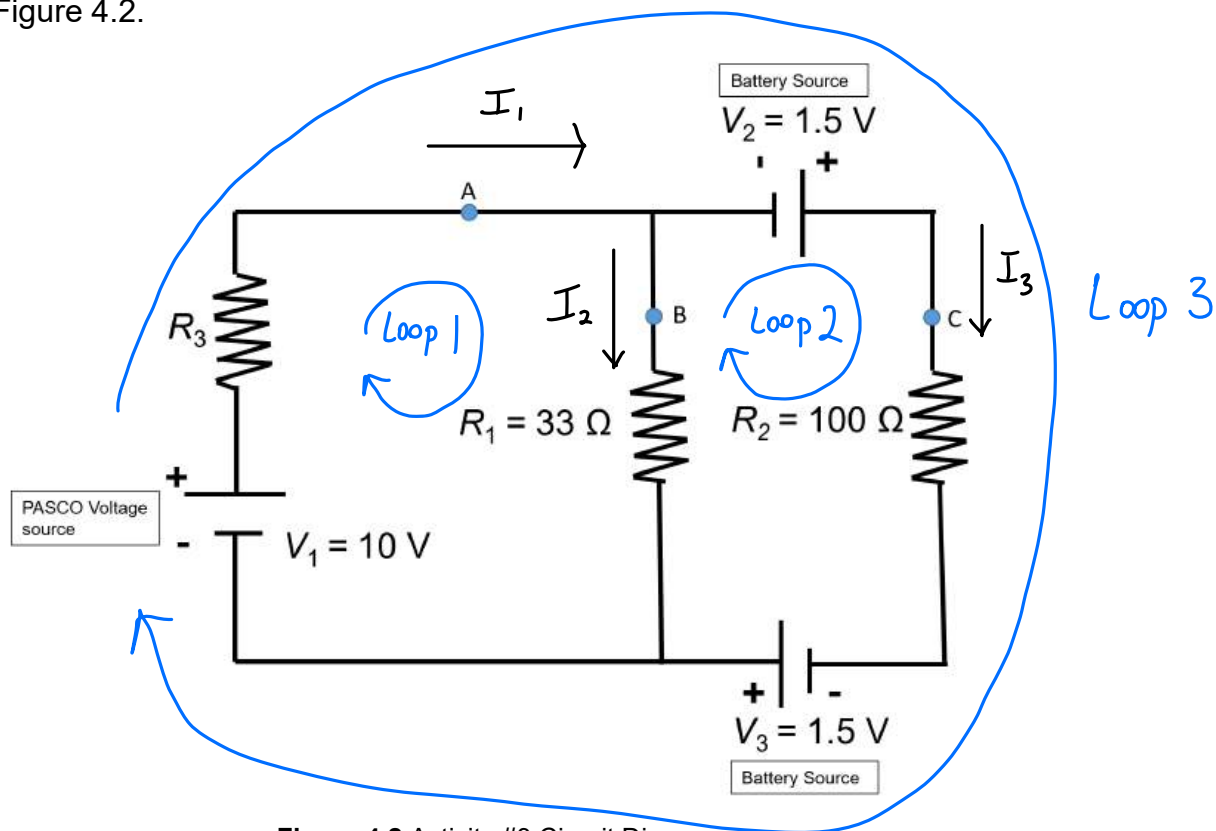


Figure 4.2 Activity #3 Circuit Diagram.

- a. **Measure** the current at either point A, B, or C (do not measure more than one) and verify the potential differences across each component already provided in Figure 4.2 to build three equations and three unknown variables. Use Table 4.4 to organize your measured values.

Parameter	Theoretical Value	Measured Value
Potential difference across source V_1	10.0 V	(9.99 ± 0.01) V
Potential difference across source V_2	1.50 V	(1.446 ± 0.001) V
Potential difference across source V_3	1.50 V	(1.446 ± 0.001) V
R_1	33 Ω	(33.1 ± 0.1) Ω
R_2	100 Ω	(99.6 ± 0.1) Ω
Current (I) measured in (A) B, or C.		(0.098 ± 0.001) A

Table 4.4 DMM Measurements for Activity #3.

△ CIs formatted properly and correct

- b. Calculate R_3 based on your current and voltage measurements. You must show all work and use only the measured potential differences from the PASCO Interface, batteries, and the measured resistances from each resistor.

$$I_1 = I_2 + I_3 \rightarrow I_2 = I_1 - I_3, I_1 = 0.098 \text{ A}$$

$$\text{Loop 1: } V_1 - I_1 R_3 - I_2 R_1 = 0$$

$$\text{Loop 2: } V_3 + I_2 R_1 + V_2 - I_3 R_2 = 0$$

$$\text{Loop 3: } V_1 - I_1 R_3 + V_2 - I_3 R_2 + V_3 = 0$$

$$V_3 + (I_1 - I_3) R_1 + V_2 - I_3 R_2 = 0$$

$$V_3 + I_1 R_1 - I_3 R_1 + V_2 - I_3 R_2 = 0$$

$$I_3 (R_1 + R_2) = V_3 + I_1 R_1 + V_2$$

$$I_3 = \frac{V_3 + I_1 R_1 + V_2}{R_1 + R_2}$$

$$I_3 = \frac{1.446 \text{ V} + (0.098 \text{ A})(33.1 \Omega) + 1.446 \text{ V}}{33.1 \Omega + 99.6 \Omega}$$

$$I_3 = 0.0462 \text{ A}$$

2 Any form of junction rule present

2 Any two loops correctly formatted
(1 pt. for each loop)

$$\rightarrow R_3 = \frac{V_1 + V_2 - I_3 R_2 + V_3}{I_1}$$

$$R_3 = \frac{9.99 \text{ V} + 1.446 \text{ V} - (0.0462 \text{ A})(99.6 \Omega) + 1.446 \text{ V}}{0.098 \text{ A}}$$

$$\underline{\underline{R_3 = 96.5 \Omega}}$$

(example based on
100 Ω resistor)

- c. Measure the resistance of R_3 and record both your calculated and measured value in the table below.

Resistor	Calculated Resistance	Measured Resistance
R_3	<u>96.5 Ω</u>	(98.2 $\pm$ 0.1) Ω

Table 4.5 DMM Measurement of R_3 .

2 R_3 correctly measured and CI properly formatted



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LAB#6 HANDOUT

Pre-Lab Tasks

I. Pre-lab Overview

In this laboratory, you will create a simple RC (resistor-capacitor) Circuit that allows you to both charge and discharge the capacitor, depending on the position of the switch as shown in Figure 6.1. The objective of this experiment is to determine the capacitance of the capacitor.

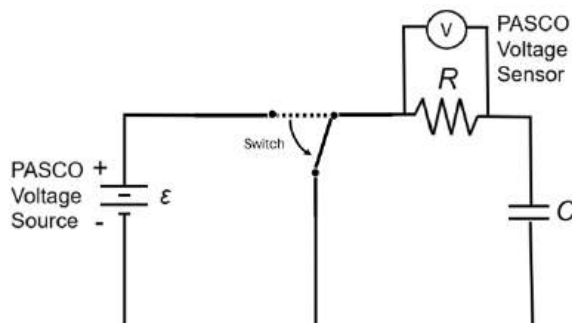


Figure 6.1 Example RC circuit that can be used to accomplish the assigned tasks. Closing the switch will charge the capacitor while opening the switch will discharge the capacitor. A voltmeter measures the electric potential across the capacitor.

You will do this by measuring the electric potential difference across the resistor, V_R , as time varies using the PASCO Voltage Sensor. Recall the equation for the charge over time of a charging capacitor, $q(t)$:

$$q(t) = C\epsilon(1 - e^{-\frac{t}{\tau}}) \quad (1)$$

where ϵ is the electromotive force in volts (V), C is the capacitance of the capacitor in farads (F), τ is the time constant in seconds (s), and time is t . Equation 2 is then derived from Equation 1 when the capacitance has the maximum amount of charge over an extended period of time:

$$C = \frac{Q}{V} \quad (2)$$

However, when analyzing V_R , you will have to find the derivative of Equation 2 through this equation:

$$I(t) = \frac{dQ}{dt} \quad (3)$$

where current with respect to time, $I(t)$, is equal to the change in charge over time, and then apply Ohm's Law.

Recall the equation for the charge over time of a discharging capacitor:

$$q(t) = q_0 e^{-\frac{t}{\tau}} \quad (4)$$

The same process as described above for the charging capacitor can be done to find the electric potential difference across the resistor, V_R .

- a. Derive an algebraic expression for the electric potential difference across the resistor, V_R , of the discharging capacitor. Express your answer in terms of R , t , C , and ϵ . (Hint: You can find the derivative of the discharge equation found on your PRC and apply Ohm's Law, or you can apply Kirchhoff's Loop Rule to the RC circuit found in Figure 6.1 to arrive at the same conclusion).

- b. The objective is to experimentally determine the Capacitance, C , of a capacitor when it is discharging. You will measure the electric potential across the resistor as a function of time. Complete the Linearization Table for a Discharging Capacitor.
 Note: There is partial information for the fundamental physics, as well as the dependent and independent quantities.

Linearization Table

APPENDIX A

Step 1: Identify fundamental physics Charge Equation for a Capacitor $Q(t) = q_0 e^{-\frac{t}{\tau}}$ $Q = CV$	Step 2: Apply physics in order to build a model equation
Step 3: Identify independent and dependent variables	Step 4: Rearrange equation to $y = mx + b$ form to build the linearized model equation.
Step 5: Identify independent and dependent quantities I.Q.: t	Step 6: Identify the objective quantity and control variables
Step 7: Write the expression for the slope of your equation	8: Rearrange your slope so that you solve for your objective quantity equation

Lab Worksheet

I. Lab Overview

You will experimentally determine/verify a capacitor's capacitance, C , defined as the ratio of the maximum charge Q that can be stored in a capacitor to the applied voltage, V , across its plates.

Use linearization to determine the capacitance of a capacitor by measuring the electric potential across the resistor, V_R , as a function of time t . The capacitance will be experimentally determined using a simple RC circuit that is **discharging**.

Use the schematic in Figure 6.1 as a baseline for your experimental setup and ensure you use 5 Volts for your PASCO Voltage source. Note: Do not "turn on" your voltage source until you begin gathering data using your Voltage Sensor.

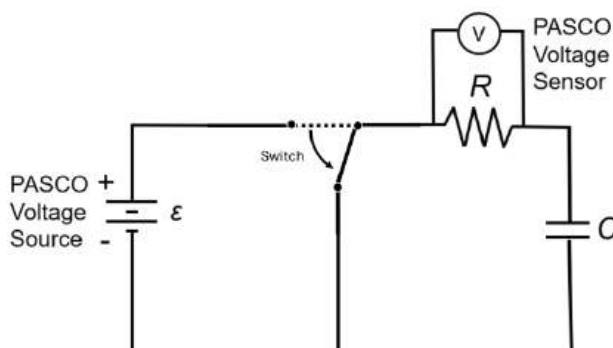


Figure 6.1 Example RC circuit that can be used to accomplish the assigned tasks. Closing the switch will charge the capacitor while opening the switch will discharge the capacitor. The PASCO Voltage Sensor measures the electric potential across the resistor.

II. Lab Equipment

- 1 x PASCO 850 Interface Box
- 1 x TENMA 72-8150/8155 Multimeter (LC Meter)
- 1 x Switch
- 1 x Resistor
- PASCO Voltage Sensor
- 1 x Capacitor
- Banana-plug patch cords as needed

III. Lab Setup

- Select a resistor and a capacitor that gives a time constant, τ , with a minimum value of 0.02 seconds.

- b. Find the accepted value of the capacitor with a measurement conducted through an LC Meter, shown in Figure 6.2. In the [TENMA Multimeter Manual](#) read page 6 for instructions on measuring capacitance and page 11 to find its absolute uncertainty.



Figure 6.2 LC Meter to measure the capacitance of the capacitor (TENMA Models: 72-8150 and 72-8155 Manual).

- c. Measure the resistance of your resistor using a Digital Multimeter (DMM) and calculate its confidence interval as shown in the [SLAM](#), pages 2-14 to 2-16.
- d. Measure the change in electric potential, V_R , across the resistor using a PASCO Voltage Sensor, shown in Figure 6.3, to record voltage over time. To begin, insert the plug-in of the PASCO Voltage Sensor in Port A of the PASCO 850 Universal Interface. In the “Hardware Setup” of the PASCO program, click on Port A of the interface and select “Voltage Sensor.” Next, go back to the main page and select “Sensor Data.” You will be presented with a graph where you need to select its x and y axes. Select “Time” on the x-axis and select “Voltage Sensor” on the y-axis. Ensure you change how many data points you collect by increasing the sampling rate from 20 Hz to at least 100 Hz at the bottom of the page.



Figure 6.3 PASCO Voltage Sensor to measure the change in electric potential V_R across the resistor (<https://www.pasco.com/products/sensors/science-workshop/ui-5100>).

- e. Hit “Record” in PASCO and conduct the experiment for discharging the capacitor. Note: you will have to charge the capacitor first before discharging. Record the plot formed by your discharging capacitor to add to your report. Only collect the data points on the slopes in your graph and as many as necessary.

IV. Submission Requirements

- a. Complete the Measured Quantities Tables (Appendix B) for all measurements conducted in this lab. This includes the accepted value for capacitance.
- b. Conduct a linear regression for the discharging capacitor data by completing a full Appendix C.
- c. Complete the confidence interval for the discharging experimental value of the capacitance using a Propagation of Uncertainty Table (Appendix D).
- d. Conduct an error analysis for the discharging capacitor using a Precision and Accuracy Table (Appendix E). Assume that a historically acceptable level of accuracy for this lab is 1.5% and a historically acceptable level of precision is 1.5%.
- e. Complete a full lab report for this lab to include Appendices A-E to your report. Submit as a single PDF and label "Lab 6_X Hour_Last Names."

As a checklist, your full lab report should include the following:

- Complete Laboratory Report
 - Cover Sheet (Signed)
 - Introduction
 - Theory
 - Procedure
 - Results
 - Discussion
 - Conclusion
 - Works Cited Page (in accordance with Encl. 4-Academic Integrity Brief)
- Appendices
 - Appendix A (Linearization Table)
 - Appendix B (Measurement Tables)
 - Appendix C (Excel table with linear regression and plot)
 - Appendix D (Propagation of Uncertainty Table)
 - Appendix E (Precision and Accuracy Table)



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LAB#6 SOLUTION

$\triangle$	14.36
18	25.76
42	606

1. Appendices (5 Points)

- Linearization Table (App. A).
- Measurement Tables (App. B).
- Linear Regression (App. C).
- Propagation of Uncertainty (App. D).
- Error Analysis (App. E).

5 all present + complete

2. Lab Report (55 Points)

I. Introduction

- Provides a section dedicated to the introduction. 1
- States the objective of the experiment is to determine the capacitance of the capacitor. 3
- Provides a succinct summary on how the objective was accomplished (varied time and measured resulting potential difference across the resistor). 1

II. Theory

- Provides a section dedicated to the theory. 1
- Addresses the discharging equation, current equation, and Ohm's Law used in the experiment. 3
- Provides and correctly explains the appropriate physics equations. 2
- Assumptions made are valid and reasonable. 1
- States model equation and explains how it was derived. 1
- States independent and dependent variables. 2
- States linearized model equation and resulting independent and dependent quantities. 2
- States objective quantity and control variables. 2
- States equation for slope. 1
- States equation for objective quantity. 1
- States method of analysis (linearization) 1

III. Procedures

- Provides a section dedicated to procedures. 1
- Explains the experimental set up/refers to a picture or diagram provided. 3
- States physical quantities being measured and methods to obtain them. 2
- Presents uncertainty of measured quantities, justification of uncertainties, and how they were determined. 2

IV. Results

- a. Provides section dedicated to results. 1
- b. Presents all measured data to include, independent and dependent variables and control variables or physical constants in a logical manner with correct formatting of confidence intervals. 3
- c. Presents plot of independent quantity vs. dependent quantity and reports slope and its uncertainty 2
- d. Shows or references App. D the objective quantity calculation. 2
- e. Shows or references App. D the calculation of the absolute uncertainty of the objective quantity. 2
- f. Values for objective quantity and its absolute uncertainty are reasonable. 1
- g. States correctly formatted final confidence interval for objective quantity. 1

V. Discussion

- a. Provides a section dedicated to discussion. 1
- b. States confidence intervals for experimental and accepted values. 2
- c. Presents a correctly formatted number line. 3
- d. States the relative uncertainty of the experimental value and compares it to the acceptable level of precision. 1
- e. States the percent difference of the experimental value and compares it to the acceptable level of accuracy. 1
- f. States dominant form of error and matches number line. 2
- g. Presents and discusses logical evidence that led to types of error. Largest quadrature term in calculation of objective quantity's absolute uncertainty for random error and high/low chart for systematic error. 1

VI. Conclusion

- a. Provides a section dedicated to the conclusion. 1
- b. Assesses whether experimental objective was met. 2
- c. Concisely communicates lessons learned from the experimental procedure. 1
- d. Provides details of how experiment could be done differently from the lessons learned. 1

1 Overall writing style, presentation, + formatting (up to instructor)



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LAB#7 SIGNATURE WRITING EVENT

Time of completion _____
30 October 2024

Cadet _____

Section _____
Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
PH2X2, Physics II
Signature Writing Event

1. Instructions:

- a. The following are authorized references: your textbook, TI-30XIIS or TI-36XPRO calculator, PRC, Lab PRC, SLAM, and all PH2X2 lab supplements. Bring only these references, your laptop, and a pen(s)/pencil(s) into the examination room. No phones, cameras, or smartwatches are allowed in the examination room.
- b. **Direct text quotations** or **images** taken from your textbook, the SLAM, or your previous laboratory reports must be cited.
- c. Any text provided within this document does not require citation if used in your submission. It is considered work completed within your lab group.
- d. Take up to two hours to complete the examination. If you leave early, record your time of completion above.

2. An instructor is in the hall or classroom.

3. Exam Security:

- a. I will not discuss the Signature Writing Event with anyone until it is released from academic security on _____ (date) at _____ (time).
 - b. Cadet Signature: _____
 - c. Cadet Name (print): _____
4. Following the completion of your work and the endorsement of the exam security statement, submit your WPWP-SWE on Canvas under the Lab 7 SWE page. The submission should be a single PDF file named ***“SWE X last_first.pdf”*** where “X” is your class hour (A, B, D, E, F) and ***“last_first”*** is your last and first name. **Instructors will not accept incorrect file types or naming conventions.**

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PH2X2 Laboratory WPWP-SWE

1. SWE Specified Tasks

- A. **Read** the provided accepted value, Introduction, Theory, Procedures, and Appendices found under Additional Information.
- B. **Complete** the lab report's **Results**, **Discussion**, and **Conclusion** sections IAW the SLAM Section 4.C.
- C. **Submit** your digital work on Canvas under the Lab 7 SWE page and turn in this packet at the end of the two-hour testing period.

2. Additional Information

- A. The accepted value for the charge per mass of an electron is
$$\frac{q}{m} = (1.8 \pm 0.1) \times 10^{11} \frac{C}{kg}.$$
- B. For your writing submission, your Instructor will provide images of tables and figures required to complete the SWE.
- C. The Introduction, Theory, Procedures, and all Appendices are shown on the following pages.

I. Introduction

The objective of this experiment is to determine the charge per mass, q/m , of an electron. This is accomplished by varying the current through a Helmholtz coil apparatus which in turn varies the strength of the magnetic field produced, and then measuring the diameter of the path of the accelerated electrons. The purpose of this experiment is to gain a better understanding of how charged particles react when moving in an external magnetic field.

II. Theory

If an electron with a charge, q , and a velocity, $\vec{v}$, experiences a magnetic field, $\vec{B}$, perpendicular to its direction, it also experiences a Lorentz force, $\vec{F}$, in a direction perpendicular to both its velocity and the magnetic field:

$$\vec{F} = q\vec{v} \times \vec{B} \quad (1)$$

This gives rise to a centripetal force on the electron in a circular path with a diameter, d . The velocity, $\vec{v}$, also depends on the accelerating voltage, ΔV , of the electron, which initially starts from rest. Therefore, the equation for the electric potential energy U_E , is

$$U_E = q\Delta V \quad (2)$$

A pair of Helmholtz coils can produce a uniform magnetic field. The two parallel, circular coils are separated by a distance equal to half the diameter, D , of the coils, and as such, the magnetic field at a point midway between the coils is given by:

$$B = \mu_o \left(\frac{4}{5}\right)^{3/2} \frac{2NI}{D} \quad (3)$$

where μ_o is the permeability of free space, N is the number of turns of wire on each coil, and I is the current through the coils. Since the magnetic field generated from the Helmholtz coils is proportional to the current through it, it is possible to vary the current to change the magnetic field affecting the electrons' circular paths.

Using the equations and concepts above, it is possible to derive a model equation for the charge per mass of an electron. Our model equation then becomes,

$$\frac{q}{m} = \frac{2D^2\Delta V}{\frac{4^3}{5}(\mu_o NId)^2} \quad (4)$$

Where the independent variable is the current, I , and the dependent variable is the diameter of the path of the electron, d . The current is varied through the Helmholtz coils and the diameter of the path of the electrons is measured. The method used to analyze

PH2X2 Laboratory WPWP-SWE

the relationship between these two variables is linearization, and equation 5 depicts the linearized model equation.

$$\frac{1}{d^2} = \frac{\frac{4^3}{5} q(\mu_0 N)^2}{2mD^2 \Delta V} (I^2) \quad (5)$$

Where I^2 is the independent quantity and $\frac{1}{d^2}$ is the dependent quantity. From this linearized equation, the slope then becomes,

$$slope = \frac{\frac{4^3}{5} q(\mu_0 N)^2}{2mD^2 \Delta V} \quad (6)$$

The objective quantity, the charge per mass, can then be solved by using the slope from the linearized equation.

$$\frac{q}{m} = \frac{(slope)2D^2 \Delta V}{\frac{4^3}{5} (\mu_0 N)^2} \quad (7)$$

One assumption made in this derivation is that our set up is a closed and isolated system, and we can apply the theory of conservation of energy where the electrons only have translational kinetic energy and electric potential energy.

III. Procedures

To experimentally determine the charge per mass of an electron, the following equipment is used: one q/m apparatus (q/m tube, Helmholtz coils, and base), one universal power supply, two digital multimeters (DMM), and eleven banana plug-patch cords. Figure 1 below shows the complete experimental set up. To begin, a pair of banana plug-patch cords were connected from the universal power supply's 6 Vac ports to the Filament ports on the side of the q/m apparatus to provide electricity to the tube. Then a pair of banana plug-patch cords were connected from the high voltage ports on the universal power supply to the accelerating voltage port in the base of the q/m apparatus. A DMM was placed in parallel to measure this voltage. Lastly, a pair of banana plug-patch cords was connected from the 0-5 V dc regulated ports to the Helmholtz coils to provide current to the coils. A DMM was placed in series to measure this current.



Figure 1: From left to right, q/m apparatus containing q/m tube, Helmholtz coils, and base; universal power supply that provides 6 V from 6 Vac ports to Filament terminals in the q/m apparatus, "High Voltage" ranging from 125 to 500 V, and "Regulated" voltage ranging from 0 to 5 V; one DMM measuring voltage in the "High Voltage" port and one DMM measuring Current in the "Regulated" port.

The current is the independent variable of this experiment. A DMM was placed in series, and the dial on the DMM was turned to the direct current setting measuring amps. The measurement was recorded when a value held steady. The current was varied and measured seven times. The most likely factor contributing to uncertainty is the quality of the measuring device. In section 2C of the SLAM, Table 2-4 gives specifications for a DMM used as an ammeter. From this table, the uncertainty was found for each of the seven measurements and recorded in Appendix C.

The diameter of the path of the electron is the dependent variable of this experiment. A ruler mounted on the back of the q/m apparatus was used to take this measurement. When the current was varied, the diameter was then measured for seven iterations. The most likely factor contributing to the uncertainty is the conditions under which the measurement was made. The width of the beam of electrons varied as current varied and this was taken into account when taking the diameter measurement. The uncertainty was found for each seven measurements and recorded in Appendix C.

The accelerating voltage is a control variable in this experiment. The dial for accelerating voltage on the universal power supply was turned to 125 V, and a DMM was placed in parallel to take the measurement. The value displayed was then recorded as a constant value in Appendix C. The most likely factor contributing to uncertainty is due to the quality of the measuring device. In section 2C of the SLAM, Table 2-3 gives specifications for a DMM used as a voltmeter. From this table, the uncertainty was found to be ± 0.1 V.

The diameter of the Helmholtz coils is another control variable in this experiment. This diameter was measured using a standard tape measurer displayed in meters. The tape measurer's end marked with 0 m was placed on the outer edge of the coils and then pulled straight and parallel to the floor across the coils. The measurement was then

PH2X2 Laboratory WPWP-SWE

recorded in Appendix C. The most likely factor contributing to the uncertainty is due to the quality of the measuring device and was found to be ± 0.01 m.

The number of turns on the Helmholtz coils is a control variable which was not directly measured but is displayed on the base of the q/m apparatus and gives the value of $N = 130$ turns. Another control variable which was not directly measured is the permeability of free space. This value can be found on the PH2X2 PRC under the physical constants table. The value is $\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$.

The data points for the independent and dependent variables stated above were later used in the linearization of the independent and dependent quantities to further determine the objective quantity.

Linearization Table

Appendix A

Step 1: Identify fundamental physics $\vec{F} = q\vec{v} \times \vec{B}$ $U_E = q\Delta V$ $B = \mu_0 \left(\frac{4}{5}\right)^{3/2} \frac{2NI}{D}$	Step 2: Apply physics in order to build a model equation $\frac{q}{m} = \frac{2D^2\Delta V}{\frac{4^3}{5} (\mu_0 N I d)^2}$
Step 3: Identify independent and dependent variables IV: I DV: d	Step 4: Rearrange equation to $y = mx + b$ form to build the linearized model equation. $\frac{1}{d^2} = \frac{\frac{4^3}{5} q(\mu_0 N)^2}{2mD^2\Delta V} (I^2)$
Step 5: Identify independent and dependent quantities IQ: I^2 DQ: $\frac{1}{d^2}$	Step 6: Identify the objective quantity and control variables OQ: $\frac{q}{m}$ CV: $N, \Delta V, D$, and μ_0
Step 7: Write the expression for the slope of your equation $slope = \frac{\frac{4^3}{5} q(\mu_0 N)^2}{2mD^2\Delta V}$	8: Rearrange your slope so that you solve for your objective quantity equation $\frac{q}{m} = \frac{(slope)2D^2\Delta V}{\frac{4^3}{5} (\mu_0 N)^2}$

Directly Measured Quantity Table

Appendix B

Measured Quantity	Confidence Interval	Relative Uncertainty	Measurement Device
d, Diameter [m] of electron's circular path	See App. C	See App. C	Ruler on Coil
	Measurement Procedure: A ruler mounted on the back of the q/m apparatus was used to take this measurement. When the current was varied, the diameter of the electrons path was then measured for seven iterations.		
	Justification of Uncertainty: The primary factor of uncertainty was the conditions under which measurement was made. The width of the beam varied so it was difficult to tell exactly where to measure from. See Appendix C for absolute uncertainty.		

Measured Quantity	Confidence Interval	Relative Uncertainty	Measurement Device
Current [A], I	See App. C	See App. C	DMM
	Measurement Procedure: A DMM was placed in series, and the dial on the DMM was turned to the direct current setting measuring amps. The measurement was recorded when a value held steady. The current was varied and measured seven times.		
	Justification of Uncertainty: The primary factor of uncertainty for the current is the quality of the measuring device. In section 2C of the SLAM, Table 2-4 gives specifications for a DMM used as an ammeter. From this table, the uncertainty was found for each of the seven measurements and recorded in Appendix C.		

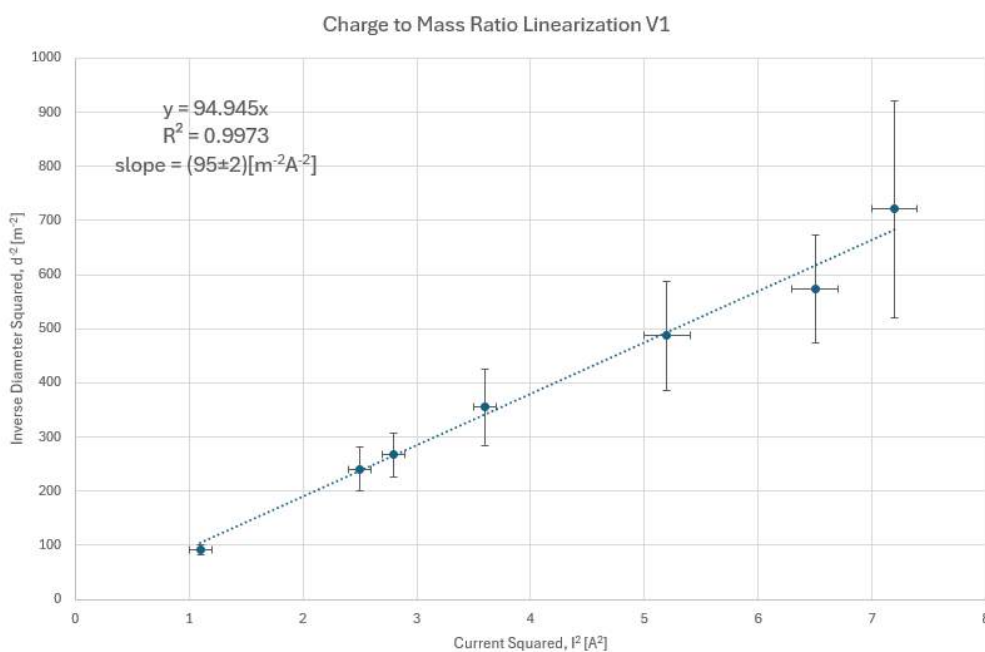
Measured Quantity	Confidence Interval	Relative Uncertainty	Measurement Device
Potential difference (Accelerating Voltage) [V]	$(129.6 \pm 0.1) \text{ V}$	0.077%	DMM
	Measurement Procedure: The dial for accelerating voltage on the universal power supply was turned to 125 V, and a DMM was placed in parallel to take the measurement. The dial on the DMM was turned to the direct current setting measuring volts. The value displayed was then recorded as a constant value in Appendix C.		
	Justification of Uncertainty: The primary factor of uncertainty for the potential difference is the quality of the measuring device. In section 2C of the SLAM, Table 2-3 gives specifications for a DMM used as a voltmeter. From this table, the uncertainty was found for the measurement and recorded in Appendix C.		

Measured Quantity	Confidence Interval	Relative Uncertainty	Measurement Device
D, Diameter [m] of the Helmholtz Coils	$(0.32 \pm 0.01) \text{ m}$	3.1%	Measuring Tape
	Measurement Procedure: This diameter was measured using a standard tape measurer displayed in meters. The tape measurer's end marked with 0 m was placed on the outer edge of the coils and then pulled straight and parallel to the floor across the coils. The measurement was then recorded in Appendix C.		
	Justification of Uncertainty: The primary factor of uncertainty was the quality of measuring device. For this measurement, we used a measuring tape, and the resolution of the measuring tape is 0.01 m.		

Linear Regression

Appendix C

Control Variables										
D [m]	δD [m]	ΔV [V]	δV [V]	N						
0.32	0.01	129.6	0.1	130						
Trial	Current I [A]	δI [A]	I rel. uncert.	diameter d [m]	δd [m]	d rel. uncert.	$ Q I^2$ [A ²]	δI^2 [A ²]	DQ $1/d^2$ [1/m ²]	$\delta 1/d^2$ [1/m ²]
1	1.03	0.03	2.9%	0.104	0.005	4.8%	1.1	0.1	92	9
2	1.58	0.03	1.9%	0.064	0.005	7.8%	2.5	0.1	241	40
3	1.68	0.03	1.8%	0.061	0.005	8.2%	2.8	0.1	267	40
4	1.90	0.03	1.6%	0.053	0.005	9.4%	3.6	0.1	355	70
5	2.27	0.04	1.8%	0.045	0.005	11%	5.2	0.2	487	100
6	2.54	0.04	1.6%	0.042	0.005	12%	6.5	0.2	574	100
7	2.68	0.04	1.5%	0.037	0.005	13%	7.2	0.2	721	200

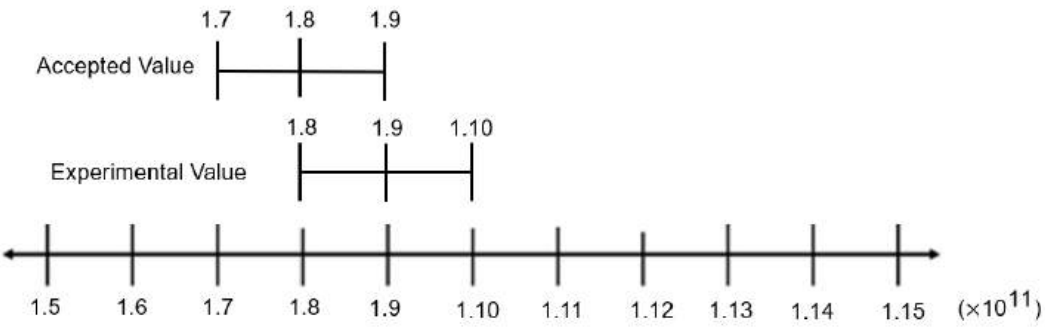


SUMMARY OUTPUT		
Regression Statistics		
R Square	0.997298	
Standard Error	24.65869	
Observations	7	
	Coefficients	Standard Error
Slope	94.94544	2.01748

Propagation of Uncertainty

Appendix D

Objective Quantity: Charge per mass, q/m		Objective Quantity Equation: $\frac{q}{m} = \frac{(slope)2D^2\Delta V}{\frac{4^3}{5}(\mu_0 N)^2}$	
Variable	Measured Value	Absolute Uncertainty	Relative Uncertainty
ΔV	129.6 V	0.1 V	0.077%
D	0.32 m	0.01 m	3.1%
$slope$	$95 \frac{CN^2}{A^4 kgVm^2}$	$2 \frac{CN^2}{A^4 kgVm^2}$	2.1%
Calculation of Objective Quantity's Magnitude: $\frac{q}{m} = \frac{(slope)2D^2\Delta V}{\frac{4^3}{5}(\mu_0 N)^2}$ $\frac{q}{m} = \frac{\left(95 \frac{CN^2}{A^4 kgVm^2}\right)2(0.32\text{ m})^2 129.6\text{ V}}{\frac{4^3}{5}(4\pi \times 10^{-7} \frac{N}{A^2})^2 (130)^2}$ $\frac{q}{m} = 1.85 \times 10^{11} \frac{C}{kg}$			
Calculation of Objective Quantity's Absolute Uncertainty: $\delta q/m = \sqrt{\left(\frac{\partial q/m}{\partial \Delta V} \delta \Delta V\right)^2 + \left(\frac{\partial q/m}{\partial D} \delta D\right)^2 + \left(\frac{\partial q/m}{\partial slope} \delta slope\right)^2}$ $\delta q/m = \sqrt{\left(\frac{slope 2D^2}{N^2 \left(\frac{4}{5}\right)^3 \mu_0^2} \delta \Delta V\right)^2 + \left(\frac{slope 4\Delta V D}{N^2 \left(\frac{4}{5}\right)^3 \mu_0^2} \delta D\right)^2 + \left(\frac{2\Delta V D^2}{N^2 \left(\frac{4}{5}\right)^3 \mu_0^2} \delta slope\right)^2}$ $\delta q/m = \sqrt{(2.03 \times 10^{18}) + (1.33 \times 10^{20}) + (1.51 \times 10^{19})}$ $\delta q/m = 1 \times 10^{10} \frac{C}{kg}$			
Objective Quantity Confidence Interval: $\frac{q}{m} = (1.9 \pm 0.1) \times 10^{11} \frac{C}{kg}$			

Experimental Value CI: $\frac{q}{m} = (1.9 \pm 0.1) \times 10^{11} \frac{C}{kg}$	Accepted Value CI: $\frac{q}{m} = (1.8 \pm 0.1) \times 10^{11} \frac{C}{kg}$
Figure 4: Number Line Physical Quantity: Charge per mass, q/m Units: C/kg 	
Dominant Form of Error Present (Random or Systematic): Random	
Acceptable Level of Precision (given): 1.5%	Acceptable Level of Accuracy (given): 1.5%
Calculation of Relative Uncertainty: $RU = \frac{\delta q/m}{ q/m }$ $RU = \frac{0.1 \times 10^{11} C/kg}{1.9 \times 10^{11} C/kg}$ $RU = 5.3\%$	Calculation of Percent Difference: $\% \text{ difference} = \frac{ \text{exp. value} - \text{acc. value} \times 100}{ \text{acc. value} }$ $\% \text{ difference} = \frac{ 1.9 \times 10^{11} C/kg - 1.8 \times 10^{11} C/kg \times 100}{ 1.8 \times 10^{11} C/kg }$ $\% \text{ difference} = 5.6\%$



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LAB#7 SOLUTION

Question 20:

Briefly explain the difference between random and systematic error. Provide an example for each type of error and explain an appropriate method to mitigate the type of error.

- Explains the difference between random and systematic error (more than just stating number lines do or do not overlap) – **4 points**
- Provides example of random error and how to mitigate – **3 points**
- Provides example of systematic error and how to mitigate – **3 points**

Question 21:

During Lab 6, cadets experimentally determined the capacitance of a capacitor using a discharging RC Circuit. Their experimental value for capacitance was $(671 \pm 9) \mu F$ and the accepted value was $(640 \pm 3) \mu F$. The acceptable level of precision and acceptable level of accuracy was 1.5%. Was their experiment within the acceptable level for precision and accuracy?

- They don't need to show calculations/equations, but they should state their calculated values for:
 - Relative Uncertainty – 1.3 % - **3 points**
 - Percent Difference – 4.8 % - **3 points**
- States that the experiment is more precise – **2 Points**
- States that the experiment is less accurate – **2 Points**

Acceptable Level of Precision (given): 1.5 %	Acceptable Level of Accuracy (given): 1.5 %
Calculation of Relative Uncertainty: $RU = \frac{\delta C}{ C }$ $RU = \frac{9 \mu F}{671 \mu F}$ $RU = 1.3 \%$	Calculation of Percent Difference: $\% \text{ diff.} = \frac{ exp. - acc. }{ acc. } \times 100$ $\% \text{ diff.} = \frac{ 671 - 640 }{ 640 } \times 100$ $\% \text{ diff.} = 4.8 \%$

$\triangle$	17.1 μ
16	22.9 μ
42	60 μ

1. Lab Report

I. Results

- Provides section dedicated to results. $\square 2$
- Presents measured data for the independent variable (current) and the dependent variable (diameter of path of particle) in logical manner with correct formatting. $\square 4$
- Presents control variables (diameter of coils and accelerating voltage) and physical constant (permeability of free space) in a logical manner with correct formatting. $\circ 3$
- Presents a plot of the independent quantity vs. dependent quantity. $\square 4$
- Reports slope of graph and its uncertainty. $\circ 3$
- Shows or references App. D for the objective quantity calculation. $\square 4$
- Shows or references App. D for the calculation of the absolute uncertainty of the objective quantity. $\square 4$
- States correctly formatted final confidence interval for objective quantity. $\triangle 1$

II. Discussion

- Provides a section dedicated to discussion. $\square 2$
- States confidence intervals for experimental value. $\square 4$
- States confidence interval for accepted value. $\square 4$
- Presents a correctly formatted number line. $\square 4$
- States the relative uncertainty of the experimental value and compares it to the acceptable level of precision. $\circ 3$
- States the percent difference of the experimental value and compares it to the acceptable level of accuracy. $\circ 3$
- States dominant form of error and matches number line (random). $\square 4$
- Presents and discusses logical evidence that led to types of error. Largest quadrature term in calculation of objective quantity's absolute uncertainty for random error. $\triangle 2$

$\triangle$ Style of writing is coherent and error discussion of results is effectively conveyed with proper formatting.

VI. Conclusion

- a. Provides a section dedicated to the conclusion. 2
- b. Assesses whether experimental objective was met. 4
- c. Concisely communicates lessons learned from the experimental procedure. 2
- d. Provides details of how the experiment could be done differently from the lessons learned. 2

△ Style of writing is coherent and assessment of objective is effectively conveyed with proper formatting

WPWP Rubric Mapped to PH2X2 SWE:

Performance Level	Exemplary	Satisfactory	Marginal	Not Proficient
Substance	Requirements stated above for each section are mostly present and in a logical manner.	Missing some requirements and lacks in clarity of writing.	Missing significant amount of requirements for each section.	Fails to include requirements for each section.
Organization	Contains three distinct sections: Results, Discussion, and Conclusion. The content is strictly appropriate for both.	While not distinct, there is a separation of referencing data/calculating obj. quantity, assessment of experimental error, and conclusion.	Required content is included but poorly organized between sections.	Fails to use any formatting for Results, Discussion, and Conclusion paragraphs.
Style and Presentation	Writing style is clear and technical in nature. Figures, tables, equations, and conf. intervals are formatted and/or referenced correctly.	Writing is clear but with minor errors in formatting and/or referencing figures, tables, equations, and conf. intervals.	Significant errors in writing style and does not present or reference all required figures, tables, equations, or conf. intervals.	Fails to write clearly and does not present or reference any figures, tables, equations, or confidence intervals.
Mechanics & Correctness	Generally free of grammatical/spelling errors. Data and analysis presented are logical and accurate.	Minor grammatical/spelling errors. Data and analysis are presented but are not logical.	Significant grammatical/spelling errors. Attempts to present data and analysis but are not complete or presented poorly.	Fails to provide any data or analysis.
Documentation (if required)	Citations are present for work completed by others.	-	-	Fails to properly cite.



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LAB#8 HANDOUT

Pre-Lab Tasks

I. Background

The impedance of an AC circuit is analogous to the resistance in a DC circuit. Both oppose the flow of current. However, impedance includes the combined effect of resistance, R , capacitive reactance, X_C , and inductive reactance, X_L .

Similar to Ohm's Law, to calculate the current root mean square (rms), I_{rms} , in an RLC circuit, the output voltage rms, V_{rms} , is divided by the impedance, Z , as shown in Equation 1:

$$I_{rms} = \frac{V_{rms}}{Z} \quad (1)$$

Impedance depends on frequency, f , and there is a specific frequency where X_C and X_L equal each other, allowing for the maximum current rms through the circuit. This frequency is known as resonant frequency, f_r .

Primary Objective: Vary the frequency in an RLC circuit driven by an AC power source, measure the potential difference across the resistor to calculate the rms current and experimentally determine its resonant frequency.

- a. Define the following terms. Provide the appropriate equations, their units, and how they relate to a series RLC circuit:
 1. Root mean square (rms) Voltage
 2. Root mean square (rms) Current
 3. Capacitive Reactance

4. Inductive Reactance

5. Impedance

6. Resonant Frequency

- b.** Model an equation that solves for the current rms, I_{rms} , through the RLC circuit in terms of f , C , L , R , and V_{rms} .

- c. Consider the following values for the series RLC circuit as seen in Figure 1: peak voltage of the driven AC power source 3.0 V, inductance 6.8 mH, capacitance 3,900 pF, and resistance 1,000. Ω . These will be the theoretical values for the components you will measure in the lab.

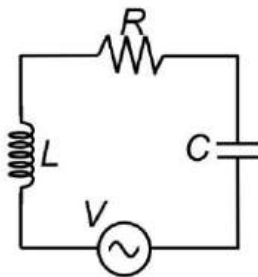


Figure 1 RLC Circuit Diagram.

1. Calculate the resonant frequency in units of Hz.
2. Open the Excel spreadsheet from Lab 9's Canvas page. It has the following terms in each column: f , V_{rms} , X_C , X_L , R , Z , and theoretical I_{rms} . It also has the following values: peak voltage of the driven AC power source 5.0 V, inductance 6.8 mH, capacitance 3,900 pF, and resistance 1,000. Ω . Notice that the capacitive reactance is already set up. Use that column as an example and update the remaining columns based on their respective equations for each term as a function of frequency.

- 4

Lab Worksheet

I. Lab Overview

Vary the frequency in an RLC Circuit driven by an AC power source and measure the rms voltage to experimentally determine its resonant frequency.

II. Activity #1: Determine the Accepted Resonant Frequency

- Select the resistor, capacitor, and inductor with the same values as in your pre-lab. Measure each component using a TENMA LC Meter and provide an absolute uncertainty per the manual's instructions.
- Based on your measurements, use Appendix D to calculate the accepted resonant frequency and its absolute uncertainty.
- Cut and paste your Excel Worksheet from your Pre-lab into a new tab (label it "Accepted") and update it with the appropriate measured values.
- Label your graphs as "Accepted Reactance/Impedance Graph" and "Accepted rms Current Graph."

III. Activity #2: Experimentally determine the Resonant Frequency

Due to activity #1, you have a better understanding of where your experimental resonant frequency will be.

- In the PASCO Capstone program, open the Signal Generator and ensure the Sine Waveform is chosen, as presented in Figure 1. Here you can also change the frequency incrementally. Choose an amplitude of 3 V, which will be the peak voltage for your system. Maintain it "Off" until ready to capture data.

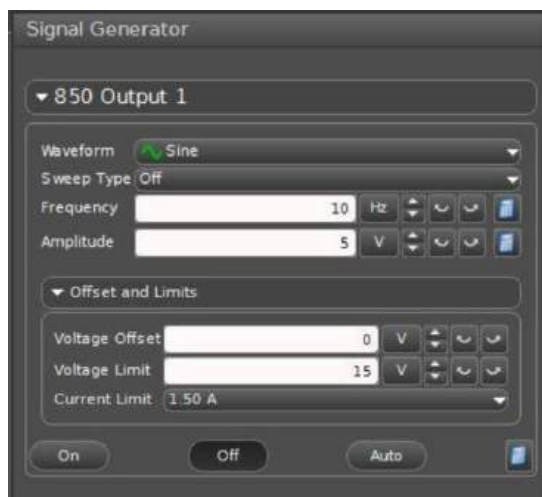


Figure 1 PASCO Signal Generator 850 Output 1.

- b. Connect the resistor, inductor, capacitor, and the AC voltage source in series as shown in Figure 2. Connect a Fluke 27/FM Digital Multimeter (DMM) across the resistor to measure its voltage rms, $V_{rms,R}$ (this is to avoid adding additional resistance to the circuit due to the DMM by measuring the current rms, I_{rms}).

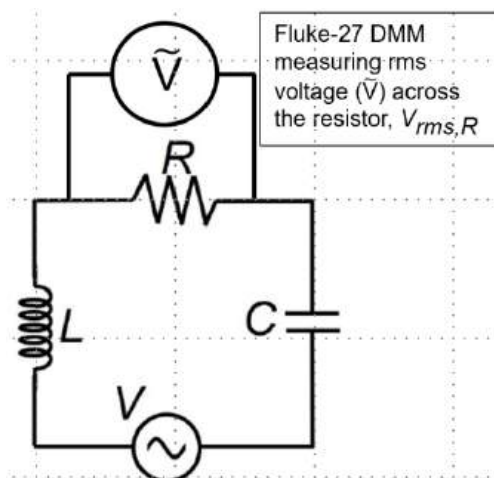


Figure 2 Setup diagram for an RLC Circuit.

- c. Create three new columns in the worksheet, one for “PASCO frequency f ,” one for “measured $V_{rms,R}$,” and the other for “experimental I_{rms} .” Note: Use Ohm’s Law to calculate the experimental I_{rms} .
- d. To determine the experimental Resonant Frequency, turn “On” the PASCO Signal Generator and begin varying the frequency (Add these in the “PASCO frequency f ” column). Measure the $V_{rms,R}$ and input your data on the excel spreadsheet. Use a step size of 1000 Hz and notice when you reach your resonant frequency. Continue capturing data until about 50,000 Hz. From your data, identify the interval where your experimental resonant frequency is, begin to bracket it, and decrease your step size until you are within a satisfactory margin of the objective. You may decrease your step size until you are within 10 Hz of your experimental Resonant Frequency.
- e. To determine the absolute uncertainty for the experimental Resonant Frequency, decrease the step size taken within the frequency interval where you recorded the maximum I_{rms} . Continue decreasing the step size until you are satisfied that you are within a small range of the experimental Resonant Frequency. As an example, if you recorded identical I_{rms} for a range of frequencies from 1150 Hz -1260 Hz with a step size of 10 Hz, your median value would become your experimental value (1205Hz). Your confidence interval will be determined by finding the range, adding the step size, and dividing by 2 ($[1260\text{Hz}-1150\text{Hz}+10\text{Hz}]/2=60\text{Hz}$). In the above example your confidence interval would be (1210 +/- 60)Hz.

- f. Plot the Accepted rms Current from Activity #1 and the Experimental rms Current from Activity #2 with respect to the driving frequency and compare them against each other. Label it as “Accepted and Experimental rms Current Graph.” As an example, see Figure 3 below.

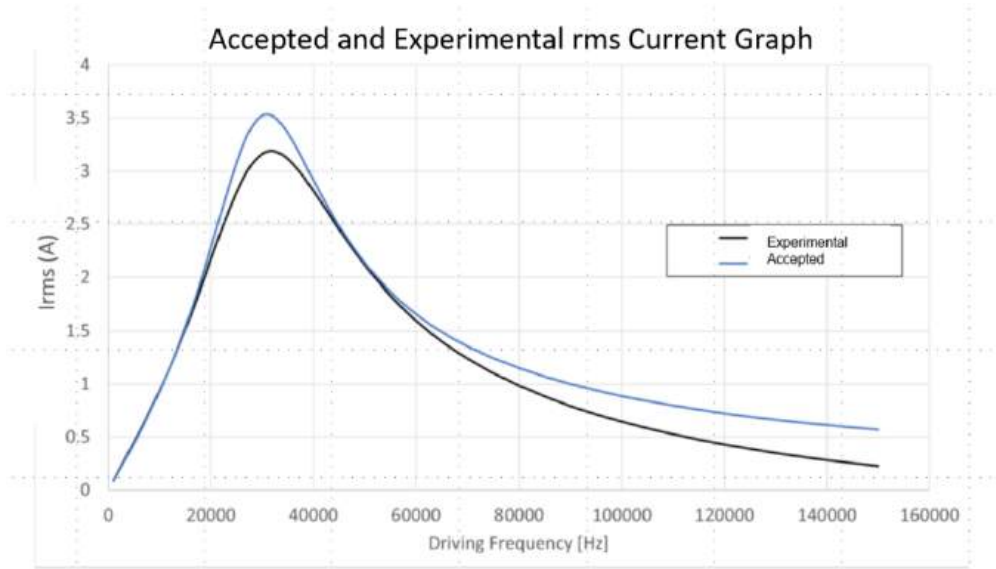


Figure 3 An example of a graph demonstrating the accepted and experimental values with respect to the frequency.

IV. Submission Requirements

- Complete the Measured Quantities Tables (Appendix B) for the following: L , C , R , and measured $V_{rms,R}$.
- Provide the “Accepted Reactance/Impedance Graph” and “Accepted and Experimental rms Current Graph” (Appendix C).
- Complete the confidence interval for the Accepted Resonant Frequency by conducting a Propagation of Uncertainty Table (Appendix D).
- Conduct an error analysis in relation to the Accepted and Experimental Resonant frequencies using a Precision and Accuracy Table (Appendix E). Assume that a historically acceptable level of accuracy for this lab is 1.0% and a historically acceptable level of precision is 1.0%.
- Discussion questions:
 - When the rms current is at a maximum what is the relationship between the capacitive reactance and the inductive reactance, and how is the overall impedance affected?
 - Why is it possible to measure the rms voltage across the resistor and still analyze the rms current through the whole circuit?
 - Compare the accepted and experimental maximum rms current values and discuss any differences. What are the contributing factors to these differences?
- Submit as a single PDF and label “Lab 8_X Hour_Last Names.”

As a checklist, your lab report should include the following:

- Cover Sheet (Signed)
- Appendices
 - Appendix B (Measurement Tables)
 - Appendix C (“Accepted Reactance/Impedance Graph” and “Accepted and Experimental rms Current Graph” on a single page)
 - Appendix D (Propagation of Uncertainty Table)
 - Appendix E (Precision and Accuracy Table)
- Answers to discussion questions stated above.
- Works Cited Page (in accordance with Encl. 5-Academic Integrity Brief)



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LAB#8 SOLUTION

Pre-Lab Tasks**I. Background**

The impedance of an AC circuit is analogous to the resistance in a DC circuit. Both oppose the flow of current. However, impedance includes the combined effect of resistance, R , capacitive reactance, X_C , and inductive reactance, X_L .

Similar to Ohm's Law, to calculate the current root mean square (rms), I_{rms} , in an RLC circuit, the output voltage rms, V_{rms} , is divided by the impedance, Z , as shown in Equation 1:

$$I_{rms} = \frac{V_{rms}}{Z} \quad (1)$$

Impedance depends on frequency, f , and there is a specific frequency where X_C and X_L equal each other, allowing for the maximum current rms through the circuit. This frequency is known as resonant frequency, f_r .

Primary Objective: Vary the frequency in an RLC circuit driven by an AC power source, measure the potential difference across the resistor to calculate the rms current and experimentally determine its resonant frequency.

- a. Define the following terms. Provide the appropriate equations, their units, and how they relate to a series RLC circuit:

1. Root mean square (rms) Voltage

$$V_{rms} = \frac{V_0}{\sqrt{2}}, \quad V_0 \rightarrow \text{voltage amplitude, units} \rightarrow V$$

The "average" of the voltage through the circuit over time

2. Root mean square (rms) Current

$$I_{rms} = \frac{I_0}{\sqrt{2}}, \quad I_0 \rightarrow \text{current amplitude, units} \rightarrow A$$

The "average" of the current through the circuit over time

3. Capacitive Reactance

$$X_C = \frac{1}{\omega C}, \quad \omega \rightarrow \text{angular frequency, } C \rightarrow \text{capacitance}$$

units $\rightarrow \Omega$

Effective resistance when alt. current passes through a capacitor.

4. Inductive Reactance

$X_L = \omega L$, $\omega \rightarrow$ angular frequency, $L \rightarrow$ inductance, units $\rightarrow \Omega$
 Effective resistance when alt. current passes through an inductor.

5. Impedance

$Z = \sqrt{R^2 + (X_L - X_C)^2}$, $R \rightarrow$ resistance, $X_L \rightarrow$ ind. react., $X_C \rightarrow$ cap. react.,
 Overall effective resistance of an RLC circuit units $\rightarrow \Omega$

6. Resonant Frequency

$\omega = \sqrt{\frac{1}{LC}}$, $L \rightarrow$ inductance, $C \rightarrow$ capacitance, units $\rightarrow \text{rad/s}$

$\omega = 2\pi f \rightarrow f_0 = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$ frequency at which RLC circuit is most efficient.
 b. Model an equation that solves for the current rms, I_{rms} , through the RLC circuit in terms of f , C , L , R , and V_{rms} .

$$V_{rms} = I_{rms} Z$$

$$I_{rms} = \frac{V_{rms}}{Z}$$

$$I_{rms} = \frac{V_{rms}}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$I_{rms} = \frac{V_{rms}}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$$I_{rms} = \frac{V_{rms}}{\sqrt{R^2 + \left(2\pi f L - \frac{1}{2\pi f C}\right)^2}}$$

- c. Consider the following values for the series RLC circuit as seen in Figure 1: peak voltage of the driven AC power source 3.0 V, inductance 6.8 mH, capacitance 3,900 pF, and resistance 1,000. Ω . These will be the theoretical values for the components you will measure in the lab.

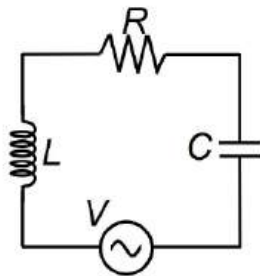


Figure 1 RLC Circuit Diagram.

1. Calculate the resonant frequency in units of Hz.

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{1}{LC}}$$

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{1}{(6.8 \times 10^{-3} \text{ H})(3,900 \times 10^{-12} \text{ F})}}$$

$$f_0 = 31 \text{ kHz}$$

2. Open the Excel spreadsheet from Lab 9's Canvas page. It has the following terms in each column: f , V_{rms} , X_C , X_L , R , Z , and theoretical I_{rms} . It also has the following values: peak voltage of the driven AC power source 5.0 V, inductance 6.8 mH, capacitance 3,900 pF, and resistance 1,000. Ω . Notice that the capacitive reactance is already set up. Use that column as an example and update the remaining columns based on their respective equations for each term as a function of frequency.

Measurement Table

APPENDIX B

Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
Resistance [Ohm]	(±) Ohm	% (2 sig fig)	TENMA Multimeter
	Measurement Procedure: Detailed description of how they measured the resistance of the resistor. Must be detailed enough that the process could be repeated using only this description and obtain a similar result. 1		
	Justification of Uncertainty: Must name one of the four factors and <u>logically explain why</u> that is the largest contributor. Most likely <u>quality of measuring device</u> . Must also include a logical <u>value</u> that must match the absolute uncertainty in the confidence interval. 1		
Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
Capacitance [F]	(±) F	% (2 sig fig)	TENMA Multimeter
	Measurement Procedure: Detailed description of how they measured the capacitance of the capacitor. Must be detailed enough that the process could be repeated using only this description and obtain a similar result. 1		
	Justification of Uncertainty: Must name one of the four factors and <u>logically explain why</u> that is the largest contributor. Most likely <u>quality of measuring device</u> . Must also include a logical <u>value</u> that must match the absolute uncertainty in the confidence interval. 1		

①.5 C.I formatted correctly ①.5 2 sig figs

Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
Inductance [Henry]	(±) H	% (2 sig fig)	TENMA Multimeter
	Measurement Procedure: Detailed description of how they measured the inductance of the inductor. Must be detailed enough that the process could be repeated using only this description and obtain a similar result. 1		
	Justification of Uncertainty: Must name one of the four factors and <u>logically explain why</u> that is the largest contributor. Most likely <u>quality of measuring device</u> . Must also include a logical <u>value</u> that must match the absolute uncertainty in the confidence interval. 1		

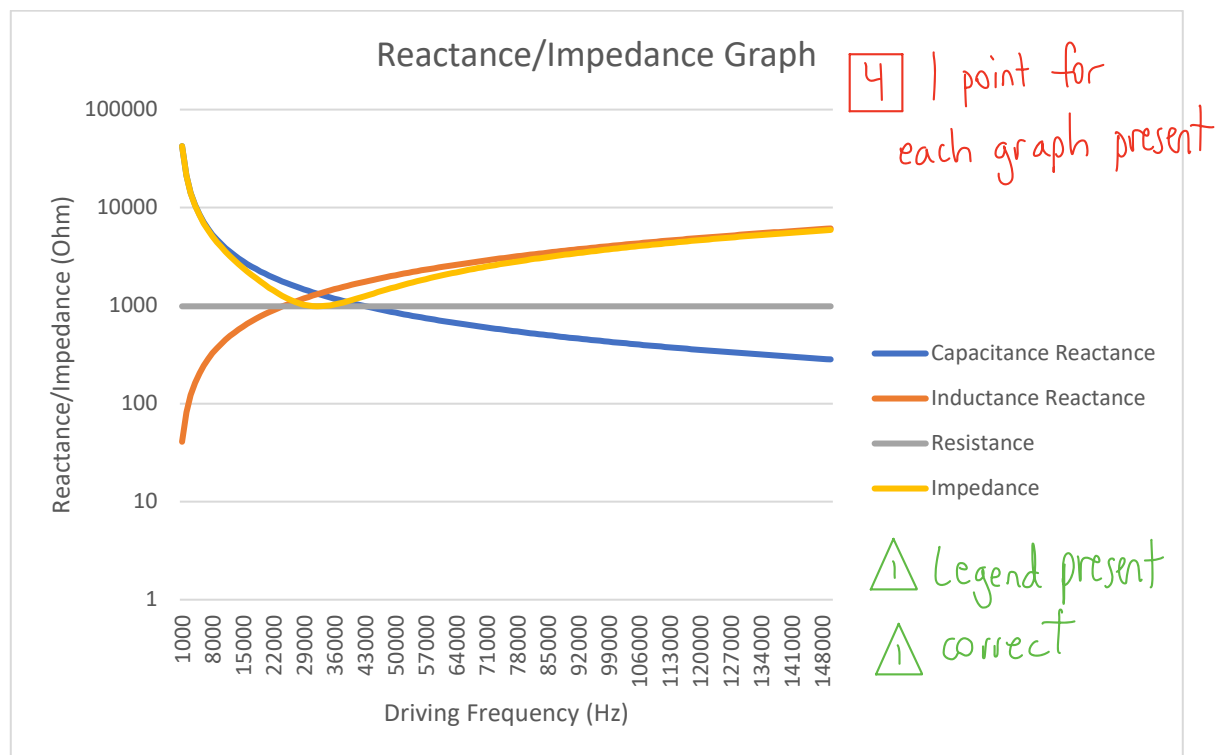
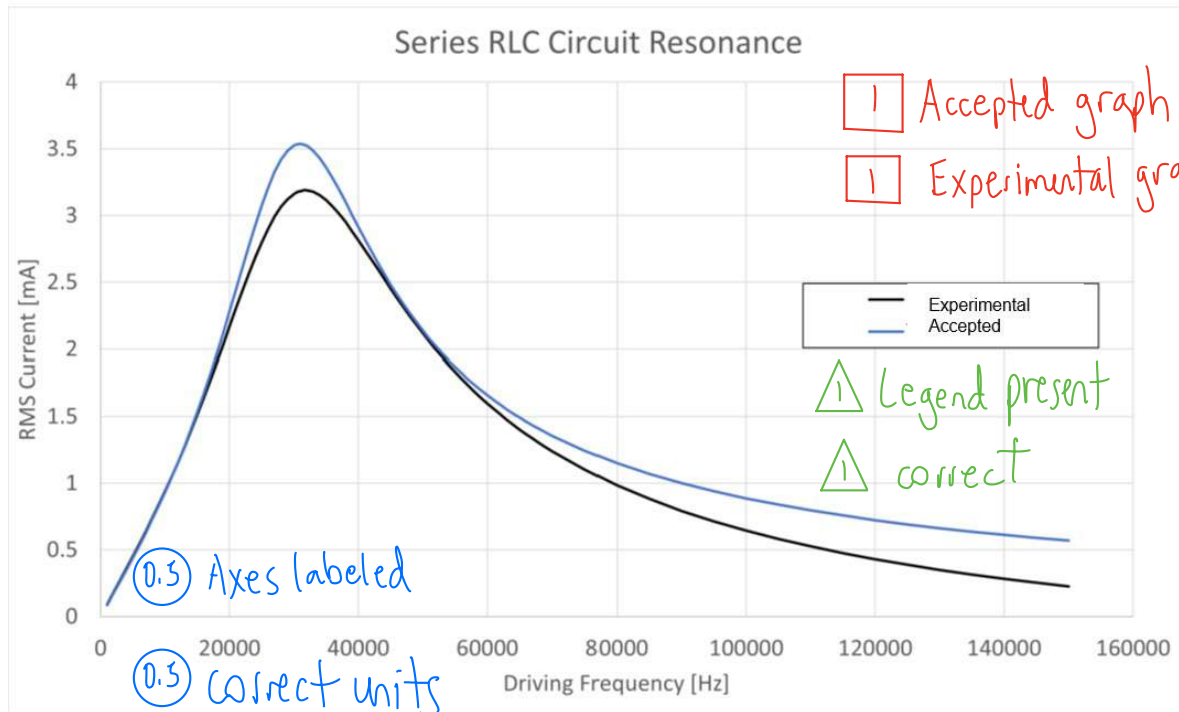
→ can use any value found (not required to be in App C.)

Measured Quantity	Confidence Interval	Relative Uncertainty	Measuring Device
rms voltage across the resistor, $V_{rms, R}$ [V]	(±) V	% (2 sig fig)	Fluke Multimeter
	Measurement Procedure: Detailed description of how they measured the voltage across the resistor. Must be detailed enough that the process could be repeated using only this description and obtain a similar result. 1		
	Justification of Uncertainty: Must name one of the four factors and <u>logically explain why</u> that is the largest contributor. Most likely <u>quality of measuring device</u> . Must also include a logical <u>value</u> that must match the absolute uncertainty in the confidence interval. 1		

4

/ 1

/ 0



Propagation of Uncertainty

APPENDIX D

Objective Quantity: f_0		Objective Formula: $f_0 = \frac{1}{2\pi} \sqrt{\frac{1}{LC}}$	
Variable	Measured Value	Absolute Uncertainty	Relative Uncertainty
1 L	0.0065 H	0.0001 H	2.0%
1 C	$3.8 \times 10^{-9} \text{ F}$	$0.1 \times 10^{-9} \text{ F}$	2.6%
Calculation of Objective Quantity's Magnitude: $f_0 = \frac{1}{2\pi} \sqrt{\frac{1}{(0.0065 \text{ H})(3.8 \times 10^{-9} \text{ F})}}$ $f_0 = 32,024 \text{ Hz} \quad \text{2 correct}$			
Calculation of Objective Quantity's Absolute Uncertainty: $\delta f = \sqrt{\left(\frac{\partial f}{\partial L} \delta L\right)^2 + \left(\frac{\partial f}{\partial C} \delta C\right)^2} \quad \text{2 Attempts intermediate method}$ $\delta f = \sqrt{\left(-\frac{1}{4\pi L^{3/2} C^{1/2}} \delta L\right)^2 + \left(-\frac{1}{4\pi L^{1/2} C^{3/2}} \delta C\right)^2}$ ① correct partial ① correct partial $\delta f = 1,000 \text{ Hz}$			
Objective Quantity Confidence Interval: $f_0 = (32,000 \pm 1,000) \text{ Hz}$			

correct variables

correct sig figs/
① units
① for values and uncertainties

① correct (reasonable and CI formatted correctly)
① units/sig figs

Experimental Value CI: $f_o = (30,800 \pm 100) \text{ Hz}$	Accepted Value CI (given): $f_o = (32,000 \pm 1,000) \text{ Hz}$
Number Line: ☐ Exp. value plot present ☐ Acc. value plot present physical quantity: f_o units: kHz Experimental value ① Formatted correctly (numbers/physical quantity/units)	
Dominant Form of Error Present (Random or Systematic): Systematic ☐ follows from numberline	
Acceptable Level of Precision (given): 1.0 %	Acceptable Level of Accuracy (given): 1.0%
Calculation of Relative Uncertainty: $\frac{\Delta f_o}{f_o} = \frac{100 \text{ Hz}}{30,800 \text{ Hz}}$ ① correct equation $RU = 0.32\%$ △ 0.5 2 sig figs	Calculation of Percent Difference: $\% \text{ diff} = \frac{ \text{exp. value} - \text{acc. value} }{ \text{acc. value} } \times 100$ $\% \text{ diff} = \frac{ 30,800 \text{ Hz} - 32,000 \text{ Hz} }{ 32,000 \text{ Hz} } \times 100$ ① correct equation $\% \text{ diff} = 3.8\%$ △ 0.5 2 sig figs

3/3/1

Discussion Questions

1). When the rms current is at a maximum, what is the relationship between the capacitive reactance and the inductive reactance, and how is the overall impedance affected? 2 relationship

At I_{rms} maximum, X_c and X_L will equal each other, and the circuit is in resonance. The overall impedance of the circuit will equal the equivalent resistance in the circuit: $Z = \sqrt{R^2 + (X_L - X_c)^2}$, where $Z = R$, when X_L and X_c are equal. 2 impedance affected

2). Why is it possible to measure the rms voltage across the resistor and still analyze rms current through the whole circuit?

Using rms voltage and current values and impedance, we know for an RLC Series circuit with alternating current, Ohm's Law states, $V_{rms} = I_{rms} Z$. When this RLC circuit is in resonance, $Z = R$, and $V_{rms} = I_{rms} R$, which is the rms voltage across the resistor and gives us rms current throughout the circuit. 2

3). Compare the accepted and experimental maximum rms current values and discuss any differences. What are the contributing factors to these differences?

The experiment max. rms current value appears to be lower than the accepted value. This could be caused by additional resistance in the inductor or in the wires used that was unaccounted for and thus lowered the exp. rms current. 2



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LAB#9 HANDOUT

Lab Worksheet

I. Lab Overview

In class you learned about refraction, where light travels through a surface that separates two different media. Refraction usually changes the light's direction of travel. In this lab, you will be using a light source and an unknown material to investigate how the index of refraction changes through different mediums.

Primary Objective: Use Snell's law of refraction to determine the index of refraction of an unknown material by varying the angle of incidence and measuring the angle of refraction.

II. Setup

For the experiment, you will have available a cylindrical lens, a semi-collimated light source, a PASCO basic optics ray table with angle markings, and an optics rail system to install your components. See Figure 9.1 for the equipment and setup.



Figure 9.1 Experimental Setup

- Adjust the slit mask on the front of the light source so one white, light ray shines across the across the top of the ray table. Set up the cylindrical lens so that light enters the flat surface of the lens at normal incidence.
- In this lab you will vary the incident angle θ_1 , and measure the refracted angle θ_2 . Ensure the flat surface of the lens is exactly centered and on the 90-degree line. Ensure the incident beam begins at the 0-degree line.
- Repeat the experiment seven times by turning the ray table so that light is incident upon its flat face, where the incident angle θ_1 occurs in the cylindrical lens, and the refracted angle θ_2 occur in the air, as shown in Figure 9.2.

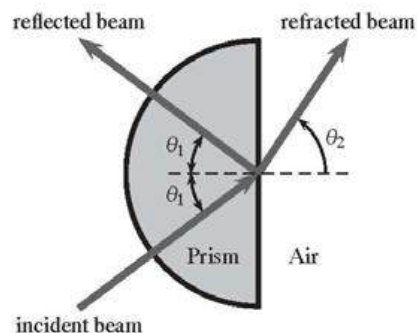


Figure 9.2 Overhead view of Lab Setup.

- You may have noticed that you could no longer gather data if you passed a certain angle. This is known as the critical angle θ_c and after you pass it, there is no refracted ray, and total internal reflection occurs. Record the critical angle.

III. Submission Requirements

- Explain** why the light ray does not refract upon entering or exiting the curved surface of the prism in the theory section.
- Clearly **explain** how the speed of light varies in different materials and **explain** Snell's Law of Refraction in the theory section.
- Explain** what total internal reflection means and why it occurs in this experiment and state the critical angle you found. Include this in the theory section.
- Compare** the accepted value of $n = 1.45 \pm 0.03$ to your experimental value for the index of refraction for the cylindrical lens.
- Complete a full lab report for this lab to include Appendices A-E to your report. Submit as a single PDF and label "Lab 9_X Hour_Last Names."

As a checklist, your complete lab report should include the following:

- Complete Laboratory Report
 - Cover Sheet (Signed)
 - Introduction
 - Theory
 - Procedure
 - Results

- Discussion
- Conclusion
- Works Cited Page and/or Acknowledgement of Assistance Page (in accordance with Encl. 5-Academic Integrity Brief)
- Appendices
 - Appendix A – Linearization Table
 - Appendix B – Measurement Tables
 - Appendix C – Excel table with plot and linear regression
 - Appendix D – Propagation of Uncertainty Table
 - Appendix E – Precision and Accuracy Table



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LAB#9 SOLUTION

	14.36
18	25.76
42	606

1. Appendices (5 Points)

- Linearization Table (App. A).
- Measurement Tables (App. B).
- Linear Regression (App. C).
- Propagation of Uncertainty (App. D).
- Error Analysis (App. E).

5 all present & complete

2. Lab Report (55 Points)**I. Introduction**

- Provides a section dedicated to the introduction. 1
- States the objective of the experiment is to determine the index of refraction of the prism. 3
- Provides a succinct summary on how the objective was accomplished (varied the angle of incidence and measured the angle of refraction). 1

II. Theory

- Provides a section dedicated to the theory. 1
- Addresses Snell's Law and index of refraction and explains how light varies in different materials. 3
- Explains total internal reflection and what the critical angle is. 2
- Assumptions made are valid and reasonable. 1
- States model equation and explains how it was derived. 1
- States independent and dependent variables. 2
- States linearized model equation and resulting independent and dependent quantities. 2
- States objective quantity and control variables. 2
- States equation for slope. 1
- States equation for objective quantity. 
- States method of analysis (linearization) 1

III. Procedures

- Provides a section dedicated to procedures. 1
- Explains the experimental set up/refers to a picture or diagram provided. 3
- States physical quantities being measured and methods to obtain them. 2
- Presents uncertainty of measured quantities, justification of uncertainties, and how they were determined. 2

IV. Results

- a. Provides section dedicated to results. 1
- b. Presents all measured data to include, independent and dependent variables and control variables or physical constants in a logical manner with correct formatting of confidence intervals. 3
- c. Presents plot of independent quantity vs. dependent quantity and reports slope and its uncertainty 2
- d. Shows or references App. D the objective quantity calculation. 2
- e. Shows or references App. D the calculation of the absolute uncertainty of the objective quantity. 2
- f. Values for objective quantity and its absolute uncertainty are reasonable. 1
- g. States correctly formatted final confidence interval for objective quantity. 1

V. Discussion

- a. Provides a section dedicated to discussion. 1
- b. States confidence intervals for experimental and accepted values. 2
- c. Presents a correctly formatted number line. 3
- d. States the relative uncertainty of the experimental value and compares it to the acceptable level of precision. 1
- e. States the percent difference of the experimental value and compares it to the acceptable level of accuracy. 1
- f. States dominant form of error and matches number line. 2
- g. Presents and discusses logical evidence that led to types of error. Largest quadrature term in calculation of objective quantity's absolute uncertainty for random error and high/low chart for systematic error. 1

VI. Conclusion

- a. Provides a section dedicated to the conclusion. 1
- b. Assesses whether experimental objective was met. 2
- c. Concisely communicates lessons learned from the experimental procedure. 1
- d. Provides details of how experiment could be done differently from the lessons learned. 1

1 Overall writing style, presentation, + formatting
(up to instructor)



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LAB#10 HANDOUT

1. Task

The purpose of this laboratory is for lab groups to brief on a topic of military relevance supported by Physics covered in PH2X1 or PH2X2. Cadets must provide a 5-7 minute presentation and be prepared to answer questions during a 2-4 minute Q&A session immediately following the brief.

2. Topics

Cadets will choose one of the following topics to present to the class:

- Rocket Propulsion (Conservation of Momentum)
- Bullet Tumbling (Kinematics and Newton's Laws)
- Helicopter Flight (Rigid Body Dynamics, Newton's Laws)
- Satellite Orbits (Newton's Laws)
- Signal Jamming (Superposition & Interference)
- DC Motors (Forces and Torque on a Current Carrying-Wire)
- Rail Gun (Faraday's Law)
- Low Pass/High Pass Filters (AC Circuits)
- Optics system (Mirrors and Lenses)
- Fiber Optics Communications (Reflection & Refraction)
- Satellite Communications (E&M Waves)
- Field Antennas (E&M Waves)
- Anti-reflection Coating (Thin Film Interference)
- Satellite Imaging Resolution (Diffraction)
- High-energy Lasers (E&M Waves & Lasers)

If Cadets wish to present another topic not listed, Instructor permission is required.

3. Format

At a minimum, the presentation needs to have the following:

- a. Agenda – List of items to be discussed in the presentation.
- b. Introduction – Present your topic and a summary of your presentation.
- c. Historical Background – Explain your topic with historical facts and how it works.
- d. Theory – Explain the Physics behind your topic with definitions and their respective equations. Use diagrams and graphs when necessary.
- e. Military Application – Explain how your topic is currently being used within the Armed Forces community.
- f. Conclusion – Explain why this topic is relevant for future leaders in the U.S. Army and how your understanding allows for better communication with your Soldiers.
- g. Works Cited – Per Encl. 5, Academic Integrity Brief, AY 25-1, Core Physics.

4. Presentation Rubric (40 Marks)

Performance Level	Excellent	Above Average	Average	Below Average	Not Adequate
Introduction	Clearly explains the purpose of the presentation.	Provides a succinct summary of the presentation.	Introduces the topic and captivates the audience. The slide is well formatted.	Introduces topic but fails to engage the audience.	Fails to communicate the topic. Slides fail to meet professional standards.
Historical Background	Clearly explains how the topic came to be with historical facts and a mastery of how it works.	Provides historical facts for the topic and an average understanding of how it works.	Slides are well formatted and with proper information but fails to communicate properly.	Provides some historical facts but fails to show the topic's historical relevance.	Fails to provide any background for the topic or how it works. Slides fail to meet professional standards.
Theory	Clearly explains the Physics behind the topic with both equations and visual tools.	Connects how equations apply to the topic and any assumptions that need to be made.	Correct explanation of appropriate Physics, but without all necessary equations.	Introduces correct Physics principles, but fails to go in depth.	Fails to communicate the correct Physics theories. Slides fail to meet professional standards.
Military Application	Clearly explains how the topic is used in the Armed Forces community with appropriate examples.	Provides information on how the topic is implemented in a military setting.	Gives examples of military application of the topic or how it is implemented.	Introduces a general overview of how the topic is applied within the military, but without examples or explanation of implementation.	Fails to present military application of the topic. Slides fail to meet professional standards.
Conclusion	Clearly explains why the topic is relevant for future leader in the U.S. Army.	Discusses how and why the information from the presentation can better inform Soldiers.	Slides are well formatted and with proper information but fails to communicate relevancy.	Provides a summary of the presentation.	Fails to provide or communicate a conclusion and relevancy for future military officers.
Works Cited (Per Encl. 5, Academic Integrity Brief, AY 24-2, Core Physics)	Citations are present in both the main slides and the Works Cited page. The citations in both correlate with each other.	-	-	-	Citations not present.



Presentation Title

Cadet Name
Cadet Name
Cadet Name

Date



- Introduction
- Historical Background
- Theory
- Military Application
- Conclusion
- Works Cited

(Note: You may add more in the agenda to add value to your presentation, but this is your baseline).



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Introduction



- Present your topic and a summary of your presentation.



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Historical Background



- Explain your topic with historical facts and how it works.



- Explain the Physics behind your topic with definitions and their respective equations. Use diagrams and graphs when necessary.



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Military Application



- Explain how your topic is currently being used within the Armed Forces community.



- Explain why this topic is relevant for future leaders in the U.S. Army and how your understanding allows for better communication with your Soldiers.



- Per Encl. 5, Academic Integrity Brief, AY 24-1, Core Physics.



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LAB#10 SOLUTION

Presentation Rubric (40 Marks)

Performance Level	Excellent	Above Average	Average	Below Average	Not Adequate
Introduction	Clearly explains the purpose of the presentation. 4	Provides a succinct summary of the presentation. 3	Introduces the topic and captivates the audience. The slide is well formatted. 2	Introduces topic but fails to engage the audience. 1	Fails to communicate the topic. Slides fail to meet professional standards. 0
Historical Background	Clearly explains how the topic came to be with historical facts and a mastery of how it works. 10	Provides historical facts for the topic and an average understanding of how it works. 8	Slides are well formatted and with proper information but fails to communicate properly. 6	Provides some historical facts but fails to show the topic's historical relevance. 4	Fails to provide any background for the topic or how it works. Slides fail to meet professional standards. 0
Theory	Clearly explains the Physics behind the topic with both equations and visual tools. 10	Connects how equations apply to the topic and any assumptions that need to be made. 8	Correct explanation of appropriate Physics, but without all necessary equations. 6	Introduces correct Physics principles, but fails to go in depth. 4	Fails to communicate the correct Physics theories. Slides fail to meet professional standards. 0
Military Application	Clearly explains how the topic is used in the Armed Forces community with appropriate examples. 10	Provides information on how the topic is implemented in a military setting. 8	Gives examples of military application of the topic or how it is implemented. 6	Introduces a general overview of how the topic is applied within the military, but without examples or explanation of implementation. 4	Fails to present military application of the topic. Slides fail to meet professional standards. 0
Conclusion	Clearly explains why the topic is relevant for future leader in the U.S. Army. 4	Discusses how and why the information from the presentation can better inform Soldiers. 3	Slides are well formatted and with proper information but fails to communicate relevancy. 2	Provides a summary of the presentation. 1	Fails to provide or communicate a conclusion and relevancy for future military officers. 0
Works Cited (Per Encl. 5, Academic Integrity Brief, AY 25-1, Core Physics)	Citations are present in both the main slides and the Works Cited page. The citations in both correlate with each other. 2	-	-	-	Citations not present. 0

4

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